Differential Topology
— *EYNTKA* Series —

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Contents

This book is an introduction to differential topology. It is part of the *Everything You Need To Know About* (EYTNKA) series, and so it will do its best to include as much fundamental material as possible. In this pre-edition, material is included when I have time to write it down.

I will assume the following prerequisites:

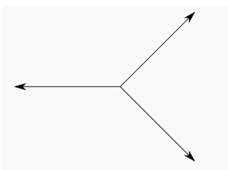
1. Elementary set theory knowledge (elements, equivalence classes, functions, etc.)
2. Knowledge of topology, including open/closed sets, continuity, connectedness and compactness, product topologies, and how these concepts work over $\mathbb{R}^n$
3. Elementary analysis knowledge such as sup and inf, lim, sequences and series, definition of differentiability and integrability in $\mathbb{R}$ and their basic properties (ex. differentiability implies continuity)
4. Elementary Category Theory (for it is very useful for intuition and seeing the big picture)

What is Geometry?

On a high-level, this book is about studying a (topological) space X satisfies the following two properties which can be considered the "definition" of being a geometric object in mathematics:

1. X locally looks like a base space B (or a collection of base spaces B_i[1]). The way we achieve this is by taking an (open) cover $\{U_i\}$ of X and map each U_i to the base space(s) B/B_i (or open subsets of the space).[2] These maps go by many names, each one presenting a different important property. For now we will call these *patch maps*.
2. The patch maps satisfies a *compatibility conditions* or *gluing conditions* on overlaps. In other words, the way in which we glue them together will determine many important properties. For example, if the glue is homeomorphic, then continuous functions are preserved. If the graph of the function is closed, then the underlying space is Hausdroff. If the gluing functions are diffeomorphic, we may try to introduce some calculus. The nature of the glue will determine the nature of the object.

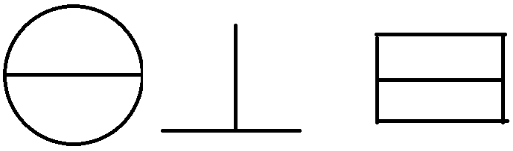
Both of these properties are important. The choice of base space and gluing condition will determine the type of object. For example, if the base space is:



and our gluing was homeomorphic, we may imagine a space that is locally the above to look like any of these:

[1]A scheme is locally Spec R for any ring R

[2]Mapping to/from X and the difference will be discussed later.



Other types of base spaces that are common are the space of solutions to polynomials (known as *varieties*) and euclidean space $\mathbb{R}^n$ or $\mathbb{C}^n$. The general study of a space E that is locally B with some regularity (smooth, continuous, etc.) is called *sheaf theory* (cover in [6]). This book will focus on the case where the base space is $\mathbb{R}^n$, and the gluing map will:

1. Be a homeomorphism or a diffeomorphism if we are defining a topological or differentiable manifold
2. be a "linear isomorphism on fibers" if we are defining a vector bundle (chapter 11 gives a precise definition)

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Different choices of gluing maps and base space will produce different fields of study (for example, the base space of solutions to polynomials[3] is the beginning of defining *schemes* a field known as *algebraic geometry*). The case of the base space being $\mathbb{R}^n$ already has plenty to explore that will fill an entire course and more, and will produce many of the shapes we are familiar with. It will also be the basis of many of our intuitions for the study of many more general theories, including in complex analytic geometry and algebraic geometry; n many ways, modern algebraic geometry starts with translating the intuition from differential geometry into algebra.

Geometric Intuition

From a classical perspective, this book is about the study of shapes and space (in a "geometric sense"). The first "space" to be well-known is that of euclidean space, named after Euclid who has postulated the rules in which shapes should reside in. For euclid, this space was inherently 3-dimensional with no particular special point (i.e. no origin), however we now have a euclidean space for every dimension, and for reference we have a point at the origin. Sometimes, euclidean space is called *euclidean affine space* to make the distinction that we have no preference for any particular point like the origin, but regard it as simply a space with the properties Euclid described.

Euclidean spaces, i.e. $\mathbb{R}^n$ for all $n \in \mathbb{N}$, have a great versatility of properties: it is the setting in which we can embed all our classical notion of euclidean/affine geometry, as well as all the geometrical shapes mathematicians have worked with since antiquity (ex. circles, regular n-ones, lines, planes, varieties, etc.). The name "euclidean" comes from the fact that the 5 axioms postulated for his theory of [now called classical] geometry (importantly, including the 5th postulate) are equivalent to the vector-space definition of $\mathbb{R}^n$ if we choose some point in euclidean space to be the origin and affix to it coordinates, i.e. a tuple of functions codify each point. While Euclid did not describe his shapes in terms of a tuple functions giving points for each part of the shape, this view has now come to

[3] really, affine schemes

dominate how shapes are perceived (ex. a circle is usually described as the set of points in $\mathbb{R}^2$ such that $x^2 + y^2 = 1$, instead of the set of points that are equidistant to a single point). If $\mathbb{A}^n$ represents affine euclidean space (i.e., no coordinates) and $\mathbb{R}^n$ represents euclidean space with coordinates, then we can write:

$$\mathbb{A}^n \mapsto \mathbb{R}^n \qquad P \mapsto (x_1(P), x_2(P), ..., x_n(P))$$

This process is called *coordinization*. There is no unique way to put coordinates on euclidean space. For example, on $\mathbb{A}^2$, we can also put the coordinates $P \mapsto (r(P), \theta(P))$ for $(r(P), \theta(P)) \in (0, \infty) \times (-\pi, \pi)$ and choose one point to map to 0. If we represent $\mathbb{A}^2$ as the standard euclidean $\mathbb{R}^2$, then

$$x = r\cos(\theta) \qquad y = r\sin(\theta)$$

Importantly these two spaces, though coordinitized differently, are still the same intrinsically, they are simply described differently. Part of the goal of differential topology is to describe the properties of shapes in a coordinate free way, and to find what properties can distinguish shapes independent of their coordinate representation.

Another more general goal is the study of shapes that "look like" $\mathbb{R}^n$ if zooming in enough. This will be the beginning of studying new "types" of spaces with properties different than $\mathbb{R}^n$. For example, if you zoom in at any point of a circle, it will start looking like a line. If you zoom in at a corner point of a triangle, it will always look like a mountain, and so never like $\mathbb{R}^1$. More generally, we will want to know how to zoom into a part of space and make it looks like a part of $\mathbb{R}^n$. The goal is to take euclidean properties and tools and try to apply them to these shapes. The nature in which we make sure a space is "locally euclidean" or "locally something" will determine the type of manifold we define. Usually, we will need the space to be diffeomorphic to an open subset of $\mathbb{R}^n$, since many results require smoothness (namely, to be locally linear), however we also sometimes care about weaker conditions such as a homeomorphism, or in the weakest case a bijection.

Categorical Intuition

We can think of the study of manifolds as the (co)completion of the right categories. For example, take the category whose objects are open intervals and whose morphisms are translations. Take the objects $(0, 4)$, $(0, 1)$ and $(3, 4)$ and have the morphism $(0, 1) \mapsto (3, 4)$ via translation. Consider the commutative diagram

$$\begin{tikzcd} & (0,4) & \\ (0,1) \arrow[ur, "\mathrm{id}"] \arrow[rr, "T"] & & (3,4) \arrow[ul, "\mathrm{id}"'] \end{tikzcd}$$

We may ask if this diagram has a colimit in this category. Without going through the formalities, the answer to this is the circle. However, the objects of this category only has intervals, and so the colimit does not exist! Hence, we will complete this category by adding all the colimits. This idea of completing a category will eventually lead us to having locally euclidean space. Naturally, we can replace the base space with anything, for example three lines joined at a point, or $\mathbb{C}$, or solutions to polynomials, and so forth. In general, this idea of completion and thinking of this as looking at things *locally* is important. Note that the fact that the map that we used to glue $(0, 1)$ and $(3, 4)$ together was a translation. Translations are diffeomorphisms that preserve linear maps, and hence linear maps are preserved by quotienting. It also preserves the distance structure too. However, if we have a general diffeomorphism, this need no be the case. If the map was just a homeomorphism, we only preserve the topological structure, i.e. continuous maps, but are not guaranteed to preserve differentiable or smooth maps on intervals.

A quick review

In this preliminary chapter, we quickly go over elementary knowledge of differentation and integration. Some notions from Topology that may be uncommon in a usual topology shall class shall quickly be defined.

0.1 Topology

I am assuming the reader is familiar with Topology, compactness, connectedness, component, local connectedness, in $\mathbb{R}^n$, secound-countable/seperable.

Definition 0.1.1: Refinement

Let $\mathcal{O}$ be an open cover for U. Then a *refinement* of $\mathcal{O}$ is another open cover $\mathcal{V}$ such that for every set $V \in \mathcal{V}$, there is a set $U \in \mathcal{O}$ such that $V \subseteq U$. if $\mathcal{V}$ is a refinement of $\mathcal{O}$, we say that $\mathcal{V}$ refines $\mathcal{O}$.
If B indexes the sets of $\mathcal{V}$ and A indexes the sets of $\mathcal{O}$, we will say there is a *refiniement map* $\varphi : \mathcal{V} \to \mathcal{O}$ satisfying $V_\beta \subseteq U_{\varphi(\beta)}$

Note that every subcover is trivially a refinement, but not all refinements are open covers.

Definition 0.1.2: Locally Finite Set

Let $\mathcal{O}$ be an open cover for a set U. Then $\mathcal{O}$ is said to be locally finite if for any point $x \in U$, only finitely many sets of $\mathcal{O}$ contain x.

Definition 0.1.3: Paracompact

Let X be a topological space. Then X is said to be *paracompact* if for any open cover $\mathcal{O}$ of X, there exists a refinement that is locally finite.

Definition 0.1.4: Component

Let X be a topological space. Then the component containing x, denoted $C(x)$ is defined as the union of all connected subsets of X containing x. A subset of a topological space X is called a [connected] component if it is equal to $C(x)$ for some $x \in X$.

Definition 0.1.5: Locally Connected

A topological space X is *locally connected* if it has a basis consisting of connected sets.

Proposition 0.1.6: Equivalence Of Paracompact

Let X be a topological space. Then the following are equivalent:

1. Every component of X is σ-compact (countable union of compact sets)
2. Every component is 2nd countable
3. X is metrizable (using Urysohn's Xetrization theorem)
4. X is paracompact
5. There exists a C^0 partition of unity subordinate to any open cover

***Proof*:**
These equivalences are point-set topology and are proved in *Topology* [5], whose paracompactness chapter establishes the implications among second countability, local compactness, metrizability and paracompactness, together with the equivalence of paracompactness with the existence of subordinate C^0 partitions of unity. They are recorded here in the combined form the manifold theory uses, as we sought to show.

(equivalence between connected and path connected components in the right situation. On a manifold, connected if and only if path connected)

(Show that there is only one norm on finite dimensional vector space)

(In general, if a function takes in a function and spits out another function, it is called an *operator*)

0.2 Differentiation

Being differentiable at a point x essentially means that if we zoom in enough at the point x, we can arbitrarily well approximate an open neighborhood of that point with some linear function, i.e., the function is locally linear. In $\mathbb{R}$, we can capture this intuition with the following: the function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a point x if

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

exists, or equivalently if we substitute $a = x + h$, we get that $h \to 0$ become $a \to x$ and

$$\lim_{a \to x} \frac{f(a) - f(x)}{a - x}$$

exists. If it exists, we usually denote the result of this limit as $f'(x)$. In higher dimensions, this formulation of differentiability doesn't quite work (for example, the composition of differentiable functions need not be differentiable). Hence, we re-formulate this the definition to be that the function f is differentiable at x if there exists a $m = f'(x)$ such that

$$\lim_{h \to 0} \frac{f(x + h) - f(x) - mh}{h} = 0$$

In this way, $f'(x)$ will in fact give us back linear maps, which in this case can always be represented as 1×1 matrices. In higher dimensions $\mathbb{R}^n$, we generalize this notion by taking the norm: a function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a point $x \in \mathbb{R}^n$ if there exists a linear transformation $A_x : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$\lim_{h \to 0} \frac{\|f(x + h) - f(x) - A_x(h)\|}{\|h\|} = 0$$

with the usual $\mathbb{R}^m$ norm in the numerator and $\mathbb{R}^n$ norm in the denominator. Recall (or re-prove) that the derivative is unique, so if two linear transformations A_x and B_x satisfy this same limit, they are in fact equal: $A_x = B_x$. If we choose the standard basis $\beta = \{e_1, ..., e_n\}$ for $\mathbb{R}^n$, then the matrix representation of any derivative at a, $Df(a)$, is will be an $m \times n$ matrix and is called the *Jacobian Matrix*. If $U \subseteq \mathbb{R}^n$ is an open set and if for every $x \in U$, f is differentiable at x, then we say that $f : U \to \mathbb{R}^m$ is a differentiable function. Usually, A_x is written as $Df(x) : \mathbb{R}^n \to \mathbb{R}^m$. If f is differentiable on U, then we can say that $Df : U \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is a map that takes in a point $x \in U \subseteq \mathbb{R}^n$ and gives a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Since $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{nm}$ as a vector-space, then it is equivalent to consider $Df : \mathbb{R}^n \to \mathbb{R}^m$, meaning we can consider the differeniability of Df. If Df is differentiable at a in $\mathbb{R}^n$, then we would write $D(Df)(a)$, or more succiently $D^2 f(a)$. If $D^2 f$ is differentiable at every $x \in \mathbb{R}^n$, then we would get a map $D^2 f : \mathbb{R}^n \to \mathbb{R}^{n^2 m}$, or in terms of linear transformation $D^2 f : \mathbb{R}^n \to \mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$. It is a useful fact to remember that $\mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$ is isomorphic to $\mathcal{L}^2(\mathbb{R}^n, \mathbb{R}^m)$ which is the set of all bilinear linear transfomrations from $\mathbb{R}^n$ to $\mathbb{R}^m$, that is all functions $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^m$ such that $f(rx, y) = f(xr, y) = rf(x, y)$ and f is linear in the first and second commponent. We can repeat this process to define $D^n f$, and if $D^n f$ exists for all $a \in \mathbb{R}^n$, then $Df^n f : \mathbb{R}^n \to \mathcal{L}^n(\mathbb{R}^n, \mathbb{R}^m)$ exists. The linear function in the codomain of DF can be "realised" as mappings between tangent spaces from U to $\mathbb{R}^m$. Recall that $\mathbb{R}^n_a$ is the vector-space centered at a. Then $Df(a)$ can be thought of having domain and codomain

$$Df(a) : \mathbb{R}^n_a \to \mathbb{R}^n_{f(a)}$$

This will become very important once we start trying to define a notion of integrating manifolds.

The composition of two differnetiable functions is differentiable: If $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^p$ are two functions such that f is differentiable at a and g is differentiale at $f(a)$, then $g \circ f(a)$ is also differentiable and

$$D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$$

given the jacobian matrix representation of these linear transofmrations, we can see this as multiplying matrices $[Dg(f(a))]_\beta \cdot [Df(a)]_\beta$. Using the chain rule, we can see that addition of two differentiable functions is differentiable (since $f + g = s(f, g)$ where $s(x, y) = x + y$ and $(f, g)(a) = (f(a), g(a))$), and $D(fg)(a) = Df(a)g(a) + f(a)Dg(a)$.

Finding the jacobian matrix $[Df(a)]_\beta$ is made easy using patial derivatives. If β is the standard basis (in particular, it is ordered), $f : \mathbb{R}^n \to \mathbb{R}$ is a function, and $a \in \mathbb{R}^n$, then f has a *ith partial derivaitve at a* if

$$\lim_{h\to 0} \frac{f(a_1, a_2, ..., a_i + h, a_{i+1}, ..., a_n)}{h}$$

exists and is denoted $D_i f(a)$ of $\frac{\partial f(a)}{\partial x_i}$. In particular, notice that this is just a function from $\mathbb{R}$ to $\mathbb{R}$ in disguise: letting $g(x) = f(a_1, a_2, ..., a_{i-1}, x, a_{i+1}, ..., a_n)$, we see that $g'(a_i) = D_i f(a)$. Hence, all standard notions and tecniques of differentiation are already available to us. As with differentiation of $f : \mathbb{R}^n \to \mathbb{R}^m$, if $D_i f(a)$ is defined for all $x \in \mathbb{R}^n$, then we have a function $D_i f : \mathbb{R}^n \to \mathbb{R}$. Hence, we can take the partial derivative of $D_i f$ in any direction: $D_j(D_i f)$, which we usually denote $D_{i,j} f$. We will soon cover how $D_{i,j} f = D_{j,i} f$ (If $D_{i,j} f$ and $D_{j,i} f$ are continuous) after we cover how we can use partial derivative information to find $[Df]_\beta$.

If Df exists, then each $D_i f$ exists. However, the converse is not true: every $D_i f$ can exist but Df might not exist. In fact, if we generalize our notion form partial derivative to directional derivative (where instead of $a_i + h$ we take $a_i + \mathbf{u}h$ for some unit vector $\mathbf{u}$), then it is possible that *every* directional derivative exists but Df still doesn't exist. However, if each $D_i f$ is *continuous*, then this in fact implies that Df exists. It would be nice if Df existing implied that each $D_i f$ was continuous, however that is not the case. Thus, since we are comfortable with working with $D_i f$, we usually don't look at differentiable functions, but the stronger class of functions with continuous partial derivatives, denoted $C^1(U, \mathbb{R}^n)$. If $n = 1$, we usually denote this as $C^1(U)$. If the partial derivatives are r-times differentiable with the partial of $D^r f$ being continuous, we denote this as $C^r(U, \mathbb{R}^n)$. If f is in C^r for all $r \geq 0$, then we say that f is *smooth* and is in C^∞. Furthermore, if f is bijective (hence invertible), then if f^{-1} is also smooth, we say that f is a *diffeomorphism*. Theorem 0.2.1 shows that if f is C^∞ and invertible, then f^{-1} is automatically C^∞, giving us a simpler criterion for finding the "isomorphisms" in the appropriate category (which we'll define soon).

One important trick to remember is that:

$$D_x f(p) = \left.\frac{d}{dt}\right|_{t=0} f(p + tx)$$

With partial derivatives defined, we can finally find $[Df(a)]_\beta$ using them. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is a functions where $f = (f^1, ..., f^m)$ where $f^i : \mathbb{R}^n \to \mathbb{R}$, then

$$[Df(a)]_\beta = \begin{pmatrix} D_1 f^1(a) & D_2 f^1(a) & \cdots & D_n f^1(a) \\ D_1 f^2(a) & D_2 f^2(a) & \cdots & D_n f^2(a) \\ \vdots & \vdots & \ddots & \vdots \\ D_1 f^m(a) & D_2 f^m(a) & \cdots & D_n f^m(a) \end{pmatrix}$$

i.e., if $M = [Df(a)]_\beta$, then $M_{ij} = D_j f^i$. We see that this matrix is indeed the derivative by first taking $m = 1$, then taking $h(x) = (a_1, ..., x, ..., a_n)$ with x in the jth place, and then noticing that

$D_j f(a) = (f \circ h)'(a_j)$ so that

$$\begin{aligned}(f \circ h)'(a_j) &= f'(a) \cdot h'(a_j) \\ &= f'(a) \cdot \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}\end{aligned}$$

We end this review with a reminder of the differentiability of f^{-1} whenever it is defined for a differentiable function f and a theorem which let's us convert "relations" into functions, in particular it is used to easily go back and forth between different coordinates:

Theorem 0.2.1: Inverse Function Theorem

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a C^1 function. $a \in U$ and $\det Df(a) \neq 0$. Then there exists an open set $V \subseteq \mathbb{R}^n$ and $W \subseteq \mathbb{R}^n$ where $a \in V$, $f(a) \in W$, such that $f : V \to W$ has a differentiable inverse $f^{-1} : W \to V$, and for every $y \in W$:

$$D(f^{-1})(y) = [Df(f^{-1}(y))]^{-1}$$

One of the great results of the inverse function theorem is that it suffices that f is C^∞ and invertible to be a diffeomorphism. First, we prove a lemma:

Lemma 0.2.2: Bounding On Closd Rectangles

Let $A \subseteq \mathbb{R}^n$ be a closed rectangle and let $f : A \to \mathbb{R}^n$ be continuously differentiable. If M exists such that for all $x \in \operatorname{int} A$, $|D_i f^i(x)| \leq M$, then:

$$|f(x) - f(y)| \leq n^2 M |x - y|$$

for all $x, y \in A$

***Proof*:**
We first create a tellescoping sum:

$$|f^i(y) - f^i(x)| = \sum_{j=1}^{n} f(y^1, y^2, ..., y^j, x^{j+1}, ..., x^n) - f(y^1, y^2, ..., y^{j-1}, x^j, ..., x^n)$$

Applying the man value to each term on the right hand side, we get:

$$\begin{aligned} f(y^1, y^2, ..., y^j, x^{j+1}, ..., x^n) - f(y^1, y^2, ..., y^{j-1}, x^j, ..., x^n) &= (y^j - x^j) D_j f^i(z_{ij}) \\ &\leq |y^j - x^j| M \end{aligned}$$

for appropraite z_{ij}, thus since $|y^i - x^i| \leq |x - y|$ by the reverse triangle inequality, $|f^i(y) - f^i(x)| \leq nM|y - x|$. Finally, with some norm manipulation and the triangle inequality, we get:

$$|f(y) - f(x)| \leq \sum_{i=1}^{n} |f^i(y) - f^i(x)| \leq n^2 M|y - x$$

completing the proof

We know proceed with the proof for the Inverse Function Theorem:

***Proof*:**
We will start by showing that an inverse exists, which will use the fact that $\det Df(a) \neq 0$ to find a neigbhorhood around a that is invertible. Showing that it's continuous will almost immediately come out of the proof of bijectivity. Then,

For notational simplicity, let λ be the linear transformation $Df(a)$. Then λ is non-singular since $\det Df(a) \neq 0$. Now, consider

$$D(\lambda^{-1} \circ f)(a) = D(\lambda^{-1})(f(a)) \circ Df(a) = \lambda^{-1} \circ Df(a) = \mathrm{id}$$

Notice that if we prove the theorem for $\mathrm{id} = \lambda^{-1} \circ f$, it is certainly true for f (since λ is non-singular, it would be the composition of two invertible functions). Thus, without loss of generality, we may assume λ is the identity.

This will now force injectivity on a local level. Let's say $f(a + h) = f(a)$. Then:

$$\frac{|f(a + h) - f(a) - \lambda(h)}{|h|} = \frac{|h|}{|h|} = 1$$

However,

$$\lim_{h \to 0} \frac{|f(a + h) - f(a) - \lambda(h)}{|h|} = \frac{|h|}{|h|} = 0$$

Thus, there must exist a clsoed rectangle, U, containing a in its interior such that $f(x) \neq f(a)$, $x \in U$ when $x \in A$. Since det is a continuous function and f is continuously differentiable, we can also assume that:

$$\det Df(x) \neq 0 \quad x \in U \tag{1}$$

$$|D_j f^i(x) - D_j f^i(a)| \leq \frac{1}{2n^2} \quad \text{for all } i, j \text{ and } x \in U \tag{2}$$

The partiular choice of bound on the partial derivaties is to employ the following trick; we see to make sure that f grows faster than the distance between two points so that we can prove that we can find an inverse. To that end, we made the choice in the 2nd bullent point so that if we combine the last point with lemma 0.2.2 and apply it to $g(x) = f(x) - x$, we get:

$$|f(x_1) - x_1 - (f(x_2) - x_2)| \leq \frac{1}{2}|x_1 - x_2|$$

Using the reverse triangle inequality, we get:

$$|x_1 - x_2| + |f(x_1) - f(x - 2)| \leq |f(x_1) - x_1 - (f(x_2) - x_2)| \leq \frac{1}{2}|x_1 - x_2|$$

Thus we also get a bound of:

$$|x_1 - x_2| \leq 2|f(x_1) - f(x_2)| \tag{3}$$

for all $x_1, x_2 \in U$, showing us that f must grow faster than the linear distance between two points.

Now, since ∂U is compact, so is $f(\partial U)$. By the the fact that $f(x) \neq f(a)$ for all $a \neq x \in U$, $f(a) \notin f(\partial U)$. Therefore, there exists a $d > 0$ such that $|f(x) - f(a)| \geq d$ for all $x \in \partial U$. Let $W = \{y \mid |y - f(a)| \geq d/2\}$. Note that for any $y \in W$, the distnce between y and $f(x)$ where $xin\partial U$ must always be greater than the distance between y and $f(a)$:

$$|y - f(a)| < |y - f(x)| \tag{4}$$

With this, we can show that for all $y \in W$, there exists a unique $t \in \int(U)$ such that $f(t) = Y$ (showing the existence of an inverse on W). To show existence, we will take advantage of the determinant information we established in equation (1) and (2). In particular, define $g : U \to \mathbb{R}$,

$$g(x) = |y - f(x)^2|^2 = \sum_{i=1}^{n} (y^i - f^i(x))^2$$

Since g is continuous on U and U is bounded, it's minimum must be achieved somehwere. Let's say the minimum as at t. Since g is differentiable, we can find the minimum using the derivative. We know that the minimum of g is achieved. Importantely, by equation (4), $g(a) < g(x)$ for all $x \in \partial U$, so the minimum is not achieved on the border (which will be importance for f^{-1} being differentiable). Then:

$$\sum_{i=1}^{n} 2(y^1 - f^i(t)) \cdot D_i f^i(t) = 0$$

however, $\det Df(x) \neq= 0$ for all $x \in \int U$! Therefore, the only possibility left for making this equation is if $y^i - f^i(t) = 0$ for all i, i.e. $y = f(t)$. For uniqueness, if two points $x_1 \neq x_2$ had $f(x_1) = f(x_2)$, then this would violate equation (3).

Thus, let $V = \int U \cap f^{-1}(W)$, so that $f : V \to W$ has inverse $f^{-1} : W \to V$. To show that f^{-1} is continuous, notice that equation (3) also implies:

$$|f^{-1}(y_1) - f^{-1}(y_2)| \leq 2|x_1 - x_2| \tag{5}$$

which immediately implies continuity of f^{-1}

Finally, we show that f^{-1} is differentiable., in particular if $\mu = Df(x)$ ($x \in V$), then $D(f^{-1})(x) = \mu^{-1}$. Similarly to how we proved the chain rule, let

$$f(x_1) = f(x) + \mu(x_1 - x) + \varphi(x_1 - x)$$

where the error term φ tends to zero faster than linearly

$$\lim_{x_1 \to x} \frac{|\varphi(x_1 - x)|}{x_1 - x} = 0$$

Manipulating the equation around, we get the equivalent equation of:

$$\mu^{-1}(f(x_1) - f(x)) = x_1 - x + \mu^{-1}(\varphi(x_1 - x))$$

$$x_1 = x + \mu^{-1}(f(x_1) - f(x)) - \mu^{-1}(\varphi(x_1 - x))$$

Since f is invertible, we can re-write $x_1 = f^{-1}(y_1)$ and $x = f^{-1}(y)$, and re-shuffle to get:

$$f^{-1}(y_1) = f^{-1}(y) + \mu^{-1}(y_1 - y) - \mu^{-1}(\varphi(f^{-1}(x_1) - f^{-1}(x)))$$

and so it suffices to show that the error term goes to zero faster than linearly:

$$\lim_{y_1 \to y} \frac{|\mu^{-1}(\varphi(f^{-1}(y_1) - f^{-1}(y)))|}{y_1 - y} = 0$$

Using the operator norm on μ^{-1}, we get that if this limit exists, then

$$\begin{aligned} \lim_{y_1 \to y} \frac{|\mu^{-1}(\varphi(f^{-1}(y_1) - f^{-1}(y)))|}{y_1 - y} &\leq \lim_{y_1 \to y} \frac{M|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y} \\ &= M \lim_{y_1 \to y} \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y} \end{aligned}$$

which will go to zero if and only if $\lim_{y_1 \to y} \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y}$ goes to zero. With this, we can do the same trick as we've done with the chain-rule:

$$\begin{aligned} &\frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{y_1 - y} \\ &= \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{f^{-1}(y_1) - f^{-1}(y)} \cdot \frac{|f^{-1}(y_1) - f^{-1}(y)|}{y_1 - y} \\ &\leq \frac{|\varphi(f^{-1}(y_1) - f^{-1}(y))|}{f^{-1}(y_1) - f^{-1}(y)} \cdot 2 && \text{by equation(5)} \end{aligned}$$

Since f^{-1} is continuous, as $y_1 \to y$, $f^{-1}(y_1) \to f^{-1}(y)$, in particular limits are preserved when taking composing with continuous functions, and hence the first factor goes to 0, and so the whole limit goes to zero. Since this is true for any $y \in W$, f^{-1} is indeed differentiable with derivative μ^{-1}, as we sought to show.

Note that it is possible that f^{-1} exists even if $\det Df(a) = 0$, (ex. x^3), howver f^{-1} cannot be differentiable at the point where $\det Df(a) = 0$

The next theorem is useful in identifying when we can write a set of equations in terms of different parameters. In particular, if we have a function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$, and take the collection of points where $F(x^1, ..., x^n, y^1, ..., y^m) = 0$ (i.e. $F^{-1}(0)$), then under appropriate conditions, there exists an $f : \mathbb{R}^n \to \mathbb{R}^m$ such that $F(x^1, ..., x^n, f(x^1, ..., x^n)) = 0$

This theorem is particularly useful if you recall that a manifold can be defned as the zero locus of a C^1 function with Df having constant rank.

Theorem 0.2.3: Implicit Function Theorem

Let $f : U \subseteq \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be a continuously differentiable function and $(a, b) \in U$, $f(a, b) = 0$. Let M be an $m \times m$ matrix such that

$$M_{ij} = D_{n+j} f^i(a, b) \qquad 1 \leq i, j \leq m$$

If $\det M \neq 0$, then there exists an open set $A \subseteq \mathbb{R}^n$ containing a and an open set $B \subseteq \mathbb{R}^m$ containing b such that for each $x \in A$, there is a unique $g(x) \in B$ such that $f(x, g(x)) = 0$. Furthermore, the function g is differentiable.

If you are visual, then M is

$$\begin{pmatrix} D_{n+1}f^1(a,b) & D_{n+2}f^1(a,b) & \cdots & D_{n+m}f^1(a,b) \\ D_{n+1}f^2(a,b) & D_{n+2}f^2(a,b) & \cdots & D_{n+m}f^2(a,b) \\ \vdots & \vdots & \ddots & \vdots \\ D_{n+1}f^m(a,b) & D_{n+2}f^m(a,b) & \cdots & D_{n+m}f^m(a,b) \end{pmatrix}$$

***Proof*:**
We will be using the Inverse Function theorem to prove this result. Let $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$, $F(x, y) = (x, f(x, y))$. Since $\det M \neq 0$, then by construction of F, if $0 = f(a, b)$ $\det DF(a, b) = \det M \neq 0$. Thus, by the Inverse Function theroem, there exists an $W \subseteq \mathbb{R}^n \times \mathbb{R}^m$ containing $F(a, b) = (a, 0)$ and an open set $V \subseteq \mathbb{R}^n \times \mathbb{R}^m$ containing (a, b), which we see can be easily written as $A \times B$, such that $F : A \times B \to W$ has a differentiable inverse function $F^{-1} : W \to A \times B$. It should be clear that by the form of F, $F^{-1}(x, y) = (x, k(x, y))$ for appropriate differentiable function k.

With this function defined, letting $\pi : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ be the natural projection function $\pi(x, y) = y$, we get the following chain of equalities:

$$\begin{aligned} f(x, k(x, y)) &= f \circ F^{-1}(x, y) \\ &= ((\pi \circ F) \circ F^{-1})(x, y) \\ &= (\pi \circ (F \circ F^{-1}))(x, y) \\ &= \pi(x, y) \\ &= y \end{aligned}$$

Thus, $f(x, k(x, y)) = y$, in particular $f(x, k(x, 0)) = 0$. Letting $g(x) = k(x, 0)$, we get $f(x, g(x)) = 0$, completing the proof.

In my mind, the key use of the Implicit function theorem is that it let's you relate a set of “inputs” to some desired output

Example 0.2.4: Implicit Function Theorem
Let $p : \mathbb{C} \to \mathbb{C}$, $p(z) = a_0 + \cdots + a_{n-1}z^{n-1}z$. Let's say all the roots of p are distinct and r is a root, $p(r) = 0$. If $\mathbb{C}^n$ is the set of all possible coefficients, is there a smooh way of relating the coefficents to the root? That is, is there a function: $f_r : \mathbb{C}^n \to \mathbb{C}$ such that given the coefficients, we can find a smooth function from $\mathbb{C}^n$ to the roots.

Going one way is easy: if $p(z) = (z - \alpha_1) \cdots (z - \alpha_n)$ are the roots, then multiplying up we get $p(z) = z^n + \sigma_{n-1} z^{n-1} + \cdots + \sigma_1 z = \sigma_0$, where σ_i are the symmetric functions. Findin an inverse of this function is really hard. However, we can find it "implicitely" by defining:

$$F(a_0, ..., a_{n-1}, z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$$

Then $f(a_0, a_1, ..., a_{n-1}, r) = 0$, and since all the roots are distinct we have that $\partial F(a, r)/\partial z \neq 0$, and hence we can find a C^1 inverse!

The key here is that there was some "input" (i.e. coefficients) that we wanted to relate to some output (in this case a root).

(A word on why it has to one root, namely the need for bijectivity).

Finding the derivative of $g(x)$ is feasible as well. Given that $f(x, g(x)) = 0$, $f^i(x, g(x)) = 0$. Taking the derivative D_j on both sides, we get:

$$D_j f^i(x, g(x))) = D_j f^i(x, g(x)) = \sum_{\alpha=1}^{m} D_{n_\alpha} f^i(x, g(x)) \cdot D_j g^\alpha(x)$$

and since $\det M \neq 0$, these equations can be solved fof $D_j g^\alpha(x)$. The answer will depend on $D_j f^i(x, g(x))$, and hence ultimately on $g(x)$. Unfortuantely, there is no more real simplification we can do here, namely since $g(x)$ is not necessarily a unique answer. For example, if we consider $f(x, y) = x^2 + y^2 - 1$, then we get $g(x) = \pm\sqrt{1 - x^2}$ as possible solutions to $f(x, g(x)) = 0$. If we differentiate f at 0, at $f(x, g(x)) = 0$, we get:

$$(D_1 f(x, g(x)) + D_2 f(x, g(x))) \cdot g'(x) = 0$$

i.e.

$$2x + 2g(x) \cdot g'(x) = 0 \quad \Leftrightarrow \quad g'(x) = \frac{-x}{g(x)}$$

which works for both $\pm$.

(when you get to Clairuts theorem show how it is a simple task to generalize form when we can swap any two indices).

0.3 Integration

Integrating over $\mathbb{R}$ and $\mathbb{R}^n$ will be taken as known (in particular, integrable functions, zero measures sets, and Fubini a). We will be reviewing what is important for understanding integrating manifolds. The first important concept to remember is that integrals are invariant under diffeomorphism as lon as the domain of integration changes properly as well:

Theorem 0.3.1: 1 Dimensional Change Of Variables

Let $u = g(x)$. Then:

$$\int_a^b f(g(x))g'(x)dx = \int_{a'}^{b'} f(u)du$$

***Proof*:**
proved in many texts

Theorem 0.3.2: Change Of Variables

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ be a diffeomorphism. If $u = g(x)$, then:

$$\int \cdots \int_a^b f(g(x)) \det(Dg(x))dx = \int \cdots \int_{a'}^{b'} f(u)du$$

***Proof*:**
In many texts

Example 0.3.3: Change Of Variables
put here all the common change of var's.

Theorem 0.3.4: Subordinate Partitions Of Unity

Let $A \subseteq \mathbb{R}^n$ and let $\mathcal{O}$ be an open cover for A. Then there exists a collection Φ of C^∞ functions φ defined defined in an open set containing A such that

1. For each $a \in A$, $0 \leq \varphi(x) \leq 1$ for all $\varphi \in \Phi$
2. For each $a \in A$, there exists an open neighborhood V_a such that $a \in V_a$ and only finitely many $\varphi \in \Phi$ are nonzero at a
3. For each $a \in A$, $\sum_{\varphi \in \Phi} \varphi(a) = 1$ (by the previous condition, this is always a finite sum).
4. For each $\varphi \in \Phi$, there is an open set $U \in \mathcal{O}$ such that $\varphi = 0$ outside some closed set V containing U.

If we just have the first 3 conditions, we call Φ a partition of unity of A, and if we add the 4th condition we say that the partition of unity is subordinate to $\mathcal{O}$. The proof follows a similar strategy to many of its kind: first we prove it on compact sets. Let U be an open set contianing A. Since A is compact, a we can choose a finite open cover automatically, satisfying the 2nd condition. Then we can take the closure of these sets, or try to refine our choice after applying compactness so that each $U_i \subseteq A$ (such an open cover is caleld admisible) and repeat a slightly more elaborate but elementary construction (Spivak calculus on Manifolds p.63). Let's label the compact sets whose interior cover A in an admissible $\{D_i\}_i$. way Then we can choose some C^∞ functions ψ_i so that they are 0 outside of D_i. Then we can define $\varphi_i = \frac{\psi_i}{\sum_i \psi_i}$. Finally, choosing some C^∞ function $f : U \to [0, 1]$ that is zero outisde of A. Then $\Phi = \{f \circ \varphi_i\}_{i=1}^n$, completing the compact case. Then we expand this in a similar way we would do for a σ-algebra, and then to general open sets by taking advantage of the fact that any open set has an "open set exaustion", and finally for arbitrary sets by taking the union of all open sets of the cover of A to be the open set containing A, then applying the previous step.

0.4 Category Theory

(If these notes expand into something publishable, I will add some elementary category theory so that I may use the terminology throughout the book. Important concecpts are universal properties, limits and colimits, isomorphisms, natural transfomrations)

0.5 Linear and Multilinear Algebra

(put dual spaces, tensor algebras, and their further derivations here like the alternating and symmetric algebra)

0.5.1 Dual Space

Recall that $V^* = V^\vee = \mathrm{Hom}_V(V, k)$ is the *dual vector space* of V, and is isomorphic to V.

Proposition 0.5.1: Basis For Dual Sapce

Let V be a vector-space with basis $(e_1, e_2, ..., e_n)$. Then the collection $(f_1, f_2, ..., f_n)$ defined by:

$$f^i(e_j) = \delta^i_j$$

where δ^j_i is the kronecker delta is a basis for V^*

***Proof*:**
exercise

Thus, if $v \in V$, then $v = \sum_i f^i(v_i)$. Since a linear functional $f : \mathbb{R}^n \to \mathbb{R}$ would have a matrix representation as a row, we sometimes write f^i in a matrix representation as:

$$f^1 = (1\ 0\ \cdots\ 0) \quad f^2 = (0\ 1\ 0\ \cdots\ 0) \quad \cdots \quad f^n = (0\ 0\ \cdots\ 1)$$

If $\omega \in V^*$, we can write it in terms of the basis as:

$$\omega = \omega_i f^i \qquad \omega(e_i) = \omega_i$$

If $f : V \to W$, then $f^* : W^* \to V^*$ is defined by taking the function $x \in W^*$ to the function $f^*(x) \in V^*$ defined by:

$$f^*(x) = x \circ f$$

Furthermore:

$$(g \circ f)^* = f^* \circ g^* \qquad (\mathrm{id}_V)^* = \mathrm{id}_{V^*}$$

showing that $(\underline{\quad})^*$ is a functor from $\mathbb{R}$-**Vect** to itself (more generally, any field k will work). Furthermore, the map $(\underline{\quad})^{**}$ is a natural isomorphism.

(If you want, add a word on angle bracket notation like John Lee doesp. 274)

0.5.2 Multilinear Algebra

Recall the following in the context of vector-spaces:

Definition 0.5.2: Multilinear Map

Let $V_1, V_2, ..., V_k$ and W be vector-spaces. Then $f : V_1 \times V_2 \times \cdots \times V_k \to W$ is called *multilinear* if it is linear in each component.

Example 0.5.3: Multi-Linear Maps

1. the dot product from $\langle \cdot, \ \cdot \rangle : \mathbb{R}^n \to \mathbb{R}$ is bilinear map that is interpreted as the "length" of vectors along with some orientation them
2. the cross product on $\cdot : \mathbb{R}^3 \to \mathbb{R}$ is a bilinear map interpreted as computing the area of a parallelogram, or finding an orthogonal vector to two vectors which with priority given to "right-handedne".
3. The determinant $\det(\_\_\_) : \mathbb{R}^{n^2} \to \mathbb{R}$ as a n-linear function giving us a notion of volume
4. The lie bracket for a lie algebra $\mathfrak{g}$ is a bilinear map (not sure how to interpret it yet)
5. The Euler form on root lattices from Lie Algebra representation is another example.

We see how multilinear maps are pretty common in geometry, and many represent geometric notions like volume or length (idk what lie brackets yet represent TBD). Recall from geometry that given a R-module M, we can define a new R-module

$$\mathcal{T}(M) = \bigoplus_{i \in \mathbb{N}} \mathcal{T}^k(M)$$

where $\mathcal{T}^k(M) = \bigotimes_i^k M$. In our case, we will let $M = V^*$ be the dual of a vector-space. Then for any $f, g \in V^*$, we define their tensor-product to be:

$$f \otimes g(a, b) = f(a)g(b)$$

notice that $f \otimes g$ is bilinear, and so by the universal property of tensors we are jusitifed in writing $f \otimes g$. We will say that $f \otimes g \in \mathcal{L}^2(V, V, R)$, or since the vector-spaces are the same, we will write $\mathcal{L}^2(V, \mathbb{R})$. Furthermore, $\mathcal{L}^2(V, \mathbb{R})$ is still a vector-space.

This process easily generalized: Let $F \in \mathcal{L}^n(V_1, V_2, ..., V_n, \mathbb{R})$ and $G \in \mathcal{L}^m(W_1, W_2, ..., W_m, \mathbb{R})$ be an n-linear and m-linear map respectively. Then:

$$F \otimes G(v_1, ..., v_n, w_1, .., w_m) = F(v_1, ..., v_n, w_1, ..., w_n)$$

is an $n + m$-linear map and belongs in $\mathcal{L}^{n+m}(V_1, V_2, ..., V_n, W_1, W_2, ..., W_m, \mathbb{R})$ (recall that tensor-products are associative, and hence this is indeed well-defined).

Recall that a basis for $\mathcal{L}^n(V_1, V_2, ..., V_n, \mathbb{R})$ is simply the k-tensor product of the basis for each V_i^*. Due to this, we have:

$$V_1^* \otimes \cdots V_k^* \cong_{k\text{-}\mathbf{Vect}} \mathcal{L}^k(V_1, ..., V_k, \mathbb{R})$$

and so since $V_1 \cong V_1^{**}$ canonically, we also have:

$$V_1 \otimes \cdots V_k \cong_{k\text{-}\mathbf{Vect}} \mathcal{L}^k(V_1^*, ..., V_k^*, \mathbb{R})$$

giving us a characterization of tensor products in terms of a k-linear map on *dual spaces.*

0.5.3 Symmetric And Alternating Tensors

The most important tensors products for us are those that have more geometric meaning. We present the two important ones here:

Symmetric Tensors

Definition 0.5.4: Symmetric Tensor

Let V be a finite dimensional vector space. Then a covariant k-tensor on V is said to be *symmetric* if it's value is unchanged if we interchange any pairs of arguments

The collection of all symmetric k-tensors is denoted $\Sigma^k(V)$

(eventually will fill)

Alternating Tensors

Definition 0.5.5: Alternating Tensor

Let V be a finite dimensional vector space. Then a covariant k-tensor on V is said to be *alternating* if it's sign changes if we interchange any pairs of arguments

The collection of all alternating k-tensors is denoted $\Lambda^k(V)$

Proposition 0.5.6: Properties Of Alternating Tensorsq

Let α be a covariant k-tensor on a finite dimensional vector-space V. Then the following are equivalent:

1. α is alternating
2. $\alpha(v_1, ..., v_n) = 0$ whenever the k-tulpe $(v_1, v_2, ..., v_n)$ is linearly independent
3. If two arguments are the same, then:

$$\alpha(v_1, ..., w, ..., w, ..., v_n) = 0$$

***Proof*:**
p.350 John Lee

0.5.4 Covariant and Contravariant tensors on Vector Spaces

The last thing we showed is that we can take the tensor product of V_i's or V_i^*'s. These behave differently, and so we will give these a name:

Definition 0.5.7: Covariant And Contravariant Tensor

Let V be a finite dimensional vector space and $k \in \mathbb{N}_{>0}$. Then a *covariant* k-tensor on V is an element of

$$V^* \otimes \cdots \otimes V^* = \mathcal{T}^k(V^*)$$

and a *contravariant* k-tensor on V is an element of

$$V \otimes \cdots \otimes V = \mathcal{T}^k(V)$$

Example 0.5.8: Covariant And Contravariant Tensors

1. Any linear functional $\omega : V \to \mathbb{R}$ is linear, and so a *covariant 1-tensor*. We also call such a function a covector, an element of $T^1(V^*)$.
2. The dot product, or more generally the inner product, is a *covariant 2-tensor*. Covariant 2-tensors are also called *bilinear forms*.
3. The determinant is a covariant n-tensor.
4. I will see if I can find a good way of adding the contravariant tensors here TBD

it is sometimes useful to mix these two:

Definition 0.5.9: Mixed Tensors

Let V be a finite dimensional vector space, and $k, l \in \mathbb{N}_{>0}$. Then ω is a *mixed tensor* of type (k, l) if it is in:

$$T^{(k,l)}(V) := \underbrace{V \otimes \cdots \otimes V}_{k \text{ times}} \otimes \underbrace{V^* \otimes \cdots \otimes V^*}_{l \text{ times}}$$

Part I

Smooth Manifolds

On a high level, a manifold is a Hausdorff topological space which is nice enough to do some useful analysis (is σ-compact) that, crucially, *locally* resembles Euclidean space. At the beginning of this book, manifolds were said to be spaces X that are copies of a base space B glued together in appropriate ways. Though this is the case, we shall be taking a more axiomatic approach to manifolds since there are multiple ways of "concretely interpreting" a manifold. This is similar to how in group theory, groups are first defined using axioms, and only some concrete "representations" of groups are introduced (ex. Caley's theorem shows that all groups embedd into $\mathrm{Sym}(G)$, Frucht's theorem which states that every finite group is the group of symmetries of a finite undirected graph, and the universal property of free groups which states that every group is the quotient of a free group).

Manifolds will be the focus of this part of the book, however to contextualize them, a quick general overview of thinking of a space as being locally another space (i.e. locally like the "base space") paints a better pictures of where manifold theory fits into mathematics. In mathematics, there are many spaces which will be defined by the "type" of locality (how subsets of a the space is isomorphic to a subset of a base space), and the "regularity" of the locality. The type of locality will depend on the choice of *base space* (ex. $\mathbb{R}^n$, $L^2(\mathbb{R})$, zero sets of multi-variate polynomials); for example a circle will be seen to be locally $\mathbb{R}$ since for any $x \in S^1$, there is a neighborhood of x, $U \subseteq S^1$, that maps isomorphically to $V \subseteq \mathbb{R}$. In the case of manifolds, they will always be locally $\mathbb{R}^n$ for some $n \in \mathbb{N}$, that is if M is a manifold, then for each $x \in M$, there exists an open neighborhood $U \subseteq M$ of x and an isomorphism $f : U \to V \subseteq \mathbb{R}^n$. For the readers awareness, if we choose a different base space, then we get different mathematical objects, for example in algebraic geometry, spaces that are locally the zero set of polynomials, are called [affine] variety. Later in this part of the book, we shall also see a slight generlization of manifolds called *orbifolds* which (for reasons that shall be clear later) shall have as base space a finite group quotient of Euclidean space.

There are two other types of space we shall study in this book that is very close to being a manifold, but will be distinct since it will have a different base space associated to it. The other two spaces we will see have the word manfold inside their name, but should not be confused with manifolds since they shall have different base spaces; these are *manifold with borders* and *manifold with corners*[4], which have respective local spaces:

$$\mathbb{H}^n = \{x \in \mathbb{R}^n \mid x_1 \geq 0\} \qquad \text{and} \qquad \mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ 1 \leq i \leq n\}$$

The type of isomorphism from U to a part of the base space will define the type of regularity; for example homeomorphic, C^k, smooth, analytic, etc. The choice of the type of map will define the type of manifold we choose, in particular for the four type of maps just mentioned we would get:

1. a Topological Manifold
2. a Differentiable Manifolds (C^1 manifold)
3. a Smooth Manifold (C^∞-manifolds)
4. an Analytic Manifold (C^ω-manifolds)

Topological manifolds can be seen as the "basic" manifold from which we may obtain other manifolds by adding other strucutres. In this book, differentiable manifolds shall be the main focus, (which is why the title of the book specifies that we are studying *differential* geometry) which shall be

[4]Though as we shall see, all manifolds are (isomorphic to) manifolds with boundary/corners, but the converse is not the case

topological manifolds with a differentiable sructure that shall be added. Given a differential structure on a manifold, we shall be able to define the the *derivative* of a map between two differentiable manifolds $f : M \to N$. Recall that a derivative is a map that is locally linear. The derivative of a map between smooth manifolds shall be a linear map at each point that will “vary smoothly”; this is a technical endeavour that shall introduce the notion of a tangent bundle to be defined and will be the focus of chapter 2.

After having defined differential manifolds and their associated tangent bundle, we shall look at the behavior of maps between differential manifolds. The key property of maps between differentaible manifolds is that they “locally” will have the property of the manifold. This will differ from the property functions usually preserve in algebra (the image $f(G)$ is another algebraic obect) or topology (we can study the local properties of codomain N via the local propeties of the domain M). The advantage of this appraoch will be evident in chapter 3 where a lot of information about a map $f : M \to N$ will be deduced by the local information of f,and lead to the generalization of theroems like the implicit and inverse function theorem, and allow for the fruitful study of the intersection of manifolds.

We shall conclude this part of the book with a chapter on vector-fields on Manifolds. Vector-fields are very useful to Physicists for studying many natural phenomena; to a mathematician, a vector-field on a manifold is a generalizaiton of the notion of solving a system of ODE’s on a manifold. We shalls see when a solution to a system of ODE’s exist, and give show how being able to define certina vector-fields on a manifold gives property of a manifold;for example, we shall be able to detect when a 2-dimensional manifold has a “hole”; or if a sphere has even or odd dimension. This exploration will naturally lead to part III

1

Manifolds

In this chapter, we define topological, differentiable, and smooth manifolds. We give many examples of each type of manifold, and give some ways of constructing and comparing these manifolds.

After having defined a nice range of manifolds, we will define the maps between them. These maps will have many nice properties we expect from morphisms in categories, however they will in other ways be lacking in important ways due to their "local" nature. The introduction of maps will allow us to ask questions like what are the subobjects of our category, the product objects, and the quotient objects. All of these will be explored in this chapter.

We will end the chapter by introducing partitions of unity for manifolds. Partitions of unity will be invaluable for finding the existence of many extensions of functions and to develop the theory of integration later on.

1.1 Topological Manifolds

A manifold is, before anything else, a space that looks locally like $\mathbb{R}^n$. Making "looks like" precise is the work of a single definition, after which two further global conditions rule out the pathologies a purely local demand cannot see.

Definition 1.1.1: Locally Euclidean Space

A topological space M is *locally Euclidean of dimension* n if every point $x \in M$ has an open neighbourhood $U \subseteq M$ homeomorphic to an open subset of $\mathbb{R}^n$.

The homeomorphisms witnessing this local structure are the coordinate systems of the theory.

Definition 1.1.2: (Topological) Chart

A *chart* on a topological space M is a homeomorphism $\varphi\colon U \to \varphi(U)$ from an open set $U \subseteq M$, the *chart domain*, onto an open subset $\varphi(U) \subseteq \mathbb{R}^n$. The chart is *centred at* $p \in U$ when $\varphi(p) = 0$.

Definition 1.1.3: Coordinate Functions

Let $\varphi\colon U \to \mathbb{R}^n$ be a chart and $\pi_i\colon \mathbb{R}^n \to \mathbb{R}$ the projection $\pi_i(a^1, \dots, a^n) = a^i$. The i-th *coordinate function* of the chart is $x^i = \pi_i \circ \varphi$, so that $\varphi = (x^1, \dots, x^n)$.

Definition 1.1.4: (Topological) Atlas

An *atlas* on a locally Euclidean space M is a collection of charts whose domains cover M.

That single local demand is by itself far too weak: it constrains nothing global, and spaces satisfying it can fail to separate their points or can be too large to be described by countably many pieces. Two global hypotheses repair this, and they are exactly the ones that place every manifold inside the class governed by the point-set criterion of *Topology* [5]: a second countable, locally compact Hausdorff space is paracompact.

Each of the two hypotheses is justified below by a space that satisfies the other two conditions and fails it, so that neither can be dropped without changing the subject.

Definition 1.1.5: Topological Manifold

Let $n \geq 0$ be an integer. A topological space M is a *topological n-manifold* if

1. M is *locally Euclidean of dimension n*: every point of M has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$;
2. M is Hausdorff (the Hausdorff axiom, [5]);
3. M is second countable (second countability, [5]).

A homeomorphism $\varphi\colon U \to \varphi(U)$ from an open set $U \subseteq M$ onto an open subset $\varphi(U) \subseteq \mathbb{R}^n$ is a *chart* on M, and U is its *chart domain*. A family of charts whose domains cover M is an *atlas*. When $M \neq \emptyset$, the integer n is the *dimension* of M; the empty space satisfies (1), (2) and (3) vacuously for every n and is assigned no dimension.

Condition (1) has an equivalent form. Requiring a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$ is the same as requiring one homeomorphic to $\mathbb{R}^n$ itself. Indeed, if $\varphi\colon U \to \varphi(U)$ is a chart around p, then $\varphi(U)$ contains an open ball $B(\varphi(p), \rho)$, and the map

$$B(0,1) \to \mathbb{R}^n, \qquad x \mapsto \frac{x}{1 - \|x\|}$$

is a homeomorphism with inverse $y \mapsto y/(1 + \|y\|)$: both formulas are continuous where written, and substituting one into the other gives

$$\frac{x/(1-\|x\|)}{1+\|x\|/(1-\|x\|)} = \frac{x/(1-\|x\|)}{1/(1-\|x\|)} = x$$

while the same manipulation with $1+\|y\|$ in place of $1-\|x\|$ returns y, so the two maps are mutually inverse and each is continuous. Composing with the translation and dilation carrying $B(0,1)$ onto $B(\varphi(p),\rho)$ produces a chart domain around p homeomorphic to all of $\mathbb{R}^n$. For nonempty M the dimension n is not an arbitrary label attached to M but is determined by the space.[1]

The sphere is the standard object satisfying all three conditions, and its charts can be written down. Let $S^n = \{x \in \mathbb{R}^{n+1} \big| \|x\| = 1\}$ carry the subspace topology from $\mathbb{R}^{n+1}$ (the subspace topology, [5]), and write $N = (0,\dots,0,1)$, so that N and $-N$ are the two poles. Stereographic projection from N is

$$\sigma_+ \colon S^n \setminus \{N\} \to \mathbb{R}^n, \qquad \sigma_+(x_1,\dots,x_{n+1}) = \frac{1}{1-x_{n+1}}(x_1,\dots,x_n)$$

which is continuous because $x_{n+1} < 1$ on its domain, so the denominator never vanishes. Its inverse is

$$\sigma_+^{-1}(u) = \frac{1}{\|u\|^2+1}\big(2u,\ \|u\|^2-1\big)$$

and three short computations confirm this. First the image lies on the sphere, since the sum of the squares of its coordinates is

$$\frac{4\|u\|^2 + (\|u\|^2-1)^2}{(\|u\|^2+1)^2} = \frac{(\|u\|^2+1)^2}{(\|u\|^2+1)^2} = 1$$

and its last coordinate $(\|u\|^2-1)/(\|u\|^2+1)$ is strictly smaller than 1, so the image avoids N. Second, writing $x = \sigma_+^{-1}(u)$ gives $1 - x_{n+1} = 2/(\|u\|^2+1)$, so $\sigma_+(x) = \frac{\|u\|^2+1}{2} \cdot \frac{2u}{\|u\|^2+1} = u$. Third, the composite in the other order is also the identity, which is what makes σ_+ injective. For $x \in S^n \setminus \{N\}$ put $x' = (x_1,\dots,x_n)$ and $u = \sigma_+(x) = x'/(1-x_{n+1})$. Then $\|x'\|^2 = 1 - x_{n+1}^2$, so

$$\|u\|^2 = \frac{1-x_{n+1}^2}{(1-x_{n+1})^2} = \frac{1+x_{n+1}}{1-x_{n+1}}$$

whence $\|u\|^2 + 1 = 2/(1-x_{n+1})$ and $\|u\|^2 - 1 = 2x_{n+1}/(1-x_{n+1})$. Therefore $2u/(\|u\|^2+1) = (1-x_{n+1})u = x'$ and $(\|u\|^2-1)/(\|u\|^2+1) = x_{n+1}$, that is, $\sigma_+^{-1}(\sigma_+(x)) = x$. The two maps are mutually inverse and both are continuous, so σ_+ is a chart with domain $S^n \setminus \{N\}$, which is open in S^n. Projection from the opposite pole, $\sigma_-(x) = (1+x_{n+1})^{-1}(x_1,\dots,x_n)$, is a chart on $S^n \setminus \{-N\}$ by the same computation with the sign of the last coordinate reversed, and the two domains cover S^n. The remaining two conditions are inherited: a subspace of a Hausdorff space is Hausdorff, since the separating open sets may be intersected with the subspace, and if $\mathcal{B}$ is a countable basis for a space Z then $\{B \cap A \big| B \in \mathcal{B}\}$ is a countable basis for a subspace $A \subseteq Z$ directly from the definition of the subspace topology. So S^n is a topological n-manifold, with an atlas of two charts.

Dropping the Hausdorff condition costs more than it might appear to, and the cheapest space that shows this is obtained by gluing two lines together along everything except their origins.

[1]What determines n is Brouwer's invariance of domain, in the form: a nonempty open subset of $\mathbb{R}^n$ is homeomorphic to an open subset of $\mathbb{R}^m$ only when $n = m$. Charts land in open subsets, so the weaker statement that $\mathbb{R}^n$ and $\mathbb{R}^m$ are homeomorphic only when $n = m$, which is a corollary of it, does not by itself settle the matter. The proof belongs to algebraic topology, where the theorem is read off from the homology of spheres, and is not available at this point. The case $n = 1$ against $n \geq 2$ is elementary, since deleting a point disconnects an open interval but leaves an open ball in $\mathbb{R}^n$ connected for $n \geq 2$ (connectedness, [5]), and connectedness is preserved by homeomorphism.

Example 1.1.6: The Line with Two Origins

Let $Y = \mathbb{R} \sqcup \mathbb{R}$, realized as $\mathbb{R} \times \{0,1\}$ with $\{0,1\}$ discrete, and let $\sim$ be the equivalence relation generated by $(x,0) \sim (x,1)$ for every $x \neq 0$. Let $X = Y/\sim$ carry the quotient topology (the quotient topology, [5]) and let $q\colon Y \to X$ be the quotient map. Write $0_0 = q(0,0)$ and $0_1 = q(0,1)$ for the two origins, and write x for the common image $q(x,0) = q(x,1)$ when $x \neq 0$.

1. *The map q is open.* Let $s\colon Y \to Y$ be the homeomorphism $s(x,i) = (x, 1-i)$ exchanging the two copies. For $U \subseteq Y$ open, a point (x,i) lies in $q^{-1}(q(U))$ exactly when $(x,i) \in U$ or both $x \neq 0$ and $s(x,i) \in U$, so

$$q^{-1}(q(U)) = U \cup s\big(U \setminus (\{0\} \times \{0,1\})\big)$$

   Both sets on the right are open, hence so is $q^{-1}(q(U))$, and therefore $q(U)$ is open in X.

2. *X is locally Euclidean of dimension* 1. Put $A_i = q(\mathbb{R} \times \{i\})$ for $i \in \{0,1\}$. Each A_i is open, since q is open and $\mathbb{R} \times \{i\}$ is open in Y, and $A_0 \cup A_1 = X$ because q is surjective. The restriction $q|_{\mathbb{R}\times\{i\}}\colon \mathbb{R} \times \{i\} \to A_i$ is injective (the relation $\sim$ never identifies two points of the same copy), it is surjective onto A_i by definition, it is continuous, and it is open because an open subset of $\mathbb{R} \times \{i\}$ is open in Y and q is open. A continuous open bijection is a homeomorphism, so $A_i \cong \mathbb{R}$. Every point of X therefore lies in an open set homeomorphic to $\mathbb{R}$.

3. *X is second countable.* The family $\mathcal{B} = \{(a,b) \times \{i\} \big| a < b \text{ rational}, \ i \in \{0,1\}\}$ is a countable basis for Y. Each $q(B)$ with $B \in \mathcal{B}$ is open. Given $W \subseteq X$ open and $p \in W$, choose $y \in q^{-1}(p)$; then $q^{-1}(W)$ is an open set containing y, so there is $B \in \mathcal{B}$ with $y \in B \subseteq q^{-1}(W)$, and applying q gives $p \in q(B) \subseteq W$. So $\{q(B) \big| B \in \mathcal{B}\}$ is a countable basis for X.

4. *X is not Hausdorff.* Let U and V be open sets with $0_0 \in U$ and $0_1 \in V$. Then $q^{-1}(U)$ is open in Y and contains $(0,0)$, so it contains $(-\epsilon,\epsilon) \times \{0\}$ for some $\epsilon > 0$; likewise $q^{-1}(V)$ contains $(-\delta,\delta) \times \{1\}$ for some $\delta > 0$. Choosing t with $0 < t < \min(\epsilon,\delta)$ gives $q(t,0) = q(t,1) \in U \cap V$, so the two origins admit no disjoint neighbourhoods.

The failure is confined to a single pair of points, and it is not a failure of the weaker separation axioms: points of X are closed, since $q^{-1}(X \setminus \{0_0\})$ is $Y \setminus \{(0,0)\}$ and $q^{-1}(X \setminus \{x\})$ is $Y \setminus \{(x,0),(x,1)\}$ for $x \neq 0$, both open. What the failure destroys is the interaction between compactness and closedness. The set $K = q([-1,1] \times \{0\})$ is compact, being the image of a compact interval under a continuous map, but it is not closed: every open neighbourhood of 0_1 contains points $q(t,1) = q(t,0)$ with $0 < t < 1$, so 0_1 lies in $\overline{K} \setminus K$. In a Hausdorff space this cannot happen (compact subspaces of Hausdorff spaces are closed, [5]), and a space in which compact sets need not be closed is one where local compactness carries no useful global consequence. That is what the Hausdorff clause in definition 1.1.5 excludes.

Second countability is the second global demand, and it too is independent of the others. The witness is the long line built, in the discussion following the ordinal space Ω [5], from the minimal uncountable well-ordered set Ω: a linearly ordered space that is locally indistinguishable from $\mathbb{R}$ and yet is uncountably long, and therefore admits no countable basis.

Example 1.1.7: The Long Line Is Not a Manifold

Let Ω be the minimal uncountable well-ordered set of the ordinal space Ω [5], with least element 0, so that every proper initial segment $S_\alpha = \{x \in \Omega \big| x < \alpha\}$ is countable, and let $\Omega \times [0,1)$ carry the lexicographic order, in which $(\alpha, s) < (\beta, t)$ when $\alpha < \beta$, and also when $\alpha = \beta$ and $s < t$. The long line is the ordered set

$$L = \big(\Omega \times [0,1)\big) \setminus \{(0,0)\}$$

introduced after the ordinal space Ω [5], with the order topology. Deleting the least element $(0,0)$ leaves a linearly ordered set with neither a least nor a greatest element, and for $\alpha \in \Omega$ with $\alpha > 0$ the set

$$L_\alpha = \{x \in L \big| x < (\alpha, 0)\}$$

is the open initial ray cut off by the α-th block.

1. *L is Hausdorff.* In any linearly ordered set with its order topology, two points $x < y$ are separated as follows: if some z satisfies $x < z < y$, the open rays $\{t \big| t < z\}$ and $\{t \big| t > z\}$ are disjoint and contain x and y respectively; if no such z exists, the rays $\{t \big| t < y\}$ and $\{t \big| t > x\}$ are disjoint, since a point in both would lie strictly between x and y, and they contain x and y. One of the two cases applies to every pair, so L is Hausdorff.

2. *L is locally Euclidean of dimension* 1. For $\beta < \alpha$ in Ω write

   $$J(\beta, \alpha) = \{x \in \Omega \times [0,1) \big| (\beta, 0) \le x < (\alpha, 0)\} = \{\gamma \in \Omega \big| \beta \le \gamma < \alpha\} \times [0,1)$$

   The claim is that $J(\beta, \alpha)$ is order-isomorphic to $[0,1)$. Two observations drive the proof. First, a single block $\{\gamma\} \times [0,1)$ is order-isomorphic to $[0,1)$ by $(\gamma, t) \mapsto t$. Second, if a linearly ordered set J is the concatenation of blocks $B_0 < B_1 < B_2 < \cdots$ indexed by $\mathbb{N}$ (each element of J lies in exactly one B_n, and every element of B_n is below every element of B_{n+1}) and each B_n is order-isomorphic to $[0,1)$, then J is order-isomorphic to $[0,1)$: compose the isomorphism of B_n with the increasing affine bijection of $[0,1)$ onto $[1-2^{-n}, 1-2^{-n-1})$ and take the union of the resulting maps. The union is a bijection onto $\bigcup_{n \in \mathbb{N}} [1-2^{-n}, 1-2^{-n-1}) = [0,1)$, and it is increasing because the blocks and their target intervals occur in the same order.

   Now argue by transfinite induction on $\alpha \in \Omega$, the inductive statement being that $J(\beta, \alpha) \cong [0,1)$ for every $\beta < \alpha$. For $\alpha = 0$ there is no such β and nothing to prove. Let $\alpha > 0$, assume the statement for every $\alpha' < \alpha$, and fix $\beta < \alpha$.

   Suppose first that α has an immediate predecessor α^-. If $\beta = \alpha^-$ then $J(\beta, \alpha) = \{\beta\} \times [0,1)$, which is order-isomorphic to $[0,1)$. If $\beta < \alpha^-$ then $J(\beta, \alpha)$ is the concatenation of $J(\beta, \alpha^-)$ and $\{\alpha^-\} \times [0,1)$; the first is order-isomorphic to $[0,1)$ by the inductive statement at $\alpha^- < \alpha$ and the second is by the first observation, so sending the first onto $[0, 1/2)$ and the second onto $[1/2, 1)$ by increasing affine maps gives $J(\beta, \alpha) \cong [0,1)$.

   Suppose instead that α has no immediate predecessor. The set $\{\gamma \in \Omega \big| \beta \le \gamma < \alpha\}$ lies inside S_α and is therefore countable; enumerate it as $\delta_0, \delta_1, \ldots$, repeating terms if it is finite. Put $\gamma_0 = \beta$ and, given $\gamma_n < \alpha$, set $\mu = \max(\gamma_n, \delta_n) < \alpha$; the set $\{x \in \Omega \big| \mu < x < \alpha\}$ is nonempty, since otherwise α would be the immediate successor of μ, so let γ_{n+1} be its least element. Then $\gamma_n < \gamma_{n+1} < \alpha$ and $\delta_n < \gamma_{n+1}$ for every n. Consequently each γ with $\beta \le \gamma < \alpha$ satisfies $\gamma_n \le \gamma < \gamma_{n+1}$ for exactly one n, namely the least n with $\gamma < \gamma_{n+1}$, which exists because $\gamma = \delta_m$ for some m and hence $\gamma < \gamma_{m+1}$. So $J(\beta, \alpha)$ is the concatenation of the blocks $J(\gamma_n, \gamma_{n+1})$, each order-isomorphic to $[0, 1)$ by the inductive statement at $\gamma_{n+1} < \alpha$, and the second observation applies. The induction is complete.

Taking $\beta = 0$ gives, for each $\alpha > 0$, an order isomorphism

$$h\colon J(0, \alpha) = \{x \in \Omega \times [0, 1) \big| x < (\alpha, 0)\} \to [0, 1)$$

An order isomorphism carries a least element to a least element, so h sends $(0, 0)$ to 0. Since L_α is exactly $J(0, \alpha) \setminus \{(0, 0)\}$, restricting h gives an order isomorphism of L_α onto $[0, 1) \setminus \{0\} = (0, 1)$.

An order isomorphism between linearly ordered sets carries open rays onto open rays and is therefore a homeomorphism for the order topologies. What remains is that the order topology of L_α is its subspace topology from L, and this holds because L_α is an interval: the trace on L_α of a ray of L is empty, or all of L_α, or a ray of L_α, and conversely every ray of L_α is such a trace. Hence L_α is homeomorphic to $(0, 1)$, an open subset of $\mathbb{R}$. Each L_α is open in L, being an open ray, and every point of L lies in one of them: a point (β, t) lies in L_α for any $\alpha > \beta$, and such an α exists because Ω has no greatest element, a greatest element g making $\Omega = S_g \cup \{g\}$ countable.

3. *L is not second countable.* The sets $\{L_\alpha\}$, for $\alpha \in \Omega$ with $\alpha > 0$, form an open cover of L by the previous item. Let $\{L_{\alpha_n}\}_{n\in\mathbb{N}}$ be a countable subfamily. The set $\{\alpha_n \big| n \in \mathbb{N}\}$ has a strict upper bound γ in Ω: otherwise every element of Ω would satisfy $x \leq \alpha_n$ for some n, giving $\Omega = \bigcup_n (S_{\alpha_n} \cup \{\alpha_n\})$, a countable union of countable sets and hence countable. Then $\gamma > 0$, so $(\gamma, 0)$ is a point of L, and $(\gamma, 0) > (\alpha_n, 0)$ for every n, so $(\gamma, 0)$ lies in no L_{α_n}. The subfamily does not cover L, and the cover has no countable subcover. A second countable space admits no such cover: if $\mathcal{B}$ is a countable basis and $\mathcal{A}$ an open cover, let $\mathcal{B}'$ consist of those $B \in \mathcal{B}$ contained in some member of $\mathcal{A}$ and choose one such $U_B \in \mathcal{A}$ for each; every x lies in some $U \in \mathcal{A}$ and hence in some $B \in \mathcal{B}$ with $x \in B \subseteq U$, so $B \in \mathcal{B}'$ and $x \in U_B$, and $\{U_B \big| B \in \mathcal{B}'\}$ is a countable subcover. Therefore L is not second countable.

The long line satisfies (1) and (2) of definition 1.1.5 and fails (3), so it is not a manifold. The failure is not a technicality about bases. The cover $\{L_\alpha\}$ exhibited above has no countable subcover, and L is not paracompact either: the transfer of the argument of the ordinal space Ω [5] from Ω to L is carried out where L is constructed, immediately after that example. So L is a space that looks locally like $\mathbb{R}$ and yet lies outside the paracompact class, which is what the third condition is there to exclude. The exclusion is not the only possible one: some treatments replace second countability by paracompactness in definition 1.1.5, which admits, for instance, the disjoint union of uncountably many copies of $\mathbb{R}$, a space that is locally Euclidean, Hausdorff and paracompact but has no countable basis. The two conventions differ only in how many components are allowed.

With the three conditions in place, the promised inclusion is a matter of producing the three hypotheses that second countable + locally compact Hausdorff implies paracompact [5] asks for. Second countability and the Hausdorff property are two of them outright; the third, local compactness, is not assumed but follows, because a chart transports the compact closed balls of $\mathbb{R}^n$ into M.

Theorem 1.1.8: Manifolds Are Paracompact

Every topological manifold is paracompact.

***Proof*:**
Let M be a topological n-manifold.

Step 1: M is locally compact. Fix $p \in M$. By definition 1.1.5 there is a chart $\varphi \colon U \to \varphi(U)$ with $p \in U$ and $\varphi(U) \subseteq \mathbb{R}^n$ open. Since $\varphi(U)$ is open and contains $\varphi(p)$, there is $\rho > 0$ with $B(\varphi(p), \rho) \subseteq \varphi(U)$; put $r = \rho/2$, so that the closed ball $\overline{B}(\varphi(p), r)$ is contained in $B(\varphi(p), \rho)$ and hence in $\varphi(U)$. That closed ball is closed and bounded in $\mathbb{R}^n$, hence compact by the Heine-Borel theorem. Set

$$C = \varphi^{-1}\big(\overline{B}(\varphi(p), r)\big)$$

Then C is compact, being the image of a compact space under the continuous map φ^{-1}, and compactness of C does not depend on whether it is viewed inside U or inside M. Moreover C contains $\varphi^{-1}(B(\varphi(p), r))$, which is open in U and therefore open in M because U is open in M, and which contains p. So C is a compact subspace of M containing a neighbourhood of p, which is what local compactness [5] requires. As p was arbitrary, M is locally compact.

Step 2: the remaining hypotheses. Clauses (2) and (3) of definition 1.1.5 say that M is Hausdorff and second countable. Combined with Step 1, M is a second countable, locally compact Hausdorff space.

Step 3: conclusion. Second countable + locally compact Hausdorff implies paracompact [5] states that every second countable locally compact Hausdorff space is paracompact. Applying it to M gives the claim, completing the proof.

Each of the three clauses does identifiable work in the argument. The locally Euclidean clause supplies compactness locally; the Hausdorff clause is what makes that local compactness usable, since without it compact sets need not be closed, as the line with two origins shows; and second countability supplies the countable exhaustion, without which the conclusion fails outright, as the long line shows. Two consequences follow at once. First, since M is locally compact Hausdorff, the local-compactness criterion in Hausdorff spaces [5] upgrades Step 1: every point of M has, inside any prescribed neighbourhood, a neighbourhood whose closure is compact, so manifolds admit neighbourhood bases of relatively compact open sets. Second, a paracompact Hausdorff space is normal (paracompact Hausdorff spaces are normal, [5]), so every manifold is normal and Urysohn's lemma (Urysohn's lemma, [5]) is available on it: disjoint closed subsets of a manifold are separated by continuous real-valued functions.

Normality separates two closed sets at a time, which is the input to the construction of a single function. The cover-indexed family of functions is what global constructions require, and theorem 1.1.8 delivers exactly the hypothesis under which such a family exists.

Corollary 1.1.9: Partitions of Unity on a Manifold

Let M be a topological n-manifold and let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M. Then M admits a partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.

***Proof*:**
By definition 1.1.5 the space M is Hausdorff, and by theorem 1.1.8 it is paracompact. So M satisfies the hypotheses of the partition-of-unity existence theorem [5], which produces for any open cover of a paracompact Hausdorff space a partition of unity subordinate to it in the sense of the definition of a partition of unity [5] and subordination to a cover [5]. Applying that theorem to the cover $\{U_\alpha\}_{\alpha\in A}$ gives the assertion, completing the proof.

The corollary is the form in which the chapter's machinery is consumed elsewhere. Suppose $\{U_\alpha\}$ is an open cover of M and a continuous function $f_\alpha\colon U_\alpha \to \mathbb{R}$ is given on each piece, with no compatibility assumed on the overlaps. Let $\{\psi_\alpha\}$ be a partition of unity subordinate to the cover and define $g_\alpha\colon M \to \mathbb{R}$ by $g_\alpha = \psi_\alpha f_\alpha$ on U_α and $g_\alpha = 0$ on $M \setminus \operatorname{supp}\psi_\alpha$. Those two sets are open and cover M, and the two formulas agree on the overlap, so each g_α is continuous on all of M. Subordination puts $\operatorname{supp}\psi_\alpha$ inside U_α, and local finiteness of the supports means every point has a neighbourhood on which all but finitely many g_α vanish, so the sum

$$f = \sum_{\alpha\in A} g_\alpha$$

is a finite sum near each point and hence a continuous function on M. Since $\sum_\alpha \psi_\alpha = 1$, the value $f(p)$ is a convex combination of the values $f_\alpha(p)$ over those α with $p \in U_\alpha$; in particular, if every f_α takes values in a fixed interval, so does f. Locally defined continuous data on a manifold can therefore be averaged into global continuous data, with no agreement on overlaps required. The partition of unity itself is written down explicitly for $M = \mathbb{R}$ in the explicit partition of unity on $\mathbb{R}$ [5].

What corollary 1.1.9 asserts is a C^0 statement, and it is a statement about the underlying space alone. A smooth structure on M is extra data, an atlas whose charts overlap by transition maps that are C^∞ with C^∞ inverses. It does not alter the topology, so the corollary applies verbatim to every smooth manifold and already yields continuous ψ_α there. What the smooth theory has to add is regularity of the functions themselves, namely that the ψ_α can be chosen C^∞, which requires replacing the Urysohn functions used here by the explicit bump functions on $\mathbb{R}^n$ constructed in [2], and then transporting those through charts. The point-set half of that upgrade is not redone in the differential topology companion, which takes it as given.

Note that it suffices to show that M can be covered by countably many charts to show that it is second-countable. Every manifold has an atlas, and most of the time, an atlas will only need finitely many charts. *Any atlas covering a manifold will henceforth assumed to be countable.* Notice that the M in example 1.1.6 is not a topological manifold. if we remove the hasudroff condition, the resulting class of sets is usually called *non-hausdorff manifolds*; these will not be studied in these notes. Note that a manifold, being second countable and regular (Hausdorff and locally Euclidean), is *metrizable* by the Urysohn metrization theorem. Putting a metric on M will be explored in the companion Riemannian Geometry volume [4], where the Riemann metric is defined.

Since the three properties of a manifold are all topological properties the collection of topological manifolds along with continuous maps forms a category:

Definition 1.1.10: The Category of Topological Manifolds

The topological manifolds, with continuous maps between them, form a category **TMan**: composition and identities are those of the underlying topological spaces.

Using Atlases, we may check for continuity of maps locally:

Lemma 1.1.11: Continuity Is Local in Charts

Let M and N be topological manifolds and $f\colon M \to N$ a map. Then f is continuous if and only if for every $p \in M$ there are a chart (U, φ) of M with $p \in U$ and a chart (V, ψ) of N with $f(p) \in V$ such that $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}\colon \varphi(U) \to \psi(V)$ is continuous.

***Proof*:**
Continuity is tested pointwise, and at each point on any neighbourhood, since charts are homeomorphisms onto open sets.

If f is continuous, fix $p \in M$ and any chart (V, ψ) of N with $f(p) \in V$. Then $f^{-1}(V)$ is open and contains p, so it contains the domain of a chart (U, φ) with $p \in U$ and $f(U) \subseteq V$, and $\psi \circ f \circ \varphi^{-1}$ is continuous as a composite of the continuous maps φ^{-1}, f, ψ.

Conversely, suppose the condition holds and fix $p \in M$ with charts (U, φ), (V, ψ) as stated. On U, $f = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$, a composite of continuous maps, so f is continuous at p. As p was arbitrary, f is continuous, as we sought to show.

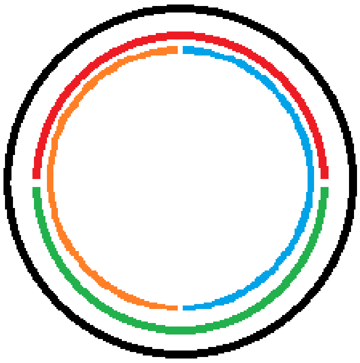
Continuity of a map between manifolds is thus a chart-local question.

Example 1.1.12: Building Topological Manifolds
The following examples include the most common manifolds and the standard ways of building new manifolds from old ones.

1. The empty space $\emptyset$ is a topological manifold of every dimension, vacuously, and is the only manifold not assigned a single dimension. A one-point space is a 0-manifold.
2. $\mathbb{R}^n$ is the n-manifold underlying all the others: the identity map is a global chart, so a single chart already forms an atlas.
3. An open subset U of a topological n-manifold X is again a topological n-manifold (exercise 1.1.1 (2)). Many manifolds arise this way, as a complement $X \setminus C$ of a closed set. For instance $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$ is open in $\mathbb{R}^{n^2}$, since the determinant is continuous, and is therefore an n^2-manifold.
4. If M_1 and M_2 are topological manifolds of dimensions n_1 and n_2, then $M_1 \times M_2$ with the product topology is a topological manifold of dimension $n_1 + n_2$ (exercise 1.1.1 (1)): a product $\varphi_p \times \psi_q$ of charts is a chart, and the Hausdorff and second countable properties pass to products.
5. The disjoint union $X \sqcup Y$ of two n-manifolds is an n-manifold, its atlas the union of the two atlases; the dimensions must agree. A manifold therefore need not be connected.
6. The first genuinely nontrivial example is the circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

with the subspace topology. It is Hausdorff and second countable as a subspace of $\mathbb{R}^2$, and compact. Covering S^1 by four open arcs and projecting each onto a coordinate axis



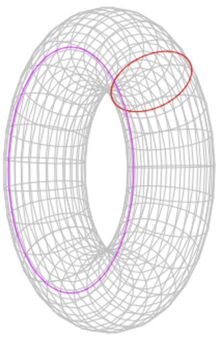
gives four charts, hence an atlas, so S^1 is a 1-manifold. It is not homeomorphic to $\mathbb{R}$: removing a point leaves S^1 connected but disconnects $\mathbb{R}$, and connectedness is a homeomorphism invariant.

7. Since a product of manifolds is a manifold, the n-fold product

$$\underbrace{S^1 \times S^1 \times \cdots \times S^1}_{n} = T^n$$

is a manifold, the *n-torus*; for $n = 2$ one says simply the torus. The 2-torus embeds in $\mathbb{R}^4$, and famously in $\mathbb{R}^3$ through the parameterization

$$x = (R + r\cos\psi)\cos\theta, \qquad y = (R + r\cos\psi)\sin\theta, \qquad z = r\sin\psi$$



Checking that this image, in the subspace topology, is homeomorphic to T^2 is a genuine computation, made routine once a smooth structure is available.

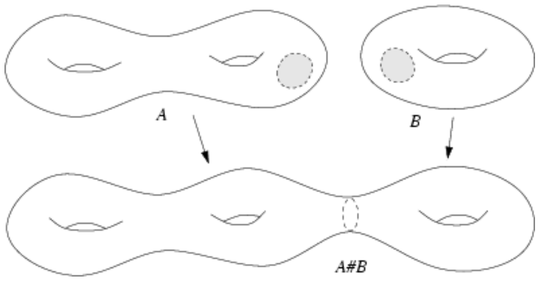
8. The triangle, being homeomorphic to the circle, is a 1-manifold in the topology induced by that homeomorphism.

9. Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}^k$ continuous. The graph

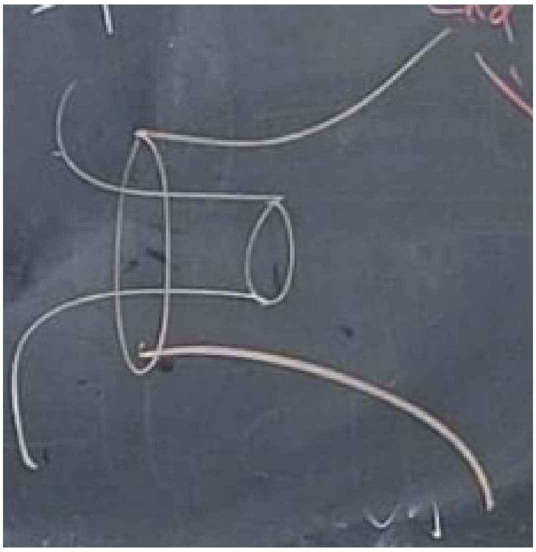
$$\Gamma(f) = \left\{(x,y) \in \mathbb{R}^n \times \mathbb{R}^k \;\middle|\; y = f(x)\right\}$$

   with the subspace topology is a topological n-manifold. The projection $\pi\colon \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ restricts to a continuous $\varphi\colon \Gamma(f) \to U$ with continuous inverse $x \mapsto (x, f(x))$, so φ is a homeomorphism exhibiting $\Gamma(f) \cong U$ as a single chart. A space that is locally a graph in this sense is thus a manifold; the converse, that every manifold is locally a graph, is settled once smooth structures and the implicit function theorem are available.

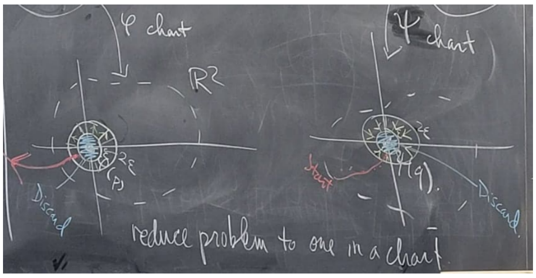
10. Two n-manifolds M_1 and M_2 can be joined at chosen points $p \in M_1$, $q \in M_2$ into their *connected sum* $M_1 \# M_2$.



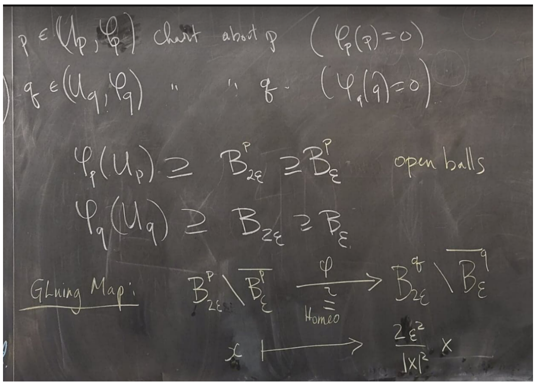
   A small chart ball is deleted around each point and the two resulting collars are glued together.



   The gluing is carried out through the charts,



by an explicit identification,



giving the definition

$$X\#Y = \left(X \setminus \overline{\varphi_p^{-1}(B_\epsilon)}\right) \coprod \left(Y \setminus \overline{\varphi_q^{-1}(B_\epsilon)}\right) \Big/ \left(\varphi_p^{-1}(B_{2\epsilon}) \ni x \sim \varphi_q^{-1}(\psi(\varphi_p(x))) \in \varphi_q^{-1}(B_{2\epsilon})\right)$$

The result is independent of the chosen points and charts, though it can depend on the orientation of the gluing; independence rests on the annulus theorem. Connected sum is the operation behind the classification of surfaces: every closed connected surface is the sphere S^2, a connected sum of tori T^2, or a connected sum of projective planes $\mathbb{R}\mathrm{P}^2$.

11. The figure below is not a topological manifold: the crossing point has no neighbourhood homeomorphic to any $\mathbb{R}^n$.

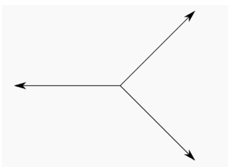


Figure 1.1: The crossing point is not locally Euclidean.

The dimension of a nonempty topological manifold is well-defined: an n-manifold cannot be homeomorphic to an m-manifold for $m \neq n$, which is why the definition fixes no particular Euclidean model, and on a connected manifold the dimension is constant. This is a theorem of Brouwer, proved through the homology of spheres in algebraic topology; it is recorded and cited here, not proved.

Theorem 1.1.13: Invariance of Domain

Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}^n$ continuous and injective. Then $f(U)$ is open in $\mathbb{R}^n$ and $f\colon U \to f(U)$ is a homeomorphism. In particular, if a nonempty open subset of $\mathbb{R}^n$ is homeomorphic to an open subset of $\mathbb{R}^m$, then $n = m$; and a homeomorphism between nonempty manifolds $M^m \to N^n$ forces $m = n$ on each connected component.

***Proof*:**
Invariance of domain is a theorem of algebraic topology, read off from the homology of spheres; its proof is given in *Algebraic Topology* [7]. The consequences are formal. Suppose $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ are nonempty, open, and homeomorphic, with $n \leq m$. Let $h\colon U \to V$ be the homeomorphism. Composing its inverse with the inclusion $\mathbb{R}^n \hookrightarrow \mathbb{R}^m$ onto the first n coordinates gives an injective continuous map $V \to \mathbb{R}^m$ on the open set $V \subseteq \mathbb{R}^m$; by invariance of domain its image is open in $\mathbb{R}^m$. But that image is $U \times \{0\} \subseteq \mathbb{R}^n \times \{0\}$, which has empty interior in $\mathbb{R}^m$ unless $n = m$, so $n = m$. For manifolds, a homeomorphism carries a chart around p to a chart around $f(p)$, matching an open subset of $\mathbb{R}^m$ with one of $\mathbb{R}^n$; the local dimensions therefore agree, and on a connected manifold the dimension is constant, completing the proof.

If the manifold has a differentiable structure (as we'll define in section 1.2.1), it is much easier to show that M must a constant dimension n using this structure (as we'll soon show).

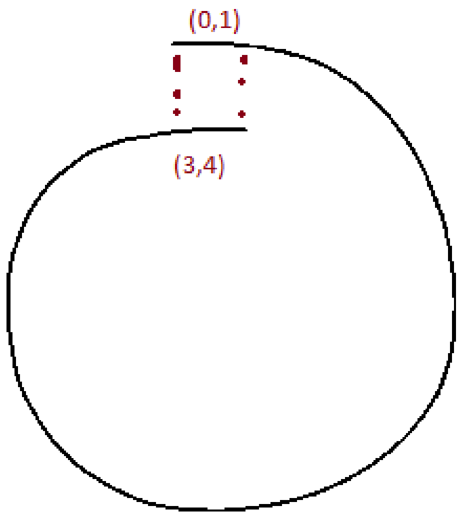
Exercise 1.1.1

1. Let M_1 and M_2 be topological manifolds of dimensions n_1 and n_2. Show that $M_1 \times M_2$ with the product topology is a topological manifold of dimension $n_1 + n_2$, verifying each clause of definition 1.1.5.
2. Show that an open subset U of a topological n-manifold is a topological n-manifold. Deduce that $\mathrm{GL}_n(\mathbb{R})$ is a topological manifold, and give its dimension.
3. Show that a topological manifold is locally path-connected, and conclude that its connected components are open and are themselves path-connected manifolds.
4. For the line with two origins (example 1.1.6), exhibit a compact subset that is not closed, and explain why this alone shows the space is not Hausdorff.

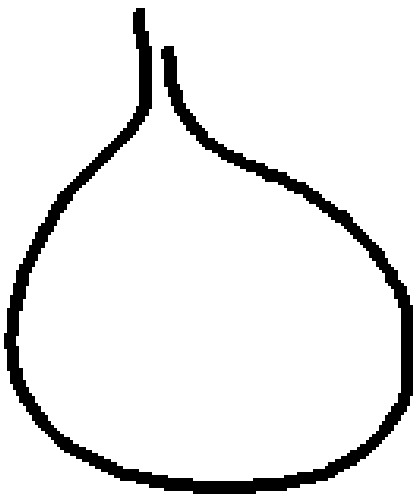
1.1.1 Constructing Manifolds Using Quotient

A manifold can be assembled from open pieces of $\mathbb{R}^n$ glued along open subsets, exactly as an atlas presents an existing one. The construction is worth isolating, since it produces manifolds that are not handed over as subspaces of a Euclidean space.

Consider first a single gluing. Let $B = (0, 4)$ and identify $(0, 1)$ with $(3, 4)$ through $\varphi(x) = x + 3$, forming $M = B/(x \sim \varphi(x))$ with the quotient topology [5].



The result is homeomorphic to S^1. The homeomorphism condition on φ is not by itself enough: gluing instead by $x \mapsto 4 - x$ brings the points 1 and 3 arbitrarily close without identifying them,



and the quotient fails to be Hausdorff, in the manner of the line with two origins (example 1.1.6). A gluing that remains a manifold must guarantee that leaving one chart's domain means entering another's; the general construction enforces this by a closedness condition on the gluing graphs, alongside the cocycle conditions that make the identifications consistent.

Proposition 1.1.14: Gluing Construction of a Manifold

Let $\{U_i\}_{i\in I}$ be a countable family of open subsets of $\mathbb{R}^n$; for each pair (i,j) let $U_{ij} \subseteq U_i$ be open and $\varphi_{ij}\colon U_{ij} \to U_{ji}$ a homeomorphism, satisfying

1. $U_{ii} = U_i$ and $\varphi_{ii} = \mathrm{id}$;
2. $\varphi_{ji} = \varphi_{ij}^{-1}$;
3. $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap \varphi_{ij}^{-1}(U_{jk})$;

and suppose each graph $\{(x, \varphi_{ij}(x)) : x \in U_{ij}\}$ is closed in $U_i \times U_j$. Then

$$M = \left(\coprod_{i\in I} U_i\right) \Big/ (U_{ij} \ni x \sim \varphi_{ij}(x) \in U_{ji})$$

with the quotient topology is a topological n-manifold, and the maps $U_i \to M$ are charts forming an atlas.

Readers who know some algebraic geometry will recognize this as the gluing of a sheaf; the same data assembles schemes from affine pieces [6].

***Proof*:**
Write $q\colon \coprod_i U_i \to M$ for the quotient map and $q_i = q|_{U_i}$.

Step 1: each q_i is an open embedding. Conditions (1) and the cocycle condition (3) make the generated relation restrict to the identity on each U_i, so q_i is injective. It is open: for $V \subseteq U_i$ open, $q^{-1}(q(V)) = \bigsqcup_j \varphi_{ij}(V \cap U_{ij})$, a union of open sets by (2), so $q(V)$ is open. A continuous open injection is an embedding; the images $q_i(U_i)$ are open and cover M, so M is locally Euclidean of dimension n with the q_i as charts.

Step 2: second countable. The index set is countable and each U_i is second countable, so M is covered by countably many charts and is second countable.

Step 3: Hausdorff. Take distinct points of M. If they share a chart they separate as in $\mathbb{R}^n$. Otherwise represent them by $x \in U_i$, $y \in U_j$ with $q(x) \neq q(y)$, so (x,y) is not on the graph of φ_{ij}; as that graph is closed, some basic box $V \times W$ about (x,y) misses it, and then $q_i(V)$ and $q_j(W)$ are disjoint neighbourhoods of the two points. So M is Hausdorff, completing the proof.

Example 1.1.15: The Circle by Gluing Two Lines
The circle is built by gluing two copies of $\mathbb{R}$ along $\mathbb{R}^\times$. The stereographic charts of S^1 have transition map $x \mapsto x^{-1}$ on $\mathbb{R}^\times$, so

$$S^1 = (\mathbb{R} \sqcup \mathbb{R})/(\mathbb{R}^\times \ni x_1 \sim x_2^{-1} \in \mathbb{R}^\times)$$

realizes S^1 through the construction, with two charts and the single transition $x \mapsto x^{-1}$. The same pattern with $\mathbb{R}^n$ and inversion $u \mapsto u/\|u\|^2$ builds S^n from two charts.

Exercise 1.1.2

1. Following the circle example, present S^n as a gluing of two copies of $\mathbb{R}^n$ along $\mathbb{R}^n \setminus \{0\}$ by the inversion $u \mapsto u/\|u\|^2$, and verify the closed-graph condition of proposition 1.1.14.
2. For the gluing of $B = (0, 4)$ by $\varphi(x) = 4 - x$ on $(0, 1) \to (3, 4)$, show that the graph of φ is not closed in $(0, 4) \times (0, 4)$, and identify the pair of points of the quotient with no disjoint neighbourhoods.
3. Show that conditions (1)-(3) of proposition 1.1.14 are exactly what make the relation generated by $x \sim \varphi_{ij}(x)$ an equivalence relation.

1.1.2 Topological Manifolds with Boundary

Many familiar shapes are locally Euclidean everywhere except along an edge, where a neighbourhood looks like a half-space rather than all of $\mathbb{R}^n$. The closed disk $D^2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ is the model case: its interior points have $\mathbb{R}^2$-neighbourhoods, but a point of the bounding circle does not. Admitting such an edge widens the theory to manifolds with boundary.

Definition 1.1.16: Topological Manifold with Boundary

Let $\mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$ be the closed upper half-space, with $\partial\mathbb{H}^n = \{x \in \mathbb{H}^n \mid x_n = 0\} \cong \mathbb{R}^{n-1}$. A Hausdorff, second countable space M is a *topological n-manifold with boundary* if every point has an open neighbourhood homeomorphic to an open subset of $\mathbb{H}^n$. A point of M is a *boundary point* if some chart carries it into $\partial\mathbb{H}^n$, and an *interior point* otherwise; the boundary points form ∂M and the interior points $\operatorname{Int} M$.

Constraining any single coordinate on either side gives a homeomorphic model, so the choice $x_n \geq 0$ is only a convention. The classification of a point as interior or boundary is, crucially, independent of the chart used to test it.

Proposition 1.1.17: The Boundary Is a Manifold

Let M be a topological n-manifold with boundary. Then ∂M and $\operatorname{Int} M$ partition M; $\operatorname{Int} M$ is an n-manifold, and ∂M is an $(n-1)$-manifold without boundary.

***Proof*:**
An interior point of $\mathbb{H}^n$ (one with $x_n > 0$) has a neighbourhood inside the open set $\{x \mid x_n > 0\}$, hence a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$; a point with $x_n = 0$ has none, since every neighbourhood in $\mathbb{H}^n$ meets $\partial\mathbb{H}^n$, which is not open in $\mathbb{R}^n$. Invariance of domain (theorem 1.1.13) makes this intrinsic: a homeomorphism between open subsets of $\mathbb{H}^n$ must send interior points to interior points, as an interior point's $\mathbb{R}^n$-neighbourhood has open image. So membership in ∂M or $\operatorname{Int} M$ does not depend on the chart, and the two sets partition M. Charts with $x_n > 0$ present $\operatorname{Int} M$ as an n-manifold; restricting boundary charts to $\partial\mathbb{H}^n \cong \mathbb{R}^{n-1}$ presents ∂M as an $(n-1)$-manifold, and it has empty boundary because $\partial\mathbb{H}^n$ does, completing the proof.

Example 1.1.18: Manifolds with Boundary

1. The closed disk $D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ is an n-manifold with boundary $\partial D^n = S^{n-1}$.
2. The interval $[0, 1]$, with the subspace topology, is a 1-manifold with boundary $\{0, 1\}$.
3. A product of two manifolds with boundary need not be one: the square $[0, 1] \times [0, 1]$ has corners where a neighbourhood looks like a quadrant, not a half-space. But the product of a boundaryless manifold X with a manifold-with-boundary Y is again a manifold with boundary, and $\partial(X \times Y) = X \times \partial Y$.

The manifold boundary ∂M is not the topological boundary of M inside an ambient space: the two coincide for the disk in $\mathbb{R}^2$, but the topological boundary depends on the embedding, whereas ∂M is intrinsic to M.

Exercise 1.1.3

1. Show that $\{x \in \mathbb{H}^n \mid x_n > 0\}$ is open in $\mathbb{R}^n$ while no neighbourhood of a point with $x_n = 0$ is homeomorphic to an open subset of $\mathbb{R}^n$; deduce from invariance of domain that a homeomorphism of open subsets of $\mathbb{H}^n$ preserves $\partial\mathbb{H}^n$.
2. Identify the boundary of the cylinder $S^1 \times [0, 1]$ and of the solid torus $S^1 \times D^2$, with their dimensions.
3. Show that gluing two copies of a manifold-with-boundary M along ∂M by the identity yields a manifold without boundary, the *double* of M, and identify the double of D^n.

1.2 Differentiable Manifolds

Topological manifolds are simply a topological space with some extra conditions. We shall now work with manifolds where we add an additional *differential structure* It shall be shown that the same manifold may have different differential structures, hence the differential structure will be considered as part of the information of a differentiable manifold[2]. In this section, we define a differential structure and define the types of maths between differential manifolds that are needed to form a category.

In most of this section, we shall be working with infinitely differentiable structures, i.e. *smooth structures*. This shall make no difference since in this section they are essentially the same, where the smooth structure is in fact simpler since there is no worry about keeping track how many times we "differentiate".

1.2.1 Smooth Charts and Smooth Structures

A topological manifold looks locally like $\mathbb{R}^n$; a *differentiable* manifold adds the data needed to do calculus on it. That data is an atlas whose charts overlap smoothly, so that differentiability transported from $\mathbb{R}^n$ through one chart agrees with the notion transported through any other. The

[2] Though as we shall see, many common manifolds shall have a unique differential structure up to isomorphism

same space can carry different such atlases, so the atlas is part of the data, not optional as it was in the topological case. The construction is set up on a bare set: the atlas *induces* a topology, on which the two global manifold conditions are then imposed.

Definition 1.2.1: (Set) Chart

A *chart* on a set M is a bijection $\varphi\colon U \to \varphi(U)$ from a subset $U \subseteq M$ onto an open subset of $\mathbb{R}^n$, written (U, φ); the integer n is its dimension. The domain U is the *patch* and φ the *coordinate map*, with components $\varphi = (x^1, \ldots, x^n)$.

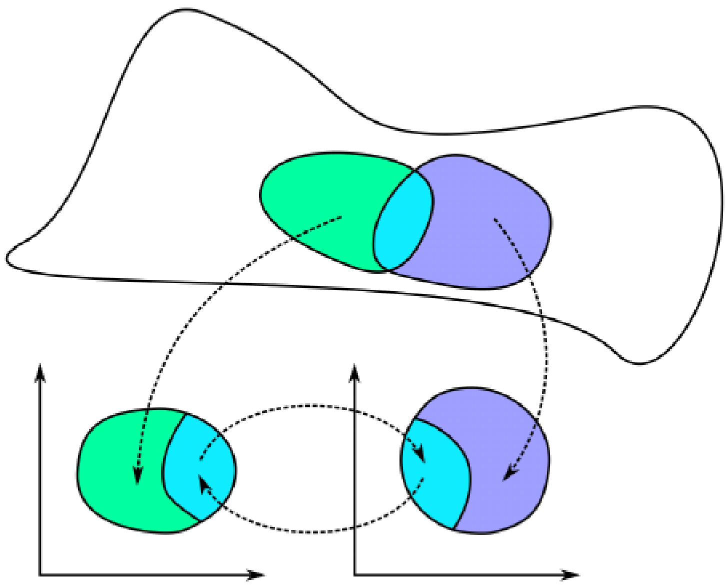
Definition 1.2.2: C^r Atlas

A C^r *atlas* of dimension n on a set M is a collection of n-charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ such that

1. the patches cover M, $\bigcup_\alpha U_\alpha = M$;
2. $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n$ for all α, β;
3. for all α, β with $U_\alpha \cap U_\beta \neq \emptyset$, the *transition map* $\varphi_\beta \circ \varphi_\alpha^{-1}\colon \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is a C^r diffeomorphism.

When $r = \infty$ the atlas is *smooth* and its charts are smooth charts.

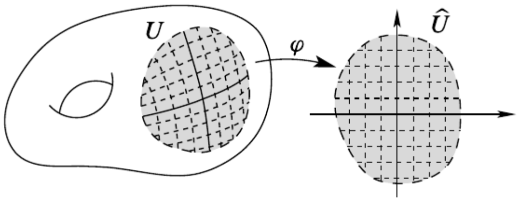
Constraining the transition maps differently selects a different category: real analytic (C^ω) if they are required analytic, and, when $n = 2m$ so that $\mathbb{R}^n \cong \mathbb{C}^m$, complex analytic if they are required holomorphic. The transition maps carry all the compatibility information and are what one computes with.



For charts (U, φ) with $\varphi = (x^1, \ldots, x^n)$ and (V, ψ) with $\psi = (y^1, \ldots, y^n)$, expressing $y^i \circ \varphi^{-1}$ as a function of $(x^1, \ldots, x^n)$ and dropping the composition gives

$$y^i = y^i(x^1, \ldots, x^n) \tag{1.1}$$

leaving deliberately ambiguous whether the x^j are numbers or coordinate functions; the same double reading appeared for the torus parameterization in example 1.1.12. A chart thus temporarily identifies its patch with a piece of $\mathbb{R}^n$, laying a coordinate grid over the set.



Example 1.2.3: Atlases on the Sphere

Regard $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ as a bare set. Split it into the four open arcs U, D, L, R (up, down, left, right)

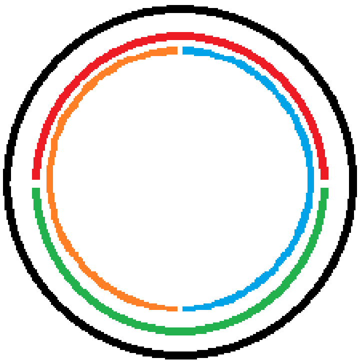


Figure 1.2: Four arcs covering the circle.

and project each to a coordinate axis: $\varphi_U(x, y) = \varphi_D(x, y) = x$ and $\varphi_L(x, y) = \varphi_R(x, y) = y$. These four injective maps are charts. Since $U \cap D = L \cap R = \emptyset$, the only overlaps are of the form $U \cap L$, where $\varphi_L^{-1}(y) = (-\sqrt{1 - y^2}, y)$ gives the transition $\varphi_U \circ \varphi_L^{-1}(y) = -\sqrt{1 - y^2}$, smooth on the open interval $\varphi_L(U \cap L) = (0, 1)$; the remaining overlaps differ only by signs. So the four charts form a smooth atlas.

The same sphere carries many atlases. The most economical uses the two stereographic charts from the poles N and S,

$$\varphi_N \colon S^n \setminus \{N\} \to \mathbb{R}^n, \quad \varphi_N(\vec{x}, t) = \frac{\vec{x}}{1 - t}, \qquad \varphi_S \colon S^n \setminus \{S\} \to \mathbb{R}^n, \quad \varphi_S(\vec{x}, t) = \frac{\vec{x}}{1 + t}$$

whose domains cover S^n. On the overlap, which φ_N maps onto $\mathbb{R}^n \setminus \{0\}$, the identity $\|\vec{x}\|^2 = 1 - t^2$

gives the transition
$$\varphi_S \circ \varphi_N^{-1}(u) = \frac{u}{\|u\|^2}$$
inversion in the unit sphere, smooth away from 0. For $n = 1$ this is $x \mapsto x^{-1}$, the transition already met in example 1.1.15.

Different atlases on a set can define the same differentiable structure, precisely when their charts overlap smoothly with one another.

Definition 1.2.4: Compatible Atlases

A chart (U, φ) is *compatible* with a C^r atlas $\mathcal{A}$ if $\mathcal{A} \cup \{(U, \varphi)\}$ is again a C^r atlas. Two atlases $\mathcal{A}$, $\mathcal{B}$ are compatible if $\mathcal{A} \cup \mathcal{B}$ is a C^r atlas.

Compatibility is an equivalence relation on atlases (exercise 1.2.1 (1)), so the atlases on a set fall into classes, each defining one differentiable structure.

Example 1.2.5: Compatible and Incompatible Atlases

1. The projection and stereographic atlases on S^1 from example 1.2.3 are compatible: their mutual transition maps are smooth (exercise 1.2.1 (2)).
2. On the set $M = \mathbb{R}$, the one-chart atlases $(\mathbb{R}, \mathrm{id})$ and $(\mathbb{R}, x \mapsto x^3)$ are *not* compatible: the transition $\mathrm{id} \circ (x \mapsto x^3)^{-1} = x^{1/3}$ is not differentiable at 0. They define two different smooth structures on the set $\mathbb{R}$.

Because compatible charts can always be adjoined, each class has a largest representative.

Definition 1.2.6: Maximal Atlas

The *maximal atlas* $\overline{\mathcal{A}}$ of a C^r atlas $\mathcal{A}$ is the collection of all charts compatible with $\mathcal{A}$. No atlas properly contains it, and it is the union of the compatibility class of $\mathcal{A}$.

Definition 1.2.7: Differentiable Structure

A C^r *differentiable structure* on a set M is a maximal C^r atlas $\mathcal{A}$, equivalently a compatibility class $[\mathcal{A}]$; when $r = \infty$ it is a *smooth structure*. Working with the class rather than a fixed atlas is what lets a structure be checked on a small, convenient atlas: for S^1 the two stereographic charts suffice.

A differentiable structure carries no topology a priori. The atlas supplies one.

Lemma 1.2.8: An Atlas Induces a Topology

Let M carry a C^r atlas $\mathcal{A}$ with maximal atlas $\overline{\mathcal{A}}$. The patches of $\overline{\mathcal{A}}$ form a basis for a topology on M, the *topology induced by the C^r structure*, in which every chart is a homeomorphism onto its open image. Any subatlas of $\overline{\mathcal{A}}$ induces the same topology.

***Proof*:**
A family of subsets is a basis for a topology when it covers M and, for members B_1, B_2 and $p \in B_1 \cap B_2$, some member B_3 satisfies $p \in B_3 \subseteq B_1 \cap B_2$. The patches cover M by the atlas axiom. Given charts $(U, \varphi), (V, \psi) \in \overline{\mathcal{A}}$ and $p \in U \cap V$, the atlas axiom makes $\varphi(U \cap V)$ open in $\mathbb{R}^n$, so $(U \cap V, \varphi|_{U \cap V})$ is a chart; its transition maps against any chart of $\overline{\mathcal{A}}$ are restrictions of transition maps of $\overline{\mathcal{A}}$, hence C^r, so it is compatible and $U \cap V$ is itself a patch of $\overline{\mathcal{A}}$, giving $p \in U \cap V \subseteq U \cap V$. The patches are therefore a basis. In the induced topology a set is open exactly when its image under each chart is open, so every chart is a homeomorphism onto its image; and a subatlas has the same patches up to this basis relation, hence the same topology, completing the proof.

With a topology in hand, the two global manifold conditions become conditions on the atlas.

Proposition 1.2.9: When the Induced Topology Is a Manifold

Let $(M, \mathcal{A})$ be a C^r structure with its induced topology.

1. If any two points of M lie in disjoint patches, M is Hausdorff.
2. If $\mathcal{A}$ has a countable subatlas, M is second countable.

When both hold, M is a topological manifold.

***Proof*:**

1. Disjoint patches are disjoint open sets, so two points lying in such patches are separated.
2. A countable subatlas covers M by countably many patches, each homeomorphic to an open subset of $\mathbb{R}^n$ and so second countable; a countable union of second countable open subspaces is second countable.

Under both hypotheses M is locally Euclidean by construction, Hausdorff and second countable by (1) and (2), and locally compact because a chart carries the compact closed balls of $\mathbb{R}^n$ into M. A second countable, locally compact Hausdorff space is paracompact [5], so M meets every clause of the definition of a topological manifold, completing the proof.

Definition 1.2.10: Differentiable Manifold

A C^r *manifold* of dimension n is a set M with a C^r structure whose induced topology is Hausdorff and second countable; by proposition 1.2.9 its underlying space is then a topological n-manifold. When $r = \infty$, M is a *smooth manifold*, and $\dim M$ denotes its dimension.

Unless stated otherwise, "manifold" means smooth manifold: Whitney showed that every maximal C^r atlas with $r \geq 1$ contains a compatible C^∞ atlas, so the smooth theory loses no generality. Requiring C^ω or holomorphic transitions instead gives a real-analytic manifold or a complex manifold.

Whether the topology alone fixes the smooth structure is a deep question. In low dimensions it does: manifolds of dimension at most 3 have a unique smooth structure, as does $\mathbb{R}^n$ for every $n \neq 4$. It fails in general, though: Kervaire and Milnor found that S^7 carries 28 distinct smooth structures, and Freedman's E_8 manifold is a topological 4-manifold with no smooth structure at all. Smoothness is genuinely additional data.

Definition 1.2.11: Product Manifold

For manifolds M, M' of dimensions n, n' with atlases $\{(U_\alpha, \varphi_\alpha)\}$ and $\{(V_\beta, \psi_\beta)\}$, the *product* $M \times M'$ is the $(n+n')$-manifold with atlas $\{(U_\alpha \times V_\beta,\ \varphi_\alpha \times \psi_\beta)\}$, where $(\varphi_\alpha \times \psi_\beta)(p,q) = (\varphi_\alpha(p), \psi_\beta(q))$.

A set may already carry a topology; the induced one agrees with it under the expected compatibility.

Proposition 1.2.12: The Induced Topology Matches a Given One

Let $(M, \mathcal{T})$ be a topological space carrying a C^r atlas $\mathcal{A}$. If every patch is $\mathcal{T}$-open and every chart is a $\mathcal{T}$-homeomorphism onto its image, then $\mathcal{T}$ is the topology induced by $\mathcal{A}$.

***Proof*:**
Write $\mathcal{T}'$ for the induced topology, with the patches as a basis. Each patch is $\mathcal{T}$-open by hypothesis, so $\mathcal{T}' \subseteq \mathcal{T}$. Conversely, let $O \in \mathcal{T}$ and $p \in O$, and choose a chart (U, φ) with $p \in U$. Then $O \cap U$ is $\mathcal{T}$-open, and since φ is a $\mathcal{T}$-homeomorphism $\varphi(O \cap U)$ is open in $\mathbb{R}^n$, so $(O \cap U, \varphi|_{O \cap U})$ is a chart and $O \cap U$ a patch, with $p \in O \cap U \subseteq O$. Thus O is a union of patches, hence $\mathcal{T}'$-open, giving $\mathcal{T} \subseteq \mathcal{T}'$. So $\mathcal{T} = \mathcal{T}'$, completing the proof.

Example 1.2.13: Smooth Manifolds
The topological manifolds of example 1.1.12, with the atlases given there, are smooth manifolds.

1. Any open subset U of a manifold M is a manifold, a *smooth open submanifold*, with the restricted charts. These are a rich source of examples: $\mathrm{GL}_n(\mathbb{R})$, open in the matrix space $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, is one.

2. A product of smooth manifolds is smooth (definition 1.2.11).

3. $\mathbb{R}^n$ is smooth with its single identity chart; the structures $(\mathbb{R}, [\mathrm{id}])$ and $(\mathbb{R}, [x^3])$ are distinct, though they will turn out isomorphic.

4. S^n is smooth by the atlas of example 1.2.3.

5. A group G that is also a smooth manifold, on which multiplication $G \times G \to G$ and inversion $G \to G$ are smooth, is a *Lie group*; the model cases are $\mathrm{GL}_n(\mathbb{R})$, the circle $S^1 \subseteq \mathbb{C}$ under multiplication, and $\mathbb{R}^n$ under addition. Lie theory proper is developed in [9].

6. For smooth $f\colon U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, the graph $\Gamma(f)$ is a smooth n-manifold, covered by the single chart $\pi\colon \Gamma(f) \to U$, $(x, f(x)) \mapsto x$. That a space is locally the graph of a smooth function characterizes smooth manifolds, as chapter 3 makes precise.

Exercise 1.2.1

1. Show that compatibility of C^r atlases (definition 1.2.4) is an equivalence relation.

2. Verify that the projection and stereographic atlases on S^1 (example 1.2.3) are compatible, hence define the same smooth structure.
3. Show that $x \mapsto x^3$ makes $\mathbb{R}$ a smooth manifold whose induced topology is the usual one, yet whose smooth structure differs from $(\mathbb{R}, [\mathrm{id}])$.
4. Let V be an n-dimensional real vector space. Show that a linear isomorphism $\mathbb{R}^n \to V$ is a global chart, and that the resulting smooth structure is independent of the isomorphism chosen.

1.2.2 Pertinent Examples of Manifolds

Points and Vector Spaces

The extremes are quick. A one-point space is a 0-manifold: its single chart maps it to $\mathbb{R}^0 = \{0\}$, and every clause holds vacuously. At the other end, any finite-dimensional real vector space V is a manifold. A linear isomorphism $E\colon \mathbb{R}^n \to V$ is a global chart (exercise 1.2.1 (4)), and the transition between the charts of two bases is the change-of-basis matrix, a linear isomorphism and hence a diffeomorphism; so the smooth structure is independent of the basis.

This already supplies many manifolds: the matrix space $M_{m\times n}(\mathbb{R})$, the space $\mathrm{Hom}_{\mathbb{R}}(V, W)$ of linear maps, and their open submanifolds. Chief among the latter is $\mathrm{GL}_n(\mathbb{R})$, open in $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$. The subgroups $\mathrm{SL}_n(\mathbb{R})$ and $\mathrm{O}(n)$ are *not* open, but will appear as regular submanifolds in chapter 3.

1.2.3 Level Sets

Let $\varphi\colon U \subseteq \mathbb{R}^N \to \mathbb{R}$ be smooth and $c \in \mathbb{R}$. The level set $M = \varphi^{-1}(c)$ is a manifold wherever φ is regular: if $D\varphi(a) \neq 0$ for each $a \in M$, the implicit function theorem writes M near a as a graph

$$x^i = f(x^1, \dots, x^{i-1}, x^{i+1}, \dots, x^N)$$

of a smooth f on an open subset of $\mathbb{R}^{N-1}$. As for the sphere, the graph charts are smooth and mutually compatible, so M is a smooth $(N-1)$-manifold. This is the special case of the regular value theorem developed in section 1.4.

Real Projective Space

The lines through the origin of a vector space form a manifold of their own, whose geometry is that of perspective, where parallel lines meet at infinity. The one-dimensional case is real projective space.

Definition 1.2.14: Real Projective Space

Real projective space is the quotient

$$\mathbb{R}\mathrm{P}^n := (\mathbb{R}^{n+1} \setminus \{0\})/\sim, \qquad x \sim y \iff x = \lambda y \text{ for some } \lambda \in \mathbb{R}^\times$$

with the quotient topology, equivalently the orbit space of the scaling action $\mathbb{R}^\times \curvearrowright (\mathbb{R}^{n+1} \setminus \{0\})$.

Every class has a representative on the unit sphere S^n, so $\mathbb{R}\mathrm{P}^n = q(S^n)$ is a continuous image of the compact S^n, hence compact and in particular paracompact. It is also Hausdorff, because the graph $\Gamma_\sim = \{(x, y) \mid x \sim y\}$ is closed: $x \sim y$ exactly when all the 2×2 minors $x_i y_j - x_j y_i$ vanish, so $\Gamma_\sim$ is the common zero set of finitely many continuous functions. A class is written in *homogeneous coordinates* $(x^0 : x^1 : \cdots : x^n) = [x]$, the notation recording that only the ratios are determined.

Proposition 1.2.15: Real Projective Space Is a Manifold

$\mathbb{R}\mathrm{P}^n$ is a smooth n-manifold.

***Proof*:**
Hausdorffness and paracompactness are shown above, so it remains to give a smooth atlas. For $0 \le i \le n$ let $U_i = \big\{(x^0 : \cdots : x^n) \;\big|\; x^i \neq 0\big\}$ and

$$\varphi_i \colon U_i \to \mathbb{R}^n, \qquad (x^0 : \cdots : x^n) \mapsto \left(\frac{x^0}{x^i}, \ldots, \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, \ldots, \frac{x^n}{x^i}\right)$$

which rescales the representative so the i-th coordinate is 1 and reads off the rest; for instance $\varphi_1(7 : 3 : 2) = (7/3, 2/3)$. Each φ_i is a bijection onto $\mathbb{R}^n$, and a homeomorphism for the quotient topology, and the U_i cover $\mathbb{R}\mathrm{P}^n$. The transition $\varphi_i \circ \varphi_j^{-1}$, defined on the set where the i-th coordinate is nonzero, again rescales to make that coordinate 1: a rational map whose only denominators are the nonvanishing i-th coordinate, hence smooth. So $(\mathbb{R}\mathrm{P}^n, \{(U_i, \varphi_i)\})$ is a smooth n-manifold, completing the proof.

Each U_i consists of the lines meeting the affine hyperplane $\big\{x \;\big|\; x^i = 1\big\} \cong \mathbb{R}^n$, and its complement is the projective space of the remaining hyperplane. This gives the cell decomposition

$$\mathbb{R}\mathrm{P}^n = \mathbb{R}^n \sqcup \mathbb{R}\mathrm{P}^{n-1} = \mathbb{R}^n \sqcup \mathbb{R}^{n-1} \sqcup \cdots \sqcup \mathbb{R}^0$$

as a set: an affine $\mathbb{R}^n$ together with the $\mathbb{R}\mathrm{P}^{n-1}$ "at infinity". The dimension of $\mathbb{R}\mathrm{P}^n$ is nonetheless the constant n.

Complex Projective Space

The same construction over $\mathbb{C}$ gives a manifold of twice the real dimension.

Definition 1.2.16: Complex Projective Space

Complex projective space is

$$\mathcal{P}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\sim, \qquad x \sim y \iff x = \lambda y \text{ for some } \lambda \in \mathbb{C}^\times$$

with the quotient topology.

Identifying $\mathbb{C}^{n+1} = \mathbb{R}^{2n+2}$, every class has a representative on the unit sphere S^{2n+1}, unique up to a unit scalar, so $\mathcal{P}^n = S^{2n+1}/\sim$ with $\|\lambda\| = 1$; Hausdorffness and paracompactness follow as for $\mathbb{R}\mathrm{P}^n$. The charts $U_i = \big\{(z^0 : \cdots : z^n) \;\big|\; z^i \neq 0\big\}$ with

$$\varphi_i \colon U_i \to \mathbb{C}^n \cong \mathbb{R}^{2n}, \qquad (z^0 : \cdots : z^n) \mapsto \left(\frac{z^0}{z^i}, \ldots, \frac{z^{i-1}}{z^i}, \frac{z^{i+1}}{z^i}, \ldots, \frac{z^n}{z^i}\right)$$

have holomorphic, hence smooth, transition maps exactly as in the real case. So $\mathcal{P}^n$ is a smooth manifold of real dimension $2n$, with the analogous cell decomposition $\mathcal{P}^n = \mathbb{C}^n \sqcup \mathbb{C}^{n-1} \sqcup \cdots \sqcup \{\mathrm{pt}\}$.

Grassmannians

Projective spaces are the lines through the origin; taking k-planes instead gives the Grassmannian.

Definition 1.2.17: Grassmannian

For an n-dimensional vector space V and $0 \leq k \leq n$, the *Grassmannian* $G_k(V)$ is the set of k-dimensional subspaces of V. When $V = \mathbb{R}^n$ one writes $G_k(\mathbb{R}^n) = G(k,n)$.

$G_k(V)$ is a smooth manifold of dimension $k(n-k)$, generalizing $G_1(\mathbb{R}^{n+1}) = \mathbb{R}\mathrm{P}^n$ and $G_1(\mathbb{C}^{n+1}) = \mathcal{P}^n$. To build charts, fix a splitting $V = P \oplus Q$ with $\dim P = k$, $\dim Q = n-k$. The graph of a linear map $T\colon P \to Q$,

$$\Gamma(T) = \{v + T(v) \mid v \in P\}$$

is a k-subspace meeting Q trivially; conversely a k-subspace S with $S \cap Q = \{0\}$ is the graph of the unique $T = \pi_Q \circ (\pi_P|_S)^{-1}$, where π_P, π_Q are the projections of V. So, writing $U_Q \subseteq G_k(V)$ for the subspaces meeting Q trivially, the map $\varphi\colon U_Q \to \mathrm{Hom}(P,Q) \cong \mathbb{R}^{k(n-k)}$, $\Gamma(T) \mapsto T$, is a bijection onto an open set.

Given a second splitting $V = P' \oplus Q'$ with chart φ', the overlap $\varphi(U_Q \cap U_{Q'})$ is the set of T whose graph meets Q' trivially. Writing $I_T(v) = v + T(v)$, so $\Gamma(T) = \mathrm{im}(I_T)$, this happens exactly when $\pi_{P'} \circ I_T$ is invertible; the entries of $\pi_{P'} \circ I_T$ are linear in T, so the condition $\det(\pi_{P'} \circ I_T) \neq 0$ is open, and $\varphi(U_Q \cap U_{Q'})$ is open. On it, the transition sends T to the T' with $\Gamma(T) = \Gamma(T')$, namely

$$T' = (\pi_{Q'} \circ I_T) \circ (\pi_{P'} \circ I_T)^{-1}$$

whose entries are rational in T with nonvanishing denominator, hence smooth. So $G_k(V)$ is a smooth $k(n-k)$-manifold.

Complex Manifolds

Requiring the charts to map to $\mathbb{C}^n$ with holomorphic transitions defines a *complex n-manifold*. Every complex manifold is in particular a real $2n$-manifold, since $\mathbb{C} \cong \mathbb{R}^2$ smoothly; $\mathcal{P}^n$ is the running example.

1.2.4 Smooth Manifolds with Boundary

The smooth counterpart of section 1.1.2 needs a notion of differentiability on the half-space $\mathbb{H}^n$, whose boundary points are not interior to $\mathbb{R}^n$.

Definition 1.2.18: Differentiable over the Half-Space

A map f on $\mathbb{H}^n$ is *smooth* if it is smooth on the interior $\mathrm{int}\,\mathbb{H}^n$ and all partial derivatives extend continuously to $\partial\mathbb{H}^n$; equivalently, if f extends to a smooth map on an open neighbourhood in $\mathbb{R}^n$.

A *smooth manifold with boundary* is then a manifold with boundary (definition 1.1.16) whose transition maps are diffeomorphisms in this sense. Boundary points are those a chart sends to $\partial\mathbb{H}^n$, and ∂M is a smooth $(n-1)$-manifold, as in the topological case. A collar gives the boundary a standard neighbourhood.

Theorem 1.2.19: Collar Neighbourhood Theorem

The boundary of a smooth manifold with boundary has a neighbourhood diffeomorphic to $\partial M \times [0,1)$, with ∂M corresponding to $\partial M \times \{0\}$.

The collar is what makes gluing along boundaries produce a smooth manifold: two copies of a bounded surface, such as a torus with an open disk removed, glue along their boundary circles into a closed surface. The theorem is proved by integrating an inward-pointing vector field, once flows are available (chapter 5).

1.2.5 Orientation

Definition 1.2.20: Oriented Manifold

A smooth manifold M is *oriented* by an atlas whose transition maps all have positive Jacobian determinant,

$$\det D(\psi \circ \varphi^{-1}) > 0$$

on every overlap; a maximal such atlas is an *orientation.* A manifold is *orientable* if it admits one.

The sign is a convention: composing every chart with a fixed orientation-reversing diffeomorphism turns a positive atlas into a negative one. What matters is that the charts be mutually consistent. A connected orientable manifold has exactly two orientations. The sphere S^2 is orientable; the Möbius band is not, since going around it reverses the sign of the Jacobian.

Exercise 1.2.2

1. Let M be a nonempty smooth manifold of dimension ≥ 1. Show that M carries uncountably many distinct smooth structures. (They need not be pairwise non-diffeomorphic; only the maximal atlases must differ.)

1.3 Smooth Maps

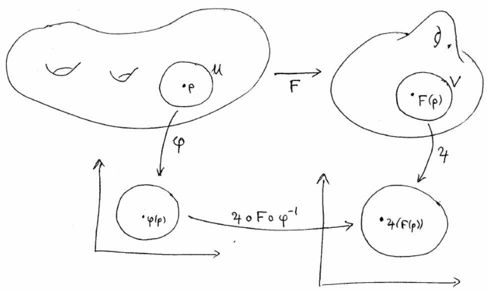
A differentiable structure lets a map be tested for differentiability through charts: read a map $f\colon M \to N$ in coordinates on both sides, and ask whether the resulting map of open subsets of Euclidean space is smooth. Because the charts overlap by diffeomorphisms, the answer does not depend on which charts are used.

Definition 1.3.1: Smooth Map

Let M, N be manifolds with C^r atlases, $f\colon M \to N$ a map, and $p \in M$. Then f is a C^r *map at* p (*smooth* when $r = \infty$) if there are a chart (U, φ) of M with $p \in U$ and a chart (V, ψ) of N with $f(p) \in V$ such that $f(U) \subseteq V$ and the *coordinate representative*

$$\psi \circ f \circ \varphi^{-1} \colon \varphi(U) \to \psi(V)$$

is C^r. It is a C^r *map* if it is one at every point.



The condition is chart-independent. If (U', φ') and (V', ψ') are other charts around p and $f(p)$ with $f(U') \subseteq V'$, then on the relevant overlap

$$\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$$

is a composite of the smooth representative with two transition diffeomorphisms, hence smooth. In particular a chart $\varphi\colon U \to \mathbb{R}^n$ is itself a smooth map, with smooth inverse – its representative $\mathrm{id} \circ \varphi \circ \varphi^{-1}$ is the identity – so a smooth manifold is *locally diffeomorphic* to $\mathbb{R}^n$, the differentiable analogue of local homeomorphism.

Definition 1.3.2: The Maps $C^r(M, N)$

For C^r manifolds M, N, write $C^r(M, N)$ for the set of C^r maps $M \to N$, and $C^r(M) = C^r(M, \mathbb{R})$.

Under pointwise operations $C^r(M)$ is an $\mathbb{R}$-algebra, and $C^r(M, \mathbb{C})$ a $\mathbb{C}$-algebra. When a map is read in coordinates on both sides, one writes $f(x^1, \ldots, x^n) = (y^1, \ldots, y^m)$ for the representative, thinking of f as a map of coordinates.

The first structural fact is that smoothness refines continuity.

Proposition 1.3.3: A Smooth Map Is Continuous

Every smooth map $f\colon M \to N$ is continuous.

***Proof*:**
Fix $p \in M$ and charts $(U,\varphi),(V,\psi)$ as in definition 1.3.1, so $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}$ is smooth, hence continuous. Then $f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$ is a composite of continuous maps, so f is continuous at p. As p was arbitrary, f is continuous, as we sought to show.

The clause $f(U) \subseteq V$ is what forces continuity; dropping it admits discontinuous maps that are smooth chart by chart.

Example 1.3.4: An "Almost Smooth" Discontinuous Map
The step function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = 1$ for $x \geq 0$ and 0 for $x < 0$, is discontinuous, hence not smooth by proposition 1.3.3. Take the identity charts $U = (-1,1)$ on the domain and $V = (\frac{1}{2}, \frac{3}{2})$ on the codomain. Here $f(U) = \{0,1\} \not\subseteq V$, so the clause $f(U) \subseteq V$ fails; but on the set $U \cap f^{-1}(V) = [0,1)$ where a representative is defined, $\psi \circ f \circ \varphi^{-1} \equiv 1$ is constant, hence smooth. Dropping the $f(U) \subseteq V$ requirement would thus admit this discontinuous f as "smooth chart by chart"; the clause is exactly what excludes it.

If instead one demands that *every* coordinate representative be smooth, the resulting notion agrees with definition 1.3.1 exactly for continuous maps.

Proposition 1.3.5: Equivalent Tests for Smoothness

Let $f\colon M \to N$ be continuous. Then f is smooth if and only if, for every chart (U,φ) of M and (V,ψ) of N, the representative

$$\psi \circ f \circ \varphi^{-1} \colon \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

(defined on the open set $U \cap f^{-1}(V)$) is smooth.

***Proof*:**
Suppose f is smooth and fix charts $(U,\varphi),(V,\psi)$. Near any $q \in U \cap f^{-1}(V)$ there are charts as in definition 1.3.1 on which f is smooth; the given representative differs from that one by transition diffeomorphisms, so is smooth near q, hence smooth. Conversely, assume the representative condition and fix p; choose (V,ψ) with $f(p) \in V$ and any (U,φ) with $p \in U$. By continuity $U \cap f^{-1}(V)$ is open and contains p; restricting φ to it gives a chart with image in V on which the representative is smooth. So f is smooth, completing the proof.

Smoothness is a local property, which is what lets maps be built piece by piece.

Proposition 1.3.6: Smoothness Is Local

A map $f\colon M \to N$ is smooth if and only if every point of M has a neighbourhood on which f restricts to a smooth map.

Corollary 1.3.7: Gluing Smooth Maps

Let $\{U_\alpha\}$ be an open cover of M and $f_\alpha\colon U_\alpha \to N$ smooth maps agreeing on overlaps, $f_\alpha|_{U_\alpha \cap U_\beta} = f_\beta|_{U_\alpha \cap U_\beta}$. Then there is a unique smooth $f\colon M \to N$ with $f|_{U_\alpha} = f_\alpha$.

Requiring the pieces to agree on overlaps is restrictive; partitions of unity (chapter 3) give a more flexible tool for assembling maps.

Proposition 1.3.8: Basic Smooth Maps

Let M, N, P be smooth manifolds.

1. Every constant map $M \to N$ is smooth.
2. The identity $\mathrm{id}\colon M \to M$ is smooth.
3. The inclusion $\iota\colon U \hookrightarrow M$ of an open submanifold is smooth.
4. If $f\colon M \to N$ and $g\colon N \to P$ are smooth, so is $g \circ f$.

***Proof*:**

1. In any charts the representative of a constant map is constant, hence smooth.
2. The representative of id in a chart and itself is the identity of an open subset of $\mathbb{R}^n$.
3. For U open in M, a chart of M restricts to a chart of U, and the representative of ι is the identity.
4. Given p, choose charts (U, φ) at p, (V, ψ) at $f(p)$ with $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}$ smooth, and (W, χ) at $g(f(p))$ with $g(V) \subseteq W$ and $\chi \circ g \circ \psi^{-1}$ smooth. Shrinking U so $f(U) \subseteq V \cap g^{-1}(W)$, the representative of $g \circ f$ is $(\chi \circ g \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1})$, a composite of smooth maps.

This completes the proof.

Note that smoothness of $g \circ f$ implies neither factor is smooth: a non-smooth bijection f with $f \circ f^{-1} = \mathrm{id}$ is a counterexample. Because composition preserves smoothness, the smooth manifolds and smooth maps form a category, written **Man**.

Proposition 1.3.9: Projections Are Smooth

Let $M_1, \ldots, M_k$ be manifolds. Each projection $\pi_i\colon M_1 \times \cdots \times M_k \to M_i$ is smooth, and a map $f\colon M \to M_1 \times \cdots \times M_k$ is smooth if and only if every $\pi_i \circ f$ is smooth.

***Proof*:**
In product charts $\varphi_1 \times \cdots \times \varphi_k$, the representative of π_i is the coordinate projection $\mathbb{R}^{n_1 + \cdots + n_k} \to \mathbb{R}^{n_i}$, which is smooth; so each π_i is smooth, and $\pi_i \circ f$ is smooth whenever f is. Conversely, in the same charts the representative of f has the representatives of the $\pi_i \circ f$ as its component blocks, and a map into a product of Euclidean spaces is smooth iff each component is; so f is smooth when all $\pi_i \circ f$ are, completing the proof.

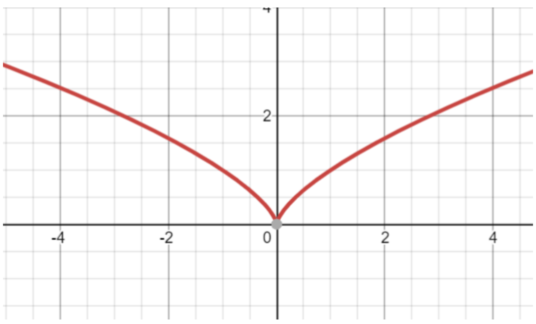
Example 1.3.10: Smooth Maps

1. Any map from a 0-manifold to a manifold is smooth, being locally constant.
2. The winding map $f\colon \mathbb{R} \to S^1$, $x \mapsto e^{2\pi i x}$, is smooth: in an angle chart its representative is $x \mapsto 2\pi x + c$. More generally $\mathbb{R}^n \to T^n$, $(x_1, \dots, x_n) \mapsto (e^{2\pi i x_1}, \dots, e^{2\pi i x_n})$, is smooth.
3. Complex conjugation $S^1 \to S^1$, $z \mapsto \bar{z}$, is smooth: in the two stereographic charts its representative is $a \mapsto 1/a$ from one pole and the identity between opposite poles.
4. The inclusion $S^n \hookrightarrow \mathbb{R}^{n+1}$ is smooth, as the stereographic representatives are rational and smooth.

An image of a manifold under a smooth map need not be a manifold – unlike the subobjects in many categories.

Example 1.3.11: The Image of a Smooth Map Need Not Be a Submanifold

The cuspidal cubic $f\colon \mathbb{R} \to \mathbb{R}^2$, $t \mapsto (t^2, t^3)$, is smooth (each component is a polynomial).



Its image is a topological manifold – f is a proper injection, hence a homeomorphism onto its image – but not a *smooth* submanifold of $\mathbb{R}^2$: f fails to be an immersion at 0 (as $f'(0) = 0$), and the image has a cusp there, so it is not locally a smooth graph. Requiring f to be an immersion does not by itself fix this: the line of irrational slope $g\colon \mathbb{R} \to T^2 = \mathbb{R}^2/\mathbb{Z}^2$, $t \mapsto [(t, \alpha t)]$ with α irrational, is an injective immersion whose image is dense in the torus, hence not a submanifold. The failure is repaired for injective maps out of *compact* manifolds, and is analyzed in general through the rank in chapter 3.

Derivations and Germs

Smooth maps deserve a notion of derivative. The full derivative of $f\colon M \to N$ is a linear map of tangent spaces, built in chapter 3; for a real-valued $f\colon M \to \mathbb{R}$ the coordinate partial derivatives are already available, read off through a chart.

Definition 1.3.12: Coordinate Derivative

Let (U, φ) be a chart of an n-manifold M with $\varphi = (x^1, \dots, x^n)$, and $f \in C^1(M)$. The *i-th coordinate derivative* of f is the function on U

$$\frac{\partial f}{\partial x^i}(p) := \partial_i(f \circ \varphi^{-1})(\varphi(p))$$

the ordinary i-th partial derivative of the representative $f \circ \varphi^{-1}$ at $\varphi(p)$.

If f is C^r then $\partial f/\partial x^i$ is C^{r-1}, and these derivatives inherit linearity and the Leibniz rule from their Euclidean counterparts. They depend on the chart: for a second chart (V, ψ) with $\psi = (y^1, \dots, y^n)$ the chain rule gives the change of variables

$$\frac{\partial f}{\partial y^i}(p) = \sum_{j=1}^{n} \frac{\partial f}{\partial x^j}(p)\,\frac{\partial x^j}{\partial y^i}(p) \tag{1.2}$$

This chart-dependence is exactly what the tangent space will resolve, by packaging the derivatives into a single intrinsic object.

Differentiability is local, so it is convenient to speak of maps defined only near a point, or on an arbitrary subset.

Definition 1.3.13: Locally C^r

Let $S \subseteq M$ be any subset of a smooth manifold and $f\colon S \to N$ continuous. Then f is *locally C^r* if every $s \in S$ has an open neighbourhood $O \subseteq M$ and a C^r map $\tilde{f}$ on O with $\tilde{f}|_{S\cap O} = f$.

Definition 1.3.14: Germs

For $p \in M$, call two smooth maps defined on neighbourhoods of p *equivalent* if they agree on some neighbourhood of p; an equivalence class is a *germ* at p, and the germs of maps into N form $C^\infty_p(M, N) = \left(\bigcup_{U \ni p} C^\infty(U, N)\right)/\sim$.

A germ has a well-defined value $[f](p) = f(p)$, and $C^\infty_p(M) = C^\infty_p(M, \mathbb{R})$ is an $\mathbb{R}$-algebra.

Diffeomorphisms

The isomorphisms of **Man** are the diffeomorphisms.

Definition 1.3.15: Diffeomorphism

A *diffeomorphism* is a smooth bijection $f\colon M \to N$ whose inverse is also smooth; then M and N are *diffeomorphic*, written $M \cong N$. The diffeomorphisms of M to itself form the group $\mathrm{Diff}(M)$.

Proposition 1.3.16: Characterizations of Diffeomorphisms

Let M, N be smooth manifolds with maximal atlases and $f\colon M \to N$ a bijection. The following are equivalent:

1. f is a diffeomorphism;
2. a chart (V, ψ) lies in the atlas of N iff $(f^{-1}(V), \psi \circ f)$ lies in that of M;
3. for every g, g is smooth on N iff $g \circ f$ is smooth on M.

***Proof*:**
(1) $\Rightarrow$ (3): if f and f^{-1} are smooth, then $g \circ f$ is smooth when g is (composition), and $g = (g \circ f) \circ f^{-1}$ is smooth when $g \circ f$ is. (3) $\Rightarrow$ (2): apply (3) to the coordinate maps, which are the smooth maps to $\mathbb{R}^n$ that are charts. (2) $\Rightarrow$ (1): condition (2) says f carries charts to charts, so both f and f^{-1} have identity representatives and are smooth. This completes the proof.

The content of (3) is that precomposing with a diffeomorphism f detects smoothness: $g \circ f$ is smooth exactly when g is. This holds only because f is invertible with smooth inverse; for a general smooth f, smoothness of $g \circ f$ says nothing about g (as proposition 1.3.8 already noted).

Example 1.3.17: Diffeomorphisms

1. The open ball and Euclidean space are diffeomorphic: $B(0,1) \to \mathbb{R}^n$, $x \mapsto x/\sqrt{1 - \|x\|^2}$, is smooth with smooth inverse $y \mapsto y/\sqrt{1 + \|y\|^2}$.
2. The two structures $(\mathbb{R}, [\mathrm{id}])$ and $(\mathbb{R}, [x^3])$ are distinct but *diffeomorphic*: the map $x \mapsto x^{1/3}$, read from the first to the second, has identity representative (the target chart x^3 composed with it is $x \mapsto x$), so it is a diffeomorphism. (Two copies of $(\mathbb{R}, [\mathrm{id}])$ joined by $x \mapsto x^3$ would not be, since $x^{1/3}$ is not differentiable at 0.) Distinct smooth structures can thus be isomorphic.
3. $\mathcal{P}^1$ is the Riemann sphere S^2. Both are covered by two charts glued by a single transition in a complex coordinate z: $z \mapsto 1/z$ for the projective charts of $\mathcal{P}^1$, while the stereographic charts of S^2 (example 1.2.3), under $\mathbb{C} \cong \mathbb{R}^2$, have transition $z \mapsto 1/\bar{z}$. The two agree after conjugating one chart, so matching them – identity on one, conjugation $z \mapsto \bar{z}$ on the other – is a diffeomorphism $\mathcal{P}^1 \cong S^2$. For $n \geq 1$, however, $\mathcal{P}^n \ncong \mathbb{R}\mathrm{P}^{2n}$: the transition Jacobians of $\mathcal{P}^n$ all have positive determinant, so $\mathcal{P}^n$ is orientable, whereas $\mathbb{R}\mathrm{P}^{2n}$ is not.

Whether a homeomorphism can always be improved to a diffeomorphism is dimension dependent. Up to dimension 3 it can, so those manifolds have a unique smooth structure. It fails in dimension 4: Donaldson and Taubes showed $\mathbb{R}^4$ carries uncountably many pairwise non-diffeomorphic smooth structures, while $\mathbb{R}^n$ for $n \neq 4$ has a unique one. In higher dimensions homeomorphic manifolds may still fail to be diffeomorphic, as the 28 smooth structures on S^7 show.

Proposition 1.3.18: Properties of Diffeomorphisms

1. A composite of diffeomorphisms is a diffeomorphism.
2. A product of diffeomorphisms $f_i \colon M_i \to N_i$ is a diffeomorphism of the product manifolds.
3. Every diffeomorphism is a homeomorphism.
4. A diffeomorphism restricts on an open submanifold to a diffeomorphism onto its image.
5. Being diffeomorphic is an equivalence relation.

***Proof*:**
(1), (4) follow from proposition 1.3.8 (composition and restriction of smooth maps are smooth), applied to f and f^{-1}; (2) from proposition 1.3.9, since a product map is smooth iff its components are; (3) because a diffeomorphism and its inverse are continuous bijections; (5) from (1), the identity being a diffeomorphism, and symmetry of the definition, completing the proof.

Invariance of dimension, deferred in the topological setting to algebraic topology (theorem 1.1.13), is elementary once smoothness is available.

Theorem 1.3.19: Invariance of Dimension for Diffeomorphisms

If $f \colon M \to N$ is a diffeomorphism of nonempty manifolds, then $\dim M = \dim N$.

***Proof*:**
In charts, f has a representative $\hat{f} \colon U \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ that is a diffeomorphism of open sets. Differentiating $\hat{f}^{-1} \circ \hat{f} = \mathrm{id}$ by the chain rule gives $D\hat{f}^{-1}(\hat{f}(a)) \circ D\hat{f}(a) = \mathrm{id}$, and likewise in the other order, so $D\hat{f}(a) \colon \mathbb{R}^n \to \mathbb{R}^m$ is a linear isomorphism, forcing $n = m$, completing the proof.

Corollary 1.3.20: Invariance of the Boundary

A diffeomorphism $f \colon M \to N$ of manifolds with boundary satisfies $f(\partial M) = \partial N$ and restricts to a diffeomorphism $\operatorname{Int} M \to \operatorname{Int} N$.

***Proof*:**
In charts f is a diffeomorphism between open subsets of $\mathbb{H}^n$. An interior point has a neighbourhood mapping to an open subset of $\mathbb{R}^n$, whose image under the diffeomorphism is again open, so an interior point maps to an interior point; applying the same to f^{-1} shows boundary maps to boundary. Hence $f(\partial M) = \partial N$ and f restricts to a diffeomorphism of interiors, completing the proof.

Finally, a bijection can carry a smooth structure across, the smooth analogue of transporting a topology.

Proposition 1.3.21: Transfer of Structure

Let M be a smooth manifold and $\varphi\colon M \to X$ a bijection of sets. There is a unique smooth structure on X making φ a diffeomorphism, obtained by declaring the charts of X to be $\varphi \circ \varphi^{-1}$ for charts φ of M.

If X already carries a topology and φ is a homeomorphism, the transferred structure induces that topology. This is how spaces with no evident smooth structure – a triangle, say, homeomorphic to S^1 – acquire one.

Exercise 1.3.1

1. Show that the quotient map $\mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n$ is smooth.
2. Show that $\mathbb{RP}^1 \cong S^1$, and locate the homeomorphism explicitly through the standard charts.
3. Show that the rotation $r_\theta(x, y, z) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta,\ z)$ is a diffeomorphism of S^2.
4. Give smooth manifolds and maps f, g with $g \circ f$ smooth but g not smooth, and explain why proposition 1.3.16 does not apply.

1.4 Regular Submanifolds

Manifolds should be closed under passing to well-behaved subsets. Restricting an atlas does not work directly – the charts of $\mathbb{R}^2$ do not restrict to charts of S^1, since no single chart covers the circle – so a submanifold is instead defined by requiring that, in suitable charts, it sits inside M the way $\mathbb{R}^k$ sits inside $\mathbb{R}^n$.

Definition 1.4.1: Regular Submanifold

A subset $S \subseteq M$ of an n-manifold is a *k-dimensional (regular) submanifold* if every $p \in S$ lies in a chart (U, φ) of M with

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$$

Such a chart is a *submanifold chart* (or *slice chart*) for S.

Proposition 1.4.2: A Submanifold Is a Manifold

A k-dimensional submanifold $S \subseteq M$ is a smooth k-manifold in the subspace topology, with charts $(U \cap S,\ \pi \circ \varphi|_{U\cap S})$ obtained from the submanifold charts by the projection $\pi\colon \mathbb{R}^n \to \mathbb{R}^k$ onto the first k coordinates; the inclusion $\iota\colon S \hookrightarrow M$ is smooth.

***Proof*:**
For submanifold charts $(U, \varphi), (V, \psi)$, the restricted charts of S have transition $\pi \circ \psi \circ \varphi^{-1} \circ \iota_k$, where $\iota_k(x) = (x, 0)$: the restriction of the smooth $\psi \circ \varphi^{-1}$ to a slice, hence smooth. So the restricted charts form a smooth atlas, and each is a homeomorphism onto an open piece of a slice, so the induced topology is the subspace topology. In a submanifold chart the representative of ι is $x \mapsto (x, 0)$, smooth, so ι is smooth, completing the proof.

Regular submanifolds are the subobjects of **Man**; henceforth "submanifold" means regular submanifold. A smooth map restricts to one.

Corollary 1.4.3: Restriction to a Submanifold

If $f: M \to N$ is smooth and $S \subseteq M$ a submanifold, then $f|_S = f \circ \iota: S \to N$ is smooth.

***Proof*:**
$f|_S = f \circ \iota$ is a composite of the smooth maps ι (proposition 1.4.2) and f, hence smooth, completing the proof.

The converse fails: a smooth map out of S need not extend to M (exercise 1.4.1 (3)).

Example 1.4.4: Submanifolds

1. The n-dimensional submanifolds of $\mathbb{R}^n$ are exactly its open subsets.
2. $\mathbb{R}^k \times \{0\} \subseteq \mathbb{R}^n$ is a k-submanifold, the standard coordinates giving a global submanifold chart.
3. $S^k \subseteq S^n$, the vanishing of the last $n-k$ coordinates, is a k-submanifold.
4. Every open subset of M is an n-submanifold.

Several equivalent tests recognize a subset of $\mathbb{R}^n$ as a submanifold, each convenient in a different situation.

Proposition 1.4.5: Characterizations of a Submanifold of $\mathbb{R}^n$

For $S \subseteq \mathbb{R}^n$, the following local conditions at each point are equivalent, and each exhibits S as a k-dimensional submanifold:

1. (*regular level set*) near the point $S = f^{-1}(0)$ for a C^r map f into $\mathbb{R}^{n-k}$ with Df of rank $n-k$;
2. (*graph*) near the point S is the graph of a C^r map $\mathbb{R}^k \to \mathbb{R}^{n-k}$, after permuting coordinates;
3. (*slice*) a C^r diffeomorphism carries S near the point onto an open piece of $\mathbb{R}^k \times \{0\}$;
4. (*embedding*) near the point S is the image of a C^r embedding of an open subset of $\mathbb{R}^k$.

1.4.6 Key Ideas The cycle (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1) and the equivalence (3) $\Leftrightarrow$ (4) run through the implicit and inverse function theorems: a regular level set is a graph by the implicit function theorem; a graph $y = g(x)$ is flattened by $(x, y) \mapsto (x, y - g(x))$; a slice is the zero set of the complementary projection; and the slice and embedding descriptions are mutually inverse. Replacing $\mathbb{R}^n$ by an arbitrary manifold, these become the rank theorem, proved in full in chapter 3, which is where the general submanifold theory is developed.

Quotients are the construction still missing. Some manifolds do arise as quotients – the Möbius band, for one – but a quotient of a manifold need not even be Hausdorff (example 1.1.6), and smooth structures behave badly under quotients in general. The conditions under which a quotient is a smooth manifold are taken up, through group actions, in chapter 4.

Exercise 1.4.1

1. Show that being a submanifold is transitive: if S is a submanifold of T and T of M, then S is a submanifold of M.
2. Show that every open subset of M is a regular submanifold of the same dimension.
3. Give a submanifold $S \subseteq M$ and a smooth $f : S \to N$ that does not extend to any smooth map $M \to N$.

1.5 Bump Functions and Partitions of Unity

Partitions of unity turn local data on a manifold into global data, and their existence in the smooth category is what sets smooth manifolds apart from complex or real-analytic ones. It rests on two inputs cited from upstream: the smooth bump function on $\mathbb{R}^n$, from [2], and the topological existence of partitions of unity on a paracompact space, from [5].

Definition 1.5.1: Support

The *support* of a function $f : X \to \mathbb{R}$ is the closure of the set on which it is nonzero,

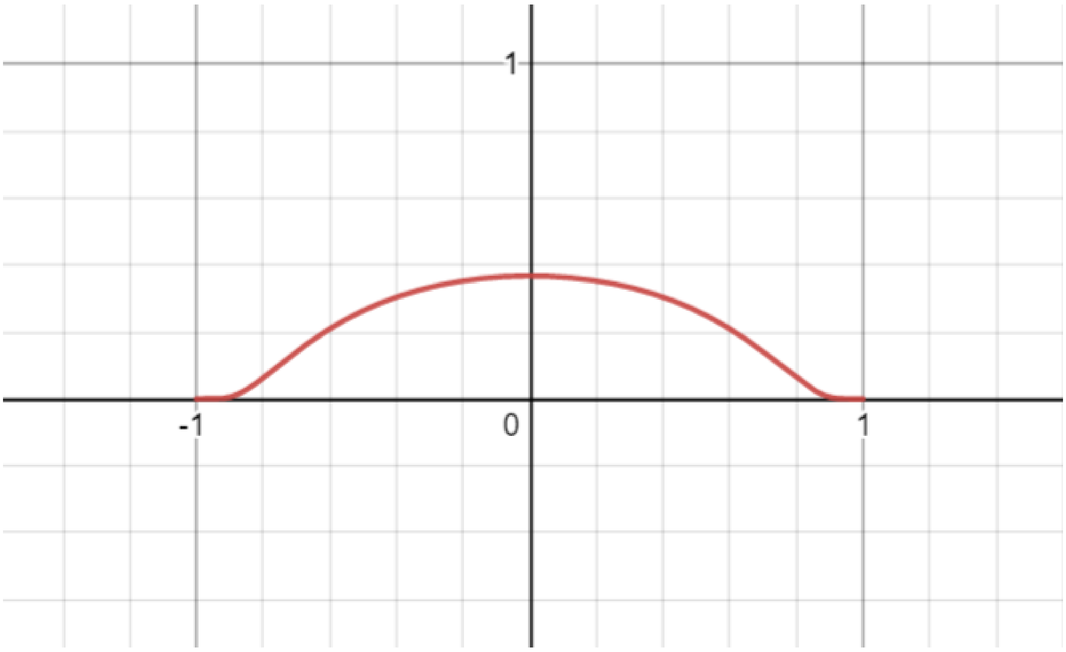
$$\operatorname{supp} f = \overline{\{x \in X \mid f(x) \neq 0\}}$$

Example 1.5.2: A Smooth Bump on $\mathbb{R}$

The function

$$\Phi(x) = \begin{cases} e^{-1/(1-x^2)} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

is smooth with support the closed unit interval, hence compactly supported.



This function and the general smooth bump on $\mathbb{R}^n$ are constructed in [2].

Definition 1.5.3: Partition of Unity Subordinate to a Cover

For an open cover $\{U_\alpha\}_{\alpha\in A}$ of a space M, a *partition of unity subordinate to it* is a family of continuous functions $\psi_\alpha\colon M \to [0,1]$ such that

1. $\text{supp}\psi_\alpha \subseteq U_\alpha$ for each α;
2. the supports are locally finite: every point has a neighbourhood meeting $\text{supp}\psi_\alpha$ for only finitely many α;
3. $\sum_{\alpha\in A} \psi_\alpha(x) = 1$ for every $x \in M$ (a finite sum near each point, by (2)).

The partition is *smooth* if every ψ_α is smooth.

On a paracompact Hausdorff space a continuous partition of unity subordinate to any open cover exists [5], and every manifold is such a space (theorem 1.1.8). The smooth refinement replaces those continuous functions by smooth ones built from bump functions.

Theorem 1.5.4: Existence of Smooth Bump Functions

Let M be a smooth manifold, $K \subseteq M$ compact, and $O \subseteq M$ open with $K \subseteq O$. There is a smooth $\beta\colon M \to [0,1]$ identically 1 on K and with compact support in O.

***Proof*:**
On $\mathbb{R}^n$, for concentric balls $B(0,r) \subseteq B(0,R)$ the smooth bump of [2] gives such a β, equal to 1 on $B(0,r)$ with support in $B(0,R)$; translating handles a ball about any point. For general M, cover the compact K by finitely many chart balls (U_i, φ_i) with $\overline{U_i} \subseteq O$, each carrying a bump β_i (through φ_i, extended by 0 off U_i) equal to 1 on a smaller compact K_i, with the interiors of the K_i still covering K. Then

$$\beta = 1 - \prod_i (1 - \beta_i)$$

is smooth, takes values in $[0,1]$, has support in $\bigcup_i U_i \subseteq O$, and equals 1 wherever some $\beta_i = 1$, in particular on K, completing the proof.

The bump functions upgrade the topological partition of unity to a smooth one; this is the result the rest of the geometry track consumes.

Theorem 1.5.5: Existence of Smooth Partitions of Unity

Every open cover of a smooth manifold admits a subordinate smooth partition of unity.

***Proof*:**
Let $\{U_\alpha\}$ cover M. By paracompactness (theorem 1.1.8) refine it to a locally finite cover $\{V_i\}$ of relatively compact chart balls with $\overline{V_i} \subseteq U_{\alpha(i)}$, and by the shrinking lemma [5] choose an open cover $\{W_i\}$ with $\overline{W_i} \subseteq V_i$. A smooth bump β_i (theorem 1.5.4) is 1 on $\overline{W_i}$ with support in V_i. Local finiteness makes $\sigma = \sum_i \beta_i$ smooth, and $\sigma > 0$ since the W_i cover M. Then $\psi_i = \beta_i / \sigma$ are smooth, sum to 1, and, gathered by α, form a smooth partition of unity subordinate to $\{U_\alpha\}$, completing the proof.

The existence of Riemannian metrics, of connections, and of global sections of vector bundles all reduce to this theorem downstream ([4], [8]). It is a genuinely C^∞ phenomenon: a nonzero real-analytic function cannot have compact support, so no analytic partition of unity subordinate to a proper cover exists (exercise 1.5.1 (3)).

The remaining corollaries separate and extend across closed sets.

Corollary 1.5.6: Smooth Bump on a Closed Set

For a closed $A \subseteq M$ and open $U \supseteq A$, there is a smooth $\psi\colon M \to [0,1]$ with $\psi \equiv 1$ on A and $\operatorname{supp}\psi \subseteq U$.

***Proof*:**
Take a smooth partition of unity $\{\psi_0, \psi_1\}$ subordinate to the cover $\{U, M \setminus A\}$ (theorem 1.5.5). Then $\operatorname{supp}\psi_0 \subseteq U$, and on A the term ψ_1 vanishes, its support avoiding A, so $\psi_0 = 1 - \psi_1 = 1$ there. Thus $\psi = \psi_0$ works, completing the proof.

Corollary 1.5.7: Smooth Urysohn Lemma

Disjoint closed sets A, B in a smooth manifold are separated by a smooth $f\colon M \to [0,1]$ with $f|_A = 0$ and $f|_B = 1$.

***Proof*:**
Apply corollary 1.5.6 to the closed set B inside the open set $M \setminus A$: the resulting f is 1 on B and, having support in $M \setminus A$, is 0 on A, completing the proof.

Corollary 1.5.8: Extending a Smooth Function off a Closed Set

Let $A \subseteq M$ be closed, $f\colon A \to \mathbb{R}^k$ smooth (definition 1.3.13), and $U \supseteq A$ open. There is a smooth $\tilde{f}\colon M \to \mathbb{R}^k$ with $\tilde{f}|_A = f$ and $\operatorname{supp}\tilde{f} \subseteq U$.

***Proof*:**
Smoothness of f on A gives a smooth extension F to an open $O \supseteq A$; shrink O into U. Let ψ be a smooth bump equal to 1 on A with $\operatorname{supp}\psi \subseteq O \cap U$ (corollary 1.5.6). Then $\tilde{f} = \psi F$, set to 0 off $\operatorname{supp}\psi$, is smooth on M, equals $F = f$ on A, and has support in U, completing the proof.

Corollary 1.5.9: A Closed Set as a Zero Set

Every closed $K \subseteq M$ is the zero set $f^{-1}(0)$ of a smooth $f\colon M \to [0, \infty)$.

***Proof*:**
The open complement $M \setminus K$ is a countable union of chart balls B_i; on each take a smooth $\beta_i \geq 0$ positive on B_i with support $\overline{B_i}$, scaled so that it and its derivatives up to order i are below 2^{-i}, so that $f = \sum_i \beta_i$ converges to a smooth function. Then $f > 0$ exactly on $\bigcup_i B_i = M \setminus K$, so $f^{-1}(0) = K$, completing the proof.

Exercise 1.5.1

1. Verify corollary 1.5.8 in the case $M = \mathbb{R}$, $A = \{0\}$, $f(0) = 1$: write down an explicit smooth extension supported in $(-1, 1)$.
2. Show the extension in corollary 1.5.8 can fail without closedness: exhibit a non-closed $A \subseteq \mathbb{R}$ and a smooth f on it with no smooth extension to $\mathbb{R}$.
3. Explain why no real-analytic partition of unity is subordinate to a proper open cover of $\mathbb{R}$.

2

Derivative and Tangent Structures

On $\mathbb{R}^n$ the derivative of a smooth map $f\colon U \to V$ at a point p is a linear map $Df(p)\colon \mathbb{R}^n \to \mathbb{R}^m$ best approximating f near p, and Df is itself a smooth map into the space of linear maps. This chapter builds the analogue on a manifold: to each $p \in M$ a vector space T_pM, the *tangent space*, and to each smooth $f\colon M \to N$ a linear $T_pf\colon T_pM \to T_{f(p)}N$ approximating f at p and varying smoothly with p. Two things must be constructed: the vector space T_pM itself – there is no ambient $\mathbb{R}^n$ whose affine subspaces could serve – and the object over which T_pf varies smoothly, the tangent bundle. The linear map also records dimensional information; it is what lets one control the rank of a smooth map, and so recognize when an image fails to be a submanifold (example 1.3.11), the theme of chapter 3.

2.1 Tangent Space

There is no single natural definition of T_pM; three constructions, each illuminating, turn out to agree. The *kinematic* one takes tangent vectors to be velocities of curves through p. The *chart* one takes them to be $\mathbb{R}^n$-vectors, one per chart, glued by the derivatives of the transition maps. The *algebraic* one takes them to be derivations of germs of functions at p – the definition that generalizes to algebraic geometry and is easiest to handle without coordinates. The three are canonically isomorphic (proposition 2.1.17); which to use is a matter of convenience.

A single mechanism underlies all three: each chart identifies a neighbourhood of p with a piece of $\mathbb{R}^n$, and the tangent space is $\mathbb{R}^n$ transported through that identification, the transition-map derivatives keeping the transport consistent. The rule

$$\overline{a}^i = \sum_{j=1}^{n} \frac{\partial \overline{x}^i}{\partial x^j}\, a^j$$

relating the components of one tangent vector in two charts is the classical *transformation law*.

The Kinematic Tangent Space

Two curves through p should determine the same tangent vector when they have the same velocity there, tested against every smooth function.

Definition 2.1.1: Tangent Curves

Curves $c_0, c_1 \colon I \to M$ with $c_0(0) = c_1(0) = p$ are *tangent at* p if $(f \circ c_0)'(0) = (f \circ c_1)'(0)$ for every smooth real-valued f defined near p.

This is an equivalence relation, and a class collects all curves through p with a common velocity.

Definition 2.1.2: Tangent Vector (via Curves)

The *tangent space* T_pM is the set of tangency classes at p; its elements are *tangent vectors*. A class is written $v_p = [c]$.

The test against real-valued functions is the same as the test against vector-valued ones.

Lemma 2.1.3: Scalar and Vector Tests Agree

Curves c_0, c_1 are tangent at p if and only if $(f \circ c_0)'(0) = (f \circ c_1)'(0)$ for every $\mathbb{R}^k$-valued smooth f near p.

***Proof*:**
If the curves are tangent, then for $f = (f^1, \dots, f^k)$ each component satisfies $(f^i \circ c_0)'(0) = (f^i \circ c_1)'(0)$, so $(f \circ c_0)'(0) = (f \circ c_1)'(0)$. Conversely, given the vector-valued equality, apply it to $f = (g, 0, \dots, 0)$ for an arbitrary real-valued g to recover $(g \circ c_0)'(0) = (g \circ c_1)'(0)$, completing the proof.

The class structure is geometric but supplies no vector-space structure directly. That comes by transporting the structure of $\mathbb{R}^n$ through a chart, consistently across charts.

Proposition 2.1.4: Consistent Transfer of a Linear Structure

Let S be a set and $\{b_\alpha \colon V_\alpha \to S\}_{\alpha \in A}$ a family of bijections from n-dimensional vector spaces. If every $b_\beta^{-1} \circ b_\alpha \colon V_\alpha \to V_\beta$ is a linear isomorphism, then S carries a unique vector-space structure making each b_α a linear isomorphism.

***Proof*:**
Transport the operations from one index: $s_1 + s_2 := b_\alpha(b_\alpha^{-1} s_1 + b_\alpha^{-1} s_2)$ and $a \cdot s := b_\alpha(a\, b_\alpha^{-1} s)$. The sum is independent of α: inserting $b_\beta b_\beta^{-1} = \mathrm{id}$ and using that $b_\alpha^{-1} b_\beta$ is linear,

$$b_\alpha(b_\alpha^{-1} s_1 + b_\alpha^{-1} s_2) = b_\alpha b_\alpha^{-1} b_\beta(b_\beta^{-1} s_1 + b_\beta^{-1} s_2) = b_\beta(b_\beta^{-1} s_1 + b_\beta^{-1} s_2)$$

and likewise for scalars. With these operations each b_α is a linear isomorphism, and any structure making some b_α linear must be this one, giving uniqueness and completing the proof.

For a chart (U, φ_α) at p, define $b_\alpha \colon \mathbb{R}^n \to T_pM$ by $v \mapsto [\gamma_v]$, where $\gamma_v(t) = \varphi_\alpha^{-1}(\varphi_\alpha(p) + tv)$ on a small interval about 0.

Lemma 2.1.5: Charts Transfer the Structure Consistently

Each $b_\alpha \colon \mathbb{R}^n \to T_pM$ is a bijection, and for a second chart (V, φ_β),

$$b_\beta^{-1} \circ b_\alpha = D(\varphi_\beta \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))$$

a linear isomorphism.

***Proof*:**
A chart recovers the defining vector: $(\varphi_\alpha \circ \gamma_v)'(0) = \frac{d}{dt}\big|_0(\varphi_\alpha(p) + tv) = v$.

Injectivity. If $b_\alpha(v) = b_\alpha(w)$ then $[\gamma_v] = [\gamma_w]$, so lemma 2.1.3 applied to the $\mathbb{R}^n$-valued φ_α gives $v = (\varphi_\alpha \circ \gamma_v)'(0) = (\varphi_\alpha \circ \gamma_w)'(0) = w$.

Surjectivity. Given $[c] \in T_pM$, set $v = (\varphi_\alpha \circ c)'(0)$. For any smooth f near p, the chain rule applied to $f \circ c = (f \circ \varphi_\alpha^{-1}) \circ (\varphi_\alpha \circ c)$ gives

$$(f \circ c)'(0) = D(f \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))\, v = (f \circ \gamma_v)'(0)$$

so c is tangent to γ_v and $b_\alpha(v) = [c]$.

Change of chart. Taking $f = \varphi_\beta$ in the surjectivity computation, $b_\beta^{-1}(b_\alpha(v)) = (\varphi_\beta \circ \gamma_v)'(0) = D(\varphi_\beta \circ \varphi_\alpha^{-1})(\varphi_\alpha(p))\, v$, a linear isomorphism since the transition map is a diffeomorphism, completing the proof.

By proposition 2.1.4 the maps b_α endow T_pM with a canonical $\mathbb{R}^n$-vector-space structure. This construction is the *kinematic tangent space*, $(T_pM)_{\text{kin}}$.

The Tangent Space via Charts

The chart construction records the same data – a vector per chart, glued by transition derivatives – without reference to curves.

Definition 2.1.6: Tangent Space via Charts

For a manifold M with maximal atlas $\mathcal{A}$ and $p \in M$, let Γ_p be the set of triples $(p, v, (U, \varphi))$ with $v \in \mathbb{R}^n$ and $(U, \varphi) \in \mathcal{A}$, $p \in U$. Declare

$$(p, v, (U, \varphi)) \sim (p, w, (V, \psi)) \iff w = D(\psi \circ \varphi^{-1})(\varphi(p))\, v$$

The quotient $(T_pM)_{\text{phys}} = \Gamma_p / \sim$ is the *tangent space via charts.*

Each chart gives a bijection $b_{(U,\varphi)} \colon \mathbb{R}^n \to \Gamma_p/\sim$, $v \mapsto [(p, v, (U, \varphi))]$: it is injective because $D(\varphi \circ \varphi^{-1}) = \text{id}$ forces $v = w$, and surjective by definition. The change of chart is again $D(\psi \circ \varphi^{-1})(\varphi(p))$, so proposition 2.1.4 transfers the vector-space structure of $\mathbb{R}^n$. This is the *physical tangent space*, the description favoured in physics, where one says $v \in \mathbb{R}^n$ represents the tangent vector in the chart (U, φ).

The Algebraic Tangent Space

The third construction takes tangent vectors to be operators on functions rather than geometric objects. On $\mathbb{R}^n$ the directional derivative $v \cdot \nabla$ in a direction v acts on smooth functions and satisfies two rules: it is linear, and it obeys the product rule. These two rules turn out to characterize directional derivatives completely, and they refer to no coordinates. Taking them as the definition gives a tangent space built from the algebra of functions alone.

The motivation is the gradient. For smooth $f : \mathbb{R}^n \to \mathbb{R}$ the partial derivatives $\partial f / \partial x^i(p)$ assemble into $\nabla f(p)$, one distinguished vector in the tangent space at p – the direction of steepest ascent. A tangent vector should be an *arbitrary* vector there, not this single one; what the gradient exposes is that the operators $\partial/\partial x^i|_p$ span the possible directions. Tracking those operators, rather than the values they produce, is what frees the construction from any chart.

Definition 2.1.7: Tangent Vector as Derivation

Let M be an n-manifold and $p \in M$. A *tangent vector at* p is a linear map $v_p : C^\infty(M) \to \mathbb{R}$ satisfying the *Leibniz rule*

$$v_p(fg) = v_p(f)\, g(p) + f(p)\, v_p(g), \qquad f, g \in C^\infty(M)$$

Such a map is a *derivation at* p: a derivation of the algebra $C^\infty(M)$ relative to the evaluation homomorphism $\mathrm{ev}_p(f) = f(p)$.

One writes $v_p f$ for $v_p(f)$. A derivation at p is in particular an element of the dual space $C^\infty(M)^*$, and the derivations at p form a linear subspace: a linear combination of maps obeying the Leibniz rule at p obeys it again. This subspace is the *algebraic tangent space* $(T_pM)_{\mathrm{Alg}}$. Smoothness is essential to finite-dimensionality: for finite r the derivations of $C^r(M)$ form an infinite-dimensional space, so C^∞ is the right regularity for a tangent space of dimension n.[1]

2.1.8 Remark The derivations carry more than linear structure. Assembling a derivation at every point into a single operator on $C^\infty(M)$ yields a *vector field*, and the commutator of two such operators is again one; this bracket makes the vector fields a Lie algebra, developed in chapter 5.

The prototype derivation is the coordinate partial. Recall (definition 1.3.12) that for a chart (U, φ) at p with $\varphi = (x^1, \ldots, x^n)$,

$$\frac{\partial f}{\partial x^i}(p) = D_i\big(f \circ \varphi^{-1}\big)(\varphi(p))$$

[1] For every finite r the space of point derivations of $C^r(\mathbb{R})$ at the origin is infinite-dimensional: functions like $x|x|^\alpha$ with $0 < \alpha < 1$ lie in C^1 yet not in $\mathfrak{m}_0^2$, giving a continuum of independent classes in $\mathfrak{m}_0/\mathfrak{m}_0^2$. Only at $r = \infty$ does it collapse to dimension 1.

Definition 2.1.9: Coordinate Derivations

For a chart (U, φ) of an n-manifold with $p \in U$, the *coordinate derivation* $\partial/\partial x^i|_p$ is

$$\left.\frac{\partial}{\partial x^i}\right|_p : C^\infty(M) \to \mathbb{R}, \qquad \left.\frac{\partial}{\partial x^i}\right|_p f = \frac{\partial f}{\partial x^i}(p)$$

Each coordinate derivation depends on the chart. Setting $c_i(t) = \varphi^{-1}(\varphi(p)+te_i)$ for the i-th standard basis vector e_i, the chain rule gives

$$\left.\frac{\partial}{\partial x^i}\right|_p f = \lim_{h\to 0} \frac{f(c_i(h)) - f(p)}{h}$$

so $\partial/\partial x^i|_p$ is the velocity of the i-th coordinate curve and is a derivation at p, an element of $(T_pM)_{\mathrm{Alg}}$. The goal of this subsection is that these n operators form a basis: for any chart at p,

$$v_p = \sum_{i=1}^n v_p(x^i) \left.\frac{\partial}{\partial x^i}\right|_p \qquad \text{for every } v_p \in (T_pM)_{\mathrm{Alg}}$$

where $x^i \in C^\infty(U)$ is the i-th coordinate function. In particular $\dim(T_pM)_{\mathrm{Alg}} = n$ independently of the chart. The first step is that a derivation sees only local behaviour.

Lemma 2.1.10: Derivations Are Local

Let $v_p \in (T_pM)_{\mathrm{Alg}}$ and $f, g, h \in C^\infty(M)$.

1. If $f = g$ on a neighbourhood of p, then $v_p(f) = v_p(g)$.
2. If h is constant on a neighbourhood of p, then $v_p(h) = 0$.

***Proof*:**
For (1), linearity reduces the claim to: if f vanishes on a neighbourhood U of p, then $v_p(f) = 0$. Choose a bump function β with $\operatorname{supp}\beta \subseteq U$ and $\beta(p) = 1$ (theorem 1.5.4). Then $\beta f \equiv 0$ on M: it vanishes on U because f does, and off U because β does. Since $f(p) = 0$, the Leibniz rule gives

$$0 = v_p(\beta f) = \beta(p)\, v_p(f) + f(p)\, v_p(\beta) = v_p(f)$$

For (2), a global constant $c = c \cdot 1$ has $v_p(1) = v_p(1 \cdot 1) = 2v_p(1)$, so $v_p(1) = 0$ and $v_p(c) = c\, v_p(1) = 0$. A locally constant h agrees with a global constant near p, so $v_p(h) = 0$ by (1), completing the proof.

Locality says v_p is determined by germs, yet the definition feeds it global functions. The next result reconciles the two: a derivation on an open $U \ni p$ carries the same information as one on all of M.

Proposition 2.1.11: Restriction to an Open Set is an Isomorphism

For open $U \subseteq M$ with $p \in U$, the map $\Phi\colon (T_pU)_{\mathrm{Alg}} \to (T_pM)_{\mathrm{Alg}}$, $\Phi(w_p)(f) = w_p(f|_U)$, is a linear isomorphism.

***Proof*:**
Each $\Phi(w_p)$ is a derivation on M because restriction $f \mapsto f|_U$ is an algebra homomorphism and evaluation at p factors through it, and Φ is linear. As neither space is yet known to be finite-dimensional, injectivity and surjectivity are shown directly.

Injectivity. Suppose $\Phi(w_p) = 0$ and let $h \in C^\infty(U)$. Choose a bump β with $\mathrm{supp}\beta \subseteq U$ and $\beta \equiv 1$ near p; then βh, extended by 0, is a global $f \in C^\infty(M)$ agreeing with h near p, so $f|_U$ agrees with h near p. Lemma 2.1.10 gives $w_p(h) = w_p(f|_U) = \Phi(w_p)(f) = 0$. As h was arbitrary, $w_p = 0$.

Surjectivity. Given $v_p \in (T_pM)_{\mathrm{Alg}}$, fix such a bump β and set $w_p(h) = v_p(\beta h)$, with βh extended by 0 to M. This is independent of β: two such bumps make βh and $\beta' h$ agree near p, so v_p gives the same value by lemma 2.1.10. It is a derivation, since $(\beta h_1)(\beta h_2) = \beta^2 h_1 h_2$ agrees with $\beta(h_1 h_2)$ near p and the Leibniz rule for v_p then transfers to w_p. Finally, $\beta \cdot f|_U$ agrees with f near p, so

$$\Phi(w_p)(f) = w_p(f|_U) = v_p\big(\beta \cdot f|_U\big) = v_p(f)$$

whence $\Phi(w_p) = v_p$, completing the proof.

The isomorphism justifies treating $\partial/\partial x^i|_p$ as living in both $(T_pU)_{\mathrm{Alg}}$ and $(T_pM)_{\mathrm{Alg}}$, and reduces the basis theorem to a local computation on $\mathbb{R}^n$. That computation rests on a first-order Taylor expansion with a smooth remainder.

Lemma 2.1.12: Hadamard's Lemma

Let $U \subseteq \mathbb{R}^n$ be a star-convex open neighbourhood of a and $f \in C^\infty(U)$. There are smooth $f_i \in C^\infty(U)$ with

$$f(x) = f(a) + \sum_{i=1}^{n} (x^i - a^i)\, f_i(x), \qquad f_i(a) = \frac{\partial f}{\partial x^i}(a)$$

***Proof*:**
With $h(t) = f(a + t(x-a))$ on $[0,1]$, the chain rule gives $h'(t) = \sum_i \frac{\partial f}{\partial x^i}(a + t(x-a))\,(x^i - a^i)$. Since $h(1) = f(x)$ and $h(0) = f(a)$,

$$f(x) - f(a) = \int_0^1 h'(t)\,dt = \sum_{i=1}^{n} (x^i - a^i) \int_0^1 \frac{\partial f}{\partial x^i}(a + t(x-a))\,dt$$

so $f_i(x) = \int_0^1 \frac{\partial f}{\partial x^i}(a + t(x-a))\,dt$ is smooth in x (differentiation under the integral) and gives the stated formula. Setting $x = a$, $f_i(a) = \int_0^1 \frac{\partial f}{\partial x^i}(a)\,dt = \frac{\partial f}{\partial x^i}(a)$, completing the proof.

Theorem 2.1.13: Coordinate Derivations Form a Basis

Let (U, φ) be a chart of an n-manifold with $p \in U$ and $\varphi = (x^1, \ldots, x^n)$. Then $\big(\partial/\partial x^1|_p, \ldots, \partial/\partial x^n|_p\big)$ is a basis of $(T_pM)_{\mathrm{Alg}}$, and every v_p expands as

$$v_p = \sum_{i=1}^{n} v_p(x^i)\ \frac{\partial}{\partial x^i}\bigg|_p$$

In particular $\dim(T_pM)_{\mathrm{Alg}} = n$.

***Proof*:**
By proposition 2.1.11 work on U, and shrinking U assume $\varphi(U)$ is star-convex; composing φ with a translation makes $\varphi(p) = 0$, so $x^i(p) = 0$. For $f \in C^\infty(U)$, apply lemma 2.1.12 to $f \circ \varphi^{-1}$ at $a = 0$ and pull back by φ: there are $g_i \in C^\infty(U)$ with

$$f = f(p) + \sum_{i=1}^{n} x^i\, g_i, \qquad g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Applying a derivation v_p, using $v_p(f(p)) = 0$ and $x^i(p) = 0$,

$$v_p(f) = \sum_{i=1}^{n} \big(v_p(x^i)\, g_i(p) + x^i(p)\, v_p(g_i)\big) = \sum_{i=1}^{n} v_p(x^i)\, \frac{\partial f}{\partial x^i}(p) = \sum_{i=1}^{n} v_p(x^i)\ \frac{\partial}{\partial x^i}\bigg|_p f$$

so the coordinate derivations span. For independence, suppose $\sum_i a_i\, \partial/\partial x^i|_p = 0$ and apply it to x^j: since $\partial x^j/\partial x^i = \delta^j_i$,

$$0 = \sum_{i=1}^{n} a_i\ \frac{\partial x^j}{\partial x^i}\bigg|_p = \sum_{i=1}^{n} a_i\, \delta^j_i = a_j$$

for every j, so the operators are independent and form a basis, completing the proof.

Coordinate charts thus supply a working basis, at the cost of one notational hazard: for $\mathbb{R}^n$ under the identity chart the coordinate functions are $x^1, \ldots, x^n$, yet a point is also written $(x_1, \ldots, x_n)$, so the tangent vector $\sum_i x_i\, \partial/\partial x^i|_p$ mixes subscripts (components) with superscripts (coordinate functions). On $\mathbb{R}$ the single coordinate is usually named t, with basis vector $\partial/\partial t|_{t_0}$ at t_0.

*The Tangent Space via Germs

The derivation construction used only local behaviour, so it descends to germs. Let $\mathcal{F}_p = C^\infty_p(M)$ be the algebra of germs at p: equivalence classes of smooth functions defined near p, two agreeing when they coincide on some neighbourhood. The value $[f](p) = f(p)$ is well defined, giving an evaluation homomorphism $\mathrm{ev}_p \colon \mathcal{F}_p \to \mathbb{R}$.

Definition 2.1.14: Germ Derivation

A *derivation* of $\mathcal{F}_p$ (relative to ev_p) is a linear $\mathcal{D}_p\colon \mathcal{F}_p \to \mathbb{R}$ with

$$\mathcal{D}_p([f][g]) = f(p)\,\mathcal{D}_p[g] + g(p)\,\mathcal{D}_p[f], \qquad [f],[g] \in \mathcal{F}_p$$

The derivations of $\mathcal{F}_p$ form a real vector space $\mathrm{Der}(\mathcal{F}_p)$. By lemma 2.1.10 and proposition 2.1.11 a derivation of $C^\infty(M)$ is determined by, and extends to, germ data, so $\mathrm{Der}(\mathcal{F}_p)$ is another model of the tangent space. Its value is that it exposes the tangent space as a purely algebraic object attached to a local ring.

Lemma 2.1.15: The Local Ring of Germs

Let $\mathcal{F}_a = C^\infty_a(M)$ be the ring of germs at a in an n-manifold. Then

1. $\mathcal{F}_a$ is a local ring with maximal ideal $\mathfrak{m}_a = \{[f] \in \mathcal{F}_a \mid f(a) = 0\}$;
2. for $M = \mathbb{R}^n$, the ideal $\mathfrak{m}_a$ is generated by $x^1 - a^1, \dots, x^n - a^n$;
3. $\mathcal{F}_a/\mathfrak{m}_a \cong \mathbb{R}$, and $\mathfrak{m}_a/\mathfrak{m}_a^2$ is an n-dimensional real vector space.

***Proof*:**
(1) The set $\mathfrak{m}_a$ is the kernel of the homomorphism ev_a, hence an ideal. If $[f] \notin \mathfrak{m}_a$ then $f(a) \neq 0$, so f is nonzero on a neighbourhood of a and $1/f$ defines a germ inverse to $[f]$. Thus every germ outside $\mathfrak{m}_a$ is a unit, so $\mathfrak{m}_a$ contains every proper ideal and is the unique maximal ideal.

(2) For $[f] \in \mathfrak{m}_a$, Hadamard's lemma (lemma 2.1.12) on a star-convex neighbourhood of a gives smooth f_i with $f = f(a) + \sum_i (x^i - a^i) f_i = \sum_i (x^i - a^i) f_i$, since $f(a) = 0$. So the germs $x^i - a^i$ generate $\mathfrak{m}_a$.

(3) Evaluation ev_a is surjective onto $\mathbb{R}$ with kernel $\mathfrak{m}_a$, so $\mathcal{F}_a/\mathfrak{m}_a \cong \mathbb{R}$. For the second, map

$$\varphi\colon \mathfrak{m}_a \to \mathbb{R}^n, \qquad [f] \mapsto \left(\tfrac{\partial f}{\partial x^1}(a), \dots, \tfrac{\partial f}{\partial x^n}(a)\right)$$

which is well defined (representatives agree near a, hence in all derivatives), linear, and surjective (the coordinate germs realize any target). The product rule gives, for $[f],[g] \in \mathfrak{m}_a$,

$$\varphi([f][g])_i = \tfrac{\partial f}{\partial x^i}(a)\,g(a) + f(a)\,\tfrac{\partial g}{\partial x^i}(a) = 0$$

so $\mathfrak{m}_a^2 \subseteq \ker\varphi$. Conversely, if $\varphi([h]) = 0$ then in a chart centred at a the germ h has vanishing value and first derivatives, so its Taylor expansion begins at order two; writing each second-order term via Hadamard as a product of two elements of $\mathfrak{m}_a$ shows $[h] \in \mathfrak{m}_a^2$. Hence $\ker\varphi = \mathfrak{m}_a^2$ and $\mathfrak{m}_a/\mathfrak{m}_a^2 \cong \mathbb{R}^n$, completing the proof.

The quotient $\mathfrak{m}_a/\mathfrak{m}_a^2$ is the *cotangent space*; a derivation kills constants and, by the Leibniz rule, kills $\mathfrak{m}_a^2$, so it factors through $\mathfrak{m}_a/\mathfrak{m}_a^2$ and $\mathrm{Der}(\mathcal{F}_a) \cong (\mathfrak{m}_a/\mathfrak{m}_a^2)^*$.

Proposition 2.1.16: Algebraic Tangent Space

The *algebraic tangent space* of M at a is $\tilde{T}_a M = (\mathfrak{m}_a/\mathfrak{m}_a^2)^*$. A smooth $f\colon M \to N$ induces a linear $\partial_a f\colon \tilde{T}_a M \to \tilde{T}_{f(a)} N$ with $\partial_a(g \circ f) = \partial_{f(a)} g \circ \partial_a f$.

***Proof*:**
Pullback $[h] \mapsto [h \circ f]$ carries $\mathfrak{m}_{f(a)}$ into $\mathfrak{m}_a$ and $\mathfrak{m}_{f(a)}^2$ into $\mathfrak{m}_a^2$: if $[h] = [h'] + [g_1 g_2]$ with $[g_1], [g_2] \in \mathfrak{m}_{f(a)}$, then $(g_1 \circ f)(g_2 \circ f) \in \mathfrak{m}_a^2$, so the composite agrees modulo $\mathfrak{m}_a^2$. Dualizing the induced map $\mathfrak{m}_{f(a)}/\mathfrak{m}_{f(a)}^2 \to \mathfrak{m}_a/\mathfrak{m}_a^2$ gives $\partial_a f$, and functoriality of pullback under composition yields the chain rule, completing the proof.

Relating the Definitions

The three constructions – kinematic $(T_pM)_{\text{kin}}$, physical $(T_pM)_{\text{phys}}$, algebraic $(T_pM)_{\text{Alg}}$ – are the same space seen three ways. Each comes equipped, for every chart (U, φ) at p, with a linear bijection $b_\varphi\colon \mathbb{R}^n \to (T_pM)_\bullet$, and in all three the change of chart is the transition derivative $D(\psi \circ \varphi^{-1})(\varphi(p))$ (lemma 2.1.5, definition 2.1.6, and theorem 2.1.13). Maps intertwining these parametrizations are therefore canonical isomorphisms.

Proposition 2.1.17: The Three Tangent Spaces Agree

There are canonical vector-space isomorphisms

$$(T_pM)_{\text{kin}} \cong (T_pM)_{\text{phys}} \cong (T_pM)_{\text{Alg}}$$

independent of the chart used to define them.

***Proof*:**
Fix a chart (U, φ) at p with coordinates x^i. Define $\Lambda\colon (T_pM)_{\text{kin}} \to (T_pM)_{\text{Alg}}$ by $\Lambda[c](f) = (f \circ c)'(0)$; this is a derivation at p by the Leibniz rule for one-variable derivatives, and depends only on the tangency class of c (definition 2.1.1). If $[c] = b_\varphi(v)$ in the kinematic space, so $v = (\varphi \circ c)'(0)$, then the chain rule gives

$$\Lambda[c](f) = (f \circ c)'(0) = D(f \circ \varphi^{-1})(\varphi(p))\, v = \sum_{i=1}^{n} v^i \left.\frac{\partial}{\partial x^i}\right|_p f$$

so $\Lambda \circ b_\varphi = b_\varphi$ as maps $\mathbb{R}^n \to (T_pM)_{\text{Alg}}$, the right-hand b_φ being the basis parametrization of theorem 2.1.13. Both parametrizations are linear bijections, so $\Lambda = b_\varphi \circ b_\varphi^{-1}$ is a linear isomorphism. It is chart-independent because in a second chart both parametrizations transform by the same transition derivative, which therefore cancels.

Identically, $[(p, v, (U, \varphi))] \mapsto \sum_i v^i\, \partial/\partial x^i|_p$ intertwines the chart parametrizations of $(T_pM)_{\text{phys}}$ and $(T_pM)_{\text{Alg}}$, giving a canonical isomorphism between them; composing yields the isomorphism with $(T_pM)_{\text{kin}}$, completing the proof.

Henceforth T_pM denotes the tangent space in any of the three descriptions, whichever is convenient, and its elements are tangent vectors. A curve carries a distinguished one.

Definition 2.1.18: Velocity Vector of a Curve

A *curve* in M is a smooth map $\gamma\colon I \to M$ on an interval $I \subseteq \mathbb{R}$. Its *velocity* at t_0 is the tangent vector $\gamma'(t_0) \in T_{\gamma(t_0)}M$ acting by

$$\gamma'(t_0)\, f = (f \circ \gamma)'(t_0), \qquad f \in C^\infty(M)$$

The map γ, not merely its image $\gamma(I)$, is part of the data; γ together with $\gamma(I)$ is a *parametrized curve*. Common notations for the velocity are $\dot\gamma(t_0)$, $\frac{d\gamma}{dt}(t_0)$, and $\frac{\partial\gamma}{\partial t}|_{t_0}$. Once the tangent map is available (section 2.2), the velocity is $\gamma'(t_0) = T_{t_0}\gamma\big(\partial/\partial t|_{t_0}\big)$, the image of the standard basis vector of $T_{t_0}\mathbb{R}$.

The Tangent Space of a Regular Submanifold

For a submanifold $M \subseteq \mathbb{R}^n$ the tangent space at a can be computed concretely, and each of the four characterizations of proposition 1.4.5 produces the same linear subspace of $T_a\mathbb{R}^n \cong \mathbb{R}^n$. Throughout, T_aM is realized inside $T_a\mathbb{R}^n$ as the velocities of curves in M through a.

Proposition 2.1.19: Tangent Space of a Regular Submanifold

Let $M \subseteq \mathbb{R}^n$ be a C^r submanifold of dimension k and $a \in M$. Under the identification $T_a\mathbb{R}^n \cong \mathbb{R}^n$, the tangent space T_aM equals each of the following, taken from the local descriptions of proposition 1.4.5:

1. $\ker Df(a)$, where $M = f^{-1}(0)$ near a for a submersion f into $\mathbb{R}^{n-k}$;
2. $\big\{(Y,Z) \in \mathbb{R}^k \times \mathbb{R}^{n-k} \;\big|\; Z = Dg(b)Y\big\}$, where M is the graph $z = g(y)$ near $a = (b, g(b))$;
3. $Dh(a)^{-1}\big(\mathbb{R}^k \times \{0\}\big)$, where h is a diffeomorphism flattening M onto a slice;
4. $D\varphi(b)(\mathbb{R}^k)$, where φ is an embedding with $\varphi(b) = a$ parametrizing M near a.

***Proof*:**
The argument computes T_aM from the embedding (4), then matches the other three by linear algebra.

$T_aM = D\varphi(b)(\mathbb{R}^k) = \ker Df(a)$. Three inclusions close the loop, with no need to invert φ. For $v \in \mathbb{R}^k$ the curve $t \mapsto \varphi(b + tv)$ lies in M, passes through a, and has velocity $D\varphi(b)v$, so $D\varphi(b)(\mathbb{R}^k) \subseteq T_aM$. Next, any smooth curve γ in M through a satisfies $f\circ\gamma \equiv 0$ (as $M = f^{-1}(0)$), so differentiating gives $Df(a)\,\gamma'(0) = 0$ and $T_aM \subseteq \ker Df(a)$. Finally $f \circ \varphi$ vanishes, so $Df(a)\,D\varphi(b) = 0$ and $D\varphi(b)(\mathbb{R}^k) \subseteq \ker Df(a)$; since $D\varphi(b)$ is injective and $Df(a)$ has rank $n-k$, both spaces have dimension k. The chain $D\varphi(b)(\mathbb{R}^k) \subseteq T_aM \subseteq \ker Df(a) = D\varphi(b)(\mathbb{R}^k)$ forces all three equal.

$= \{(Y,Z) \mid Z = Dg(b)Y\}$. The graph is the level set of $f(y,z) = z - g(y)$, a submersion with

$Df(a) = (-Dg(b),\, I)$. Hence $\ker Df(a) = \{(Y, Z) \mid Z - Dg(b)Y = 0\}$, the stated subspace.

$= Dh(a)^{-1}(\mathbb{R}^k \times \{0\})$. The flattening h satisfies $M \cap U = h^{-1}(\mathbb{R}^k \times \{0\})$, so $f = \pi \circ h$ with $\pi\colon \mathbb{R}^n \to \mathbb{R}^{n-k}$ the last-coordinate projection is a defining submersion. Then $\ker Df(a) = \ker(\pi \circ Dh(a)) = Dh(a)^{-1}(\ker \pi) = Dh(a)^{-1}(\mathbb{R}^k \times \{0\})$, completing the proof.

For the sphere $S^{n-1} = f^{-1}(1)$ with $f(x) = |x|^2$ one has $Df(a) = 2a^\top$, so $T_a S^{n-1} = \ker Df(a) = a^\perp$, the hyperplane orthogonal to a – the familiar picture of the tangent plane.

The Tangent Space of a Vector Space

A finite-dimensional real vector space V is its own tangent space at every point. Concretely, define $\iota_p\colon V \to T_pV$ by sending v to the velocity at 0 of the line $t \mapsto p + tv$; in the three descriptions this is

$$\iota_p(v) = [\, t \mapsto p + tv\,] = \big[(p,\, E(v),\, (V, E))\big] = \Big(f \mapsto \tfrac{d}{dt}\big|_0 f(p + tv)\Big)$$

for any linear chart $E\colon V \to \mathbb{R}^n$. The map ι_p is a linear isomorphism, so $T_pV \cong V$ canonically, independently of p and of the chart. This identification is used silently whenever the tangent space of $\mathbb{R}^n$ is treated as $\mathbb{R}^n$ again.

Exercise 2.1.1

1. Let (U, φ) and (V, ψ) be charts at p with coordinates x^j and $\bar{x}^i$. For a tangent vector $v_p = \sum_j a^j\, \partial/\partial x^j|_p$, use theorem 2.1.13 to show its components in the barred chart are $\bar{a}^i = \sum_j \frac{\partial \bar{x}^i}{\partial x^j}(p)\, a^j$, the transformation law.
2. Show that $T_A\, \mathrm{GL}_n(\mathbb{R}) \cong M_n(\mathbb{R})$ for every A, and that the curve $t \mapsto A + tB$ realizes $B \in M_n(\mathbb{R})$ as a velocity at A.
3. Show that $v_p(f)$ depends only on the first-order Taylor data of f at p: if f and g have equal value and equal first partial derivatives at p in some chart, then $v_p(f) = v_p(g)$ for every $v_p \in T_pM$.

2.2 Tangent Map

Each description of the tangent space produces a description of the map a smooth f induces on tangent spaces, and the three agree. Some authors write f_{*p}, but that notation is reserved here for the *pushforward* of vector fields, so the tangent map is written T_pf.

The geometric version pushes velocities forward along f.

Definition 2.2.1: Tangent Map (via Curves)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, the *tangent map* $T_pf\colon T_pM \to T_qN$ sends $v_p = [c]$ to

$$T_pf(v_p) = [f \circ c]$$

the tangency class of the image curve $f \circ c$.

This is well defined: if $c \sim c'$ then for every $g \in C^\infty(N)$ one has $g \circ f \in C^\infty(M)$, so $(g \circ f \circ c)'(0) = (g \circ f \circ c')'(0)$ and $f \circ c \sim f \circ c'$. Geometrically T_pf carries the velocity $c'(0)$ to the velocity $(f \circ c)'(0)$, the infinitesimal image of the curve.

In coordinates this is the Jacobian acting on components.

Definition 2.2.2: Tangent Map (via Charts)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, choose charts (U, φ) at p and (V, ψ) at q. If v_p is represented by $(p, v, (U, \varphi))$, then $T_pf(v_p)$ is represented in (V, ψ) by $(q, w, (V, \psi))$ with

$$w = D(\psi \circ f \circ \varphi^{-1})(\varphi(p))\, v$$

The chain rule makes this independent of the charts. This description is best when the coordinate representation of f is explicit; for instance the inclusion $\iota\colon M \to M \times N$ at a fixed second point has $T_p\iota(v) = (v, 0)$, read off from the product charts.

Example 2.2.3: Nondegenerate Fixed Points

Let $f\colon M \to M$ fix p, so $f(p) = p$ and $T_pf\colon T_pM \to T_pM$. Call p *nondegenerate* if $T_pf - \mathrm{id}$ is invertible, i.e. 1 is not an eigenvalue of T_pf. In a chart centred at p the coordinate map $\hat{f} = \varphi \circ f \circ \varphi^{-1}$ fixes 0 with $D\hat{f}(0)$ representing T_pf (definition 2.2.2), so $D\hat{f}(0) - I$ is invertible. Then $x \mapsto \hat{f}(x) - x$ has invertible derivative at 0, hence is a local diffeomorphism (theorem 0.2.1), so its only zero near 0 is 0 itself: p is an *isolated* fixed point. These nondegenerate fixed points are the ones counted by the Lefschetz and Poincaré-Hopf theorems (section 8.6).

The algebraic version is coordinate-free and the most convenient for computation: T_pf precomposes a derivation with f.

Definition 2.2.4: Tangent Map (via Derivations)

For a smooth $f\colon M \to N$ and $p \in M$ with $q = f(p)$, the *tangent map* $T_pf\colon T_pM \to T_qN$ sends v_p to the derivation acting by

$$(T_pf\, v_p)(g) = v_p(g \circ f), \qquad g \in C^\infty(N)$$

Since $g \circ f \in C^\infty(M)$, the value $v_p(g \circ f)$ makes sense, and $T_pf\, v_p$ is a derivation at q: it is linear in g, and the Leibniz rule for v_p gives $(T_pf\, v_p)(gh) = (T_pf\, v_p)(g)\, h(q) + g(q)\, (T_pf\, v_p)(h)$, so $T_pf\, v_p \in T_qN$. One notational trap: although written with subscript p, the vector $T_pf\, v_p$ lives in T_qN. When a construction needs the tangent map distinguished in notation from the tangent space, it is written D_pf or df_p.

Proposition 2.2.5: Properties of the Tangent Map

Let $f\colon M \to N$ and $g\colon N \to P$ be smooth. Then

1. $T_p f\colon T_p M \to T_{f(p)} N$ is linear;
2. $T_p(g \circ f) = T_{f(p)} g \circ T_p f$;
3. $T_p \mathrm{id} = \mathrm{id}_{T_p M}$;
4. if f is a diffeomorphism, $T_p f$ is a linear isomorphism with $(T_p f)^{-1} = T_{f(p)}(f^{-1})$.

***Proof*:**
(1) For $g \in C^\infty(N)$ and scalars a, b, $T_p f(av_p + bw_p)(g) = (av_p + bw_p)(g \circ f) = a\,T_p f(v_p)(g) + b\,T_p f(w_p)(g)$, so $T_p f$ is linear.

(2) For $h \in C^\infty(P)$, both sides act by $v_p \mapsto v_p(h \circ g \circ f)$: on the left, $T_p(g \circ f)(v_p)(h) = v_p(h \circ (g \circ f))$; on the right, $\big(T_{f(p)} g \circ T_p f\big)(v_p)(h) = T_p f(v_p)(h \circ g) = v_p((h \circ g) \circ f)$. These agree.

(3) $T_p \mathrm{id}(v_p)(h) = v_p(h \circ \mathrm{id}) = v_p(h)$.

(4) By (1) $T_p f$ is linear, and f^{-1} is smooth. Applying (2) and (3) to $f^{-1} \circ f = \mathrm{id}_M$ and $f \circ f^{-1} = \mathrm{id}_N$ gives $T_{f(p)} f^{-1} \circ T_p f = \mathrm{id}_{T_p M}$ and $T_p f \circ T_{f(p)} f^{-1} = \mathrm{id}_{T_{f(p)} N}$, so $T_p f$ is invertible with the stated inverse, completing the proof.

The converse of (4) fails: $f\colon \mathbb{R} \to S^1$, $x \mapsto e^{2\pi i x}$, is not a diffeomorphism (it is not injective) yet $T_p f$ is an isomorphism at every point. It is a *local* diffeomorphism, a notion central to chapter 3. The following partial converse, stated now to show how $T_p f$ captures f locally, is the inverse function theorem for manifolds.

Theorem 2.2.6: Inverse Mapping Theorem for Manifolds

Let $f\colon M \to N$ be smooth with $T_p f$ an isomorphism. Then f restricts to a diffeomorphism of an open neighbourhood of p onto an open neighbourhood of $f(p)$. If $T_p f$ is an isomorphism for every p, then f is a local diffeomorphism.

***Proof*:**
In charts $T_p f$ is represented by $D(\psi \circ f \circ \varphi^{-1})(\varphi(p))$ (definition 2.2.2), which is therefore invertible; the Euclidean inverse function theorem (theorem 0.2.1) then makes $\psi \circ f \circ \varphi^{-1}$ a local diffeomorphism, and composing with the charts gives the claim. The full argument is theorem 3.1.10 in chapter 3, completing the proof.

Properties (1)–(3) say T is a functor.

Definition 2.2.7: Pointed Tangent Functor

On the category Man_* of pointed manifolds, the *pointed tangent functor* T sends (M, p) to T_pM and a smooth $f\colon (M,p) \to (N,q)$ to the linear map $T_pf\colon T_pM \to T_qN$. Its basepoint-free counterpart on whole manifolds is the tangent functor of corollary 2.4.7.

Property (4) gives the dimension of a tangent space at once.

Corollary 2.2.8: Dimension of the Tangent Space

For an m-manifold M, $\dim T_pM = m$ at every p.

***Proof*:**
A chart (U, φ) at p is a diffeomorphism onto $\varphi(U) \subseteq \mathbb{R}^m$, so $T_p\varphi\colon T_pU \to T_{\varphi(p)}\varphi(U)$ is a linear isomorphism by proposition 2.2.5. With $T_pU \cong T_pM$ and $T_{\varphi(p)}\varphi(U) \cong T_{\varphi(p)}\mathbb{R}^m \cong \mathbb{R}^m$ (proposition 2.1.11), transitivity gives $T_pM \cong \mathbb{R}^m$, completing the proof.

Every tangent space of M is thus isomorphic to $\mathbb{R}^m$, hence to every other; the vector-space structure is the same at each point. The isomorphisms are not canonical, though – the tangent spaces sit differently in space, as the tangent lines around a circle already show. Section 2.4 assembles them into a single object that records this placement, and chapter 11 settles when they can be globally identified by translation, the question of parallelizability.

The open-submanifold identification $T_pU \cong T_pM$ (proposition 2.1.11) will be used silently from now on. In particular $T_p\mathbb{R}^n \cong \mathbb{R}^n$ and, for any finite-dimensional V, $T_pV \cong V$, compatibly with linear maps.

Proposition 2.2.9: Tangent Space of a Vector Space

Let V be a finite-dimensional vector space with its standard smooth structure. The canonical isomorphism $\iota_p\colon V \to T_pV$ sending v to the velocity of $t \mapsto p + tv$ is natural in V: for a linear $L\colon V \to W$ the square

$$\begin{tikzcd} T_pV \arrow[r, "T_pL"] & T_{Lp}W \\ V \arrow[u, "\iota_p"] \arrow[r, "L"] & W \arrow[u, "\iota_{Lp}"'] \end{tikzcd}$$

commutes.

***Proof*:**
For $v \in V$, $\iota_p(v)$ is the velocity of $t \mapsto p + tv$, so $T_pL(\iota_p v)$ is the velocity of $t \mapsto L(p+tv) = Lp + t\,Lv$, which is exactly $\iota_{Lp}(Lv)$. Thus $T_pL \circ \iota_p = \iota_{Lp} \circ L$, completing the proof.

The tangent map between manifolds with boundary needs the tangent spaces at boundary points handled with care, through the constant rank theorem; that discussion is deferred to chapter 3, where the theorem is proved.

2.2.1 Computing the Tangent Map

Since $T_pf\colon T_pM \to T_{f(p)}N$ is linear, it has a matrix once bases are chosen, and the coordinate basis makes that matrix the Jacobian. Fix a chart (U,φ) at p. By the proof of corollary 2.2.8, $T_p\varphi\colon T_pM \to T_{\varphi(p)}\mathbb{R}^n$ is an isomorphism, and $T_{\varphi(p)}\mathbb{R}^n$ has the coordinate basis $\partial/\partial x^i|_{\varphi(p)}$. Its preimage is a basis of T_pM,

$$\left.\frac{\partial}{\partial x^i}\right|_p = T_{\varphi(p)}(\varphi^{-1})\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right)$$

written with the same symbol to stress the identification with the coordinate representation. By definition 2.2.4, for $f \in C^\infty(M)$ with $\hat{f} = f\circ\varphi^{-1}$ and $\hat{p} = \varphi(p)$,

$$\left.\frac{\partial}{\partial x^i}\right|_p (f) = \left.\frac{\partial}{\partial x^i}\right|_{\hat{p}} (\hat{f}) = \frac{\partial \hat{f}}{\partial x^i}(\hat{p})$$

so $\partial/\partial x^i|_p$ differentiates the coordinate representation of f. These are the *coordinate vectors* at p; on $\mathbb{R}^n$ they are the partial derivative operators.

To confirm T_pf generalizes the derivative, take $f\colon U\subseteq\mathbb{R}^n \to V\subseteq\mathbb{R}^m$ with standard coordinates x^i on $\mathbb{R}^n$ and y^j on $\mathbb{R}^m$. For $h\in C^\infty(V)$, the chain rule gives

$$T_pf\left(\left.\frac{\partial}{\partial x^i}\right|_p\right)(h) = \left.\frac{\partial}{\partial x^i}\right|_p (h\circ f) = \sum_{j=1}^m \frac{\partial h}{\partial y^j}(f(p))\,\frac{\partial f^j}{\partial x^i}(p) = \left(\sum_{j=1}^m \frac{\partial f^j}{\partial x^i}(p)\;\left.\frac{\partial}{\partial y^j}\right|_{f(p)}\right)(h)$$

so, as a map of coordinate bases,

$$T_pf\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \sum_{j=1}^m \frac{\partial f^j}{\partial x^i}(p)\;\left.\frac{\partial}{\partial y^j}\right|_{f(p)} \tag{2.1}$$

whose matrix is the Jacobian

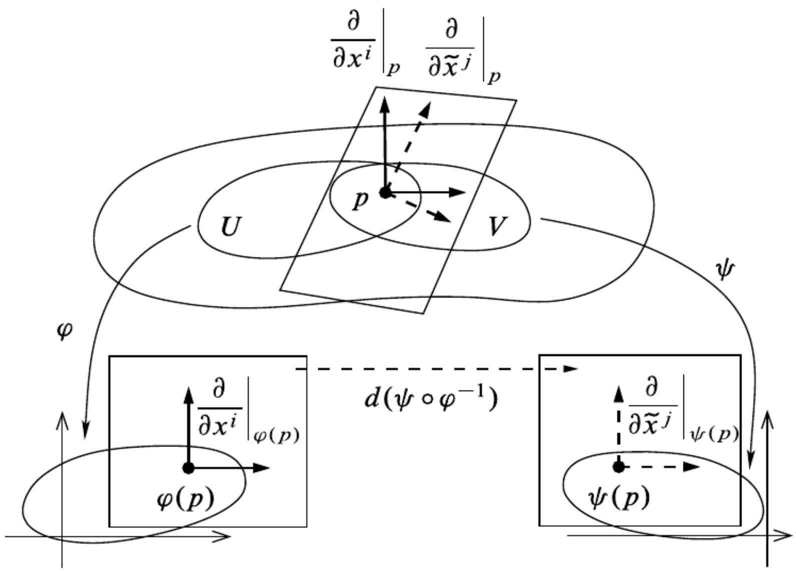
$$\begin{pmatrix} \frac{\partial f^1}{\partial x^1}(p) & \cdots & \frac{\partial f^1}{\partial x^n}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1}(p) & \cdots & \frac{\partial f^m}{\partial x^n}(p) \end{pmatrix}$$

Thus for $T_pf\colon T_p\mathbb{R}^n \to T_{f(p)}\mathbb{R}^m$ one recovers the total derivative $Df(p)$. For general M and N, choosing charts at p and $f(p)$ reduces T_pf to this Jacobian of the coordinate representation, so the tangent map genuinely extends the derivative – built without reference to a basis, but recovering the classical one whenever a basis is chosen.

2.2.2 Change of Coordinates

It remains to record how the coordinate basis transforms between two charts (U,φ) and (V,ψ) with $U\cap V \neq \emptyset$, coordinates x^i and y^j. At $p\in U\cap V$ a tangent vector expands in either $\partial/\partial x^i|_p$ or $\partial/\partial y^j|_p$. The transition map is $\psi\circ\varphi^{-1}$, whose j-th component at $\varphi(p)$ is y^j read in the x-coordinates. By eq. (2.1),

$$T_{\varphi(p)}(\psi\circ\varphi^{-1})\left(\left.\frac{\partial}{\partial x^i}\right|_{\varphi(p)}\right) = \sum_{j=1}^n \frac{\partial y^j}{\partial x^i}(\varphi(p))\;\left.\frac{\partial}{\partial y^j}\right|_{\psi(p)}$$



Transporting through $T(\psi^{-1})$, the coordinate basis transforms as

$$\frac{\partial}{\partial x^i}\bigg|_p = \sum_{j=1}^{n} \frac{\partial y^j}{\partial x^i}(\hat{p})\, \frac{\partial}{\partial y^j}\bigg|_p, \qquad \hat{p} = \varphi(p)$$

and applying this to $v = \sum_i v^i\, \partial/\partial x^i|_p$ gives its components in the y-chart,

$$v = \sum_j \tilde{v}^j\, \frac{\partial}{\partial y^j}\bigg|_p, \qquad \tilde{v}^j = \sum_{i=1}^{n} \frac{\partial y^j}{\partial x^i}(\hat{p})\, v^i$$

the transformation law again, now derived from the tangent map.

Exercise 2.2.1

1. Using definition 2.2.2, verify that the inclusion $\iota\colon M \to M \times N$, $x \mapsto (x, q_0)$ at a fixed q_0, has $T_p\iota(v) = (v, 0)$ under the product-chart identification $T_{(p,q_0)}(M \times N) \cong T_pM \times T_{q_0}N$.
2. For the projection $\pi\colon \mathbb{R}^2 \to \mathbb{R}$, $\pi(x, y) = x$, and a curve $c(t) = (a(t), b(t))$, compute $T_{c(0)}\pi(c'(0))$ directly from definition 2.2.1 and check it against the Jacobian (eq. (2.1)).
3. Let $f\colon M \to N$ be a diffeomorphism and (U, φ) a chart at p. Show that $\big(T_pf(\partial/\partial x^i|_p)\big)_i$ is a basis of $T_{f(p)}N$, and identify the chart on N for which it is the coordinate basis.

2.3 Tangents of Products

Having found the tangent space of a regular submanifold and of an open submanifold, one case remains: a product. Its tangent space is the product of the factors' tangent spaces, matched by the projections.

Proposition 2.3.1: Tangent Space of a Product

Let $M_1, \ldots, M_k$ be smooth manifolds with projections $\pi_i\colon M_1 \times \cdots \times M_k \to M_i$. For $p = (p_1, \ldots, p_k)$ the map

$$\alpha\colon T_p(M_1 \times \cdots \times M_k) \to T_{p_1}M_1 \times \cdots \times T_{p_k}M_k, \qquad \alpha(v) = (T_p\pi_1(v), \ldots, T_p\pi_k(v))$$

is a linear isomorphism. The same holds if one factor is a manifold with boundary.

***Proof*:**
Each $T_p\pi_i$ is linear (proposition 2.2.5), so α is linear. Both sides have dimension $\sum_i \dim M_i$: the product manifold has that dimension, and so does the product of the tangent spaces. It therefore suffices that α is injective. A product chart $\varphi_1 \times \cdots \times \varphi_k$ gives $T_p(M_1 \times \cdots \times M_k)$ a coordinate basis split into k blocks, the i-th being the coordinates from M_i; the coordinate representation of π_i is the projection onto that block, so by eq. (2.1) $T_p\pi_i$ returns the i-th block of a vector. If $\alpha(v) = 0$ then every block of v vanishes, so $v = 0$, completing the proof.

The identification is natural in the factors.

Corollary 2.3.2: Tangent Map of a Product Map

For smooth $f_i\colon M_i \to N_i$, the product $f_1 \times \cdots \times f_k$ is smooth, and under α

$$T_p(f_1 \times \cdots \times f_k) = T_{p_1}f_1 \times \cdots \times T_{p_k}f_k$$

***Proof*:**
Smoothness is checked in product charts, where the coordinate representation of $f_1 \times \cdots \times f_k$ is the product of those of the f_i. For the tangent map, $\pi_i^N \circ (f_1 \times \cdots \times f_k) = f_i \circ \pi_i^M$; applying T and proposition 2.2.5 gives $T\pi_i^N \circ T(f_1 \times \cdots \times f_k) = Tf_i \circ T\pi_i^M$, which is the i-th component of the block-diagonal form under α, completing the proof.

2.4 Tangent Bundle

The tangent spaces have been treated one at a time. Yet the change-of-coordinate computation showed that the coordinate bases of nearby tangent spaces are related smoothly, so the collection of all tangent spaces should itself carry a smooth structure. Assembling them gives the tangent bundle, the object over which the tangent map varies smoothly and the setting for higher derivatives.

Definition 2.4.1: Tangent Bundle

The *tangent bundle* of a manifold M is the disjoint union

$$TM = \coprod_{p \in M} T_pM$$

with the *projection* $\pi : TM \to M$ sending $v_p \in T_pM$ to p.

The projection is part of the data: it is the glue that will make TM more than a disjoint set of spaces.

Example 2.4.2: The Tangent Bundle of $\mathbb{R}^n$

For $M = \mathbb{R}^n$ each $T_p\mathbb{R}^n$ is identified with $\{p\} \times \mathbb{R}^n$, so

$$T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$$

an element being a pair (p, v). This product form is special: TS^1 admits a product description but no canonical one, and TS^2 is not a product at all (chapter 11).

A smooth map lifts to the bundles fibrewise.

Definition 2.4.3: Tangent Lift

For smooth $f : M \to N$, the *tangent lift* $Tf : TM \to TN$ is the map restricting to $T_pf : T_pM \to T_{f(p)}N$ on each fibre.

As a bare set TM is only a disjoint union, and Tf only a set map that is linear on each fibre. The projection π is what promotes it: much as an initial topology is pulled back along a map, the smooth structure on TM is the one making the chart data of M lift. It is *not* the disjoint-union topology, which would record no closeness between different tangent spaces.

Example 2.4.4: Tangent Lift on $\mathbb{R}^n$

For open $U_1 \subseteq \mathbb{R}^n$, $U_2 \subseteq \mathbb{R}^m$ and smooth $f : U_1 \to U_2$, the identifications $TU_1 \cong U_1 \times \mathbb{R}^n$ and $TU_2 \cong U_2 \times \mathbb{R}^m$ (example 2.4.2) turn Tf into

$$(p, v) \mapsto (f(p), Df(p)\, v)$$

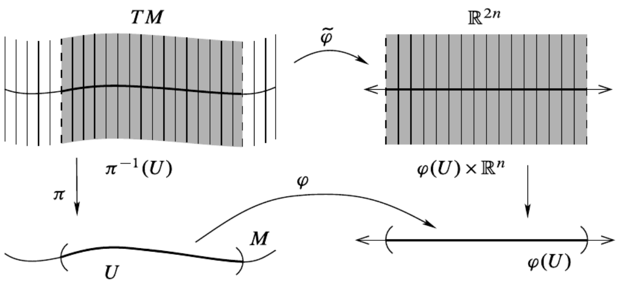
Not every TM is a global product $M \times \mathbb{R}^n$, but over a small chart domain it is, so Tf looks like $(p, v) \mapsto (f(p), Df(p)v)$ in local coordinates. The precise statement is that TM is itself a manifold.

Proposition 2.4.5: The Tangent Bundle is a Manifold

For an m-manifold M, the tangent bundle TM carries a natural topology and smooth structure making it a $2m$-manifold, and the projection $\pi : TM \to M$ is smooth.

***Proof*:**
A chart (U, φ) of M with $\varphi = (x^1, \dots, x^m)$ gives a map $\tilde{\varphi}\colon \pi^{-1}(U) \to \mathbb{R}^{2m}$,

$$\tilde{\varphi}\left(\sum_i v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = (x^1(p), \dots, x^m(p), v^1, \dots, v^m)$$



Its image is $\varphi(U) \times \mathbb{R}^m$, open in $\mathbb{R}^{2m}$, and it is a bijection with inverse $(x, v) \mapsto \sum_i v^i\, \partial/\partial x^i|_{\varphi^{-1}(x)}$. For a second chart (V, ψ), $\psi = (y^1, \dots, y^m)$, the change-of-coordinates rule $\partial/\partial x^i|_p = \sum_j \frac{\partial y^j}{\partial x^i}(\varphi(p))\, \partial/\partial y^j|_p$ gives the transition

$$\tilde{\psi} \circ \tilde{\varphi}^{-1}(x, v) = \left(\psi \circ \varphi^{-1}(x),\ \sum_{i=1}^m \frac{\partial y^1}{\partial x^i}(x)\, v^i, \dots, \sum_{i=1}^m \frac{\partial y^m}{\partial x^i}(x)\, v^i\right)$$

on $\varphi(U \cap V) \times \mathbb{R}^m$. This is smooth, its components being the transition map $\psi \circ \varphi^{-1}$ and its partial derivatives, so the charts $\tilde{\varphi}$ form a smooth atlas.

The induced topology – $W \subseteq TM$ open iff $\tilde{\varphi}(W \cap \pi^{-1}(U))$ is open for every chart – is second countable, since a countable atlas $\{U_i\}$ of M gives the countable cover $\{\pi^{-1}(U_i)\}$ of TM. It is Hausdorff: two vectors over distinct $p \neq q$ are separated by $\pi^{-1}(U), \pi^{-1}(V)$ for disjoint $U \ni p$, $V \ni q$; two distinct vectors over the same p lie in the one chart $\pi^{-1}(U)$ and are separated there. Finally π reads $(x, v) \mapsto x$ in these charts, hence is smooth, completing the proof.

Corollary 2.4.6: Single-Chart Bundles are Trivial

If a smooth m-manifold M is covered by a single chart, then $TM \cong M \times \mathbb{R}^m$.

***Proof*:**
For the global chart (M, φ), the map $\tilde{\varphi}$ of proposition 2.4.5 is a diffeomorphism $TM \to \varphi(M) \times \mathbb{R}^m \cong M \times \mathbb{R}^m$, completing the proof.

The converse fails: $TS^1 \cong S^1 \times \mathbb{R}$ although S^1 needs more than one chart (**??** 2.4.1). As with $T_p f$ detecting only local diffeomorphism, triviality of TM carries only local information – being locally a product is automatic, being globally one is not.

Corollary 2.4.7: The Tangent Functor

For smooth $f\colon M \to N$, the tangent lift $Tf\colon TM \to TN$ is smooth, with coordinate representation

$$Tf(x,v) = \left(\hat{f}(x),\ \sum_i \frac{\partial \hat{f}^1}{\partial x^i}(x)\, v^i, \dots, \sum_i \frac{\partial \hat{f}^n}{\partial x^i}(x)\, v^i\right), \qquad \hat{f} = \psi \circ f \circ \varphi^{-1}$$

Hence $T\colon$ Man $\to$ Man, $M \mapsto TM$, $f \mapsto Tf$, is a functor.

***Proof*:**
In the bundle charts of proposition 2.4.5, Tf is the stated map, smooth because $\hat{f}$ and its partials are. Functoriality $T(g \circ f) = Tg \circ Tf$ and $T\mathrm{id} = \mathrm{id}$ hold fibrewise by proposition 2.2.5, completing the proof.

Proposition 2.4.8: Properties of the Tangent Lift

Let $f\colon M \to N$ and $g\colon N \to P$ be smooth. Then

1. $T(g \circ f) = Tg \circ Tf$;
2. $T\mathrm{id} = \mathrm{id}_{TM}$;
3. if f is a diffeomorphism, Tf is a diffeomorphism, linear on each fibre, with $(Tf)^{-1} = T(f^{-1})$.

***Proof*:**
Statements (1) and (2) hold on each fibre by proposition 2.2.5 and, both sides being smooth (corollary 2.4.7), hold as maps of bundles. For (3), f^{-1} is smooth, so Tf and $T(f^{-1})$ are smooth and, by (1)–(2), mutually inverse; thus Tf is a diffeomorphism, linear on each fibre by proposition 2.2.5, completing the proof.

Property (3) makes Tf a *bundle isomorphism*. In general Tf carries the fibre $\pi_M^{-1}(p)$ into $\pi_N^{-1}(f(p))$, so the square

$$\begin{tikzcd} TM \arrow[r, "Tf"] \arrow[d, "\pi_M"'] & TN \arrow[d, "\pi_N"] \\ M \arrow[r, "f"] & N \end{tikzcd}$$

commutes; a fibre-respecting map of this kind is a *bundle homomorphism*, the notion generalized to arbitrary fibre bundles in chapter 11.

The bundle can also be built by gluing, as manifolds themselves were. For a countable, locally finite atlas $\{(U_i, \varphi_i)\}$ of an m-manifold, set $\coprod_i \varphi_i(U_i) \times \mathbb{R}^m$ (Euclidean opens) and identify

$$(x,v) \sim (y,w) \iff y = \varphi_j \varphi_i^{-1}(x), \quad w = D(\varphi_j \varphi_i^{-1})(x)\, v$$

All atlases give canonically isomorphic results, recovering TM.

Exercise 2.4.1

1. Show that $TS^1 \cong S^1 \times \mathbb{R}$, and conclude that the converse of corollary 2.4.6 fails, since S^1 is not covered by a single chart.

3

Smooth Maps and Submanifolds

The tangent map Tf linearizes a smooth f at every point, and its one diffeomorphism-invariant – the rank – controls the local shape of f. This chapter extracts that local structure. The constant rank theorem puts a map of locally constant rank into a linear normal form, and the maximal-rank cases (immersions, submersions, embeddings) are read off from it. That machinery then settles the questions left open about submanifolds in section 1.4: which subsets $S \subseteq M$ are submanifolds, when the image of a smooth map is one, what the tangent space of a submanifold is, and how level sets cut them out. The measure-zero refinement of the rank – Sard's theorem – and its consequences (embeddings into $\mathbb{R}^N$, smooth approximation, transversality, intersection theory) open Part II.

3.1 Rank

The rank is the one feature of a linear map invariant under change of basis, so it passes to T_pf without reference to coordinates.

Definition 3.1.1: Rank

The *rank* of a smooth $f\colon M \to N$ at p is the rank of the linear map T_pf. If it is the same at every point, f has *constant rank*. If it is the largest possible, $\operatorname{rank} T_pf = \min(\dim M, \dim N)$, then f has *full rank at p*; if f has full rank at every point, f has *full rank*.

Rank is a diffeomorphism invariant, so it can be computed from any coordinate representative. The two full-rank cases behave very differently and are named separately.

Definition 3.1.2: Immersion and Submersion

Let $f\colon M \to N$ be smooth.

1. f is an *immersion at p* if T_pf is injective; an *immersion* if it is one at every point (so $\operatorname{rank} f \equiv \dim M$).

2. f is a *submersion at p* if T_pf is surjective; a *submersion* if it is one at every point (so $\operatorname{rank} f \equiv \dim N$).

An immersion may still self-intersect, retrace itself, or accumulate onto itself, since injectivity of T_pf is purely local; controlling the global image needs the extra hypothesis that f is a homeomorphism onto its image, taken up in section 3.1.2. The full-rank locus is open, a fact packaged in the following matrix lemma.

Lemma 3.1.3: Full-Rank Matrices

Let $M(m,n)$ be the space of $m \times n$ real matrices and $M(m,n;k)$ the set of those of rank k. Then $M(m,n;k)$ is a submanifold of $M(m,n)$ of dimension $k(m+n-k)$. In particular, for $k = \min(m,n)$ it is an open submanifold.

***Proof*:**
Fix $X_0 \in M(m,n;k)$. Since X_0 has rank k, some $k \times k$ minor is nonsingular; permutation matrices P, Q move it to the top left, so

$$PX_0Q = \begin{pmatrix} A_0 & B_0 \\ C_0 & D_0 \end{pmatrix}, \qquad A_0 \in M(k,k) \text{ nonsingular}$$

As det is continuous and $\det A_0 \neq 0$, the top-left block A stays nonsingular on a neighbourhood Ω of X_0 (after the fixed permutation); on Ω write $Y = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.

Rank criterion. For such Y with A nonsingular, the row operation

$$\begin{pmatrix} I_k & 0 \\ -CA^{-1} & I_{m-k} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$$

preserves rank, and the right-hand side has rank $k + \operatorname{rank}(D - CA^{-1}B)$ since A is nonsingular. Hence $\operatorname{rank} Y = k \iff D = CA^{-1}B$.

Submanifold. On Ω define the smooth map (its entries are rational with nonvanishing denominator $\det A$)

$$g\colon \Omega \to M(m-k, n-k) \cong \mathbb{R}^{(m-k)(n-k)}, \qquad g(Y) = D - CA^{-1}B$$

By the criterion, $M(m,n;k) \cap \Omega = g^{-1}(0)$. Moreover g is a submersion: holding A, B, C fixed, $D \mapsto D - CA^{-1}B$ is an affine isomorphism onto $M(m-k, n-k)$, so Dg is surjective everywhere. Thus 0 is a regular value, and by the regular-level-set characterization (proposition 1.4.5) $g^{-1}(0)$ is a submanifold of codimension $(m-k)(n-k)$, i.e. of dimension $mn-(m-k)(n-k) = k(m+n-k)$. For $k = \min(m,n)$ one factor of $(m-k)(n-k)$ vanishes, so g has target $\mathbb{R}^0$ and $M(m,n;k) \cap \Omega = \Omega$ is open, completing the proof.

The lemma turns pointwise full rank into an open condition, which is what makes the immersion and submersion properties locally stable.

Proposition 3.1.4: Immersion or Submersion is Locally Stable

Let $f\colon M \to N$ be smooth and $p \in M$. If T_pf is injective (resp. surjective), then f is an immersion (resp. submersion) on a neighbourhood of p.

***Proof*:**
Injectivity or surjectivity of T_pf means f has full rank at p. In coordinate charts the Jacobian of f is a matrix-valued smooth function, and by lemma 3.1.3 the full-rank matrices form an open set; by continuity the Jacobian remains of full rank on a neighbourhood of p, so f is an immersion (resp. submersion) there, completing the proof.

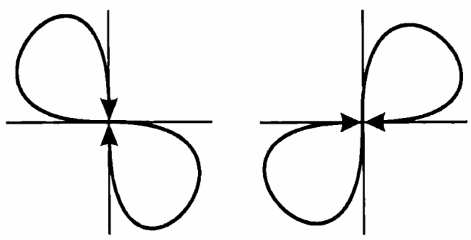
Example 3.1.5: Immersions and Submersions

1. The standard immersion of the torus $T^2 = S^1 \times S^1$ into $\mathbb{R}^3$,

$$(e^{i\theta_1}, e^{i\theta_2}) \mapsto \big((a + b\cos\theta_1)\cos\theta_2,\ (a + b\cos\theta_1)\sin\theta_2,\ b\sin\theta_1\big)$$

is an injective immersion, and hence – as T^2 is compact – an embedding, exactly when $a > b > 0$, when the tube of radius b clears the axis; for $a \leq b$ the tube reaches the axis and the parametrization fails to be an immersion where $a + b\cos\theta_1 = 0$.

2. The figure-eight $\gamma\colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion whose image is a figure-eight. It is *not* an embedding: the image with the subspace topology is not homeomorphic to an interval, as the crossing point has no interval neighbourhood.



3. The projection $\pi\colon \mathbb{R}^n \to \mathbb{R}^k$ onto the first k coordinates is a submersion, as is any surjective linear map; the quotient $\mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{RP}^n$ is a submersion onto a manifold.

The case where T_pf is a linear isomorphism deserves its own name; the next section proves that it is exactly the infinitesimal form of a local diffeomorphism.

Definition 3.1.6: Local Diffeomorphism

A smooth map $f: M \to N$ is a *local diffeomorphism* if every $p \in M$ has an open neighbourhood U with $f(U)$ open and $f|_U: U \to f(U)$ a diffeomorphism (and similarly with C^r in place of smooth).

Example 3.1.7: Local Diffeomorphism

The quotient $\pi: S^2 \to \mathbb{R}\mathrm{P}^2$ sending a point to the line through it and the origin is a local diffeomorphism but not a global one: it is two-to-one, while a diffeomorphism is injective.

Proposition 3.1.8: Properties of Local Diffeomorphisms

Let $f: M \to N$ and $g: N \to P$ be smooth.

1. A composition of local diffeomorphisms is a local diffeomorphism.
2. A finite product of local diffeomorphisms is a local diffeomorphism.
3. Every local diffeomorphism is an open map and a local homeomorphism.
4. The restriction of a local diffeomorphism to an open submanifold is a local diffeomorphism.
5. Every diffeomorphism is a local diffeomorphism, and a bijective local diffeomorphism is a diffeomorphism.
6. f is a local diffeomorphism if and only if each point has a coordinate representative that is a diffeomorphism onto an open set.

***Proof*:**
(1)–(2): near a point, each factor restricts to a diffeomorphism of open sets, and compositions and finite products of diffeomorphisms of open sets are again such. (3): f is locally a diffeomorphism onto an open set, hence locally open, hence open, and a local homeomorphism. (4): an open submanifold meets each neighbourhood in an open set, on which f still restricts to a diffeomorphism. (6): restated locally, "$f|_U$ a diffeomorphism onto an open set" is exactly the existence of a diffeomorphism coordinate representative.

(5) A diffeomorphism restricts to one on any open set, so it is a local diffeomorphism. Conversely let f be a bijective local diffeomorphism; it is a continuous open bijection, hence a homeomorphism, and its inverse is smooth because on each $f(U)$ it agrees with the smooth $(f|_U)^{-1}$ (proposition 1.3.6). Thus f is a diffeomorphism, completing the proof.

Immersion and submersion together pin down the isomorphism case.

Proposition 3.1.9: Local Diffeomorphism via Rank

For smooth $f\colon M \to N$:

1. f is a local diffeomorphism if and only if it is both an immersion and a submersion;
2. if $\dim M = \dim N$ and f is an immersion or a submersion, then f is a local diffeomorphism.

***Proof*:**
T_pf is a linear isomorphism if and only if it is both injective and surjective, and (proved in the next subsection) f is a local diffeomorphism near p exactly when T_pf is an isomorphism. This gives (1). When $\dim M = \dim N$ a linear map between the tangent spaces is injective if and only if surjective, so either hypothesis in (2) forces T_pf to be an isomorphism at every point, completing the proof.

3.1.1 Rank Theorem

A linear map of rank r can be row- and column-reduced to the block $\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)$. The rank theorem is the nonlinear analogue: a smooth map of constant rank r becomes, in suitable charts, exactly the linear map $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0)$. The bijective case is the inverse function theorem, cheap and worth stating first.

Theorem 3.1.10: Inverse Function Theorem for Manifolds

Let $f\colon M \to N$ be smooth and $p \in M$ with T_pf invertible. Then f restricts to a diffeomorphism of an open neighbourhood of p onto an open neighbourhood of $f(p)$.

***Proof*:**
Choose charts (U, φ) at p and (V, ψ) at $f(p)$ with $f(U) \subseteq V$, and set $\hat{f} = \psi \circ f \circ \varphi^{-1}$. By definition 2.2.2 the Jacobian $D\hat{f}(\varphi(p))$ represents T_pf in the coordinate bases, so it is invertible. The Euclidean inverse function theorem (theorem 0.2.1) gives an open $\hat{U} \ni \varphi(p)$ on which $\hat{f}$ is a diffeomorphism onto the open set $\hat{f}(\hat{U})$. Then $U_0 = \varphi^{-1}(\hat{U})$ and $V_0 = \psi^{-1}(\hat{f}(\hat{U}))$ are open, and

$$f|_{U_0} = \psi^{-1} \circ \hat{f} \circ \varphi$$

is a composition of diffeomorphisms, hence a diffeomorphism $U_0 \to V_0$, completing the proof.

The general statement drops invertibility for constant rank; the inverse function theorem is the case $r = m = n$.

Theorem 3.1.11: Constant Rank Theorem

Let $f\colon M \to N$ have constant rank r, with $\dim M = m$, $\dim N = n$. Then each $p \in M$ has charts (U, φ) centred at p and (V, ψ) centred at $f(p)$ with $f(U) \subseteq V$ in which

$$\psi \circ f \circ \varphi^{-1}(x^1, \dots, x^r, x^{r+1}, \dots, x^m) = (x^1, \dots, x^r, 0, \dots, 0)$$

In particular a submersion is locally $(x^1, \dots, x^m) \mapsto (x^1, \dots, x^n)$, and an immersion is locally $(x^1, \dots, x^m) \mapsto (x^1, \dots, x^m, 0, \dots, 0)$.

***Proof*:**
Working in charts, take M open in $\mathbb{R}^m$ and N open in $\mathbb{R}^n$, and after permuting coordinates and translating assume $p = 0$, $f(0) = 0$, with the top-left $r \times r$ block of $Df(0)$ nonsingular. Split coordinates as (x, y) on the domain ($x \in \mathbb{R}^r$, $y \in \mathbb{R}^{m-r}$) and (u, v) on the codomain ($u \in \mathbb{R}^r$, $v \in \mathbb{R}^{n-r}$), and write $f(x, y) = (Q(x, y), R(x, y))$ with $Q \in \mathbb{R}^r$, so $\partial Q/\partial x$ is nonsingular at 0.

The map $\varphi(x, y) = (Q(x, y), y)$ has derivative $\begin{pmatrix} \partial Q/\partial x & \partial Q/\partial y \\ 0 & I \end{pmatrix}$, nonsingular at 0, so by theorem 0.2.1 it has a local smooth inverse $\varphi^{-1}(x, y) = (A(x, y), B(x, y))$. From $\varphi \circ \varphi^{-1} = \mathrm{id}$ comes $B(x, y) = y$, hence

$$f \circ \varphi^{-1}(x, y) = (x,\, S(x, y)), \qquad S(x, y) = R(A(x, y), y)$$

Its derivative is $\begin{pmatrix} I_r & 0 \\ \partial S/\partial x & \partial S/\partial y \end{pmatrix}$. Composing with a diffeomorphism does not change rank, so $f \circ \varphi^{-1}$ still has rank r; the first r columns already contribute r, forcing $\partial S/\partial y = 0$. Thus $S = S(x)$ is independent of y, and $f \circ \varphi^{-1}(x, y) = (x, S(x))$.

Finally $\sigma(u, v) = (u, v - S(u))$ is a diffeomorphism of the codomain, and $\sigma \circ f \circ \varphi^{-1}(x, y) = (x, 0)$. Taking $\varphi = \varphi$ and $\psi = \sigma$ (composed with the initial charts) gives the stated normal form, completing the proof.

Restated invariantly: constant rank is exactly the condition for f to look linear in suitable coordinates.

Corollary 3.1.12: Constant Rank Means Locally Linear

For a smooth $f\colon M \to N$ with M connected, the following are equivalent:

1. every point has charts in which f has a linear coordinate representative;
2. f has constant rank.

***Proof*:**
If f has constant rank, the constant rank theorem gives the linear (indeed projection) representative near each point. Conversely a linear representative has a constant rank on its chart, and the rank, being locally constant, is constant on the connected M, completing the proof.

The theorem localizes injectivity and surjectivity of f from those of $T_p f$.

Theorem 3.1.13: Local Injectivity and Surjectivity

Let $f\colon M^m \to N^n$ be smooth and $p \in M$.

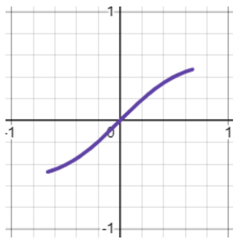
1. If T_pf is injective, then f is injective on a neighbourhood of p.
2. If T_pf is surjective, then f is surjective onto a neighbourhood of $f(p)$.
3. If T_pf is bijective, then f restricts to a bijection near p.

***Proof*:**
If T_pf is injective, f is an immersion near p (proposition 3.1.4), hence of constant rank m there; the constant rank theorem represents it as $x \mapsto (x, 0)$, manifestly injective, so f is injective near p. If T_pf is surjective, the same theorem represents f as the projection $(x, y) \mapsto x$, which is surjective onto a neighbourhood of $f(p)$. Bijectivity follows from both, completing the proof.

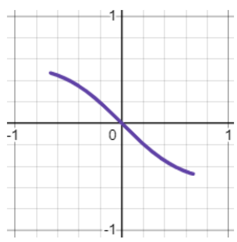
Local injectivity is not global injectivity. The map $f\colon S^1 \to \mathbb{R}^2$ tracing the lemniscate,

$$f(t) = \left(\frac{a\sqrt{2}\cos t}{1 + \sin^2 t}, \frac{a\sqrt{2}\cos t \sin t}{1 + \sin^2 t} \right)$$

is an immersion and is injective on each of the arcs $(\frac{\pi}{4}, \frac{3\pi}{4})$ and $(\frac{5\pi}{4}, \frac{7\pi}{4})$,



(a) $\hat{f}$ on $(\pi/4, 3\pi/4)$



(b) $\hat{f}$ on $(5\pi/4, 7\pi/4)$

yet not globally injective, since $f(\frac{\pi}{2}) = f(\frac{3\pi}{2})$ at the crossing. Surjectivity has no such defect: if $f|_U$ is surjective onto $f(U)$ then f is too. This asymmetry – injectivity needing a global hypothesis, surjectivity not – is pursued in section 3.1.2.

Global constant rank upgrades these local statements to global ones.

Theorem 3.1.14: Global Rank Theorem

Let $f\colon M \to N$ have constant rank. Then:

1. if f is injective, it is an immersion;
2. if f is surjective, it is a submersion;
3. if f is bijective, it is a diffeomorphism.

***Proof*:**
Let $\operatorname{rank} f \equiv r$, $\dim M = m$, $\dim N = n$.

(1) If f were not an immersion then $r < m$, and the constant rank theorem represents f near a point as $(x^1, \ldots, x^r, x^{r+1}, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0)$, which identifies $(0, \ldots, 0, \varepsilon)$ with $(0, \ldots, 0)$ for small ε – contradicting injectivity. So $r = m$.

(2) If f were not a submersion then $r < n$. Near each p the constant rank theorem represents f as $(x^1, \ldots, x^m) \mapsto (x^1, \ldots, x^r, 0, \ldots, 0)$; shrinking to a coordinate ball U with $f(\overline{U}) \subseteq V$, the image $f(\overline{U})$ is a compact subset of the slice $\{y \in V \mid y^{r+1} = \cdots = y^n = 0\}$, hence closed with empty interior, i.e. nowhere dense in N. Covering M by countably many such U_i writes $f(M)$ as a countable union of the nowhere-dense $f(\overline{U}_i)$, so by the Baire category theorem $f(M)$ has empty interior and cannot be all of N – contradicting surjectivity. So $r = n$.

(3) A bijection is by (1) and (2) an immersion and a submersion, hence a local diffeomorphism (proposition 3.1.9); a bijective local diffeomorphism is a diffeomorphism (proposition 3.1.8), completing the proof.

This answers the question left open earlier: a bijective smooth map is a diffeomorphism provided it has constant rank. Without that hypothesis it can fail.

Example 3.1.15: A Bijective Non-Diffeomorphism
The map $x \mapsto x^3$ of $\mathbb{R}$ (with the standard structure) is a smooth bijection but not a diffeomorphism: its inverse $x^{1/3}$ is not differentiable at 0, where $T_0 f$ drops rank. The rank is 1 everywhere except at the single point 0, so constant rank fails on a measure-zero set – exactly enough to break the conclusion of theorem 3.1.14.

3.1.2 Embeddings

As example 3.1.5 shows, an immersion may fail to be a submanifold: injectivity of $T_p f$ is local, and even a globally injective immersion need not carry the subspace topology onto its image. Demanding that f be a homeomorphism onto its image removes both defects.

Definition 3.1.16: Smooth Embedding

A *smooth embedding* is a smooth immersion $f\colon M \to N$ that is a homeomorphism onto its image $f(M)$ with the subspace topology.

A smooth embedding is in particular a topological embedding, and its image is a topological manifold; the next section shows it is a smooth submanifold. Several common hypotheses upgrade an injective immersion to an embedding.

Proposition 3.1.17: Sufficient Conditions for an Embedding

An injective smooth immersion $f\colon M \to N$ is a smooth embedding if any of the following holds:

1. f is an open or a closed map;
2. f is proper;
3. M is compact;
4. M has empty boundary and $\dim M = \dim N$.

***Proof*:**
It suffices in each case that f is a homeomorphism onto its image. (1) An injective continuous map that is open or closed onto its image is a homeomorphism onto it. (2) A proper map into a locally compact Hausdorff space is closed, so (1) applies. (3) A continuous map from a compact space to a Hausdorff space is closed, so (1) applies. (4) With $\dim M = \dim N$ and empty boundary, the immersion f is a local diffeomorphism (proposition 3.1.9), hence an open map, so (1) applies, completing the proof.

Immersions are exactly the maps that are embeddings locally.

Theorem 3.1.18: Immersions are Local Embeddings

A smooth map $f\colon M \to N$ is an immersion if and only if every $p \in M$ has a neighbourhood U on which $f|_U$ is a smooth embedding.

***Proof*:**
If $f|_U$ is an embedding for a neighbourhood of each point, then f is an immersion at each point, hence an immersion. Conversely let f be an immersion. By the constant rank theorem (theorem 3.1.11) with rank $m = \dim M$, in suitable charts near p the map is the linear inclusion $x \mapsto (x, 0)$. Restrict to a coordinate ball U with compact closure carried into the chart: on it f is a continuous injection from a set with compact closure into a Hausdorff space, hence a homeomorphism onto its image, and an immersion, so $f|_U$ is a smooth embedding, completing the proof.

3.1.3 Submersions

A submersion is the smooth counterpart of a topological quotient map. Recall that a surjective continuous $\pi\colon X \to Y$ presents Y as a quotient of X when it carries saturated open sets to open sets – equivalently, when a set in Y is open iff its preimage is; being an open or closed surjection suffices. Quotient maps satisfy a universal factorization property, and open ones additionally admit continuous local right inverses; both features reappear in the smooth category. A surjective submersion realizes both smoothly, with one caveat – a smooth submersion need not admit a *global* smooth section, only local ones.

Definition 3.1.19: Local Section

A *section* of a smooth $\pi\colon M \to N$ is a smooth right inverse, a smooth $\sigma\colon N \to M$ with $\pi \circ \sigma = \mathrm{id}_N$; a *local section* at p is one defined on a neighbourhood of $\pi(p)$ with $\sigma(\pi(p)) = p$.

Theorem 3.1.20: Submersions and Local Sections

A smooth $\pi\colon M \to N$ is a submersion if and only if every $p \in M$ lies in the image of a smooth local section.

***Proof*:**
If $p = \sigma(\pi(p))$ for a local section σ, then differentiating $\pi \circ \sigma = \mathrm{id}$ gives $T_p\pi \circ T_{\pi(p)}\sigma = \mathrm{id}$, so $T_p\pi$ is surjective; as p was arbitrary, π is a submersion. Conversely let π be a submersion. By the constant rank theorem, near p it is the projection $(x, y) \mapsto x$ in suitable charts; the map $x \mapsto (x, 0)$ is then a smooth local section through p, completing the proof.

Proposition 3.1.21: A Submersion is an Open Map

Every smooth submersion $f\colon M \to N$ is an open map; if it is also surjective, it is a topological quotient map.

***Proof*:**
By the constant rank theorem f is, near each point and in charts, the projection $(x, y) \mapsto x$, which is open; since the charts are diffeomorphisms (open maps), f is locally open, hence open. A surjective open continuous map is a quotient map, completing the proof.

Corollary 3.1.22: Submersions off a Compact Manifold

A smooth submersion $f\colon M \to N$ with M compact and N connected is surjective.

***Proof*:**
The image $f(M)$ is open (proposition 3.1.21) and, being the continuous image of a compact set, closed; a nonempty clopen subset of the connected N is all of N, completing the proof.

Local sections give surjective submersions their defining feature: smoothness of a map out of the base is detected upstairs.

Proposition 3.1.23: Characteristic Property of Submersions

Let $\pi\colon M \to N$ be a surjective smooth submersion and P a smooth manifold (with or without boundary). A map $f\colon N \to P$ is smooth if and only if $f \circ \pi$ is smooth.

***Proof*:**
If f is smooth, so is $f \circ \pi$. Conversely suppose $f \circ \pi$ is smooth and fix $q \in N$. Choose p with $\pi(p) = q$ (surjectivity) and a smooth local section σ through p (theorem 3.1.20), so $\pi \circ \sigma = \mathrm{id}$ near q. Then $f = (f \circ \pi) \circ \sigma$ near q is a composition of smooth maps, hence smooth near q; as q was arbitrary, f is smooth, completing the proof.

Theorem 3.1.24: Universal Property of Submersions

Let $\pi\colon M \to N$ be a surjective smooth submersion and P a smooth manifold (with or without boundary). If $f\colon M \to P$ is smooth and constant on the fibres of π, then there is a unique smooth $\tilde{f}\colon N \to P$ with $\tilde{f} \circ \pi = f$:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ N \arrow[r, dashed, "\tilde{f}"] & P \end{tikzcd}$$

***Proof*:**
Since f is constant on fibres and π is surjective, setting $\tilde{f}(q) = f(p)$ for any $p \in \pi^{-1}(q)$ is well defined and is the unique map with $\tilde{f} \circ \pi = f$. It is smooth: $\tilde{f} \circ \pi = f$ is smooth, so proposition 3.1.23 makes $\tilde{f}$ smooth, completing the proof.

Exercise 3.1.1

1. For $f\colon \mathbb{R}^2 \to \mathbb{R}^2$, $f(x, y) = (x^2 - y^2, 2xy)$, compute $\mathrm{rank} T_p f$ at each p and determine where f is an immersion, a submersion, and a local diffeomorphism.
2. Show that the figure-eight $\gamma\colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion but not a smooth embedding.
3. Reprove that a smooth submersion $\pi\colon M \to N$ is an open map using local sections (theorem 3.1.20) rather than the constant rank theorem.

3.2 Submanifolds

Section 1.4 defined a regular submanifold by the slice condition and left two things unproved: that the slice condition really makes S a smooth manifold with the inclusion an embedding (and conversely), and that the four descriptions of proposition 1.4.5 agree. The rank theorem settles both, and then delivers the level-set constructions and the tangent space of a submanifold. Recall (definition 1.4.1) that $S \subseteq M$ in an m-manifold is a *regular k-submanifold* if every $p \in S$ lies in a chart (U, φ) of M with $\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$; such charts are *submanifold* (or *slice*) charts, and a submanifold of codimension one is a *hypersurface*. The precise slice vocabulary is as follows.

Definition 3.2.1: Local Slice

Let $U \subseteq \mathbb{R}^n$ be open.

1. A *k-slice* of U is a subset $\{(x^1, \dots, x^n) \in U \mid x^{k+1} = c^{k+1}, \dots, x^n = c^n\}$ for constants c^i (usually 0; the positions of the constants may be permuted).
2. For a smooth chart (U, φ) and $S \subseteq U$ with $\varphi(S)$ a k-slice of $\varphi(U)$, one says S is a k-slice of U.
3. A subset $S \subseteq M$ satisfies the *local k-slice condition* if each of its points lies in a smooth chart (U, φ) of M with $S \cap U$ a single k-slice of U. Such a chart is a *slice chart*, its coordinates *slice coordinates*, and a covering family a *slice atlas*.

A submanifold atlas is a slice atlas. The next lemma proves that the slice condition is equivalent to being a smoothly embedded subset – discharging the slice $\Leftrightarrow$ embedding half of proposition 1.4.5.

Lemma 3.2.2: Slice Condition Equals Embedding

Let M be a smooth m-manifold and $S \subseteq M$. If S satisfies the local k-slice condition, then with the subspace topology S is a topological k-manifold carrying a smooth structure that makes it a k-dimensional regular submanifold and the inclusion $\iota \colon S \to M$ a smooth embedding. Conversely, if $\iota \colon S \to M$ is a smooth embedding, then S satisfies the local k-slice condition.

***Proof*:**
Suppose S satisfies the local k-slice condition, and give it the subspace topology, inheriting Hausdorff and second countable. Let $\pi \colon \mathbb{R}^m \to \mathbb{R}^k$ be projection onto the first k coordinates. For a slice chart (U, φ) set $V = U \cap S$, $\hat{V} = \pi(\varphi(V))$, and $\psi = \pi \circ \varphi|_V \colon V \to \hat{V}$. Since $\varphi(V)$ is $\varphi(U)$ intersected with a k-slice A (setting $x^{k+1} = c^{k+1}, \dots$), it is open in A, and $\pi|_A \colon A \to \mathbb{R}^k$ is a diffeomorphism, so $\hat{V}$ is open in $\mathbb{R}^k$. The map ψ is a homeomorphism, with continuous inverse $\varphi^{-1} \circ j$, $j(x^1, \dots, x^k) = (x^1, \dots, x^k, c^{k+1}, \dots, c^m)$. Thus S is a topological k-manifold.

For smoothness, two slice charts $(U, \varphi), (U', \varphi')$ give charts $(V, \psi), (V', \psi')$ of S with transition $\psi' \circ \psi^{-1} = \pi \circ \varphi' \circ \varphi^{-1} \circ j$, a composition of smooth maps; so the induced atlas is smooth. In a slice chart the inclusion is $(x^1, \dots, x^k) \mapsto (x^1, \dots, x^k, c^{k+1}, \dots, c^n)$, a smooth immersion, and it is a topological embedding, hence a smooth embedding.

Conversely let $\iota \colon S \to M$ be a smooth embedding. As an immersion, by the constant rank theorem it has, near each $p \in S$, charts (U, φ) of S and (V, ψ) of M in which $\iota|_U$ is $(x^1, \dots, x^k) \mapsto (x^1, \dots, x^k, 0, \dots, 0)$. Shrinking to a ball $B_\varepsilon(p) \subseteq U$ with $B_\varepsilon(\iota(p)) \subseteq V$, the image $\iota(B_\varepsilon(p))$ is a single slice of $B_\varepsilon(\iota(p))$. Since S carries the subspace topology, $B_\varepsilon(p) = W \cap S$ for some open $W \subseteq M$; then $V' = B_\varepsilon(\iota(p)) \cap W$ is a slice chart with $V' \cap S = B_\varepsilon(p)$, so S meets the slice condition, completing the proof.

Uniqueness of the smooth structure induced on S is taken up with the general uniqueness of subspace structures in chapter 4. A second equivalent description cuts S out as a level set, the form closest to algebraic and complex geometry.

Proposition 3.2.3: Submanifold as a Local Zero Set

A subset $S \subseteq M$ with a submanifold chart (U, φ) at p (so $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$) is, near p, the zero set $U \cap S = f^{-1}(0)$ of the submersion $f = \pi \circ \varphi \colon U \to \mathbb{R}^{m-k}$, where π projects onto the last $m-k$ coordinates. Conversely such a zero set is a submanifold chart. Thus S has codimension $m-k$.

***Proof*:**
With $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$ (the slice $x^{k+1} = \cdots = x^m = 0$),

$$f^{-1}(0) = \varphi^{-1}(\pi^{-1}(0)) = \varphi^{-1}(\varphi(U) \cap \mathbb{R}^k) = \varphi^{-1}(\varphi(U \cap S)) = U \cap S$$

Conversely, if $U \cap S = f^{-1}(0)$ for $f = \pi \circ \varphi$, then $\varphi(U \cap S) = \varphi(U) \cap \mathbb{R}^k$, a submanifold chart, completing the proof.

Together with lemma 3.2.2 this discharges the equivalences of proposition 1.4.5: slice charts, smooth embeddings, and regular level sets describe the same submanifolds, and the graph description joins them – a graph is an embedding (example 3.2.6), and a regular level set is a graph by the implicit function theorem. Restriction of a smooth map to a submanifold is smooth, a fact already recorded in corollary 1.4.3.

With embeddings and the rank theorem in hand, the image of an embedding is a submanifold.

Theorem 3.2.4: The Image of an Embedding is a Submanifold

If $f \colon M \to N$ is a smooth embedding, then $f(M)$ is a regular submanifold of N of dimension $\dim M$.

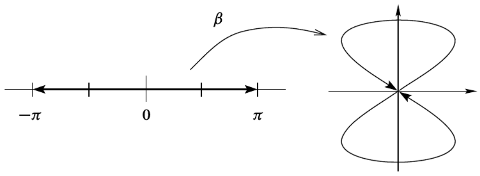
***Proof*:**
Let $\dim M = k$, $\dim N = n$, and $q = f(p) \in f(M)$; injectivity gives a unique such p. By the constant rank theorem (theorem 3.1.11) there are charts (U, φ) at p and (V, ψ) at q with $\psi \circ f \circ \varphi^{-1}(x^1, \ldots, x^k) = (x^1, \ldots, x^k, 0, \ldots, 0)$. As f is a homeomorphism onto its image, $f(U)$ is open in $f(M)$, so $f(U) = f(M) \cap O_0$ for some open $O_0 \subseteq V$. Shrinking to $O = O_0 \cap \psi^{-1}(\varphi(U) \times \mathbb{R}^{n-k})$ preserves $f(M) \cap O = f(U)$ and gives $\psi(O) \cap (\mathbb{R}^k \times \{0\}) = \varphi(U) \times \{0\} = \psi(f(U))$, so $(O, \psi|_O)$ is a submanifold chart for $f(M)$ at q. Since q was arbitrary, $f(M)$ is a regular k-submanifold, completing the proof.

Dropping either hypothesis – immersion or homeomorphism onto the image – breaks the conclusion.

Example 3.2.5: Non-Embeddings

1. The cuspidal cubic $t \mapsto (t^2, t^3)$ (example 1.3.11) is a topological embedding and a smooth map, but not an immersion (rank 0 at $t = 0$), hence not a smooth embedding: its image is not a smooth submanifold. So a smooth embedding is more than a topological embedding that happens to be smooth.
2. The figure-eight $\beta \colon (-\pi, \pi) \to \mathbb{R}^2$, $t \mapsto (\sin 2t, \sin t)$, is an injective immersion but not an embedding: its image is compact (the ends $t \to \pm\pi$ return to $\beta(0) = 0$), while the domain is

not, so β cannot be a homeomorphism onto its image.



3. The irrational-slope line $\gamma\colon \mathbb{R} \to T^2$, $\gamma(t) = (e^{2\pi it}, e^{2\pi i\alpha t})$ with α irrational, is an injective immersion whose image is dense in the torus; it is not an embedding and its image is not a submanifold.

Conversely, many familiar constructions are embeddings.

Example 3.2.6: Smooth Embeddings

1. An open subset $U \subseteq M$ is a regular submanifold of the same dimension.
2. For a fixed $q \in N$, the slice $x \mapsto (x, q)$ embeds M as the regular submanifold $M \times \{q\}$ of $M \times N$.
3. (Graphs) For open $U \subseteq M$ and smooth $f\colon U \to N$, the graph map $\gamma_f\colon U \to M \times N$, $x \mapsto (x, f(x))$, is a smooth embedding, so the graph $\Gamma(f) = \{(x, f(x)) \mid x \in U\}$ is a regular m-submanifold of $M \times N$. Indeed γ_f is an immersion (its coordinate Jacobian $\binom{I}{Df}$ has injective top block) and a homeomorphism onto its image, with continuous inverse the projection $M \times N \to M$ restricted to $\Gamma(f)$.

A smooth embedding is *proper* when it is a proper map; a submanifold is properly embedded if and only if it is a closed subset, so every compact submanifold is properly embedded, as is every graph $\Gamma(f)$ in $U \times N$ (though not in $M \times N$ unless $U = M$).

Level sets generalize the $\mathbb{R}^n$ construction of section 1.2.3 to maps between manifolds. Not every level set is a submanifold – $f_1(x, y) = x^2 - y^2$ has for $f_1^{-1}(0)$ a crossed pair of lines, and $f_2(x, y) = x^2 - y^3$ a cusp – and indeed every closed subset of M is the zero set of a smooth function (corollary 1.5.9), so an unconstrained level set is arbitrary. A rank hypothesis is what forces a submanifold.

Proposition 3.2.7: Constant-Rank Level Set Theorem

If $f\colon M \to N$ has constant rank r, then for each $q \in f(M)$ the level set $f^{-1}(q)$ is a regular submanifold of M of codimension r.

***Proof*:**
Fix $x \in f^{-1}(q)$. By the constant rank theorem there are charts (U, φ) at x and (V, ψ) at q, the latter recentred so $\psi(q) = 0$, in which

$$\psi \circ f \circ \varphi^{-1}(x^1, \ldots, x^m) = (x^1, \ldots, x^r, 0, \ldots, 0)$$

Then $f(y) = q$ for $y \in U$ exactly when the first r coordinates of $\varphi(y)$ vanish, so $\varphi(U \cap f^{-1}(q)) = \varphi(U) \cap \{x^1 = \cdots = x^r = 0\}$, an $(m - r)$-slice. Thus (U, φ) is a submanifold chart, and $f^{-1}(q)$ is a regular submanifold of codimension r, completing the proof.

Corollary 3.2.8: Submersion Level Set Theorem

If Df has full rank at every point of $f^{-1}(q)$, then $f^{-1}(q)$ is a regular submanifold of M.

***Proof*:**
The full-rank locus is open (proposition 3.1.4), so f has constant full rank on an open $U \supseteq f^{-1}(q)$; applying proposition 3.2.7 to $f|_U$ gives the claim, completing the proof.

This is the manifold form of the linear-algebra fact that the kernel of a surjective linear map $\mathbb{R}^m \to \mathbb{R}^r$ is a subspace of codimension r. It needs the rank hypothesis only on the level set itself, which the following vocabulary isolates.

Definition 3.2.9: Critical and Regular Points and Values

For a smooth $f\colon M \to N$, a point $p \in M$ is a *regular point* if $T_p f$ is surjective, and a *critical* (or *singular*) point otherwise. A point $q \in N$ is a *regular value* if every point of $f^{-1}(q)$ is regular (vacuously so when $f^{-1}(q) = \emptyset$), and a *critical value* otherwise; $f^{-1}(q)$ is then a *regular level set*.

A submersion has only regular points, and when $\dim M < \dim N$ every point is critical. The set of regular points is open (proposition 3.1.4).

Corollary 3.2.10: Regular Level Set Theorem

Every regular level set of a smooth $f\colon M \to N$ is a properly embedded submanifold of codimension $\dim N$.

***Proof*:**
Let c be a regular value. Every $x \in f^{-1}(c)$ is regular, so Df has full rank on $f^{-1}(c)$, and corollary 3.2.8 makes $f^{-1}(c)$ an embedded submanifold of codimension $\dim N$. It is closed in M by continuity, hence properly embedded, completing the proof.

Not every submanifold is a global regular level set ($\mathcal{P}^1$ is not, and a torus in S^3 is only awkwardly one), but every submanifold is *locally* one – the converse of proposition 3.2.3 without assuming the cutting function has the special form $\pi \circ \varphi$.

Theorem 3.2.11: Submanifolds are Locally Level Sets

A subset $S \subseteq M$ is an embedded k-submanifold if and only if every $x \in S$ has a neighbourhood U on which $U \cap S$ is a level set of a smooth submersion $f\colon U \to \mathbb{R}^{m-k}$.

***Proof*:**
If S is an embedded submanifold, then by proposition 3.2.3 one may take $f = \pi \circ \varphi$ in a submanifold chart, a submersion with $U \cap S = f^{-1}(0)$. Conversely, if $U \cap S$ is a level set of a

submersion $f\colon U \to \mathbb{R}^{m-k}$, then corollary 3.2.8 makes $U \cap S$ an embedded submanifold of U, so it meets the local slice condition (lemma 3.2.2); as this holds near each point, S is an embedded submanifold, completing the proof.

Definition 3.2.12: Defining Map

A *defining map* for an embedded submanifold $S \subseteq M$ is a smooth $\varphi\colon M \to N$ having S as a regular level set; when $N = \mathbb{R}^{m-k}$ it is a *defining function.* A *local defining map* is one defined on an open U with $S \cap U$ a regular level set.

The sphere S^n has defining function $f(x) = |x|^2$, and by theorem 3.2.11 every embedded submanifold admits a local defining map near each of its points.

3.2.1 Immersed Submanifolds

An embedded submanifold carries the subspace topology, but the image of an injective immersion need not: the figure-eight and the dense torus line (example 3.2.5) have images that, with the subspace topology, are not manifolds. Remembering the parametrizing map rather than only the image repairs this.

Definition 3.2.13: Immersed Submanifold

An *immersed submanifold* of M is a subset $S \subseteq M$ with a manifold topology (possibly finer than the subspace topology) and a smooth structure making the inclusion $\iota\colon S \hookrightarrow M$ an injective immersion.

Every embedded submanifold is immersed, with the subspace topology; the converse fails, as the figure-eight shows. Conversely, the image of any injective immersion $f\colon P \to M$ is an immersed submanifold when given the topology and smooth structure transported from P (so $f\colon P \to f(P)$ is a diffeomorphism), and it is embedded exactly when f is an embedding. Immersed submanifolds are the natural home of Lie subgroups and of the integral manifolds of the Frobenius theorem (chapter 6), whose leaf topology is genuinely finer than the subspace one.

3.2.2 Tangent Space of a Submanifold

For an immersed or embedded submanifold $S \subseteq M$ the inclusion $\iota\colon S \hookrightarrow M$ is an injective immersion, so $T_p\iota\colon T_pS \to T_pM$ is injective and identifies T_pS with a subspace of T_pM. On derivations $T_p\iota(v)$ acts by $T_p\iota(v)(f) = v(f \circ \iota) = v(f|_S)$: a tangent vector to S sees only the restriction of f to S. The three descriptions of the Euclidean case (proposition 2.1.19) now take their general form.

Proposition 3.2.14: Tangent Space via Curves

Let $S \subseteq M$ be a submanifold and $p \in S$. A vector $v \in T_pM$ lies in T_pS if and only if $v = \gamma'(0)$ for a smooth curve γ into M whose image lies in S, smooth as a map into S, with $\gamma(0) = p$.

***Proof*:**
If $\gamma = \iota \circ \gamma_S$ for a smooth curve γ_S into S, then $v = \gamma'(0) = T_p\iota(\gamma_S'(0)) \in T_pS$. Conversely, if $v \in T_pS$, write $v = T_p\iota(w)$; realize $w = \gamma_S'(0)$ for a smooth curve γ_S in S through p (definition 2.1.18), and set $\gamma = \iota \circ \gamma_S$, completing the proof.

Proposition 3.2.15: Tangent Space via Vanishing Functions

For a regular submanifold $S \subseteq M$ and $p \in S$,

$$T_pS = \{v \in T_pM \mid v(f) = 0 \text{ whenever } f \in C^\infty(M),\ f|_S = 0\}$$

***Proof*:**
If $v = T_p\iota(w) \in T_pS$ and $f|_S = 0$, then $v(f) = w(f \circ \iota) = w(f|_S) = 0$. Conversely, suppose v kills every f vanishing on S. In a slice chart at p, S is $\{x^{k+1} = \cdots = x^m = 0\}$; for $i > k$ the function βx^i (with β a bump equal to 1 near p, supported in the chart) is a global smooth function vanishing on S and agreeing with x^i near p, so $v(x^i) = v(\beta x^i) = 0$. By the coordinate expansion $v = \sum_i v(x^i)\, \partial/\partial x^i|_p = \sum_{i \le k} v(x^i)\, \partial/\partial x^i|_p$, which lies in $T_pS = \operatorname{span}\{\partial/\partial x^i|_p : i \le k\}$, completing the proof.

Proposition 3.2.16: Tangent Space via a Defining Map

If $\varphi\colon U \to N$ is a local defining map for a regular submanifold $S \subseteq M$, then $T_pS = \ker T_p\varphi$ for each $p \in S \cap U$.

***Proof*:**
Identify T_pS with $T_p\iota(T_pS) \subseteq T_pM$. Since $\varphi \circ \iota$ is constant on $S \cap U$, $T_p\varphi \circ T_p\iota = 0$, so $\mathrm{im} T_p\iota \subseteq \ker T_p\varphi$. As φ is a submersion, $T_p\varphi$ is surjective, so by rank–nullity $\dim\ker T_p\varphi = \dim T_pM - \dim T_{\varphi(p)}N = k = \dim T_pS = \dim \mathrm{im} T_p\iota$. Equal dimensions force equality, completing the proof.

Corollary 3.2.17: Tangent Space via a Defining Function

If $S = \varphi^{-1}(0)$ for a submersion $\varphi = (\varphi^1, \dots, \varphi^k)\colon M \to \mathbb{R}^k$, then $v \in T_pM$ is tangent to S if and only if $v(\varphi^1) = \cdots = v(\varphi^k) = 0$.

***Proof*:**
By proposition 3.2.16, $v \in T_pS$ iff $T_p\varphi(v) = 0$. The j-th component of $T_p\varphi(v)$ is $T_p\varphi(v)(y^j) = v(y^j \circ \varphi) = v(\varphi^j)$, so $T_p\varphi(v) = 0$ iff every $v(\varphi^j) = 0$, completing the proof.

3.2.3 Maps and Submanifolds

A smooth $f\colon M \to N$ restricts to a smooth map $f|_S = f \circ \iota\colon S \to N$ on any submanifold $S \subseteq M$, with $T_p(f|_S) = T_pf \circ T_p\iota$ on tangent spaces. If $f(M)$ lies in a regular submanifold $S \subseteq N$, then f

corestricts to a smooth map $M \to S$, using the subspace topology of S. Corestriction to an *immersed* submanifold is subtler, since its topology may be finer than the subspace topology; the criterion is taken up with the Frobenius theorem (chapter 6).

3.2.4 Submanifolds with Boundary

The level-set constructions extend to manifolds with boundary once the boundary is required to be regular as well.

Proposition 3.2.18: Regular Preimage with Boundary

Let $f\colon M \to N$ be smooth from a manifold with boundary, and $q \in N$ a regular value of both f and $f|_{\partial M}$. Then $f^{-1}(q)$ is a k-manifold with boundary, where $k = \dim M - \dim N$, and $\partial(f^{-1}(q)) = f^{-1}(q) \cap \partial M$.

***Proof*:**
On the interior, q is a regular value of $f|_{\operatorname{int} M}$, so $f^{-1}(q) \cap \operatorname{int} M$ is a k-manifold (corollary 3.2.10). At a boundary point $p \in f^{-1}(q) \cap \partial M$, take a boundary chart identifying a neighbourhood with $\mathbb{H}^m$. Regularity of f at p and of $f|_{\partial M}$ at p make f a submersion whose restriction to $\partial\mathbb{H}^m$ is still a submersion, so in the submersion normal form $f^{-1}(q)$ is a k-slice of $\mathbb{H}^m$ meeting $\partial\mathbb{H}^m$ in a $(k-1)$-slice. Thus $f^{-1}(q)$ is a manifold with boundary $f^{-1}(q) \cap \partial M$, completing the proof.

Proposition 3.2.19: Preimage of a Regular Interval

Let $f\colon M \to \mathbb{R}$ be smooth and proper on a boundaryless manifold, and $a < b$ regular values. Then $f^{-1}([a, b])$ is a compact manifold with boundary $f^{-1}(a) \sqcup f^{-1}(b)$.

***Proof*:**
As $[a, b]$ is compact and f proper, $f^{-1}([a, b])$ is compact, and $f^{-1}((a, b))$ is an open submanifold. At $p \in f^{-1}(a)$, regularity makes f a submersion, so by the constant rank theorem f is the coordinate x^m in suitable charts (after recentering); there $f^{-1}([a, b])$ is $\{x^m \geq 0\}$, a boundary chart into $\mathbb{H}^m$, and similarly $\{x^m \leq 0\}$ at $p \in f^{-1}(b)$. Hence $f^{-1}([a, b])$ is a manifold with boundary $f^{-1}(a) \sqcup f^{-1}(b)$, completing the proof.

This is the basic building block of cobordism theory (chapter 10): the level sets $f^{-1}(a)$ and $f^{-1}(b)$ together bound the compact region between them.

Exercise 3.2.1

1. Using corollary 3.2.17 and the defining function $f(x) = |x|^2$, show that the tangent space to the sphere is $T_pS^{n-1} = p^\perp = \{v \in \mathbb{R}^n \mid \langle v, p \rangle = 0\}$.
2. Show that $\mathrm{O}(n) = \left\{A \in M(n, n) \;\middle|\; A^\top A = I\right\}$ is a regular submanifold of $M(n, n)$ and that $T_I \,\mathrm{O}(n)$ is the space of skew-symmetric matrices, using the defining map $F(A) = A^\top A$ into the symmetric matrices.

3. For a smooth $f\colon \mathbb{R}^n \to \mathbb{R}^m$, show that the graph $\Gamma(f) = \{(x, f(x)) \mid x \in \mathbb{R}^n\}$ is a regular submanifold of $\mathbb{R}^{n+m}$ diffeomorphic to $\mathbb{R}^n$.

4

Quotient Manifolds and Homogeneous Spaces

Quotients are the third way of building manifolds, after charts (chapter 1) and level sets (chapter 3). Collapsing a manifold by an equivalence relation, identifying the points of each orbit of a group action or gluing along a symmetry, is how projective spaces, Grassmannians, and the homogeneous spaces of geometry arise. But a quotient of a manifold need not be a manifold, or even Hausdorff, so the construction demands hypotheses. This chapter isolates them: an action that is *free* and *proper* yields a smooth quotient with the projection a submersion, and the characteristic property of submersions (theorem 3.1.24) then supplies the smooth maps out of it.

Section 4.1 sets up group actions, orbits, and the orbit space; section 4.2 isolates proper actions and treats the discrete case, where the quotient is a covering manifold; section 4.3 proves the quotient manifold theorem; and section 4.4 specializes to homogeneous spaces G/H and builds the examples of projective spaces, Grassmannians, Stiefel and flag manifolds, and the classical matrix groups as regular-value examples. Throughout, a Lie group means a smooth manifold with smooth group operations; their structure theory is developed in [9], and only the manifold structure is used here.

4.1 Group Actions

Every quotient construction in this chapter begins with a group moving the points of a manifold. This section fixes the language for that movement, the action map and the maps it induces, and reads off the two invariants a point carries under it: its orbit and its stabilizer. The orbit space M/G is then assembled with the quotient topology, and its one structural feature available before any smoothness is proved, the universal property, is the tool used throughout to identify a quotient with a manifold one already knows.

Throughout, G is a Lie group or a discrete group Γ, meaning a smooth manifold whose multiplication and inversion are smooth (its structure theory is developed in [9]; only the underlying manifold is used here), and a discrete group is given the discrete topology.

Definition 4.1.1: Group Action

Let G be a Lie group and M a smooth manifold. A (left) *action* of G on M, written $G \curvearrowright M$, is a map

$$\theta\colon G \times M \to M, \qquad (g, p) \mapsto g \cdot p$$

satisfying $e \cdot p = p$ for all $p \in M$ and $g \cdot (h \cdot p) = (gh) \cdot p$ for all $g, h \in G$ and $p \in M$. The action is *continuous* if θ is continuous (G a topological group), and *smooth* if θ is smooth (G a Lie group). For a discrete group Γ the product $\Gamma \times M$ is the disjoint union $\coprod_{g\in\Gamma} M$, so a smooth action is precisely an action of Γ by diffeomorphisms.[a]

[a] A *right* action is a map $M \times G \to M$, $(p, g) \mapsto p \cdot g$, with $p \cdot (gh) = (p \cdot g) \cdot h$; it becomes a left action on setting $g \cdot p := p \cdot g^{-1}$, so no generality is lost by working with left actions.

Fixing $g \in G$ and letting the point vary produces the map $\theta_g\colon M \to M$, $p \mapsto g \cdot p$. The two axioms say exactly that $\theta_e = \mathrm{id}_M$ and $\theta_{gh} = \theta_g \circ \theta_h$, so $\theta_g \circ \theta_{g^{-1}} = \theta_e = \mathrm{id}_M$ and likewise in the other order: each θ_g is a bijection with inverse $\theta_{g^{-1}}$. When the action is smooth, θ_g is smooth with smooth inverse, hence a diffeomorphism, and $g \mapsto \theta_g$ is a group homomorphism

$$G \to \mathrm{Diff}(M)$$

into the diffeomorphism group of M. An action is thus the same data as such a homomorphism: a rule assigning to each g a symmetry of M, compatibly with the group law. The simplest example is the defining action of $\mathrm{GL}_n(\mathbb{R})$ on $\mathbb{R}^n$ by $(A, x) \mapsto Ax$, which is smooth because matrix-vector multiplication is polynomial in the entries; another is the group $\mathbb{Z}$ acting on $\mathbb{R}$ by translation, $n \cdot x = x + n$, each θ_n being the diffeomorphism $x \mapsto x + n$.

These maps describe the action globally; fixing instead a single point p and asking how the group meets it isolates two pieces of data, the points p can be moved to and the elements that hold it fixed.

Definition 4.1.2: Orbit and Stabilizer

Let $G \curvearrowright M$ and $p \in M$. The *orbit* of p is the set of points reachable from p,

$$G \cdot p = \{g \cdot p \mid g \in G\} \subseteq M$$

and the *stabilizer* (or isotropy group) of p is the subgroup of elements fixing p,

$$\mathrm{Stab}(p) = G_p = \{g \in G \mid g \cdot p = p\} \leq G$$

The orbits partition M: they are the equivalence classes of the relation $p \sim q$ iff $q = g \cdot p$ for some g, which is reflexive (e), symmetric (g^{-1}), and transitive (the second axiom). A point's orbit and stabilizer are not independent. Sending a group element to where it moves p collapses along cosets of G_p and gives the *orbit-stabilizer correspondence*

$$G/G_p \to G \cdot p, \qquad g\,G_p \mapsto g \cdot p \tag{4.1}$$

The map (4.1) is well defined and injective, since $g\,G_p = g'\,G_p$ holds exactly when $g^{-1}g' \in G_p$, that is when $g^{-1}g' \cdot p = p$, that is when $g \cdot p = g' \cdot p$; it is surjective onto the orbit by definition; and it intertwines the left G-actions, $h\cdot(g\cdot p) = (hg)\cdot p$. So an orbit is a homogeneous copy of the coset space G/G_p. When G is a Lie group acting smoothly this set bijection is upgraded to a diffeomorphism, the content of the homogeneous-space theorem (theorem 4.4.2) developed in section 4.4. For a concrete reading, take $\mathrm{SO}(n)$ acting on $\mathbb{R}^n$: the orbits are the spheres $\{|x| = r\}$ of each radius $r \geq 0$ together with the fixed origin, and the stabilizer of a point $p \neq 0$ consists of the rotations of the hyperplane $p^\perp$, a copy of $\mathrm{SO}(n-1)$.

The action's global shape is captured by three adjectives, according to how large the orbits and stabilizers are.

Definition 4.1.3: Free, Transitive, and Effective Actions

An action $G \curvearrowright M$ is

1. *free* if every stabilizer is trivial: $G_p = \{e\}$ for all p, equivalently $g\cdot p = p$ for some p forces $g = e$;
2. *transitive* if there is a single orbit: $G\cdot p = M$ for one (hence every) p; and
3. *effective* (or faithful) if only the identity acts as the identity map: $g \cdot p = p$ for all p forces $g = e$, that is, $g \mapsto \theta_g$ is injective.

Free is the strongest of the three fixed-point conditions and effective the weakest: a free action is effective, since a trivial stabilizer at even one point already rules out a nonidentity g fixing everything, but not conversely. The kernel of the homomorphism $g \mapsto \theta_g$ is $\bigcap_p G_p$, so effectiveness is the statement that this kernel is trivial and G embeds in $\mathrm{Diff}(M)$. Transitivity is a condition of a different kind, on the orbits rather than the stabilizers: it says M is a single orbit, so by (4.1) a transitive action exhibits M as a coset space G/G_p. The three run independently. The translation action of $\mathbb{Z}$ on $\mathbb{R}$ is free but far from transitive (its orbits are the countable cosets $x+\mathbb{Z}$); the action of $\mathrm{GL}_n(\mathbb{R})$ on $\mathbb{R}^n$ is effective but neither free (every matrix fixes 0) nor transitive (no matrix moves 0); and $\mathrm{SO}(n)$ restricted to the sphere S^{n-1} is transitive, with each stabilizer a copy of $\mathrm{SO}(n-1)$, so it is not free (for $n \geq 3$); it is the transitive but non-free case, in contrast to the two-orbit $\mathrm{GL}_n(\mathbb{R})$ action above.

With orbits understood, the object of interest is the space of orbits itself. Collapsing each orbit to a point produces the set M/G, and the topology it should carry is forced by the demand that the collapsing map be continuous and carry as many open sets as possible.

Definition 4.1.4: Orbit Space

Let $G \curvearrowright M$. The *orbit space* M/G is the set of orbits, and the *quotient map* $\pi\colon M \to M/G$ sends p to its orbit $G\cdot p$. The orbit space carries the *quotient topology*: a set $V \subseteq M/G$ is open exactly when its preimage $\pi^{-1}(V) \subseteq M$ is open. For a discrete group Γ the orbit space is also written M/Γ.

The quotient topology is the finest topology on M/G making π continuous, and by construction the open sets of M/G are the images of the π-saturated open sets of M. Nothing here uses smoothness or even continuity of the action, only that the orbits partition M; this is the same construction the reader met assembling $\mathbb{RP}^n$ and other quotients in chapter 1. The orbit space of $\mathbb{Z}$ acting on $\mathbb{R}$ by translation is the circle, computed below in example 4.1.6. What makes the quotient topology usable is not its definition but the property it was built to have: maps out of M/G are detected upstairs, on M.

Proposition 4.1.5: Universal Property of the Orbit Space

Let G act on a topological space M, give the orbit space M/G the quotient topology, and let $\pi\colon M \to M/G$ be the quotient map. Let N be any topological space.

1. A map $h\colon M/G \to N$ is continuous if and only if $h \circ \pi\colon M \to N$ is continuous.
2. If $f\colon M \to N$ is continuous and constant on orbits, that is $f(g \cdot p) = f(p)$ for all $g \in G$ and $p \in M$, then there is a unique continuous map $\overline{f}\colon M/G \to N$ with $\overline{f} \circ \pi = f$:

$$\begin{tikzcd} M \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ M/G \arrow[r, dashed, "\overline{f}"'] & N \end{tikzcd}$$

***Proof*:**

1. Since π is continuous, if h is continuous then so is $h \circ \pi$. Conversely suppose $h \circ \pi$ is continuous and let $V \subseteq N$ be open. Then
$$\pi^{-1}\big(h^{-1}(V)\big) = (h \circ \pi)^{-1}(V)$$
is open in M, so by the definition of the quotient topology $h^{-1}(V)$ is open in M/G. As V was arbitrary, h is continuous.
2. The orbits are exactly the fibres of π, so f is constant on orbits if and only if it is constant on the fibres of π. Since π is surjective, setting $\overline{f}(G \cdot p) = f(p)$ is well defined and is the only map with $\overline{f} \circ \pi = f$. This map is continuous: $\overline{f} \circ \pi = f$ is continuous, so part (1) applies to $h = \overline{f}$, completing the proof.

The universal property is the device by which a quotient is recognized. To identify M/G with a familiar space N, one produces a continuous map $f\colon M \to N$ constant on orbits, and proposition 4.1.5 descends it to $\overline{f}\colon M/G \to N$ at no further cost; the descended map is automatically continuous. Whether $\overline{f}$ is a homeomorphism, however, is a separate question the universal property does not settle, and in practice it is closed under a compactness argument: a continuous bijection from a compact space to a Hausdorff space is a homeomorphism [5]. This division of labour, descent for free and homeomorphism by compactness, recurs in every identification below.

Example 4.1.6: Orbit Spaces of Group Actions

1. *The circle as* $\mathbb{R}/\mathbb{Z}$. The group $\mathbb{Z}$ acts on $\mathbb{R}$ by translation, $n \cdot x = x + n$. Each stabilizer is trivial, so the action is free; it is not transitive, its orbits being the cosets $x + \mathbb{Z}$. The map $x \mapsto e^{2\pi i x}$ from $\mathbb{R}$ to the unit circle $S^1 \subseteq \mathbb{C}$ is continuous, surjective, and constant on orbits since $e^{2\pi i(x+n)} = e^{2\pi i x}$, so proposition 4.1.5 descends it to a continuous bijection $\mathbb{R}/\mathbb{Z} \to S^1$ (injective because $e^{2\pi i x} = e^{2\pi i y}$ forces $x - y \in \mathbb{Z}$). Because π maps the compact interval $[0,1]$ onto $\mathbb{R}/\mathbb{Z}$, the orbit space is compact, and a continuous bijection from a compact space onto the Hausdorff S^1 is a homeomorphism; hence $\mathbb{R}/\mathbb{Z} \cong S^1$.

2. *Projective space as $S^n/(\mathbb{Z}/2)$.* The group $\mathbb{Z}/2 = \{\pm 1\}$ acts on the sphere $S^n \subseteq \mathbb{R}^{n+1}$ by the antipodal map, $(-1) \cdot x = -x$. Since $0 \notin S^n$, no point equals its antipode, so the stabilizers are trivial and the action is free; the orbits are the antipodal pairs $\{x, -x\}$, so it is not transitive. The orbit space is the real projective space $\mathbb{RP}^n$ of chapter 1, and $\pi\colon S^n \to \mathbb{RP}^n$ is its standard two-to-one quotient. This action is properly discontinuous, the hypothesis put to work in section 4.2.

3. *The Hopf action on S^3.* The circle $G = S^1 = \{\lambda \in \mathbb{C} \mid |\lambda| = 1\}$ acts on $S^3 \subseteq \mathbb{C}^2$ by $\lambda \cdot (z_1, z_2) = (\lambda z_1, \lambda z_2)$, a smooth action since scalar multiplication is smooth. It is free: if $(\lambda z_1, \lambda z_2) = (z_1, z_2)$ with $(z_1, z_2) \neq (0, 0)$, then $\lambda z_i = z_i$ for some nonzero z_i, forcing $\lambda = 1$. The orbits are the circles $\{(\lambda z_1, \lambda z_2) \mid |\lambda| = 1\}$, so the action is not transitive. Two points of S^3 share an orbit exactly when they differ by a unit complex scalar, and rescaling any nonzero $(z_1, z_2) \in \mathbb{C}^2$ to unit norm shows every complex line meets S^3 in one such orbit; hence the orbit space is the complex projective line $\mathcal{P}^1$ of chapter 1, diffeomorphic to S^2. The quotient map $\pi\colon S^3 \to S^2$ is the Hopf map.

4. *A linear group on $\mathbb{R}^n$.* The group $\mathrm{GL}_n(\mathbb{R})$ acts on $\mathbb{R}^n$ by $(A, x) \mapsto Ax$. The action is smooth and effective, since $Ax = x$ for all x forces $A = I$, but it is not free (every A fixes 0) and not transitive on all of $\mathbb{R}^n$ (no A moves 0); it has exactly two orbits, $\{0\}$ and $\mathbb{R}^n \setminus \{0\}$, and is transitive on the latter. Restricting to the orthogonal group $\mathrm{O}(n)$ refines the picture: the orbits are the spheres $\{|x| = r\}$, and on a single sphere S^{n-1} the action is transitive with the stabilizer of a point a copy of $\mathrm{O}(n-1)$. By (4.1) this presents S^{n-1} as a coset space $\mathrm{O}(n)/\mathrm{O}(n-1)$, an identification made smooth in section 4.4.

Three of these four actions are free but not transitive, and they are the ones that will yield smooth quotients in the sections to follow: translation, the antipodal action, and the Hopf action each collapse a manifold onto a manifold of strictly lower dimension, the fibres of π being the orbits. The linear action fails both conditions, and the failure shows what they exclude. It is not free at the origin, and the origin is exactly the point whose orbit is a single point rather than a copy of the others, so the orbit space $\mathbb{R}^n/\mathrm{GL}_n(\mathbb{R})$ is the two-point set $\{[0], [\text{rest}]\}$ with a non-Hausdorff topology. Freeness excludes such collapsed orbits, and a second condition, properness, will exclude the topological pathologies that freeness alone leaves open; both are taken up in section 4.2.

Exercise 4.1.1

1. Let $G \curvearrowright M$ be a smooth action. Show that each $\theta_g\colon M \to M$ is a diffeomorphism and that $g \mapsto \theta_g$ is a group homomorphism $G \to \mathrm{Diff}(M)$. Deduce that the action is effective if and only if this homomorphism is injective, and that its kernel is $\bigcap_{p \in M} G_p$.

2. Let $\mathbb{Z}$ act on $\mathbb{R}^2$ by $n \cdot (x, y) = (x + n, y)$. Determine the orbits and stabilizers, decide whether the action is free and whether it is transitive, and identify the orbit space $\mathbb{R}^2/\mathbb{Z}$ with the cylinder $S^1 \times \mathbb{R}$.

3. Let $\mathrm{SO}(2)$ act on $\mathbb{R}^2$ by rotation about the origin. Describe the orbits and the stabilizer of each point, and decide whether the action is free, effective, or transitive. Using proposition 4.1.5, identify the orbit space $\mathbb{R}^2/\mathrm{SO}(2)$ with the half-line $[0, \infty)$.

4. Let a group G act on itself by conjugation, $g \cdot x = gxg^{-1}$. Identify the orbits and stabilizers in group-theoretic terms, and identify the kernel of the homomorphism $G \to \mathrm{Diff}(G)$. For which groups G is this action free, and for which is it effective?

5. Give a complete proof, via proposition 4.1.5, that $\mathbb{R}/\mathbb{Z}$ is homeomorphic to S^1, making explicit where compactness of the source and the Hausdorff property of the target are used.

4.2 Proper Actions and Covering Quotients

An action furnishes a candidate quotient, the orbit space M/G of definition 4.1.4, but the quotient topology alone controls nothing: M/G can fail to be Hausdorff, and even when the underlying set is a manifold the projection π need not be smooth in any usable sense. This section isolates the two conditions that tame the orbit space. *Properness* of the action is what forces M/G to be Hausdorff, and in the discrete case it specializes to a local separation condition under which $\pi\colon M \to M/\Gamma$ is a covering map. The general quotient manifold theorem, where G is a positive-dimensional Lie group, is taken up in section 4.3; here the payoff is the self-contained discrete case, which already builds the tori, the projective spaces, and the mapping tori.

The obstruction to a Hausdorff quotient is that two distinct orbits can be arbitrarily close: a sequence lying in one orbit may converge to a point of another. Properness rules this out by constraining how the group can move a compact set.

Definition 4.2.1: Proper Action

Let G be a topological group acting continuously on a manifold M. The action is *proper* if the map

$$\Theta\colon G \times M \to M \times M, \qquad (g,p) \mapsto (g \cdot p,\, p)$$

is a proper map, that is, $\Theta^{-1}(K)$ is compact for every compact $K \subseteq M \times M$.

The image of Θ is the *orbit relation*

$$R = \{(g \cdot p,\, p) \mid g \in G,\ p \in M\} \subseteq M \times M$$

the set of pairs sharing an orbit; R is exactly the equivalence relation whose quotient is M/G. Properness is a compactness bound on how far G can carry a compact set back onto itself: unwinding the definition, the action is proper if and only if for every compact $K \subseteq M$ the return set $G_K = \{g \in G \mid (g \cdot K) \cap K \neq \emptyset\}$ has compact closure, which for a discrete group says G_K is finite. Two families of proper actions recur below. Every action of a compact group G is proper, since $\Theta^{-1}(K)$ is a closed subset of $G \times \mathrm{pr}_2(K)$, a product of compact sets; in particular every finite group acts properly. The translation action of $\mathbb{Z}^n$ on $\mathbb{R}^n$, $k \cdot x = x + k$, is proper: a compact K lies in a ball of radius ρ, and $(k \cdot K) \cap K \neq \emptyset$ forces $|k| \leq 2\rho$, leaving finitely many k.

Whether M/G is Hausdorff is decided entirely by the orbit relation, because the projection π is an open map. This is the tool that converts the geometric condition of properness into the topological one of separation.

Proposition 4.2.2: Hausdorff Criterion for Orbit Spaces

Let G act continuously on a manifold M, with orbit relation $R = \{(g \cdot p,\, p) \mid g \in G,\ p \in M\}$.

1. M/G is Hausdorff if and only if R is closed in $M \times M$.
2. If the action is proper, then R is closed, so a proper action has Hausdorff orbit space.

***Proof*:**
First, π is open. For open $U \subseteq M$,

$$\pi^{-1}(\pi(U)) = \bigcup_{g \in G} g \cdot U$$

because $\pi(x) = \pi(y)$ holds iff $y = g \cdot x$ for some g. Each $g \cdot U$ is open, as $p \mapsto g \cdot p$ is a homeomorphism with inverse $p \mapsto g^{-1} \cdot p$; so the union is open, and by definition of the quotient topology $\pi(U)$ is open.

1. Suppose R is closed, and take $[p] \neq [q]$ in M/G. Then p and q lie in different orbits, so $(p, q) \notin R$. As R is closed there is a basic open box $U \times V \ni (p, q)$ with $(U \times V) \cap R = \emptyset$. The sets $\pi(U)$ and $\pi(V)$ are open (since π is open) and contain $[p]$ and $[q]$; they are disjoint, for if $[u] = [v]$ with $u \in U$, $v \in V$ then $(u, v) \in R \cap (U \times V) = \emptyset$. Thus M/G is Hausdorff. Conversely, suppose M/G is Hausdorff, so its diagonal Δ is closed in $(M/G) \times (M/G)$. Since $(a, b) \in R$ iff $\pi(a) = \pi(b)$, one has $R = (\pi \times \pi)^{-1}(\Delta)$, the preimage of a closed set under a continuous map, hence closed.

2. The product $M \times M$ is locally compact Hausdorff, and a proper continuous map into a locally compact Hausdorff space is closed ([5]). Thus Θ is a closed map, and $R = \Theta(G \times M)$ is closed. Part (1) then gives that M/G is Hausdorff, completing the proof.

Properness is therefore not merely sufficient for a Hausdorff quotient; it delivers closedness of R through one compactness argument, the mechanism theorem 4.3.1 will exploit. For the separation question only the closedness of R matters, and the sharpest way to see how it can fail is an action whose orbit relation is not closed.

Example 4.2.3: A Non-Hausdorff Orbit Space
Let $M = \mathbb{R}^2 \setminus \{0\}$ and let $\mathbb{Z}$ act smoothly by

$$n \cdot (x, y) = (2^n x,\ 2^{-n} y)$$

each n acting by the linear diffeomorphism $\operatorname{diag}(2^n, 2^{-n})$. The action is free: $2^n x = x$ and $2^{-n} y = y$ with $(x, y) \neq 0$ force $n = 0$. Its orbit relation is not closed. For $k \geq 1$ set $p_k = (2^{-k}, 1)$; then $k \cdot p_k = (2^{-k} \cdot 2^k,\ 2^{-k}) = (1, 2^{-k})$, so

$$(k \cdot p_k,\ p_k) = \big((1, 2^{-k}),\ (2^{-k}, 1)\big) \in R$$

As $k \to \infty$ one has $p_k \to (0, 1)$ and $k \cdot p_k \to (1, 0)$, both points of M, so $\big((1, 0), (0, 1)\big) \in \overline{R}$. Yet $(1, 0)$ and $(0, 1)$ share no orbit: $g \cdot (0, 1) = (0, 2^{-g})$ has vanishing first coordinate and so never equals $(1, 0)$. Hence $\big((1, 0), (0, 1)\big) \notin R$, the relation R is not closed, and by proposition 4.2.2 the orbit space $M/\mathbb{Z}$ is not Hausdorff: the points $[(1, 0)]$ and $[(0, 1)]$ have no disjoint neighbourhoods. This $\mathbb{Z}$-action is free and, as definition 4.2.4 makes checkable, properly discontinuous, while it is not proper. Proper discontinuity alone is thus not enough for a manifold quotient; Hausdorffness has to be secured separately.

For the discrete quotients the properness condition takes a local form that can be tested one point at a time, by asking that the group carry a small neighbourhood entirely off itself.

Definition 4.2.4: Properly Discontinuous Action

An action of a discrete group Γ on a manifold M is *properly discontinuous* if every $p \in M$ has a neighbourhood U such that

$$(g \cdot U) \cap U = \emptyset \quad \text{for all } g \in \Gamma,\ g \neq e$$

The condition forces the action to be free: if $g \cdot p = p$ for some $g \neq e$ then $p \in (g \cdot U) \cap U$, against the definition. So a properly discontinuous action is automatically free, and "free and properly discontinuous" names one hypothesis twice. Such a U is disjoint from all its nontrivial translates, and this is precisely the local picture of a covering: the translates $\{g \cdot U\}_{g \in \Gamma}$ are pairwise disjoint, since $(g \cdot U) \cap (h \cdot U) \neq \emptyset$ gives $(h^{-1}g) \cdot U \cap U \neq \emptyset$, hence $g = h$, and Γ permutes them freely. Proper discontinuity is the local part of properness: it is implied by a free proper action but is strictly weaker, and the action of example 4.2.3 is properly discontinuous yet not proper, which is exactly why its quotient loses Hausdorffness. What proper discontinuity does control unaided is the projection π. The translation action of $\mathbb{Z}$ on $\mathbb{R}$ is properly discontinuous: for $U = (p - \frac{1}{2}, p + \frac{1}{2})$ the translate $U + n$ misses U whenever $n \neq 0$; balls of radius $\frac{1}{2}$ give the same for $\mathbb{Z}^n$ on $\mathbb{R}^n$.

The quotient map of a properly discontinuous action is the differential-topological form of a covering. Recording the target notion first turns the theorem into an identification.

Definition 4.2.5: Smooth Covering Map

A *smooth covering map* is a smooth surjection $\pi \colon E \to M$ of smooth manifolds such that every $p \in M$ has a connected open neighbourhood U that is *evenly covered*: the preimage $\pi^{-1}(U)$ is a disjoint union of open subsets of E, the *sheets*, each carried diffeomorphically onto U by π. Then E is a *covering manifold* of the *base* M; when E is connected and simply connected it is the *universal cover*.

Restricted to a sheet, π is a diffeomorphism, so a smooth covering map is a surjective local diffeomorphism, in particular a surjective submersion. It therefore carries the characteristic property of proposition 3.1.23: a map out of the base is smooth iff its composite with π is. The topological theory of coverings, path lifting and the classification of covers by subgroups of the fundamental group, is developed in [7]; what is added here is that all of it runs in the smooth category. The exponential $\pi \colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, is a smooth covering: any proper open arc U has $\pi^{-1}(U)$ a disjoint union of intervals, each mapped diffeomorphically onto U. The map $z \mapsto z^k$ on S^1 is a k-sheeted smooth covering.

These pieces assemble into the discrete quotient theorem: a properly discontinuous action produces a covering, and the covering is smooth because π is a local diffeomorphism onto whose base the smooth structure of M descends.

Theorem 4.2.6: Covering Quotient Theorem

Let Γ be a discrete group acting smoothly and properly discontinuously on a smooth n-manifold M, and suppose the orbit space M/Γ is Hausdorff (by proposition 4.2.2, this holds whenever the action is proper, in particular whenever Γ is finite). Then M/Γ carries a unique smooth structure for which $\pi \colon M \to M/\Gamma$ is a smooth covering map, and with it M/Γ is a smooth n-manifold.

***Proof*:**
Write $\theta_g : M \to M$ for the diffeomorphism $p \mapsto g \cdot p$, with inverse $\theta_{g^{-1}}$; smoothness of the action makes each θ_g a diffeomorphism (definition 4.1.1).

Step 1: π is open. For open $U \subseteq M$,

$$\pi^{-1}(\pi(U)) = \bigcup_{g\in\Gamma} \theta_g(U)$$

since $\pi(x) = \pi(y)$ iff $y = \theta_g(x)$ for some g. Each $\theta_g(U)$ is open, so $\pi^{-1}(\pi(U))$ is open and $\pi(U)$ is open. Thus π is an open continuous surjection.

Step 2: even covering. Fix $p \in M$ and, by proper discontinuity, a connected neighbourhood $U \ni p$ with $\theta_g(U) \cap U = \emptyset$ for $g \neq e$. The translates $\{\theta_g(U)\}_{g\in\Gamma}$ are pairwise disjoint: $\theta_g(U) \cap \theta_h(U) \neq \emptyset$ yields $\theta_{h^{-1}g}(U) \cap U \neq \emptyset$, forcing $h^{-1}g = e$. The restriction $\pi|_U : U \to \pi(U)$ is a continuous open bijection (surjective by construction, and injective because $\pi(x) = \pi(y)$ with $x, y \in U$ gives $y = \theta_g(x) \in \theta_g(U) \cap U$, so $g = e$ and $x = y$), hence a homeomorphism onto the open set $\pi(U)$, which is connected as a continuous image of U. For each g the map $\pi|_{\theta_g(U)} = \pi|_U \circ \theta_{g^{-1}}$ is likewise a homeomorphism onto $\pi(U)$, and $\pi^{-1}(\pi(U)) = \coprod_{g\in\Gamma} \theta_g(U)$. So $\pi(U)$ is evenly covered and π is a local homeomorphism.

Step 3: M/Γ is a topological n-manifold. It is Hausdorff by hypothesis. It is second countable: for a countable basis $\{B_i\}$ of M, the open sets $\{\pi(B_i)\}$ form a countable basis, because any open $V \subseteq M/\Gamma$ has $\pi^{-1}(V) = \bigcup \{B_i \mid B_i \subseteq \pi^{-1}(V)\}$ and therefore $V = \pi(\pi^{-1}(V)) = \bigcup \{\pi(B_i) \mid B_i \subseteq \pi^{-1}(V)\}$. It is locally Euclidean: with U as in Step 2 chosen inside a chart domain, $(\pi|_U)^{-1}$ carries $\pi(U)$ homeomorphically onto the locally Euclidean U.

Step 4: descent of the smooth structure. For each p take U as in Step 2 inside a smooth chart (U, φ) of M (intersecting with a chart domain preserves the disjointness) and set

$$\widetilde{\varphi} = \varphi \circ (\pi|_U)^{-1} : \pi(U) \to \varphi(U) \subseteq \mathbb{R}^n$$

a homeomorphism onto an open set, hence a chart on M/Γ. Let (U, φ) and (U', φ') be two such with $\pi(U) \cap \pi(U') \neq \emptyset$. Fix $[q]$ in the overlap, and let $x = (\pi|_U)^{-1}([q]) \in U$ and $x' = (\pi|_{U'})^{-1}([q]) \in U'$; since $[x] = [x']$ there is a unique $g \in \Gamma$ with $x' = \theta_g(x)$, the uniqueness by freeness. For $[r]$ near $[q]$ the points $(\pi|_{U'})^{-1}([r])$ and $\theta_g\big((\pi|_U)^{-1}([r])\big)$ both lie in $\theta_g(U)$ and map to $[r]$; as $\pi|_{\theta_g(U)}$ is injective they agree. Hence, on a neighbourhood of $[q]$,

$$\widetilde{\varphi}' \circ \widetilde{\varphi}^{-1} = \varphi' \circ (\pi|_{U'})^{-1} \circ \pi|_U \circ \varphi^{-1} = \varphi' \circ \theta_g \circ \varphi^{-1}$$

a composition of smooth maps, so the transition is smooth and the charts $\{(\pi(U), \widetilde{\varphi})\}$ form a smooth atlas.

Step 5: π is a smooth covering. In the charts (U, φ) and $(\pi(U), \widetilde{\varphi})$ the projection reads as $\widetilde{\varphi} \circ \pi \circ \varphi^{-1} = \varphi \circ (\pi|_U)^{-1} \circ \pi \circ \varphi^{-1} = \mathrm{id}$ on $\varphi(U)$, so π is a local diffeomorphism. By Step 2 each $\pi(U)$ is evenly covered, its sheets $\theta_g(U)$ each carried onto $\pi(U)$ by the diffeomorphism $\pi|_U \circ \theta_{g^{-1}}$. Thus π is a smooth covering map, and M/Γ, Hausdorff and second countable and locally Euclidean with this structure, is a smooth n-manifold. For uniqueness, any smooth structure making π a local diffeomorphism has each $\widetilde{\varphi} = \varphi \circ (\pi|_U)^{-1}$ as a smooth chart, so its atlas is compatible with the one built here and defines the same structure, completing the proof.

The theorem reduces a range of manifold constructions to a checklist: produce a discrete group acting smoothly and freely, verify the local disjointness of proper discontinuity, and confirm Hausdorffness, most often by proving the action proper or by noting that Γ is finite. The quotient is then a manifold of the *same* dimension as M, and π is a covering, so every local computation downstairs lifts to M. The standard flat and elliptic quotients are all instances.

Example 4.2.7: Tori, Projective Spaces, and Mapping Tori

1. The lattice $\mathbb{Z}^n$ acts on $\mathbb{R}^n$ by $k \cdot x = x + k$. The action is smooth, free, and properly discontinuous (balls of radius $\frac{1}{2}$), and proper, since for compact $K \subseteq \mathbb{R}^n$ only finitely many k satisfy $(K + k) \cap K \neq \emptyset$; so $\mathbb{R}^n/\mathbb{Z}^n$ is Hausdorff. By theorem 4.2.6 it is a smooth n-manifold, the torus T^n of chapter 1, and $\pi \colon \mathbb{R}^n \to T^n$ is a smooth universal covering. The map $x \mapsto (e^{2\pi i x^1}, \dots, e^{2\pi i x^n})$ descends to a diffeomorphism $T^n \cong (S^1)^n$.

2. The group $\mathbb{Z}/2\mathbb{Z} = \{e, a\}$ acts on S^n by the antipodal map $a \cdot x = -x$. It is smooth and free (no $x \in S^n$ has $x = -x$), and properly discontinuous: the open hemisphere $U_x = \{y \in S^n \mid \langle y, x\rangle > 0\}$ contains x, while $a \cdot U_x = \{y \mid \langle y, x\rangle < 0\}$ is disjoint from it. Being finite the action is proper, so $S^n/(\mathbb{Z}/2\mathbb{Z})$ is Hausdorff; by theorem 4.2.6 it is a smooth n-manifold, real projective space $\mathbb{R}\mathrm{P}^n$, with $\pi \colon S^n \to \mathbb{R}\mathrm{P}^n$ a smooth double cover. This recovers the smooth structure on $\mathbb{R}\mathrm{P}^n$ from chapter 1, now presented as a covering quotient.

3. For a diffeomorphism $\varphi \colon N \to N$ of a smooth manifold, let $\mathbb{Z}$ act on $N \times \mathbb{R}$ by $k \cdot (p, t) = (\varphi^k(p),\, t+k)$. The action is smooth, free (the coordinate t is shifted by k, so $k \neq 0$ moves every point), properly discontinuous (take $U = N \times (t - \frac{1}{2}, t + \frac{1}{2})$, whose translates shift the interval off itself), and proper (a compact K lies in $N \times [-\rho, \rho]$, and $(k \cdot K) \cap K \neq \emptyset$ forces $|k| \leq 2\rho$). By theorem 4.2.6 the *mapping torus* $N_\varphi = (N \times \mathbb{R})/\mathbb{Z}$ is a smooth manifold of dimension $\dim N + 1$, and π is a smooth covering. The $\mathbb{Z}$-invariant map $(p, t) \mapsto [t] \in \mathbb{R}/\mathbb{Z} \cong S^1$ descends (proposition 4.1.5) to a smooth surjection $N_\varphi \to S^1$ realizing N_φ as fibred over the circle with fibre N. With $N = \mathbb{R}$ and $\varphi(x) = -x$ this produces the open Möbius band; with $\varphi = \mathrm{id}$, the trivial cylinder $N \times S^1$.

Exercise 4.2.1

1. Let a finite group Γ act smoothly and freely on a smooth n-manifold M. Show that the action is proper and properly discontinuous, and conclude that M/Γ is a smooth n-manifold and π an $|\Gamma|$-sheeted smooth covering.

2. For the antipodal action of $\mathbb{Z}/2\mathbb{Z}$ on S^n, write down an explicit evenly covered neighbourhood of an arbitrary point of $\mathbb{R}\mathrm{P}^n$ and verify that $\pi \colon S^n \to \mathbb{R}\mathrm{P}^n$ restricts to a diffeomorphism on each sheet.

3. Show that if G acts properly on M, then for every compact $K \subseteq M$ the return set $G_K = \{g \in G \mid (g \cdot K) \cap K \neq \emptyset\}$ is compact. (This is the compact-return form of properness quoted after definition 4.2.1.)

4. Let $\varphi \colon N \to N$ be a diffeomorphism. Verify that the $\mathbb{Z}$-action $k \cdot (p, t) = (\varphi^k(p),\, t + k)$ on $N \times \mathbb{R}$ is free, properly discontinuous, and proper, and deduce that the mapping torus N_φ is a smooth manifold. Show that the second-coordinate projection descends to a smooth surjection $N_\varphi \to S^1$.

5. Let $\pi \colon E \to M$ be a smooth covering map with M connected. Show that π is a submersion

and that the number of sheets $|\pi^{-1}(q)|$ is the same for every $q \in M$.

4.3 The Quotient Manifold Theorem

Section 4.2 settled the case of a discrete group, where a free proper action makes the quotient map a covering and the orbit space a manifold of the same dimension. Dropping discreteness is the task of this section. When a Lie group G acts freely and properly on M, the orbit space M/G is again a smooth manifold, now of dimension $\dim M - \dim G$, and $\pi\colon M \to M/G$ is a submersion. The general construction rests on two facts of Lie-group structure theory, stated and cited to [9]; the descent of smooth data across π is the submersion machinery of chapter 3, applied without change.

The discrete case built charts on M/Γ from evenly covered neighbourhoods, one for each point upstairs. For a positive-dimensional G the orbits are themselves positive-dimensional, so a chart on M/G can no longer come from an open set of M mapped bijectively down; it must instead be cut out *transverse* to the orbits, meeting each nearby orbit in a single point. Freeness forces every orbit to have the same dimension $\dim G$, so a transverse slice has a well-defined codimension; properness makes M/G Hausdorff and lets the slices patch. The theorem packages these two hypotheses into a single statement.

Theorem 4.3.1: Quotient Manifold Theorem

Let G be a Lie group acting smoothly, freely, and properly on a smooth manifold M. Then the orbit space M/G carries a smooth manifold structure of dimension $\dim M - \dim G$ for which the quotient map $\pi\colon M \to M/G$ is a smooth submersion.

***Proof*:**
The discrete case is proved in full; the positive-dimensional case is reduced to two inputs from Lie-group structure theory, cited to [9], followed by the submersion descent of chapter 3.

The discrete case ($\dim G = 0$). Here G is a discrete group Γ acting smoothly and freely on M, and properness is the assertion that the map $\Gamma \times M \to M \times M$, $(g, p) \mapsto (g \cdot p, p)$, is proper. Two consequences follow.

First, the action is properly discontinuous. Fix $p \in M$. Since M is locally compact Hausdorff, choose a compact neighbourhood K of p. Properness makes $\{g \in \Gamma \mid gK \cap K \neq \emptyset\}$ finite; call its nonidentity members $g_1, \dots, g_k$. Freeness gives $g_i \cdot p \neq p$ for each i, so, using the Hausdorff property and continuity of $x \mapsto g_i \cdot x$, pick an open $V_i \ni p$ with $g_i V_i \cap V_i = \emptyset$. Set $U = \operatorname{int} K \cap \bigcap_i V_i$, an open neighbourhood of p. For any $g \neq e$: if $gU \cap U \neq \emptyset$ then $gK \cap K \neq \emptyset$, so $g = g_i$ for some i, contradicting $g_i U \cap U \subseteq g_i V_i \cap V_i = \emptyset$. Thus $gU \cap U = \emptyset$ for all $g \neq e$, which is the properly discontinuous condition (definition 4.2.4).

Second, M/Γ is Hausdorff. A proper map into a locally compact Hausdorff space is closed, so the image of $(g, p) \mapsto (g \cdot p, p)$, which is the orbit relation $\{(g \cdot p, p) \mid g \in \Gamma,\ p \in M\}$, is closed in $M \times M$. By proposition 4.2.2, M/Γ is Hausdorff.

The action is now free and properly discontinuous, so theorem 4.2.6 makes $\pi\colon M \to M/\Gamma$ a smooth covering map and equips M/Γ with an atlas pulled back through the evenly covered charts; the Hausdorff property just established upgrades this locally Euclidean space to a manifold. A covering map is a local diffeomorphism (definition 4.2.5), hence a submersion, and preserves dimension, so $\dim M/\Gamma = \dim M = \dim M - \dim G$. This settles the discrete case.

The positive-dimensional case. Two facts about the action of a positive-dimensional Lie group are the input from [9], where the structure theory of Lie groups and their actions is developed; only the smooth-manifold consequences are used here.

1. Because G is a Lie group and the action is smooth, free, and proper, each orbit $G \cdot p$ is a properly embedded submanifold of M and the orbit map $g \mapsto g \cdot p$ is a diffeomorphism of G onto it; in particular every orbit has dimension $\dim G$.

2. *(Slice.)* Each p has an embedded submanifold $S \ni p$ of dimension $k = \dim M - \dim G$, transverse to the orbit, together with a G-invariant open neighbourhood of $G \cdot p$ and a diffeomorphism $G \times S \to G \cdot S$ under which the action becomes translation in the first factor, $g \cdot (h, s) = (gh, s)$.

Granting these, the remaining work is the submersion descent, which is in scope. Read off the local model $G \times S \cong G \cdot S$: the restriction of π to this set is the projection $(h, s) \mapsto s$ followed by $S \hookrightarrow M/G$, so π carries the open set $G \cdot S$ onto the image of S and is an open map near $G \cdot p$; as p was arbitrary, π is open. The slice S meets each orbit of $G \cdot S$ exactly once, so $\pi|_S : S \to M/G$ is a continuous open injection, hence a homeomorphism onto the open set $\pi(S)$; composing $(\pi|_S)^{-1}$ with a chart of S gives a chart on M/G, and these cover M/G with k-dimensional charts. On an overlap $\pi(S) \cap \pi(S')$ the transition sends $s \in S$ to the unique point of S' on its orbit, namely $(\pi|_{S'})^{-1}(\pi(s))$; in the two local models this is a composition of the smooth trivializations, hence smooth, so the charts form a smooth atlas. Properness makes the orbit relation closed exactly as in the discrete case, so M/G is Hausdorff by proposition 4.2.2, and M/G is second countable as the image of the second-countable M under the open surjection π. Thus M/G is a smooth k-manifold. In the slice charts π is the projection $(h, s) \mapsto s$, a smooth submersion, so $\pi : M \to M/G$ is a smooth submersion of the asserted dimension $k = \dim M - \dim G$, completing the proof.

The theorem adds the third and last of the book's manifold-building constructions to charts (chapter 1) and level sets (chapter 3). Its output is a submersion, not a bare topological space, and that is what makes it usable: every statement proved about surjective smooth submersions in chapter 3 now applies to π. In particular a map out of M/G is smooth if and only if its composition with π is (proposition 3.1.23), so smooth functions on the orbit space are exactly the G-invariant smooth functions on M; and a G-invariant smooth map $M \to P$ descends uniquely to a smooth map $M/G \to P$ (theorem 3.1.24). The quotient is thereby characterized by how maps leave it, not by any chart used to build it.

The discrete case is not a curiosity kept for the proof: it already produces the two most familiar quotients. The antipodal action of $\mathbb{Z}/2$ on S^n is free and, being an action of a finite group, proper, so $\mathbb{RP}^n = S^n/(\mathbb{Z}/2)$ is a smooth n-manifold with π a double cover; and the translation action of $\mathbb{Z}^n$ on $\mathbb{R}^n$ is free and proper, so the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ is a smooth n-manifold with π the universal covering. Both sit inside the theorem as the case $\dim G = 0$.

The atlas built in the proof depended on choices: a slice S at each point and a chart of S. A different choice of slices and charts would build a different atlas. That the resulting smooth structure does not depend on these choices is what justifies speaking of *the* quotient manifold, and it follows from the submersion requirement alone.

Proposition 4.3.2: Uniqueness of the Quotient Smooth Structure

Let G act smoothly, freely, and properly on M. There is at most one smooth manifold structure on M/G for which $\pi\colon M \to M/G$ is a smooth submersion.

***Proof*:**
Suppose two such structures, giving smooth manifolds N_1 and N_2 with the same underlying topological space M/G, and write $\pi_i\colon M \to N_i$ for the common map π regarded as landing in N_i; each π_i is a surjective smooth submersion. Consider the set-theoretic identity $\mathrm{id}\colon N_1 \to N_2$. Its composition with π_1 is $\mathrm{id} \circ \pi_1 = \pi_2$, which is smooth. Since π_1 is a surjective smooth submersion, proposition 3.1.23 makes $\mathrm{id}\colon N_1 \to N_2$ smooth. By the symmetric argument $\mathrm{id}\colon N_2 \to N_1$ is smooth, so id is a diffeomorphism and the two smooth structures coincide, completing the proof.

Uniqueness converts the theorem into a recognition principle. Because the quotient smooth structure is pinned down by the single requirement that π be a submersion, any smooth manifold N equipped with a surjective smooth submersion $M \to N$ whose fibres are exactly the G-orbits must be M/G: the universal property (theorem 3.1.24) produces mutually inverse smooth maps between N and M/G. To identify an orbit space with a familiar manifold, then, it suffices to exhibit one submersion off M that separates the orbits and lands on that manifold. The Hopf fibration (example 4.3.3) is identified with S^2 in exactly this way.

Both hypotheses are necessary, and each fails in a recognizable manner. Freeness controls the dimension of the fibres. Without it the orbit through p has dimension $\dim G - \dim \mathrm{Stab}(p)$, which jumps as the stabilizer grows: the rotation action of $\mathrm{SO}(2)$ on $\mathbb{R}^2$ fixes the origin, where $\mathrm{Stab}(0) = \mathrm{SO}(2)$ and the orbit is a point, while every other orbit is a circle, so the orbit dimension is not constant. The orbit space $\mathbb{R}^2/\mathrm{SO}(2) \cong [0, \infty)$, coordinatized by $r = |x|$, is a manifold with boundary rather than a manifold; the image of the origin is a boundary point with no Euclidean neighbourhood. Properness controls separation. Its failure is precisely the failure of M/G to be Hausdorff, by proposition 4.2.2: a non-proper action can have orbits that accumulate on one another, so no two of them can be prised apart in the quotient. The non-Hausdorff quotient of example 4.2.3 in section 4.2 is the discrete instance, and the dense-orbit example of exercise 4.3.1 (5) below is the continuous one.

One example repays a full computation: the action producing the Hopf fibration, the first fibre bundle whose total space, base, and fibre are all spheres.

Example 4.3.3: The Hopf Fibration $S^3 \to S^2$

Regard $S^3 \subseteq \mathbb{C}^2$ as the unit sphere, $S^3 = \big\{(w_1, w_2) \in \mathbb{C}^2 \bigm| |w_1|^2 + |w_2|^2 = 1\big\}$, and let the circle $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ act by scalar multiplication,

$$z \cdot (w_1, w_2) = (zw_1, zw_2)$$

This is smooth, being the restriction of a smooth action on $\mathbb{C}^2$. It is free: if $(zw_1, zw_2) = (w_1, w_2)$ with $(w_1, w_2) \neq 0$, then $z = 1$. It is proper because S^1 is compact: the action map $S^1 \times S^3 \to S^3 \times S^3$ has compact domain and Hausdorff codomain, so it is closed, and the preimage of a compact set is a closed subset of the compact $S^1 \times S^3$, hence compact. By theorem 4.3.1 the orbit space S^3/S^1 is a smooth manifold of dimension $3 - 1 = 2$ and $\pi\colon S^3 \to S^3/S^1$ is a smooth submersion.

To name the quotient, note that the orbit of (w_1, w_2) is $S^3 \cap \mathbb{C}(w_1, w_2)$, the intersection of S^3 with the complex line through (w_1, w_2). The orbit space is therefore the set of complex lines through the origin, which is $\mathcal{P}^1$. Precisely, the quotient map $q\colon \mathbb{C}^2 \setminus \{0\} \to \mathcal{P}^1$ is a surjective smooth submersion (chapter 1) whose fibres are the punctured complex lines. Its restriction $q|_{S^3}$ is constant on the S^1-orbits and has those orbits as its fibres, and it is a submersion: writing $T_wS^3 = \{v \in \mathbb{C}^2 \mid \mathrm{Re}\langle v, w\rangle = 0\}$ for the real orthogonal complement, the kernel of T_wq is the real 2-plane $\mathbb{R}w \oplus \mathbb{R}(iw)$ tangent to the complex line, of which only the line $\mathbb{R}(iw)$ lies in T_wS^3; hence T_wq maps the 3-dimensional T_wS^3 onto the 2-dimensional $T_{q(w)}\mathcal{P}^1$. By the universal property (theorem 3.1.24) $q|_{S^3}$ descends to a smooth bijection $S^3/S^1 \to \mathcal{P}^1$, and being a bijective submersion between 2-manifolds it is a local diffeomorphism, hence a diffeomorphism. Since $\mathcal{P}^1 \cong S^2$ (chapter 1),

$$\pi\colon S^3 \to S^2$$

is the *Hopf fibration*. Under the identification $\mathcal{P}^1 \cong S^2 \subseteq \mathbb{C} \times \mathbb{R} \cong \mathbb{R}^3$ it is the *Hopf map*

$$h(w_1, w_2) = \left(2w_1\overline{w_2},\ |w_1|^2 - |w_2|^2\right)$$

whose image lies in S^2 because $|h(w)|^2 = 4|w_1|^2|w_2|^2 + (|w_1|^2 - |w_2|^2)^2 = (|w_1|^2 + |w_2|^2)^2 = 1$, and which is constant on orbits since $zw_1\,\overline{zw_2} = |z|^2w_1\overline{w_2} = w_1\overline{w_2}$.

The fibres of the Hopf fibration are the great circles $\pi^{-1}(\mathrm{pt})$, and any two of them are linked once in S^3, which is why the bundle is nontrivial: were S^3 diffeomorphic to $S^2 \times S^1$ over S^2, the fibres would be unlinked. The number recording this linking is the first invariant this book computes that distinguishes a bundle from a product, and it returns later in the guise of a degree and of a de Rham class. For now the example makes the theorem concrete: a free proper action of a positive-dimensional group has produced the 2-sphere as the base of a submersion off the 3-sphere.

Exercise 4.3.1

1. Let a compact Lie group G act smoothly on a smooth manifold M. Show that the action is proper. Conclude that if the action is also free, then M/G is a smooth manifold with π a submersion.

2. The antipodal action of $\mathbb{Z}/2 = \{\pm 1\}$ on S^n sends x to $-x$. Show it is free and proper, and deduce from the discrete case of theorem 4.3.1 that $\mathbb{RP}^n = S^n/(\mathbb{Z}/2)$ is a smooth n-manifold with $\pi\colon S^n \to \mathbb{RP}^n$ a smooth covering map. For which n is it the universal cover?

3. Let G act smoothly, freely, and properly on M, and let $F\colon M \to N$ be a surjective smooth submersion onto a smooth manifold N whose fibres are exactly the G-orbits. Show that N is diffeomorphic to M/G.

4. Fix coprime integers $p \geq 1$ and q, and let $\zeta = e^{2\pi i/p}$. Let $\mathbb{Z}/p$ act on $S^3 \subseteq \mathbb{C}^2$ by $m \cdot (w_1, w_2) = (\zeta^m w_1, \zeta^{qm} w_2)$. Show that the action is free and proper, and conclude that the *lens space* $L(p; q) = S^3/(\mathbb{Z}/p)$ is a smooth 3-manifold with $\pi\colon S^3 \to L(p; q)$ a smooth covering map with p sheets.

5. Fix an irrational α and let $\mathbb{R}$ act on the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t \cdot [x, y] = [x + t, y + \alpha t]$. Show that the action is free but not proper, and that $T^2/\mathbb{R}$ is not Hausdorff. Which hypothesis of theorem 4.3.1 does this violate?

4.4 Homogeneous Spaces and Examples

The quotient manifold theorem (theorem 4.3.1) turns a free proper action into a smooth quotient. This section takes up the opposite extreme of the orbit structure: an action with a *single* orbit. Such an action recovers its manifold as a coset space, and the coset spaces of the classical groups $\mathrm{O}(n)$ and $\mathrm{U}(n)$ furnish a large supply of examples: projective spaces, Grassmannians, Stiefel and flag manifolds.

A transitive action makes every point look like every other, so a manifold carrying one has no distinguished point and its global shape is pinned down by the symmetry group together with a single stabilizer. This is the situation to name.

Definition 4.4.1: Homogeneous Space

Let G be a Lie group. A *homogeneous space* (or *homogeneous G-space*) is a smooth manifold M with a smooth transitive action $G \curvearrowright M$. Transitivity (definition 4.1.3) means $M = G \cdot p$ is a single orbit for any $p \in M$.

The sphere is the first instance. The orthogonal group $\mathrm{O}(n)$ acts on $S^{n-1} \subseteq \mathbb{R}^n$ by $A \cdot x = Ax$; since an orthogonal transformation carries any unit vector to any other, the action is transitive and S^{n-1} is a homogeneous $\mathrm{O}(n)$-space. Fixing the pole e_n, an orthogonal A with $Ae_n = e_n$ also preserves $e_n^\perp = \mathbb{R}^{n-1}$ and acts on it by some $C \in \mathrm{O}(n-1)$, so the stabilizer $\mathrm{Stab}(e_n)$ is the copy $\{\mathrm{diag}(C, 1) \mid C \in \mathrm{O}(n-1)\}$ of $\mathrm{O}(n-1)$. Theorem 4.4.2 reads this off as a diffeomorphism $S^{n-1} \cong \mathrm{O}(n)/\mathrm{O}(n-1)$.

To make the identification $M \cong G/G_p$ precise, two things are needed: that the coset space G/H is a manifold for a closed subgroup H, and that the orbit map through p descends from G to a diffeomorphism of G/G_p. Both are supplied at once. The first is a descent along the submersion machinery of chapter 3; the Lie-theoretic inputs (that G is a manifold, that closed subgroups are embedded Lie subgroups, and that the relevant action is proper) are cited to [9].

Theorem 4.4.2: Homogeneous Space Theorem

Let G be a Lie group.

1. *(Coset spaces.)* For a closed subgroup $H \leq G$, the coset space G/H carries a unique smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient $\pi\colon G \to G/H$ is a smooth submersion. The left action $G \curvearrowright G/H$, $g \cdot (g'H) = (gg')H$, is smooth and transitive, making G/H a homogeneous space.
2. *(Orbit-stabilizer.)* If $G \curvearrowright M$ is a smooth transitive action and $p \in M$, then the stabilizer $G_p = \mathrm{Stab}(p)$ is a closed subgroup, and the orbit map $g \mapsto g \cdot p$ descends to a G-equivariant diffeomorphism
$$F\colon G/G_p \to M, \qquad F(gG_p) = g \cdot p$$

***Proof*:**

1. Because H is closed, it is an embedded Lie subgroup of G, and the right-translation action $G \times H \to G$, $(g, h) \mapsto gh$, is smooth, free, and proper ([9]). Freeness is immediate, since $gh = g$ forces $h = e$; the orbits of this action are exactly the left cosets gH, so its orbit space is G/H. The quotient manifold theorem (theorem 4.3.1) then makes G/H a smooth manifold of dimension $\dim G - \dim H$ with $\pi\colon G \to G/H$ a smooth submersion, and proposition 4.3.2 makes this structure the unique one with π a submersion.

   For the left action, consider the smooth map $G \times G \to G/H$, $(g, g') \mapsto \pi(gg')$. It is constant on the fibres of the surjective submersion $\mathrm{id}_G \times \pi\colon G \times G \to G \times G/H$, so by the universal property of submersions (theorem 3.1.24) it descends to a smooth map $G \times G/H \to G/H$, which is the action $g \cdot (g'H) = (gg')H$. Any two cosets $g'H$ and $g''H$ are exchanged by left translation by $g''(g')^{-1}$, so the action is transitive, completing the proof of the first part.

2. The orbit map $\theta_p\colon G \to M$, $g \mapsto g \cdot p$, is the restriction of the smooth action to $G \times \{p\}$, hence smooth. Since M is Hausdorff, $\{p\}$ is closed, so $G_p = \theta_p^{-1}(p)$ is a closed subgroup, and part (1) makes G/G_p a smooth manifold with $\pi\colon G \to G/G_p$ a submersion. As $\theta_p(gh) = (gh) \cdot p = g \cdot p = \theta_p(g)$ for $h \in G_p$, the orbit map is constant on the fibres of π, so theorem 3.1.24 descends it to a smooth map $F\colon G/G_p \to M$ with $F(gG_p) = g \cdot p$.

   The map F is a bijection: it is surjective because the action is transitive, and injective because $g \cdot p = g' \cdot p$ gives $g^{-1}g' \in G_p$, hence $gG_p = g'G_p$. It is G-equivariant, since $F(g \cdot (g'G_p)) = F(gg'G_p) = (gg') \cdot p = g \cdot F(g'G_p)$.

   Equivariance forces F to have constant rank. Write ℓ_g for left translation by g on G/G_p and τ_g for the action of g on M, both diffeomorphisms (with inverses $\ell_{g^{-1}}$ and $\tau_{g^{-1}}$). Equivariance is the identity $F \circ \ell_g = \tau_g \circ F$; differentiating at the base point $o = eG_p$ gives $T_{gG_p}F \circ T_o\ell_g = T_{F(o)}\tau_g \circ T_oF$. The outer maps $T_o\ell_g$ and $T_{F(o)}\tau_g$ are isomorphisms, so $\operatorname{rank} T_{gG_p}F = \operatorname{rank} T_oF$; as every point of G/G_p has the form gG_p, the rank is constant. A smooth bijection of constant rank is a diffeomorphism (chapter 3): injectivity forces the rank to equal $\dim(G/G_p)$, so F is an immersion, surjectivity forces it to equal $\dim M$, so F is a submersion, and a bijective local diffeomorphism is a diffeomorphism. Thus F is a G-equivariant diffeomorphism, as we sought to show.

The theorem is a dictionary between symmetry and shape, read in two directions. Read forward, it manufactures manifolds: any closed subgroup H of a Lie group produces the smooth quotient G/H, whose dimension is the bookkeeping difference $\dim G - \dim H$. Read backward, it recognizes a manifold already in hand: a transitive action exhibits M as G/G_p, so computing $\dim M$ reduces to computing a single stabilizer. The base point p is a choice, and different choices give conjugate stabilizers $G_{g \cdot p} = gG_pg^{-1}$, so the coset space is well defined up to the induced diffeomorphism. Everything below is one application of the backward reading, with G a classical group.

The Grassmannian was built by hand in chapter 1 as an atlas of graph charts; the homogeneous-space description recovers it in a line and, as a bonus, exhibits its compactness.

Example 4.4.3: The Grassmannian as $\mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$
Let $\mathrm{Gr}(k, n) = G_k(\mathbb{R}^n)$ be the Grassmannian of k-planes in $\mathbb{R}^n$ (definition 1.2.17). The orthogonal group $\mathrm{O}(n)$ acts on it by $A \cdot W = A(W)$; in the graph charts this action is given by rational formulas in the entries of A, so it is smooth. Here $\mathrm{O}(n)$ is used as a Lie group: its manifold structure is the regular-value computation of exercise 3.2.1 (2) (revisited in example 4.4.5 below),

and its full group structure is developed in [9].

The action is transitive: given a k-plane W, an orthonormal basis of W extends to an orthonormal basis of $\mathbb{R}^n$, and the matrix sending the standard basis to it is orthogonal and carries the standard plane $\mathbb{R}^k = \mathrm{Span}(e_1, \dots, e_k)$ to W. To compute the stabilizer of $\mathbb{R}^k$, note that an orthogonal A with $A(\mathbb{R}^k) = \mathbb{R}^k$ also preserves the orthogonal complement $(\mathbb{R}^k)^\perp = \mathbb{R}^{n-k}$, so in the splitting $\mathbb{R}^n = \mathbb{R}^k \oplus \mathbb{R}^{n-k}$ it is block-diagonal $A = \mathrm{diag}(B, C)$ with $B \in \mathrm{O}(k)$ and $C \in \mathrm{O}(n-k)$. Hence

$$\mathrm{Stab}(\mathbb{R}^k) = \mathrm{O}(k) \times \mathrm{O}(n-k)$$

embedded block-diagonally. By the orbit-stabilizer diffeomorphism (theorem 4.4.2),

$$\mathrm{Gr}(k, n) \cong \mathrm{O}(n)/\big(\mathrm{O}(k) \times \mathrm{O}(n-k)\big)$$

The dimension count reproduces the chart computation of chapter 1:

$$\dim \mathrm{Gr}(k, n) = \frac{n(n-1)}{2} - \frac{k(k-1)}{2} - \frac{(n-k)(n-k-1)}{2} = k(n-k)$$

Because $\mathrm{Gr}(k, n)$ is the continuous image of the compact group $\mathrm{O}(n)$, it is compact. The special case $k = 1$ is real projective space: the 1-planes in $\mathbb{R}^n$ are the lines through the origin, so

$$\mathbb{R}\mathrm{P}^{n-1} = \mathrm{Gr}(1, n) \cong \mathrm{O}(n)/\big(\mathrm{O}(1) \times \mathrm{O}(n-1)\big)$$

with $\mathrm{O}(1) = \{\pm 1\}$ and dimension $1 \cdot (n-1) = n - 1$.

The presentation of $\mathrm{Gr}(k, n)$ as a compact homogeneous space is the one that later chapters use: compactness is what makes the Grassmannian a legitimate target for degree and intersection arguments, and the symmetry group acts by diffeomorphisms, so no point of the Grassmannian is special. Remembering a little more than the k-plane (an ordered orthonormal basis of it) gives the next family.

Example 4.4.4: Stiefel and Flag Manifolds

The *Stiefel manifold* $V_k(\mathbb{R}^n)$ is the set of orthonormal k-frames $(v_1, \dots, v_k)$ in $\mathbb{R}^n$. Regarding a frame as the $n \times k$ matrix with columns v_i, one has $V_k(\mathbb{R}^n) = \{A \in M(n, k) \mid A^\top A = I_k\}$, and the map $G(A) = A^\top A$ into the symmetric $k \times k$ matrices $\mathrm{Sym}(k)$ has derivative $DG(A)H = H^\top A + A^\top H$; at $A \in V_k(\mathbb{R}^n)$ this is onto $\mathrm{Sym}(k)$, since $H = \frac{1}{2}AS$ yields $H^\top A + A^\top H = S$ for any $S = S^\top$. So I_k is a regular value and $V_k(\mathbb{R}^n)$ is a properly embedded submanifold of $M(n, k)$ (corollary 3.2.10) of dimension $nk - \frac{k(k+1)}{2}$.

The group $\mathrm{O}(n)$ acts smoothly on $V_k(\mathbb{R}^n)$ by $A \cdot (v_1, \dots, v_k) = (Av_1, \dots, Av_k)$, the restriction of the linear action. It is transitive, because any orthonormal k-frame extends to an orthonormal basis of $\mathbb{R}^n$, giving an orthogonal matrix carrying the standard frame $(e_1, \dots, e_k)$ to it. The stabilizer of $(e_1, \dots, e_k)$ consists of the orthogonal A fixing $e_1, \dots, e_k$; such an A fixes $\mathbb{R}^k$ pointwise and preserves $\mathbb{R}^{n-k}$, so $A = \mathrm{diag}(I_k, C)$ with $C \in \mathrm{O}(n-k)$. Hence $\mathrm{Stab}(e_1, \dots, e_k) = \mathrm{O}(n-k)$ and, by theorem 4.4.2,

$$V_k(\mathbb{R}^n) \cong \mathrm{O}(n)/\,\mathrm{O}(n-k)$$

of dimension $\frac{n(n-1)}{2} - \frac{(n-k)(n-k-1)}{2} = nk - \frac{k(k+1)}{2}$, in agreement with the regular-value count. The extremes are $V_1(\mathbb{R}^n) = S^{n-1}$ and $V_n(\mathbb{R}^n) = \mathrm{O}(n)$. Forgetting the frame and keeping only its span sends $V_k(\mathbb{R}^n) \to \mathrm{Gr}(k, n)$; in coset terms this is $\mathrm{O}(n)/\,\mathrm{O}(n-k) \to \mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$, whose fibre $(\mathrm{O}(k) \times \mathrm{O}(n-k))/\,\mathrm{O}(n-k) \cong \mathrm{O}(k)$ is the set of orthonormal frames of a fixed k-plane.

Iterating the block decomposition gives the *flag manifolds*. Fix a composition $n = n_1 + \cdots + n_r$ into positive integers, and let $\mathrm{Fl}(n_1, \ldots, n_r)$ be the set of orthogonal decompositions $\mathbb{R}^n = W_1 \oplus \cdots \oplus W_r$ with $\dim W_i = n_i$. The group $\mathrm{O}(n)$ acts transitively, and the stabilizer of the standard decomposition into coordinate blocks is the block-diagonal subgroup $\mathrm{O}(n_1) \times \cdots \times \mathrm{O}(n_r)$, so

$$\mathrm{Fl}(n_1, \ldots, n_r) \cong \mathrm{O}(n)/\big(\mathrm{O}(n_1) \times \cdots \times \mathrm{O}(n_r)\big)$$

The Grassmannian is the two-block case $\mathrm{Fl}(k, n-k) = \mathrm{Gr}(k, n)$.

Each of these spaces is compact, being a continuous image of $\mathrm{O}(n)$, and each is manufactured from the same group by choosing which subgroup to quotient by. The acting group has so far been treated as a Lie group on faith; the last example makes its manifold structure explicit and records the classical groups as regular-value examples, the form in which they appear throughout the book.

Example 4.4.5: The Classical Matrix Groups as Regular Level Sets

Each of the classical groups is cut out of a matrix space as the regular level set of a symmetry condition, which is what makes it a manifold; the Lie-group structure (the bracket, the exponential map, and the correspondence with a Lie algebra) is developed in [9].

1. *Orthogonal group.* As in exercise 3.2.1 (2), the map $F(A) = A^\top A$ from $M(n,n)$ to the symmetric matrices $\mathrm{Sym}(n)$ has I as a regular value, so $\mathrm{O}(n) = F^{-1}(I)$ is a submanifold of dimension $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$. By proposition 3.2.16 its tangent space at the identity is

$$T_I\,\mathrm{O}(n) = \ker DF(I) = \big\{H \in M(n,n) \;\big|\; H^\top + H = 0\big\}$$

   the space of skew-symmetric matrices, of dimension $\frac{n(n-1)}{2}$.

2. *Unitary group.* Over $\mathbb{C}$, view $M(n,n;\mathbb{C}) \cong \mathbb{R}^{2n^2}$ and let $G(A) = A^*A$ land in the Hermitian matrices $\mathrm{Herm}(n)$, a real vector space of dimension n^2, where A^* is the conjugate transpose. Its derivative $DG(A)H = H^*A + A^*H$ is onto $\mathrm{Herm}(n)$ at any $A \in \mathrm{U}(n)$: for a Hermitian S, taking $H = \frac{1}{2}AS$ gives $H^*A + A^*H = \frac{1}{2}(SA^*A + A^*AS) = S$. So I is a regular value and $\mathrm{U}(n) = G^{-1}(I)$ is a submanifold of dimension $2n^2 - n^2 = n^2$, with

$$T_I\,\mathrm{U}(n) = \{H \in M(n,n;\mathbb{C}) \mid H^* + H = 0\}$$

   the skew-Hermitian matrices, of real dimension n^2.

3. *Special orthogonal group.* On $\mathrm{O}(n)$ the determinant takes the values ± 1 and is continuous, hence locally constant, so $\mathrm{SO}(n) = \mathrm{O}(n) \cap \{\det = 1\}$ is open and closed in $\mathrm{O}(n)$: it is the identity component, an open submanifold of the same dimension $\frac{n(n-1)}{2}$, with the same tangent space $T_I\,\mathrm{SO}(n) = T_I\,\mathrm{O}(n)$ of skew-symmetric matrices.

The tangent space at the identity of each of these groups is its Lie algebra: skew-symmetric matrices for $\mathrm{O}(n)$ and $\mathrm{SO}(n)$, skew-Hermitian matrices for $\mathrm{U}(n)$. That the identity component alone carries this infinitesimal data, and that the group is reconstructed from it by the exponential map, is the content of the Lie theory picked up in [9]. What this book needs is only what has been proved: these are smooth manifolds, cut out by regular values, and they act smoothly on the projective spaces, Grassmannians, and frame manifolds that populate the rest of the geometry track.

Exercise 4.4.1

1. For $n \geq 2$, show that $\mathrm{SO}(n)$ acts transitively on S^{n-1} with stabilizer $\mathrm{SO}(n-1)$, so that $S^{n-1} \cong \mathrm{SO}(n)/\mathrm{SO}(n-1)$, and check that the dimension count gives $n-1$.
2. Show that the unitary group acts transitively on complex projective space $\mathcal{P}^n$ with stabilizer $\mathrm{U}(1) \times \mathrm{U}(n)$, giving $\mathcal{P}^n \cong \mathrm{U}(n+1)/(\mathrm{U}(1) \times \mathrm{U}(n))$, and verify that the real dimension is $2n$.
3. Show that orthogonal complementation $W \mapsto W^\perp$ is a diffeomorphism $\mathrm{Gr}(k,n) \to \mathrm{Gr}(n-k,n)$, and explain how this is visible in the coset description of example 4.4.3.
4. Real projective space has two descriptions: the coset space $\mathrm{O}(n)/(\mathrm{O}(1) \times \mathrm{O}(n-1))$ of example 4.4.3, and the quotient $S^{n-1}/\{\pm 1\}$ of the sphere by the antipodal action. Verify that the antipodal action is free and properly discontinuous, so that theorem 4.2.6 makes $S^{n-1} \to \mathbb{R}\mathrm{P}^{n-1}$ a smooth double cover, and reconcile the two descriptions.
5. Let $\mathrm{SU}(n) = \mathrm{U}(n) \cap \{\det = 1\}$. Using that $\det\colon \mathrm{U}(n) \to \mathrm{U}(1)$ is a submersion, show that $\mathrm{SU}(n)$ is a regular submanifold of dimension $n^2 - 1$ and that $T_I\,\mathrm{SU}(n)$ is the space of traceless skew-Hermitian matrices.

5

Vector Fields and Flows

A vector field assigns a tangent vector to each point of a manifold, smoothly. Two readings run through the chapter. Analytically, a vector field is an ordinary differential equation without coordinates: its integral curves are the trajectories of the equation, and their assembly into a flow is how a manifold moves under an infinitesimal prescription, the setting for geometric evolution equations like the Ricci flow. Structurally, a vector field is an infinitesimal symmetry, the derivative of a one-parameter family of diffeomorphisms; the vector fields form a Lie algebra under a bracket, and on a Lie group this is the Lie algebra whose exponential recovers the group.

Section 5.2 bases vector fields locally through frames; section 5.3 identifies them with derivations of $C^\infty(M)$, section 5.4 tracks them under smooth maps, and section 5.5 builds their bracket; sections 5.6 to 5.8 develop integral curves, flows, and completeness, with the Lie derivative (section 5.10) tying the bracket to the flow.

5.1 Vector Fields

A vector field is a section of the tangent bundle. The section language is set up in general, so that it transfers unchanged to the vector bundles of chapter 11.

Definition 5.1.1: Section

A *section* of a smooth map $\pi\colon E \to M$ is a map $\sigma\colon M \to E$ with $\pi \circ \sigma = \mathrm{id}_M$; a *local section* over an open $U \subseteq M$ satisfies $\pi \circ \sigma = \mathrm{id}_U$. It is a *smooth* (or C^r) section when σ is smooth (or C^r).

$$\begin{tikzcd} E \arrow[r, "\pi"] & M \arrow[l, bend right, dashed, "\sigma"'] \end{tikzcd}$$

Definition 5.1.2: Vector Field

A *vector field* on a manifold M is a smooth section $X\colon M \to TM$ of the tangent bundle $\pi\colon TM \to M$, so $X_p := X(p) \in T_pM$ for each p. A section that is not required to be smooth is a *rough vector field*.

The section formulation pays off in chapter 11, where TM is replaced by a general vector bundle and smoothness by other regularity. The *support* of X is the closure of $\{p \in M \mid X_p \neq 0\}$; X is *compactly supported* when its support is compact. Vector fields are read in coordinates through the coordinate basis.

Definition 5.1.3: Component Functions

For a chart (U, φ) with coordinates x^i, a rough vector field X is written at each $p \in U$ as

$$X_p = \sum_{i=1}^{n} X^i(p)\, \frac{\partial}{\partial x^i}\bigg|_p$$

The functions $X^i\colon U \to \mathbb{R}$ are the *component functions* of X in the chart, and one writes $X = \sum_i X^i\, \partial/\partial x^i$.

Proposition 5.1.4: Smoothness in Coordinates

A rough vector field X is smooth on a chart (U, φ) if and only if its component functions X^i are smooth on U.

***Proof*:**
In the natural coordinates (x^i, v^i) on $\pi^{-1}(U) \subseteq TM$ (proposition 2.4.5) the coordinate representation of X is

$$\widehat{X}(x) = (x^1, \ldots, x^n, X^1(x), \ldots, X^n(x))$$

whose smoothness is equivalent to that of the X^i, completing the proof.

A vector field given on one chart transports to another by the transition derivative, the transformation law of tangent vectors applied pointwise.

Lemma 5.1.5: Change of Chart

Let (U, φ) and (V, ψ) be charts with $U \cap V \neq \emptyset$ and X a smooth vector field on $U \cap V$. In the two charts $X = \sum_k a^k\, \partial/\partial x^k = \sum_k b^k\, \partial/\partial y^k$, with the components related by

$$b^k = \sum_l \frac{\partial y^k}{\partial x^l}\, a^l, \qquad \text{equivalently} \qquad X_{\psi\text{-rep}} = D(\psi \circ \varphi^{-1})\, X_{\varphi\text{-rep}}$$

***Proof*:**
Apply the change-of-coordinates law for the coordinate basis (section 2.2): $\partial/\partial x^l = \sum_k (\partial y^k/\partial x^l)\,\partial/\partial y^k$, so $\sum_l a^l\,\partial/\partial x^l = \sum_k(\sum_l(\partial y^k/\partial x^l)a^l)\,\partial/\partial y^k$, giving $b^k = \sum_l(\partial y^k/\partial x^l)a^l$. This is $D(\psi\circ\varphi^{-1})$ applied to the component vector, and being a smooth function of a smooth field it is again smooth, completing the proof.

The lemma is what lets a field defined chart by chart be glued into a global field when the local pieces agree on overlaps, and it is the pointwise form of the pushforward studied in section 5.4.

Example 5.1.6: Vector Fields

1. *(Coordinate fields.)* On a chart $(U,(x^i))$, $p \mapsto \partial/\partial x^i|_p$ is a smooth local vector field $\partial/\partial x^i$ (its components are constant), defined on U, not on all of M.
2. *(Euler field.)* On $\mathbb{R}^n$, $E = \sum_i x^i\,\partial/\partial x^i$ is smooth (linear components), vanishes only at the origin, and points radially outward; it appears in Euler's homogeneous-function theorem.
3. *(Angle field on S^1.)* In the angle coordinate θ, the field $\partial/\partial\theta$ is a nowhere-vanishing smooth global field on S^1 (the velocity of the standard parametrization), the global frame of corollary 2.4.6.
4. *(Angle fields on T^n.)* On $T^n = (S^1)^n$ the fields $\partial/\partial\theta^1,\ldots,\partial/\partial\theta^n$ are nowhere-zero and independent at each point, a global frame; T^n is parallelizable.
5. *(Gluing on S^2.)* Take the stereographic charts φ_N, φ_S with transition $\varphi_{NS}(s,t) = (s,t)/(s^2+t^2)$. Setting $X_N = \partial/\partial s$ on the N-chart and pushing forward by φ_{NS}, the derivative
$$D\varphi_{NS}(s,t) = \frac{1}{(s^2+t^2)^2}\begin{pmatrix} t^2-s^2 & -2st \\ -2st & s^2-t^2 \end{pmatrix}$$
evaluated at $(s,t) = \varphi_{NS}^{-1}(u,v)$ (the involution gives $s^2+t^2 = (u^2+v^2)^{-1}$) becomes $\begin{pmatrix} v^2-u^2 & -2uv \\ -2uv & u^2-v^2 \end{pmatrix}$, so
$$X_S = (v^2-u^2)\,\frac{\partial}{\partial u} - 2uv\,\frac{\partial}{\partial v}$$
which extends smoothly across $(u,v)=0$; the two pieces agree on the overlap and define a smooth field on S^2 vanishing only at the north pole, a single zero of index 2.

Every such field on S^2 vanishes somewhere: no smooth vector field on S^2 is nowhere zero, the *hairy ball theorem* (section 8.6).

Because T_pU is identified with T_pM, a vector field on U is read as a map $U \to TU$ or $U \to TM$ interchangeably, and $X|_U$ is a smooth field on U whenever X is smooth on M. Compactly supported fields extend from closed sets, exactly as smooth functions do.

Proposition 5.1.7: Extension of Vector Fields

Let $A \subseteq M$ be closed, $U \supseteq A$ open, and X a smooth vector field on a neighbourhood of A. There is a smooth vector field $\tilde{X}$ on all of M with $\tilde{X} = X$ on A and $\mathrm{supp}\tilde{X} \subseteq U$. In particular a tangent vector at a point extends to a compactly supported field.

***Proof*:**
Take a smooth bump β equal to 1 on A with $\mathrm{supp}\beta \subseteq U$ contained in the domain of X (corollary 1.5.6). Then $\tilde{X} = \beta X$, extended by 0 off $\mathrm{supp}\beta$, is smooth, agrees with X on A, and has support in U, completing the proof.

Definition 5.1.8: The Space of Vector Fields

$\mathfrak{X}(M)$ denotes the set of all smooth vector fields on M, with the zero field $p \mapsto 0 \in T_pM$.

Proposition 5.1.9: Vector Fields as a Module

$\mathfrak{X}(M)$ is a module over $C^\infty(M)$ under pointwise operations, $(fX + gY)_p = f(p)X_p + g(p)Y_p$.

***Proof*:**
Pointwise sum and scaling by functions produce rough vector fields, and in any chart the components of $fX + gY$ are $fX^i + gY^i$, smooth when X, Y are smooth and $f, g \in C^\infty(M)$; so $\mathfrak{X}(M)$ is closed under these operations, and the module axioms hold pointwise from those of each T_pM, completing the proof.

5.2 Local and Global Frames

The coordinate vector fields $(\partial/\partial x^i)$ give a basis of T_pM at each point of a chart. A collection of vector fields with this property carries a name.

Definition 5.2.1: Frame

An ordered tuple $(X_1, \ldots, X_n)$ of vector fields on an open $U \subseteq M$ is a *local frame* on U if $(X_1|_p, \ldots, X_n|_p)$ is a basis of T_pM for every $p \in U$; it is a *global frame* when $U = M$, and a *smooth frame* when each X_i is smooth. A smooth local frame is classically a *moving frame*.

A frame is abbreviated (X_i) with the index range understood.

Example 5.2.2: Frames

1. The standard coordinate fields form a smooth global frame on $\mathbb{R}^n$.
2. On any chart $(U, (x^i))$ the coordinate fields $(\partial/\partial x^i)$ form a smooth local frame, the *coordinate frame*; so every point of M lies in the domain of a smooth local frame.

3. The angle field $\partial/\partial\theta$ is a smooth global frame on S^1, and $(\partial/\partial\theta^1, \dots, \partial/\partial\theta^n)$ one on T^n (example 5.1.6).

Independent vector fields can always be completed to a frame nearby, because independence is an open condition.

Proposition 5.2.3: Completion of Frames

Let $(Y_1, \dots, Y_k)$ be smooth vector fields on an open set, linearly independent at each point. Then every point p has a neighbourhood on which there are smooth vector fields $Y_{k+1}, \dots, Y_n$ making $(Y_1, \dots, Y_n)$ a smooth local frame.

***Proof*:**
At p the vectors $Y_1|_p, \dots, Y_k|_p$ are independent in T_pM; complete them to a basis by adjoining coordinate vectors $\partial/\partial x^{j_{k+1}}|_p, \dots, \partial/\partial x^{j_n}|_p$ from a chart at p. The n smooth fields $Y_1, \dots, Y_k, \partial/\partial x^{j_{k+1}}, \dots, \partial/\partial x^{j_n}$ are then independent at p. In the coordinate frame their components form an $n \times n$ matrix whose determinant is a smooth function, nonzero at p, hence nonzero on a neighbourhood; there the fields remain a basis, giving a smooth local frame, completing the proof.

Local frames are thus abundant, but global ones are not: a smooth global frame gives M a strong triviality.

Definition 5.2.4: Parallelizable

A smooth manifold is *parallelizable* if it admits a smooth global frame.

A parallelizable manifold has trivial tangent bundle, $TM \cong M \times \mathbb{R}^n$ (chapter 11), the frame supplying the trivialization. Among the examples, $\mathbb{R}^n$, S^1, and T^n are parallelizable, as are S^3 and S^7; but S^2 is not, by the hairy ball theorem (section 8.6), and in fact S^1, S^3, S^7 are the only parallelizable spheres, a theorem of algebraic topology tied to the real division algebras. Every Lie group is parallelizable, its frame built from a basis of the tangent space at the identity by left translation, so among the spheres a Lie-group structure can occur only for S^1, S^3, S^7; in fact only S^1 and S^3 are Lie groups, as S^7 is parallelizable through octonion multiplication but non-associative. Parallelizability is not the same as being covered by one chart: the torus T^n is parallelizable but needs several charts.

5.3 Vector Fields as Derivations

A vector field acts on functions: for $X \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, define Xf pointwise by $(Xf)(p) = X_p f$, using that $X_p \in T_pM$ is a derivation at p. The action is local, $(Xf)|_V = X(f|_V)$ for open V, since a tangent vector reads only the germ of f. Smoothness of X is detected through this action.

Proposition 5.3.1: Smoothness via the Action on Functions

For a rough vector field X on M, the following are equivalent:

1. X is smooth;
2. $Xf \in C^\infty(M)$ for every $f \in C^\infty(M)$;
3. $Xf \in C^\infty(U)$ for every open $U \subseteq M$ and every $f \in C^\infty(U)$.

***Proof*:**
(1) ⇒ (2). In a chart $Xf = \sum_i X^i \, \partial f/\partial x^i$; the X^i are smooth (proposition 5.1.4) and so is each $\partial f/\partial x^i$, so Xf is smooth there, hence on M.

(2) ⇒ (3). For $f \in C^\infty(U)$ and $p \in U$, a bump β equal to 1 near p with $\operatorname{supp}\beta \subseteq U$ makes βf a global smooth function agreeing with f near p, so $Xf = X(\beta f)$ near p by locality, smooth near p; as p was arbitrary, $Xf \in C^\infty(U)$.

(3) ⇒ (1). On a chart with coordinates x^j, apply (3) to $f = x^j$: then $Xx^j = X^j$, the j-th component of X, is smooth. As all components are smooth, X is smooth by proposition 5.1.4, completing the proof.

A smooth X thus gives a map $C^\infty(M) \to C^\infty(M)$ that is $\mathbb{R}$-linear and, by the Leibniz rule at each point, satisfies $X(fg) = (Xf)g + f(Xg)$: it is a *derivation* of the algebra $C^\infty(M)$. The correspondence is exact.

Proposition 5.3.2: Vector Fields are the Derivations of $C^\infty(M)$

A map $D\colon C^\infty(M) \to C^\infty(M)$ is a derivation if and only if $Df = Xf$ for a unique smooth vector field $X \in \mathfrak{X}(M)$.

***Proof*:**
A smooth X gives such a derivation, as above. Conversely let D be a derivation. It is local: if f vanishes on a neighbourhood of p, take a bump β with $\beta(p) = 1$ and support in that neighbourhood, so $\beta f \equiv 0$; then $0 = D(\beta f)(p) = D\beta(p)\, f(p) + \beta(p)\, Df(p) = Df(p)$ by the Leibniz rule and $f(p) = 0$. Hence $X_p\colon C^\infty(M) \to \mathbb{R}$, $X_p f = (Df)(p)$, is well defined and, inheriting linearity and the Leibniz rule from D, is a derivation at p, so $X_p \in T_pM$. This defines a rough vector field X with $Xf = Df$; since $Df \in C^\infty(M)$ for all f, X is smooth by proposition 5.3.1. Uniqueness holds because $Xf = X'f$ for all f forces $X_p = X'_p$ at each p, completing the proof.

The identification $\mathfrak{X}(M) \cong \operatorname{Der}(C^\infty(M))$ is the algebraic home of vector fields, and the source of the Lie bracket (section 5.5): the commutator of two derivations is again one.

5.4 Vector Fields and Smooth Maps

A smooth $f\colon M \to N$ sends X_p to $T_pf(X_p) \in T_{f(p)}N$, but these need not assemble into a vector field on N: a non-injective f can send different X_p to the same point, and a non-surjective f leaves points uncovered. The vector fields that do correspond are named.

Definition 5.4.1: f-Related Vector Fields

For a smooth $f\colon M \to N$, fields $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are *f-related* if $T_pf(X_p) = Y_{f(p)}$ for every p, that is $Tf \circ X = Y \circ f$.

Relatedness is exactly compatibility of the two derivations through pullback.

Proposition 5.4.2: f-Relatedness on Functions

X and Y are f-related if and only if $X(g \circ f) = (Yg) \circ f$ for every $g \in C^\infty(N)$.

***Proof*:**
For $p \in M$ and $g \in C^\infty(N)$, $X(g \circ f)(p) = X_p(g \circ f) = T_pf(X_p)(g)$ by the definition of the tangent map (definition 2.2.4), while $((Yg) \circ f)(p) = Y_{f(p)}(g)$. These agree for all g exactly when $T_pf(X_p) = Y_{f(p)}$, completing the proof.

For an arbitrary f a given X may have no f-related field on N; a diffeomorphism removes the obstruction.

Proposition 5.4.3: Pushforward along a Diffeomorphism

If $f\colon M \to N$ is a diffeomorphism, then each $X \in \mathfrak{X}(M)$ has a unique f-related field on N.

***Proof*:**
Relatedness $T_pf(X_p) = Y_{f(p)}$ determines Y at $q = f(p)$ as $Y_q = T_{f^{-1}(q)}f\big(X_{f^{-1}(q)}\big)$, so Y is unique. This $Y = Tf \circ X \circ f^{-1}$ is a composition of smooth maps, hence a smooth vector field, and is f-related to X by construction, completing the proof.

The two directions of transport have names. Functions pull back along any smooth map.

Definition 5.4.4: Pullback

The *pullback* along a smooth $f\colon M \to N$ is $f^*\colon C^\infty(N) \to C^\infty(M)$, $f^*h = h \circ f$.

Pullback is an algebra homomorphism, making $C^\infty(-)$ a contravariant functor from manifolds to $\mathbb{R}$-algebras. Vector fields push forward, but only along a diffeomorphism.

Definition 5.4.5: Pushforward

The *pushforward* along a diffeomorphism $f\colon M \to N$ is $f_*\colon \mathfrak{X}(M) \to \mathfrak{X}(N)$, $f_*X = Tf \circ X \circ f^{-1}$, the unique field f-related to X.

By construction f_*X is f-related to X, so proposition 5.4.2 gives $(f_*X)(h) = \big(X(h \circ f)\big) \circ f^{-1}$ for $h \in C^\infty(N)$; and pushforward is functorial, $(f \circ g)_* = f_* \circ g_*$ and $(\mathrm{id})_* = \mathrm{id}$, from the chain rule. These are the transformations under which the Lie bracket is natural (section 5.5).

5.5 The Lie Bracket

The composition XY of two vector fields, read as derivations, is a second-order operator, not a derivation: applied to a product it produces the extra term $(Xf)(Yg)+(Yf)(Xg)$. The antisymmetric combination removes it.

Definition 5.5.1: Lie Bracket

The *Lie bracket* of $X, Y \in \mathfrak{X}(M)$ is the operator $[X,Y] = XY - YX$ on $C^\infty(M)$, $[X,Y]f = X(Yf) - Y(Xf)$.

Proposition 5.5.2: The Bracket is a Vector Field

For $X, Y \in \mathfrak{X}(M)$ the operator $[X,Y]$ is a derivation of $C^\infty(M)$, hence a smooth vector field. In a chart its components are

$$[X,Y]^j = \sum_i \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right)$$

***Proof*:**
$[X,Y]$ is $\mathbb{R}$-linear, and the Leibniz rule holds because the two second-order terms cancel: $X(Y(fg)) - Y(X(fg))$ expands, and the pieces $(Xf)(Yg) + (Yf)(Xg)$ appear symmetrically in both $X(Y(fg))$ and $Y(X(fg))$, so they subtract to zero, leaving $[X,Y](fg) = ([X,Y]f)g + f([X,Y]g)$. Thus $[X,Y]$ is a derivation and, by proposition 5.3.2, a smooth vector field. For the components, with $X = \sum_i X^i \partial/\partial x^i$ and $Y = \sum_j Y^j \partial/\partial x^j$, the second derivatives $\partial^2 f/\partial x^i \partial x^j$ cancel by symmetry of mixed partials, and collecting the first-order terms gives the stated formula, completing the proof.

The bracket makes $\mathfrak{X}(M)$ a Lie algebra, the infinitesimal counterpart of the diffeomorphism group.

Proposition 5.5.3: The Lie Algebra of Vector Fields

The bracket on $\mathfrak{X}(M)$ is $\mathbb{R}$-bilinear and antisymmetric, $[X,Y] = -[Y,X]$, and satisfies the Jacobi identity

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0$$

Over functions it is not bilinear: for $f, g \in C^\infty(M)$,

$$[fX, gY] = fg\,[X,Y] + f(Xg)\,Y - g(Yf)\,X$$

***Proof*:**
Bilinearity and antisymmetry are immediate from $[X,Y] = XY - YX$. Writing each bracket as a commutator of operators, the Jacobi sum expands into twelve terms $XYZ, XZY, \ldots$ that cancel in pairs, the standard commutator identity. For the last, the Leibniz rule gives $(fX)(gYh) = f(Xg)(Yh) + fg\,X(Yh)$, and subtracting the symmetric expression collects into the stated three terms, completing the proof.

The bracket is natural under f-relatedness, which is what lets it descend along maps and, on a Lie group, single out the left-invariant fields.

Proposition 5.5.4: Naturality of the Bracket

If X is f-related to X' and Y to Y' under a smooth $f\colon M \to N$, then $[X,Y]$ is f-related to $[X',Y']$.

***Proof*:**
By proposition 5.4.2, for $g \in C^\infty(N)$, $X(g \circ f) = (X'g) \circ f$ and likewise for Y. Then

$$[X,Y](g \circ f) = X\big((Y'g) \circ f\big) - Y\big((X'g) \circ f\big) = (X'Y'g) \circ f - (Y'X'g) \circ f = ([X',Y']g) \circ f$$

so $[X,Y]$ is f-related to $[X',Y']$ by proposition 5.4.2, completing the proof.

In particular a diffeomorphism intertwines brackets, $f_*[X,Y] = [f_*X, f_*Y]$, so pushforward is a Lie-algebra isomorphism. Two brackets vanish for a reason worth naming: $[\partial/\partial x^i, \partial/\partial x^j] = 0$ for coordinate fields, since their components are constant, so a coordinate frame commutes. The converse, that a commuting frame is locally a coordinate frame, is the flow-box theorem of section 5.10, and vanishing brackets are exactly the integrability condition of the Frobenius theorem (chapter 6).

Exercise 5.5.1

1. On $\mathbb{R}^2$ compute $[X,Y]$ for $X = \partial/\partial x$ and $Y = x\,\partial/\partial y$, and for $X = y\,\partial/\partial x$, $Y = x\,\partial/\partial y$.
2. Show that $[\partial/\partial x^i, \partial/\partial x^j] = 0$ for all i,j, and deduce that any two constant-coefficient fields on $\mathbb{R}^n$ commute.
3. Deduce the Jacobi identity for the Lie bracket from the associativity of composition of operators, by expanding the three double commutators.

5.6 Integral Curves

An integral curve of a vector field follows the field: its velocity at each point is the field's value there. Finding integral curves is solving a system of ordinary differential equations in a chart, and the standard existence theory transfers to manifolds unchanged.

Definition 5.6.1: Integral Curve

An *integral curve* of a vector field X on M is a smooth curve $\gamma\colon I \to M$ with $\gamma'(t) = X_{\gamma(t)}$ for all $t \in I$. If $0 \in I$, the point $\gamma(0)$ is the *starting point*.

Example 5.6.2: Integral Curves

1. For $V = \partial/\partial x$ on $\mathbb{R}^2$, the integral curves are the horizontal lines $\gamma(t) = (a + t, b)$.
2. For $W = x\,\partial/\partial y - y\,\partial/\partial x$ on $\mathbb{R}^2$, writing $\gamma(t) = (x(t), y(t))$ the condition $\gamma'(t) = W_{\gamma(t)}$ is the system $x' = -y$, $y' = x$, with solution

$$\gamma(t) = (a \cos t - b \sin t,\ a \sin t + b \cos t)$$

Starting at (a, b), this is the constant curve when $(a, b) = 0$ and a circle traversed counter-clockwise otherwise; the integral curves are the origin and the circles about it.

In a chart, writing $\gamma(t) = (\gamma^1(t), \ldots, \gamma^n(t))$, the condition $\gamma'(t) = X_{\gamma(t)}$ is the autonomous system

$$\dot{\gamma}^i(t) = X^i(\gamma^1(t), \ldots, \gamma^n(t)), \qquad i = 1, \ldots, n$$

whose existence, uniqueness, and smooth dependence on initial data are the content of the fundamental theorem of ordinary differential equations ([3]); the name *integral curve* recalls that solving such a system is "integrating" it. The manifold consequences follow at once.

Proposition 5.6.3: Local Existence

Let X be a smooth vector field on M. Each $p \in M$ has an $\varepsilon > 0$ and a smooth integral curve $\gamma\colon (-\varepsilon, \varepsilon) \to M$ of X starting at p.

***Proof*:**
In a chart at p the integral-curve condition is the autonomous system above, whose right-hand side X^i is smooth; the existence theorem for ODEs ([3]) provides a smooth solution on a small interval about 0 with $\gamma(0) = p$, completing the proof.

Affine reparametrizations rescale and translate integral curves in the expected way.

Lemma 5.6.4: Rescaling

If $\gamma\colon I \to M$ is an integral curve of X, then for $a \in \mathbb{R}$ the curve $\tilde{\gamma}(t) = \gamma(at)$ on $\tilde{I} = \{t \mid at \in I\}$ is an integral curve of aX.

Proof:
For f smooth near $\tilde{\gamma}(t_0)$, the chain rule and $\gamma' = X_\gamma$ give

$$\tilde{\gamma}'(t_0)f = \frac{d}{dt}\bigg|_{t_0} f(\gamma(at)) = a\,(f\circ\gamma)'(at_0) = a\,\gamma'(at_0)f = a\,X_{\tilde{\gamma}(t_0)}f$$

so $\tilde{\gamma}'(t_0) = (aX)_{\tilde{\gamma}(t_0)}$, completing the proof.

Lemma 5.6.5: Translation

If $\gamma\colon I \to M$ is an integral curve of X, then for $b \in \mathbb{R}$ the curve $\hat{\gamma}(t) = \gamma(t+b)$ on $\hat{I} = \{t \mid t+b \in I\}$ is also an integral curve of X.

Proof:
By the chain rule $\hat{\gamma}'(t) = \gamma'(t+b) = X_{\gamma(t+b)} = X_{\hat{\gamma}(t)}$, completing the proof.

Integral curves transport along f-related fields, which is the pointwise root of the naturality of flows.

Proposition 5.6.6: Naturality of Integral Curves

For a smooth $f\colon M \to N$, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are f-related if and only if f carries integral curves of X to integral curves of Y: for every integral curve γ of X, $f\circ\gamma$ is an integral curve of Y.

Proof:
If X, Y are f-related and γ an integral curve of X, then $\sigma = f\circ\gamma$ satisfies

$$\sigma'(t) = T_{\gamma(t)}f(\gamma'(t)) = T_{\gamma(t)}f(X_{\gamma(t)}) = Y_{f(\gamma(t))} = Y_{\sigma(t)}$$

so σ is an integral curve of Y. Conversely, given $p \in M$, an integral curve γ of X starting at p has $f\circ\gamma$ an integral curve of Y starting at $f(p)$, so $Y_{f(p)} = (f\circ\gamma)'(0) = T_pf(X_p)$; as p was arbitrary, X and Y are f-related, completing the proof.

5.7 Flow

A vector field moves the whole manifold: run every point along its integral curve for time t, and the resulting maps assemble into a one-parameter family of diffeomorphisms, the *flow*. When all integral curves are defined for all time, this family is an $\mathbb{R}$-action; in general it is defined only on part of $\mathbb{R}\times M$.

Suppose first that the integral curve $\theta^{(p)}\colon \mathbb{R} \to M$ starting at each p is defined for all time. Define $\theta_t\colon M \to M$ by $\theta_t(p) = \theta^{(p)}(t)$, sliding each point t units along its curve. By the translation lemma (lemma 5.6.5), $t \mapsto \theta^{(p)}(t+s)$ is the integral curve starting at $\theta^{(p)}(s)$, so uniqueness gives $\theta^{(q)}(t) = \theta^{(p)}(t+s)$ with $q = \theta^{(p)}(s)$, that is

$$\theta_t \circ \theta_s = \theta_{t+s}, \qquad \theta_0 = \mathrm{id}$$

making $\theta\colon \mathbb{R}\times M \to M$ an action of $\mathbb{R}$.

Definition 5.7.1: Global Flow

A *global flow* (or one-parameter group action) on M is a smooth left $\mathbb{R}$-action $\theta\colon \mathbb{R}\times M \to M$: for all $s,t \in \mathbb{R}$ and $p \in M$,

$$\theta(t,\theta(s,p)) = \theta(s+t,p), \qquad \theta(0,p) = p$$

Writing $\theta_t(p) = \theta(t,p)$, each θ_t is a diffeomorphism with inverse θ_{-t}. Every smooth global flow comes from a vector field.

Definition 5.7.2: Infinitesimal Generator

The *infinitesimal generator* of a global flow θ is the rough vector field $X_p = (\theta^{(p)})'(0)$.

Proposition 5.7.3: Generators of Global Flows

The infinitesimal generator X of a smooth global flow θ is a smooth vector field, and each $\theta^{(p)}$ is an integral curve of X.

***Proof*:**
For smoothness, by proposition 5.3.1 it suffices that Xf is smooth for each $f \in C^\infty(M)$; and

$$Xf(p) = X_pf = (\theta^{(p)})'(0)f = \left.\frac{\partial}{\partial t}\right|_{(0,p)} f(\theta(t,p))$$

is a partial derivative of the smooth function $f\circ\theta$, hence smooth in p. For the integral-curve property, fix t_0 and set $q = \theta_{t_0}(p)$. The group law gives $\theta^{(q)}(t) = \theta^{(p)}(t+t_0)$, so for f smooth near q,

$$X_qf = (\theta^{(q)})'(0)f = \left.\frac{d}{dt}\right|_0 f(\theta^{(p)}(t+t_0)) = (\theta^{(p)})'(t_0)f$$

i.e. $(\theta^{(p)})'(t_0) = X_{\theta^{(p)}(t_0)}$, completing the proof.

Example 5.7.4: Global Flows

The fields of example 5.6.2 generate global flows:

$$V = \frac{\partial}{\partial x}\colon\ \tau_t(x,y) = (x+t,y), \qquad W = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\colon\ \theta_t(x,y) = (x\cos t - y\sin t,\ x\sin t + y\cos t)$$

Dropping the all-time hypothesis, every global flow still has a generator, but a vector field need not generate a *global* one: an integral curve can run off the manifold or to infinity in finite time.

Example 5.7.5: Fields Without Global Flows

1. On $M = \mathbb{R}^2\setminus\{0\}$ with $V = \partial/\partial x$, the integral curve from $(-1,0)$ is $\gamma(t) = (t-1,0)$, which cannot extend past $t=1$: as $t \nearrow 1$ it would have to reach $(0,0) \notin M$, so no integral curve is

defined at $t = 1$.

2. On $M = \mathbb{R}^2$ with $W = x^2\,\partial/\partial x$, the integral curve from $(1, 0)$ solves $x' = x^2$, $x(0) = 1$, giving $x(t) = 1/(1-t)$, which blows up as $t \nearrow 1$; the curve escapes to infinity in finite time.

The right domain is a flow domain: open, and an interval about 0 in each time slice.

Definition 5.7.6: Flow Domain and Local Flow

A *flow domain* is an open $\mathcal{D} \subseteq \mathbb{R} \times M$ such that $\mathcal{D}^{(p)} = \{t \mid (t, p) \in \mathcal{D}\}$ is an open interval containing 0 for each p. A *local flow* is a smooth $\theta\colon \mathcal{D} \to M$ on a flow domain with $\theta(0, p) = p$ and, whenever $s \in \mathcal{D}^{(p)}$, $t \in \mathcal{D}^{(\theta(s,p))}$, and $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(s + t, p)$$

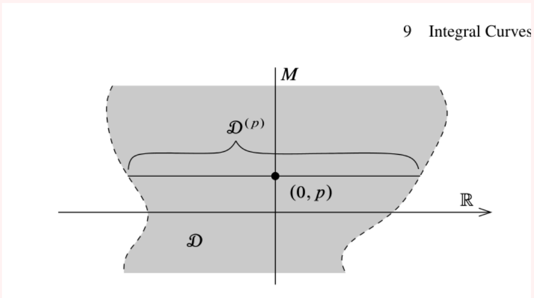


Figure 5.1: A flow domain.

The partial group law makes this a groupoid action rather than a group action. Writing $\theta_t(p) = \theta(t, p)$ and $M_t = \{p \mid (t, p) \in \mathcal{D}\}$, the generator is again $X_p = (\theta^{(p)})'(0)$ when θ is smooth.

Proposition 5.7.7: Generators of Local Flows

The infinitesimal generator of a smooth local flow is a smooth vector field, and each $\theta^{(p)}$ is an integral curve of it.

***Proof*:**
Both claims are local in t and p, and the argument of proposition 5.7.3 uses the group law only for s, t small enough to lie in the flow domain, so it applies verbatim, completing the proof.

Maximal integral curves and flows are those admitting no extension.

Definition 5.7.8: Maximal Integral Curve and Flow

A *maximal integral curve* extends to no integral curve on a larger interval; a *maximal flow* extends to no flow on a larger flow domain.

Assembling the local integral curves gives the central theorem: each vector field has a unique maximal flow.

Theorem 5.7.9: Fundamental Theorem of Flows

Every smooth vector field X on M has a unique smooth maximal flow $\theta\colon \mathcal{D} \to M$ with infinitesimal generator X. Moreover:

1. each $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of X from p;
2. if $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))} = \mathcal{D}^{(p)} - s$;
3. each M_t is open, and $\theta_t\colon M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

***Proof*:**
In a chart the ODE $\dot\gamma = X(\gamma)$ has, by the existence–uniqueness theorem ([3]), a unique solution through each point; uniqueness is chart independent, so any two integral curves of X agreeing at one time agree on their common domain. Hence the integral curves from p glue to a unique maximal one $\theta^{(p)}\colon \mathcal{D}^{(p)} \to M$ on the union $\mathcal{D}^{(p)}$ of their domains, an open interval about 0; this gives (1). Set $\mathcal{D} = \left\{(t,p) \;\middle|\; t \in \mathcal{D}^{(p)}\right\}$ and $\theta(t,p) = \theta^{(p)}(t)$. The translation lemma and uniqueness give, for $s \in \mathcal{D}^{(p)}$, that $t \mapsto \theta^{(p)}(t+s)$ is the maximal integral curve from $\theta(s,p)$, whence $\mathcal{D}^{(\theta(s,p))} = \mathcal{D}^{(p)} - s$ and the group law $\theta_t \circ \theta_s = \theta_{t+s}$ where defined, proving (2). Smooth dependence on initial conditions ([3]) makes $\mathcal{D}$ open and θ smooth, so θ is a smooth flow with generator X; the group law then makes each θ_t a diffeomorphism $M_t \to M_{-t}$ with inverse θ_{-t}, proving (3). Maximality and uniqueness of θ follow from those of the $\theta^{(p)}$, completing the proof.

Definition 5.7.10: Flow of a Vector Field

The maximal flow of the previous theorem is the *flow of X*.

Flows are natural under f-related fields, the flow-level form of proposition 5.6.6.

Proposition 5.7.11: Naturality of Flows

Let $f\colon M \to N$ be smooth with $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ f-related, and let θ, η be their flows. Then for each t, $f(M_t) \subseteq N_t$ and $\eta_t \circ f = f \circ \theta_t$ on M_t:

$$\begin{tikzcd} M_t \arrow[r, "f"] \arrow[d, "\theta_t"'] & N_t \arrow[d, "\eta_t"] \\ M_{-t} \arrow[r, "f"'] & N_{-t} \end{tikzcd}$$

***Proof*:**
For $p \in M_t$, the curve $f \circ \theta^{(p)}$ is an integral curve of Y (proposition 5.6.6) starting at $f(p)$ and defined on $\mathcal{D}^{(p)} \ni t$, so by uniqueness and maximality it is a restriction of $\eta^{(f(p))}$; thus $t \in N^{(f(p))}$, i.e. $f(p) \in N_t$, and $\eta_t(f(p)) = f(\theta^{(p)}(t)) = f(\theta_t(p))$, completing the proof.

Corollary 5.7.12: Diffeomorphism Invariance of Flows

If $f\colon M \to N$ is a diffeomorphism and θ is the flow of X, then the flow of f_*X is $\eta_t = f\circ\theta_t\circ f^{-1}$ on $N_t = f(M_t)$.

***Proof*:**
As f_*X is f-related to X (definition 5.4.5), proposition 5.7.11 gives $\eta_t \circ f = f \circ \theta_t$, i.e. $\eta_t = f \circ \theta_t \circ f^{-1}$, and applying it to f^{-1} shows the domains match, $N_t = f(M_t)$, completing the proof.

5.8 Complete Vector Fields

A vector field is complete when its flow is global: no integral curve escapes in finite time. Completeness cannot always be read off a field, but two large classes are complete for structural reasons.

Definition 5.8.1: Complete Vector Field

A smooth vector field is *complete* if it generates a global flow, equivalently if every maximal integral curve is defined on all of $\mathbb{R}$.

The examples of example 5.7.5 are incomplete because a curve reaches an edge or blows up. A uniform lower bound on the existence times rules this out.

Lemma 5.8.2: Uniform Time Lemma

Let X have flow θ. If there is $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(p)}$ for every p, then X is complete.

***Proof*:**
Suppose some $\mathcal{D}^{(p)}$ has a finite upper bound b. Choose $t_0 \in (b-\varepsilon, b)$ and set $q = \theta_{t_0}(p)$; the integral curve from q is defined on $(-\varepsilon, \varepsilon)$, so by the translation lemma (lemma 5.6.5) $\theta^{(p)}$ extends to $t_0 + \varepsilon > b$, contradicting maximality. Thus each $\mathcal{D}^{(p)}$ is unbounded above, and likewise below, so X is complete, completing the proof.

Theorem 5.8.3: Compact Support Implies Complete

Every compactly supported smooth vector field on a manifold is complete.

***Proof*:**
Let $K = \operatorname{supp} X$. Each $p \in K$ lies in a flow-box neighbourhood on which the flow is defined for $|t| < \varepsilon_p$ (proposition 5.6.3 with smooth dependence); finitely many cover K, giving a uniform $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for all $q \in K$. For $q \notin K$ one has $X_q = 0$, so $\theta^{(q)} \equiv q$ is defined for all time. Hence $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for every q, and X is complete by lemma 5.8.2, completing the proof.

Corollary 5.8.4: Fields on Compact Manifolds are Complete

Every smooth vector field on a compact manifold is complete.

***Proof*:**
On a compact M every field has support contained in the compact M, hence is compactly supported and complete by theorem 5.8.3, completing the proof.

The obstruction to completeness is always escape from compact sets, a fact worth isolating.

Lemma 5.8.5: Escape Lemma

If $\gamma\colon I \to M$ is a maximal integral curve of X whose domain has a finite upper bound b, then for each $t_0 \in I$ the image $\gamma([t_0, b))$ lies in no compact subset of M.

***Proof*:**
If $\gamma([t_0, b)) \subseteq K$ for a compact K, the uniform-time argument of theorem 5.8.3 gives $\varepsilon > 0$ with $(-\varepsilon, \varepsilon) \subseteq \mathcal{D}^{(q)}$ for all $q \in K$. Taking $t_1 \in (\max(t_0, b - \varepsilon), b)$ and $q = \gamma(t_1) \in K$, the translation lemma extends γ past b, contradicting maximality; so no such K exists, completing the proof.

Incompleteness is thus a feature of the manifold, not repairable in general: $x^2\, \partial/\partial x$ is incomplete on $\mathbb{R}$ but complete on the one-point compactification S^1, yet a compactification adding singular points is rarely available or well-behaved. On a Lie group the left-invariant vector fields are complete, their flows the one-parameter subgroups; this belongs to the structure theory of [9].

5.9 Flowout

The flow straightens a vector field. Starting from a hypersurface transverse to X, the flow sweeps out a neighbourhood in which X is a single coordinate field; this is the canonical local form of a nonvanishing vector field.

Theorem 5.9.1: Flowout Theorem

Let X be a smooth vector field on M, $S \subseteq M$ an embedded hypersurface, and suppose X is nowhere tangent to S. Then there is a flow domain $O \subseteq \mathbb{R} \times S$ containing $\{0\} \times S$ and a diffeomorphism $\Phi\colon O \to \Phi(O)$ onto an open subset of M, $\Phi(t, s) = \theta_t(s)$, carrying $\partial/\partial t$ to X.

***Proof*:**
Let $\Phi(t, s) = \theta(t, s)$ be the flow restricted to $\mathbb{R} \times S$, defined on the flow domain $O = (\mathbb{R} \times S) \cap \mathcal{D}$. At $(0, s)$ its differential sends $\partial/\partial t$ to X_s and T_sS identically into T_sM; since $X_s \notin T_sS$ by transversality, these span T_sM, so $T_{(0,s)}\Phi$ is an isomorphism. By the inverse function theorem Φ is a local diffeomorphism near $\{0\} \times S$, and shrinking O makes it a diffeomorphism onto an open set. As $t \mapsto \Phi(t, s)$ is an integral curve of X, Φ carries $\partial/\partial t$ to X, completing the proof.

Corollary 5.9.2: Canonical Form of a Nonvanishing Field

If $X_p \neq 0$, there is a chart $(s^1, \ldots, s^n)$ centred at p in which $X = \partial/\partial s^1$.

***Proof*:**
Choose a chart at p in which $X_p = \partial/\partial x^1|_p$ and let S be the slice $\{x^1 = 0\}$, a hypersurface with $X_p \notin T_pS$; shrinking S to a neighbourhood of p, X stays transverse to it by continuity. The flowout $\Phi(t, s)$ of theorem 5.9.1 is a diffeomorphism carrying $\partial/\partial t$ to X; its inverse is a chart in which $X = \partial/\partial s^1$ (with $s^1 = t$), completing the proof.

5.10 The Lie Derivative

The flow of X lets one differentiate another vector field along X: transport Y back by the flow and take the t-derivative at 0. The result is a vector field, and it is exactly the Lie bracket, so the algebraic bracket and the dynamical flow are two faces of one object.

Definition 5.10.1: Lie Derivative

Let $X \in \mathfrak{X}(M)$ have flow θ. The *Lie derivative* of $Y \in \mathfrak{X}(M)$ along X is the vector field

$$(L_X Y)_p = \left.\frac{d}{dt}\right|_{t=0} \big((\theta_{-t})_* Y\big)_p = \left.\frac{d}{dt}\right|_{t=0} T\theta_{-t}\big(Y_{\theta_t(p)}\big)$$

the derivative at 0 of the curve $t \mapsto T\theta_{-t}(Y_{\theta_t(p)})$ in T_pM.

Theorem 5.10.2: The Lie Derivative is the Bracket

For all $X, Y \in \mathfrak{X}(M)$, $L_X Y = [X, Y]$.

***Proof*:**
Work in a chart, writing θ_t and X, Y by their coordinate expressions. Since $\theta_{-t} \circ \theta_t = \mathrm{id}$, the chain rule gives $D\theta_{-t}(\theta_t(p)) = [D\theta_t(p)]^{-1}$, so

$$\big((\theta_{-t})_* Y\big)_p = [D\theta_t(p)]^{-1}\, Y(\theta_t(p))$$

Differentiate at $t = 0$, where $\theta_0 = \mathrm{id}$ and $D\theta_0 = I$. From $\partial_t \theta_t|_0 = X$ come $\frac{d}{dt}\big|_0 Y(\theta_t(p)) = DY(p)\, X(p)$ and $\frac{d}{dt}\big|_0 D\theta_t(p) = DX(p)$, hence $\frac{d}{dt}\big|_0 [D\theta_t(p)]^{-1} = -DX(p)$. By the product rule,

$$(L_X Y)_p = -DX(p)\, Y(p) + DY(p)\, X(p)$$

whose i-th component is $\sum_j \big(X^j\, \partial Y^i/\partial x^j - Y^j\, \partial X^i/\partial x^j\big) = [X, Y]^i$ by proposition 5.5.2, so $L_X Y = [X, Y]$, completing the proof.

The identification makes the bracket dynamical: $[X, Y]$ measures the failure of Y to be carried to itself by the flow of X, and its vanishing is the commuting of flows.

Theorem 5.10.3: Commuting Fields and Commuting Flows

For $X, Y \in \mathfrak{X}(M)$ with flows θ, η, the following are equivalent:

1. $[X, Y] = 0$;
2. Y is invariant under the flow of X: $(\theta_t)_* Y = Y$ wherever defined;
3. the flows commute: $\theta_t \circ \eta_s = \eta_s \circ \theta_t$ where both sides are defined.

***Proof*:**
The family $t \mapsto (\theta_{-t})_* Y$ satisfies $\frac{d}{dt}(\theta_{-t})_* Y = (\theta_{-t})_*(L_X Y)$: differentiating at $t = t_0$ and using the group law $\theta_{-t} = \theta_{-t_0} \circ \theta_{-(t-t_0)}$ reduces it to the $t_0 = 0$ case, which is definition 5.10.1. If $[X, Y] = 0$ then $L_X Y = 0$ (theorem 5.10.2), so this derivative vanishes and $(\theta_{-t})_* Y$ is constant, equal to its value Y at $t = 0$; thus (1) $\Rightarrow$ (2). Conversely (2) gives $L_X Y = \frac{d}{dt}\big|_0 (\theta_{-t})_* Y = 0$, so (2) $\Rightarrow$ (1). Finally $(\theta_t)_* Y = Y$ says θ_t carries Y to itself, hence carries integral curves of Y to integral curves of Y; by naturality of flows (proposition 5.7.11) applied to the diffeomorphism θ_t, this is $\theta_t \circ \eta_s = \eta_s \circ \theta_t$, giving (2) $\Leftrightarrow$ (3), completing the proof.

A commuting frame is therefore a coordinate frame, the multi-field companion of the straightening corollary 5.9.2.

Theorem 5.10.4: Canonical Form of a Commuting Frame

If $X_1, \ldots, X_k \in \mathfrak{X}(M)$ are independent at p and pairwise commute, $[X_i, X_j] = 0$, then there is a chart $(s^1, \ldots, s^n)$ centred at p in which $X_i = \partial/\partial s^i$ for $i = 1, \ldots, k$.

***Proof*:**
Let θ^i be the flow of X_i; by theorem 5.10.3 the θ^i commute. Complete $X_1|_p, \ldots, X_k|_p$ to a basis of T_pM with coordinate vectors, and let S be the $(n-k)$-slice they span through p. Define

$$\Phi(t^1, \ldots, t^k, s) = \theta^1_{t^1} \circ \cdots \circ \theta^k_{t^k}(s), \qquad s \in S$$

near 0. Its differential at 0 sends $\partial/\partial t^i$ to $X_i|_p$ and $\partial/\partial s^j$ to the completing vectors, a basis, so Φ is a local diffeomorphism. Because the flows commute, θ^i_t can be moved to the front, showing $\Phi(\cdots + \varepsilon \text{ in slot } i \cdots)$ is the flow of X_i, so $\Phi_*(\partial/\partial t^i) = X_i$. The inverse of Φ is the required chart with $s^i = t^i$, completing the proof.

When the fields do not commute the frame cannot be straightened, and the obstruction is measured by the brackets $[X_i, X_j]$. Whether a k-plane field spanned by such fields is nonetheless tangent to a family of submanifolds is the integrability question answered by the Frobenius theorem (chapter 6).

5.11 Application: The Classification of 1-Manifolds

The flow of a nowhere-vanishing field parametrizes a manifold by time, and in dimension one that parametrization exhausts the manifold: every compact connected 1-manifold is an interval or a circle.

This is the base case of the classification of manifolds, and its corollary (a compact 1-manifold has an even number of boundary points) is the counting principle behind the intersection theory of chapter 8.

Theorem 5.11.1: Classification of 1-Manifolds

A smooth compact connected 1-manifold with boundary is diffeomorphic to the circle S^1 if its boundary is empty, and to the interval $[0, 1]$ otherwise.

***Proof*:**
Equip M with a Riemannian metric, patched from the Euclidean metrics of finitely many charts by a partition of unity (theorem 1.5.5). Call a smooth curve $\gamma\colon J \to M$ *unit-speed* if $|\gamma'(t)| = 1$ throughout. Such a curve has nonvanishing velocity, so in dimension one it is an immersion, hence a local diffeomorphism onto an open subset of M, carrying interior points of J to $M \setminus \partial M$ and an endpoint of J to ∂M.

A unit tangent vector at a point determines a unit-speed curve through it, unique wherever two are both defined, as both solve the same autonomous first-order equation with the same initial data (theorem 5.7.9). Let $\gamma\colon J \to M$ be a maximal unit-speed curve, J an interval. Its image is open; it is also closed, for if $\gamma(t_k) \to q$ with $t_k \to \sup J$ (a limit exists by compactness), a unit-speed curve through q agrees with γ near $\sup J$, so either γ extends past $\sup J$ (impossible by maximality unless $q \in \partial M$ and $\sup J \in J$) or γ returns to an earlier value. As M is connected and $\gamma(J)$ is nonempty, open, and closed, γ is onto.

If γ is injective it is a bijective local diffeomorphism, hence a diffeomorphism $J \xrightarrow{\sim} M$; compactness forces $J = [0, L]$ closed, its endpoints the two boundary points, so $M \cong [0, 1]$. Otherwise let $t_2 = \inf \{t \mid \gamma(t) \in \gamma(J_{<t})\}$ be the first return, with $\gamma(t_1) = \gamma(t_2) =: q$ for some $t_1 < t_2$. The velocities $\gamma'(t_1), \gamma'(t_2)$ are unit vectors of the line T_qM; were they opposite, the images of small arcs about t_1 and t_2 would coincide, producing a return before t_2 and contradicting its minimality, so $\gamma'(t_1) = \gamma'(t_2)$. Uniqueness then gives $\gamma(t + t_2 - t_1) = \gamma(t)$ for all t: the curve is periodic of least period $\ell = t_2 - t_1$ and descends to a diffeomorphism $\mathbb{R}/\ell\mathbb{Z} \xrightarrow{\sim} M$, so $M \cong S^1$ with $\partial M = \emptyset$, completing the classification.

Corollary 5.11.2: Boundary Parity of Compact 1-Manifolds

A compact 1-manifold with boundary has an even number of boundary points.

***Proof*:**
A compact 1-manifold has finitely many components, each a compact connected 1-manifold; by theorem 5.11.1 each is a circle, contributing no boundary points, or an interval, contributing exactly two, so the total is even.

This parity is the entire secret of intersection theory: the parameter space of a homotopy is one dimension up, and the two ends of the homotopy sit on its boundary, where they must balance.

6

Distributions and the Frobenius Theorem

A vector field prescribes a line at each point, and its integral curves knit those lines into curves filling the manifold (section 5.7). Raising the dimension, a *distribution* prescribes a k-plane at each point, and one asks the same question: is there a family of k-dimensional submanifolds tangent to it everywhere? For a single field the answer is always yes; for a plane field it is not, and the Frobenius theorem identifies exactly when. The criterion is bracket-closure: the plane field must be closed under the Lie bracket, and this *involutivity* is both necessary and sufficient for tangent submanifolds to exist. The theorem packages the commuting-frame straightening of section 5.10 into a coordinate-free integrability condition; it is the manifold input on which the integration of Lie subalgebras to subgroups rests ([9]).

Section 6.1 sets up distributions, involutivity, and integral manifolds and proves the easy half; section 6.2 proves the Frobenius theorem via a flat chart; and section 6.3 assembles the integral manifolds into a foliation.

6.1 Distributions

A distribution is a field of tangent planes, the plane-field analogue of a vector field.

Definition 6.1.1: Distribution

A *distribution* of rank k on M assigns to each p a k-dimensional subspace $D_p \subseteq T_pM$, varying smoothly: each point has a neighbourhood on which D is spanned by k smooth vector fields $X_1, \ldots, X_k$, independent at every point. Such a tuple is a *local frame* for D. A vector field X is a *section* of D, written $X \in \Gamma(D)$, if $X_p \in D_p$ for all p.

The bracket of two sections need not be a section; when it always is, the distribution is involutive.

Definition 6.1.2: Involutive Distribution

A distribution D is *involutive* if $[X, Y] \in \Gamma(D)$ whenever $X, Y \in \Gamma(D)$.

Involutivity need only be checked on a local frame, because the bracket is tensorial in its failure to be linear over functions.

Proposition 6.1.3: Involutivity on a Frame

A distribution D is involutive if and only if every local frame $(X_1, \ldots, X_k)$ satisfies $[X_i, X_j] \in \Gamma(D)$ for all i, j.

***Proof*:**
Necessity is immediate. For sufficiency, any two sections are locally $X = \sum_i f^i X_i$ and $Y = \sum_j g^j X_j$, and by the bracket expansion (proposition 5.5.3)

$$[X, Y] = \sum_{i,j} f^i g^j \, [X_i, X_j] + \sum_j (X g^j) \, X_j - \sum_i (Y f^i) \, X_i$$

The last two sums are sections of D (combinations of the X_i), and the first is too if each $[X_i, X_j] \in \Gamma(D)$; hence $[X, Y] \in \Gamma(D)$, completing the proof.

The submanifolds a distribution might be tangent to are its integral manifolds.

Definition 6.1.4: Integral Manifold

An *integral manifold* of a distribution D is a connected immersed k-dimensional submanifold $N \subseteq M$ with $T_pN = D_p$ for every $p \in N$. The distribution D is *integrable* if every point of M lies in an integral manifold.

Integrability forces involutivity: on an integral manifold the sections of D are tangent fields, and tangent fields bracket-close.

Proposition 6.1.5: Integrable Implies Involutive

Every integrable distribution is involutive.

***Proof*:**
Let $X, Y \in \Gamma(D)$ and $p \in M$, lying in an integral manifold $\iota\colon N \hookrightarrow M$. Since $D_q = T_qN$ for $q \in N$, the restrictions $X|_N, Y|_N$ are sections of TN, i.e. smooth vector fields $\tilde{X}, \tilde{Y}$ on N to which X, Y are ι-related. By naturality of the bracket (proposition 5.5.4), $[X, Y]$ is ι-related to $[\tilde{X}, \tilde{Y}]$, so $[X, Y]_p = T_p\iota\big([\tilde{X}, \tilde{Y}]_p\big) \in T_pN = D_p$. As p was arbitrary, $[X, Y] \in \Gamma(D)$, completing the proof.

Two examples frame the theorem to come: one integrable, one not.

Example 6.1.6: Integrable and Non-Integrable Plane Fields

1. *(Slices.)* On $\mathbb{R}^n$, $D = \operatorname{span}(\partial/\partial x^1, \ldots, \partial/\partial x^k)$ is integrable: through each point the k-slice $\{x^{k+1} = c^{k+1}, \ldots, x^n = c^n\}$ is an integral manifold. It is involutive, the coordinate fields commuting (exercise 5.5.1 (2)).
2. *(A contact plane field.)* On $\mathbb{R}^3$, $D = \operatorname{span}(\partial/\partial x,\ \partial/\partial y + x\,\partial/\partial z)$ is a rank-2 distribution with
$$[\partial/\partial x,\ \partial/\partial y + x\,\partial/\partial z] = \partial/\partial z$$
which at each point is not in $D_p = \operatorname{span}((1, 0, 0), (0, 1, x))$. So D is not involutive, hence not integrable (proposition 6.1.5): no surface is tangent to D everywhere.

The Frobenius theorem is the converse of proposition 6.1.5: the bracket obstruction of the second example is the only one.

6.2 The Frobenius Theorem

Involutivity is not only necessary for integrability but sufficient. The proof turns an involutive frame into a commuting one, which the commuting-frame canonical form of section 5.10 straightens to coordinate fields; the coordinate slices are then the integral manifolds.

Theorem 6.2.1: Frobenius Theorem

A smooth distribution is integrable if and only if it is involutive. Moreover, an involutive rank-k distribution D is *flat*: each point has a chart $(s^1, \ldots, s^n)$ in which $D = \operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k)$, so the slices $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$ are integral manifolds.

***Proof*:**
Integrability implies involutivity by proposition 6.1.5, and the flat form exhibits integral manifolds, so it remains to derive the flat chart from involutivity.

A commuting frame. Fix p and a chart $(x^1, \ldots, x^n)$ at p chosen so that D_p is complementary to $\operatorname{span}(\partial/\partial x^{k+1}|_p, \ldots, \partial/\partial x^n|_p)$. Then the projection $\pi\colon D_q \to \operatorname{span}(\partial/\partial x^1, \ldots, \partial/\partial x^k)$ onto the first k coordinates is an isomorphism for q near p, and there are unique sections $Y_i \in \Gamma(D)$ with

$\pi(Y_i) = \partial/\partial x^i$, namely

$$Y_i = \frac{\partial}{\partial x^i} + \sum_{a>k} b_i^a \frac{\partial}{\partial x^a}$$

for smooth b_i^a. The Y_i span D near p. Their brackets have no $\partial/\partial x^l$ component for $l \leq k$: since $Y_i^l = \delta_i^l$ and $Y_j^l = \delta_j^l$ are constant, $[Y_i, Y_j]^l = Y_i(Y_j^l) - Y_j(Y_i^l) = 0$. But D is involutive, so $[Y_i, Y_j] \in \Gamma(D)$, and a section of D is determined by its first k components (where π is an isomorphism); one with all of them zero is zero. Hence $[Y_i, Y_j] = 0$: the frame commutes.

The flat chart. The $Y_1|_p, \ldots, Y_k|_p$ are independent and pairwise commute, so by the canonical form for a commuting frame (theorem 5.10.4) there is a chart $(s^1, \ldots, s^n)$ centred at p with $Y_i = \partial/\partial s^i$ for $i \leq k$. Then $D = \operatorname{span}(Y_1, \ldots, Y_k) = \operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k)$, and each slice $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$ has tangent space $\operatorname{span}(\partial/\partial s^1, \ldots, \partial/\partial s^k) = D$ at each of its points, so it is an integral manifold through the corresponding point. Thus D is integrable and flat, completing the proof.

The theorem reduces a nonlinear question, the existence of tangent submanifolds, to the linear-algebraic test of bracket-closure on a frame. The contact plane field of example 6.1.6 fails the test, and so no surface is tangent to it: the planes twist too fast to be integrated, the prototype of a genuinely non-integrable field.

6.3 Foliations

The flat charts of an involutive distribution overlap consistently, and the slices through a point chain together into a single maximal integral manifold. The manifold is thereby partitioned into these *leaves*, a foliation.

Definition 6.3.1: Foliation

A *foliation* of dimension k on M is a partition $\mathcal{F}$ of M into connected immersed k-dimensional submanifolds, the *leaves*, such that each point has a flat chart $(s^1, \ldots, s^n)$ in which each leaf meets the chart domain in a union of slices $\{s^{k+1} = c^{k+1}, \ldots, s^n = c^n\}$.

Theorem 6.3.2: Global Frobenius Theorem

An involutive rank-k distribution D on M determines a k-dimensional foliation whose leaves are the maximal connected integral manifolds of D; through each point passes exactly one leaf.

***Proof*:**
By the Frobenius theorem each point has a flat chart, in which the slices are integral manifolds; call these the *plaques*. On an overlap of two flat charts, a plaque of one is an integral manifold tangent to D, hence meets each plaque of the other in a subset open-and-closed in their overlap (the last $n-k$ coordinates of the second chart are constant along the first plaque), so plaques match up along overlaps. Declare $q \sim q'$ if a finite chain of plaques joins them; this is an equivalence relation, and each class L, given the topology and smooth structure in which the

plaques are open submanifolds, is a connected immersed k-dimensional integral manifold. It is maximal: any connected integral manifold through a point of L is a union of plaques (an integral manifold is locally a plaque in each flat chart), hence contained in L. Distinct classes are disjoint, so the leaves partition M with one through each point, and the flat charts are exactly the foliation charts, completing the proof.

Example 6.3.3: Foliations

1. *(Fibres of a submersion.)* For a submersion $f\colon M \to N$, the kernels $D_p = \ker T_p f$ form an involutive distribution whose leaves are the connected components of the fibres $f^{-1}(q)$ (corollary 3.2.10); the product $\mathbb{R}^k \times \mathbb{R}^{n-k}$ foliated by $\mathbb{R}^k \times \{c\}$ is the model.
2. *(Kronecker foliation.)* On the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, the field $X = \partial/\partial x + \alpha\,\partial/\partial y$ spans a rank-1 distribution; for irrational α each leaf is the image of a line of slope α, which is dense in T^2 (exercise 4.3.1 (5)). The leaves are immersed but not embedded, and the leaf space $T^2/\mathcal{F}$ is not Hausdorff: a foliation need not fibre the manifold.

Exercise 6.3.1

1. Show that every rank-1 distribution is involutive, and hence integrable, recovering the local existence of integral curves.
2. Show that for a submersion $f\colon M \to N$ the kernel distribution $D_p = \ker T_p f$ is involutive.
3. For $D = \operatorname{span}(\partial/\partial x,\, \partial/\partial y + x\,\partial/\partial z)$ on $\mathbb{R}^3$, verify D is non-involutive and conclude no embedded surface is tangent to D at every point.

Part II

Differential Topology

Part I built the local theory of smooth manifolds: charts, the rank theorem, vector fields and their flows. Part II turns to the global question it makes possible, what smooth maps reveal about the *shape* of a manifold. The tools are two genericity principles. Sard's theorem says that the critical values of a smooth map are negligible, so regular values, where preimages are submanifolds, are the rule; transversality says that two submanifolds can always be perturbed to meet cleanly. From these come the invariants of the subject: the *degree* of a map, counting preimages with sign; the *index* of a vector field and the Poincaré-Hopf theorem, equating it with the Euler characteristic; Morse theory, reading the topology of a manifold off the critical points of a function; and cobordism, the equivalence relation under which manifolds bound. Each rests on the measure-zero and transversality arguments of the first two chapters.

7

Genericity and the Whitney Theorems

The failure of a smooth map to be a submersion is confined to a small set: Sard's theorem says the critical values, the images of the points where the rank drops, form a set of measure zero. Regular values are therefore dense, and their preimages are submanifolds (corollary 3.2.10), so the level-set constructions apply to almost every value. The measure-zero language also underlies integration on manifolds (chapter 13) and Whitney's approximation theorems.

The chapter's second theme, *transversality*, is the companion genericity principle: a map is transversal to a submanifold when, wherever the two meet, the image of the derivative and the tangent space of the submanifold together span the ambient tangent space, and transversal preimages are again submanifolds. Sard's theorem makes transversality *generic*: any map can be perturbed to meet a fixed submanifold transversally, and this genericity is the engine behind the intersection and degree invariants of the chapters that follow.

7.1 Measure Zero

Measure zero is the right notion of negligible for this chapter: it is preserved by smooth maps between equidimensional manifolds and needs no measure to define, only covers by cubes of small total volume.

Definition 7.1.1: Measure Zero

A subset $A \subseteq \mathbb{R}^n$ has *measure zero* if for every $\varepsilon > 0$ there is a countable cover $A \subseteq \bigcup_i W_i$ by cubes with $\sum_i \mathrm{vol}(W_i) < \varepsilon$.

A C^1 map cannot enlarge a measure-zero set, because it is locally Lipschitz.

Lemma 7.1.2: Smooth Image of a Measure-Zero Set

If $U \subseteq \mathbb{R}^n$ is open, $f\colon U \to \mathbb{R}^n$ is C^1, and $A \subseteq U$ has measure zero, then $f(A)$ has measure zero.

***Proof*:**
By second countability U is a countable union of open cubes with compact closure in U, so it suffices to show $f(A \cap Q)$ has measure zero for one such cube Q. On the compact $\overline{Q}$ the derivative is bounded, so the mean value inequality makes f Lipschitz there with some constant c: a set of diameter d maps to one of diameter $\le cd$. Given $\varepsilon > 0$, cover $A \cap Q$ by cubes W_i of side s_i with $\sum_i \operatorname{vol}(W_i) < \varepsilon$. Each W_i has diameter $s_i\sqrt{n}$, so $f(A \cap Q \cap W_i)$ (which lies in $\overline{Q}$, where f is c-Lipschitz) sits in a cube of side $cs_i\sqrt{n}$, of volume $(c\sqrt{n})^n \operatorname{vol}(W_i)$. Hence $f(A \cap Q)$ is covered with total volume $\le (c\sqrt{n})^n \sum_i \operatorname{vol}(W_i) < (c\sqrt{n})^n\varepsilon$; as ε was arbitrary, $f(A \cap Q)$ has measure zero, completing the proof.

Measure zero passes to manifolds through charts, consistently because transitions are C^1.

Definition 7.1.3: Measure Zero on a Manifold

A subset A of a smooth n-manifold M has *measure zero* if $\varphi(A \cap U)$ has measure zero in $\mathbb{R}^n$ for every chart (U, φ).

Proposition 7.1.4: One Atlas Suffices

If $A \subseteq M$ has $\varphi_\alpha(A \cap U_\alpha)$ of measure zero for every chart of some atlas $\{(U_\alpha, \varphi_\alpha)\}$, then A has measure zero.

***Proof*:**
Let (V, ψ) be any chart. For each α, $\psi(A \cap U_\alpha \cap V) = (\psi \circ \varphi_\alpha^{-1})\big(\varphi_\alpha(A \cap U_\alpha \cap V)\big)$ is the C^1 image of a measure-zero set, hence measure zero (lemma 7.1.2). By second countability the cover $\{U_\alpha\}$ of $A \cap V$ has a countable subcover, so $\psi(A \cap V)$ is a countable union of measure-zero sets, hence measure zero; since (V, ψ) was arbitrary, A has measure zero, completing the proof.

Proposition 7.1.5: Measure Zero under Smooth Maps

If $f\colon M \to N$ is smooth with $\dim M \le \dim N$ and $A \subseteq M$ has measure zero, then $f(A)$ has measure zero.

***Proof*:**
Cover M by countably many charts $(U_\alpha, \varphi_\alpha)$ and N by charts (V_β, ψ_β). Fix charts and consider the coordinate representation $F = \psi_\beta \circ f \circ \varphi_\alpha^{-1}\colon \mathbb{R}^{\dim M} \to \mathbb{R}^{\dim N}$. View the measure-zero set $\varphi_\alpha(A \cap U_\alpha \cap f^{-1}(V_\beta))$ inside $\mathbb{R}^{\dim N}$ through the inclusion into the first $\dim M$ coordinates – still measure zero, as $\dim M \le \dim N$ – and extend F to a C^1 endomorphism of $\mathbb{R}^{\dim N}$ by precomposing with the projection onto those coordinates. This endomorphism carries the (measure-zero) embedded set to ψ_β of the corresponding piece of $f(A)$, which is therefore measure zero (lemma 7.1.2). Countably many such pieces cover $f(A)$, so it has measure zero by

proposition 7.1.4, completing the proof.

7.2 Sard's Theorem

Sard's theorem is proved by reducing to a map of Euclidean domains and inducting on the dimension of the source, stratifying the critical set by the order to which the derivatives vanish. Two analytic inputs are cited: Taylor's inequality ([2]) and the measure-zero form of Fubini's theorem ([1]) – a σ-compact subset of $\mathbb{R}^m$ every slice $\{t\} \times \mathbb{R}^{m-1}$ of which has measure zero is itself measure zero. The critical-value sets below are continuous images of σ-compact critical sets, hence σ-compact, so this form applies.

Theorem 7.2.1: Sard's Theorem

Let $f\colon M \to N$ be a smooth map of manifolds. Then the set of critical values of f has measure zero in N; in particular its complement, the regular values, is dense.

***Proof*:**
Covering M and N by countably many charts reduces the claim, by proposition 7.1.4, to the Euclidean statement: for smooth $f\colon U \to \mathbb{R}^m$ on open $U \subseteq \mathbb{R}^n$ with critical set $C = \{x \mid \operatorname{rank} Df(x) < m\}$, the image $f(C)$ has measure zero. Proceed by induction on n; the case $n = 0$ is trivial, $f(C)$ being at most a point. Stratify C by

$$C_i = \{x \in U \mid \text{all partial derivatives of } f \text{ of order} \leq i \text{ vanish at } x\}$$

so $C \supseteq C_1 \supseteq C_2 \supseteq \cdots$, and split $f(C)$ into $f(C \setminus C_1)$, the $f(C_i \setminus C_{i+1})$, and $f(C_k)$ for a large k.

Step 1: $f(C \setminus C_1)$ has measure zero. At $p \in C \setminus C_1$ some first partial is nonzero, say $\partial f^1/\partial x^1(p) \neq 0$. Then $h(x) = (f^1(x), x^2, \ldots, x^n)$ has nonsingular Jacobian at p, hence is a diffeomorphism near p, and $g = f \circ h^{-1}$ preserves the first coordinate: $g(t, y) = (t, g^t(y))$ with $g^t\colon \mathbb{R}^{n-1} \to \mathbb{R}^{m-1}$ smooth. A point (t, y) is critical for g iff y is critical for g^t, so the critical values of g over the slice $\{t\} \times \mathbb{R}^{m-1}$ are the critical values of g^t, which have measure zero by the inductive hypothesis in dimension $n-1$. By the measure-zero Fubini theorem the critical values of g, and so $f(C \setminus C_1)$ near p, have measure zero; countably many such neighbourhoods cover $C \setminus C_1$.

Step 2: $f(C_i \setminus C_{i+1})$ has measure zero, $i \geq 1$. At $p \in C_i \setminus C_{i+1}$ some $(i+1)$-st partial is nonzero, so some i-th partial w has $\partial w/\partial x^1(p) \neq 0$. Then $h(x) = (w(x), x^2, \ldots, x^n)$ is a diffeomorphism near p carrying C_i into the hyperplane $\{x^1 = 0\}$, since w vanishes on C_i. The restriction of $f \circ h^{-1}$ to that hyperplane is a smooth map of an $(n-1)$-dimensional domain whose critical set contains $h(C_i)$, so by induction its critical values, hence $f(C_i \setminus C_{i+1})$ near p, have measure zero.

Step 3: $f(C_k)$ has measure zero for $k \geq n/m$. On C_k all partials up to order k vanish, so Taylor's inequality ([2]) gives, on a compact cube $Q \subseteq U$, a constant c with $|f(x+h) - f(x)| \leq c|h|^{k+1}$ for $x \in C_k \cap Q$ and small h. Subdivide Q (side δ) into l^n subcubes of side δ/l. A subcube meeting C_k has its image in a cube of side $c'(1/l)^{k+1}$, of volume $\leq c''\, l^{-m(k+1)}$; there are at most l^n such subcubes, so $f(C_k \cap Q)$ is covered with total volume $\leq c''\, l^{\,n-m(k+1)}$. For $k+1 > n/m$ this exponent is negative, so the bound tends to 0 as $l \to \infty$; hence $f(C_k \cap Q)$, and so $f(C_k)$, has measure zero.

Taking $k \geq n/m$, $C = (C \setminus C_1) \cup \bigcup_{i=1}^{k-1}(C_i \setminus C_{i+1}) \cup C_k$, and each image has measure zero by Steps 1–3, so $f(C)$ has measure zero. This completes the induction and the proof; density of the regular values follows since the complement of a measure-zero set is dense.

Corollary 7.2.2: Regular Values are Dense

The regular values of a smooth $f: M \to N$ are dense in N; if f is proper, they are also open, hence a dense open set.

***Proof*:**
Density is the last line of theorem 7.2.1. If f is proper, the critical value set is closed (the image of the closed critical set under a proper, hence closed, map), so the regular values are open, completing the proof.

Sard also bounds the size of an image whenever the source is too small to fill the target.

Corollary 7.2.3: Small-Dimensional Images are Negligible

If $f: M \to N$ is smooth with $\dim M < \dim N$, then $f(M)$ has measure zero; in particular f is not surjective.

***Proof*:**
Every $T_pf: T_pM \to T_{f(p)}N$ has rank at most $\dim M < \dim N$, so is not surjective; thus every point of M is critical and $f(M)$ is the set of critical values, of measure zero by theorem 7.2.1. A measure-zero set is not all of N, so f is not surjective, completing the proof.

A C^1 curve $\mathbb{R} \to \mathbb{R}^2$ therefore cannot fill a square: a space-filling curve is necessarily not continuously differentiable.

Exercise 7.2.1

1. Show that the set of critical *points* need not have measure zero, even though the critical *values* do: give a smooth $f: \mathbb{R} \to \mathbb{R}$ whose critical set is all of $\mathbb{R}$.
2. Deduce from corollary 7.2.3 that no smooth map $S^1 \to S^2$ is surjective, and that every smooth $f: S^1 \to S^2$ misses an open set.
3. Let $f: M \to \mathbb{R}$ be smooth. Show that for almost every $c \in \mathbb{R}$ the level set $f^{-1}(c)$ is either empty or an embedded hypersurface of M.

7.3 Transversality

Sard's theorem says a smooth map meets a chosen point cleanly for almost every point. Transversality is the same principle for a submanifold in place of a point: two submanifolds, or a map and a submanifold, meet *cleanly* when their tangent data together fill the ambient tangent space, and theorem 7.2.1 will show that clean meeting is the generic case. Clean intersections are exactly the ones that are again submanifolds, with codimensions adding.

The linear model comes first. Subspaces $U, W \subseteq V$ are *transverse* in V if $U + W = V$. The sum need not be direct: in $\mathbb{R}^3$ the planes $\langle e_1, e_2\rangle$ and $\langle e_2, e_3\rangle$ are transverse yet meet in the line $\langle e_2\rangle$. For transverse U, W, dimension counting gives $\dim(U \cap W) = \dim U + \dim W - \dim V$ (in general $\dim(U + W)$ replaces $\dim V$), which in codimensions ($\mathrm{codim} U = \dim V - \dim U$) reads

$$\mathrm{codim}(U \cap W) = \mathrm{codim} U + \mathrm{codim} W.$$

Codimension counts constraints, and a transverse intersection imposes the constraints of each factor independently, so the counts add. Two planes in $\mathbb{R}^3$ (codim1 each) meet transversally in a line (codim2); two lines in $\mathbb{R}^3$ (codim2 each) would need codim4, which is impossible in a 3-space, so lines in $\mathbb{R}^3$ can meet only degenerately, never transversally except when disjoint.

Imposing this on every tangent space is the definition.

Definition 7.3.1: Transverse Submanifolds

Regular submanifolds $K, L \subseteq M$ are *transverse*, written $K \pitchfork L$, if

$$T_pK + T_pL = T_pM \qquad \text{for every } p \in K \cap L.$$

The condition is vacuous where K and L are disjoint, so disjoint submanifolds count as transverse. Where they meet, tangential transversality forces the intersection to be a submanifold of the predicted codimension.

Proposition 7.3.2: A Transverse Intersection Is a Submanifold

If $K, L \subseteq M$ are transverse regular submanifolds, then $K \cap L$ is a regular submanifold with $\mathrm{codim}(K \cap L) = \mathrm{codim} K + \mathrm{codim} L$, and $T_p(K \cap L) = T_pK \cap T_pL$ at each $p \in K \cap L$.

***Proof*:**
Being a submanifold is local and detected by defining maps (theorem 3.2.11): near $p \in K \cap L$ choose neighbourhoods U, V and submersions $f\colon U \to \mathbb{R}^{\mathrm{codim} K}$, $g\colon V \to \mathbb{R}^{\mathrm{codim} L}$ with $K \cap U = f^{-1}(0)$ and $L \cap V = g^{-1}(0)$. Set $(f, g)\colon U \cap V \to \mathbb{R}^{\mathrm{codim} K + \mathrm{codim} L}$, $(f, g)(x) = (f(x), g(x))$. Then

$$(f, g)^{-1}(0) = f^{-1}(0) \cap g^{-1}(0) = K \cap L \cap U \cap V,$$

so it suffices to show 0 is a regular value of (f, g). At $q \in (f, g)^{-1}(0)$ the derivative $D_q(f, g) = \begin{pmatrix} D_qf \\ D_qg \end{pmatrix}$ has kernel $\ker D_qf \cap \ker D_qg = T_qK \cap T_qL$ (proposition 3.2.16). Transversality gives $T_qK + T_qL = T_qM$, so

$$\dim(T_qK \cap T_qL) = \dim T_qK + \dim T_qL - \dim T_qM = \dim M - \mathrm{codim} K - \mathrm{codim} L.$$

The kernel therefore has dimension $\dim M - (\mathrm{codim} K + \mathrm{codim} L)$, so $D_q(f, g)$ has full rank $\mathrm{codim} K + \mathrm{codim} L$ onto $\mathbb{R}^{\mathrm{codim} K + \mathrm{codim} L}$; thus 0 is a regular value. Hence $K \cap L$ is a regular submanifold of codimension $\mathrm{codim} K + \mathrm{codim} L$, with tangent space the kernel $T_qK \cap T_qL$, completing the proof.

Transverse intersections manufacture manifolds, sometimes exotic ones.

Example 7.3.3: Brieskorn's Exotic Spheres
In $\mathbb{C}^5 \setminus \{0\}$, intersect a Brieskorn variety with the unit sphere:

$$\Sigma_k = \left\{ z \in \mathbb{C}^5 \;\middle|\; z_1^2 + z_2^2 + z_3^2 + z_4^3 + z_5^{6k-1} = 0 \right\} \cap \left\{ z \in \mathbb{C}^5 \;\middle|\; \textstyle\sum_{i=1}^5 |z_i|^2 = 1 \right\}, \qquad k \in \{1, \ldots, 28\}.$$

The two real hypersurfaces meet transversally: the field $\frac{1}{2}z_1\partial_{z_1} + \frac{1}{2}z_2\partial_{z_2} + \frac{1}{2}z_3\partial_{z_3} + \frac{1}{3}z_4\partial_{z_4} + \frac{1}{6k-1}z_5\partial_{z_5}$ is tangent to the first variety but transverse to the sphere, so their tangent spaces span. Each Σ_k is thus a smooth 7-manifold, and each is homeomorphic to S^7; but the 28 of them realize the 28 distinct oriented diffeomorphism classes of the 7-sphere, no two diffeomorphic. These are the *exotic 7-spheres* of Brieskorn. (In real coordinates $z = x + iy$, $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$.)

7.4 Transversal Maps and Preimages

Replacing the two submanifolds by smooth maps costs nothing and gains the generality the rest of the chapter needs. The inclusion of a submanifold is a special case, and the inverse-image constructions of chapter 3 become the special case where one map is a constant.

Definition 7.4.1: Transverse Maps

Smooth maps $f\colon K \to M$ and $g\colon L \to M$ are *transverse*, written $f \pitchfork g$, if

$$\operatorname{im}(D_a f) + \operatorname{im}(D_b g) = T_pM \qquad \text{whenever } f(a) = g(b) = p.$$

The two derivatives $D_af\colon T_aK \to T_pM$ and $D_bg\colon T_bL \to T_pM$ share the codomain T_pM, so the sum makes sense. The most frequent case is a single map $f\colon N \to M$ against a submanifold $S \subseteq M$, meaning f transverse to the inclusion $\iota\colon S \hookrightarrow M$; there $\operatorname{im} D_b\iota = T_bS$, and the condition reads $\operatorname{im}(D_af) + T_{f(a)}S = T_{f(a)}M$ at each $a \in f^{-1}(S)$. When $S = \{y\}$ is a point, $T_yS = 0$ and transversality becomes $\operatorname{im} D_af = T_yM$ for all $a \in f^{-1}(y)$, i.e. y is a regular value: *regularity is transversality to a point.*

The reduction that proves the level-set theorem proves the general statement. Near $p \in S$ pick a submersion $\varphi\colon U \to \mathbb{R}^{\operatorname{codim} S}$ with $S \cap U = \varphi^{-1}(0)$; then near a point $a \in f^{-1}(S)$,

$$f^{-1}(S) = (\varphi \circ f)^{-1}(0),$$

so $f^{-1}(S)$ is cut out by $\varphi \circ f$, and it is a submanifold as soon as 0 is a regular value of $\varphi \circ f$. Since $D_a(\varphi \circ f) = D_{f(a)}\varphi \circ D_af$ and $\ker D_{f(a)}\varphi = T_{f(a)}S$, the composite $\varphi \circ f$ is a submersion at a exactly when $\operatorname{im} D_af$ surjects onto $T_{f(a)}M/T_{f(a)}S$, which is the transversality condition

7.4.2 Transversality Criterion

$$\operatorname{im}(D_af) + T_{f(a)}S = T_{f(a)}M.$$

Proposition 7.4.3: The Preimage of a Transverse Submanifold

Let $S \subseteq M$ be an embedded submanifold and $f\colon N \to M$ a smooth map transverse to S. Then $f^{-1}(S)$ is an embedded submanifold of N with $\operatorname{codim} f^{-1}(S) = \operatorname{codim} S$.

***Proof*:**
Let $k = \operatorname{codim} S$. Fix $a \in f^{-1}(S)$ and, using that S is embedded, a neighbourhood U of $f(a)$ and a submersion $\varphi\colon U \to \mathbb{R}^k$ with $S \cap U = \varphi^{-1}(0)$. On $f^{-1}(U)$,

$$(\varphi \circ f)^{-1}(0) = f^{-1}(\varphi^{-1}(0)) = f^{-1}(S \cap U) = f^{-1}(S) \cap f^{-1}(U),$$

so it suffices that 0 be a regular value of $\varphi \circ f$. Let $p \in (\varphi \circ f)^{-1}(0)$ and $z \in \mathbb{R}^k = T_0\mathbb{R}^k$. As φ is a submersion there is $y \in T_{f(p)}M$ with $D_{f(p)}\varphi(y) = z$; transversality writes $y = y_0 + D_p f(v)$ with $y_0 \in T_{f(p)}S$ and $v \in T_pN$. Since $\varphi \equiv 0$ on $S \cap U$ and $T_{f(p)}S = \ker D_{f(p)}\varphi$, we get $D_{f(p)}\varphi(y_0) = 0$, so

$$D_p(\varphi \circ f)(v) = D_{f(p)}\varphi\big(D_p f(v)\big) = D_{f(p)}\varphi(y) = z.$$

Thus $D_p(\varphi \circ f)$ is surjective, 0 is a regular value, and theorem 3.2.11 makes $f^{-1}(S)$ an embedded submanifold of codimension k, completing the proof.

A submersion is transverse to everything, so the corollary needs no separate argument.

Corollary 7.4.4: Submersions Pull Back Submanifolds

If $f\colon N \to M$ is a submersion and $S \subseteq M$ an embedded submanifold, then f is transverse to S and $f^{-1}(S)$ is an embedded submanifold with $\operatorname{codim} f^{-1}(S) = \operatorname{codim} S$.

***Proof*:**
A submersion has $\operatorname{im} D_p f = T_{f(p)}M$, so $\operatorname{im} D_p f + T_{f(p)}S = T_{f(p)}M$ trivially; apply proposition 7.4.3.

The tangent space of the preimage is read off the same criterion.

Corollary 7.4.5: Tangent Space of a Transverse Preimage

With $f \pitchfork S$ and $W = f^{-1}(S)$ as above, $T_pW = (D_p f)^{-1}(T_{f(p)}S)$ at each $p \in W$.

***Proof*:**
Locally $W = (\varphi \circ f)^{-1}(0)$ with φ a submersion cutting out S, so $T_pW = \ker D_p(\varphi \circ f) = \ker\big(D_{f(p)}\varphi \circ D_p f\big) = (D_p f)^{-1}(\ker D_{f(p)}\varphi) = (D_p f)^{-1}(T_{f(p)}S)$.

For two general maps there is no ambient submanifold to invert. The remedy is to build the space the intersection ought to live on. If $f\colon K \to M$ and $g\colon L \to M$, the set of matched pairs

$$K\,{}_f\!\times_g L = \{(a,b) \in K \times L \mid f(a) = g(b)\}$$

is the *fibered product* of f and g; when the maps are understood it is written $K \times_M L$. It reduces to a preimage in the previous sense by cutting the product $K \times L$ with the diagonal.

Proposition 7.4.6: The Fibered Product of Transverse Maps

If $f\colon K \to M$ and $g\colon L \to M$ are transverse, then $K \times_M L$ is a regular submanifold of $K \times L$ of codimension $\dim M$, and it carries smooth projections to K and L making

$$\begin{tikzcd} K \times_M L \arrow[r, "\pi_L"] \arrow[d, "\pi_K"'] & L \arrow[d, "g"] \\ K \arrow[r, "f"'] & M \end{tikzcd}$$

commute.

***Proof*:**
Consider $f \times g\colon K \times L \to M \times M$ and the diagonal $\Delta_M \subseteq M \times M$. A pair (a, b) lies in $(f \times g)^{-1}(\Delta_M)$ iff $f(a) = g(b)$, so $(f \times g)^{-1}(\Delta_M) = K \times_M L$. It suffices, by proposition 7.4.3, to show $f \times g$ is transverse to Δ_M; since $\operatorname{codim}\Delta_M = \dim M$, the codimension statement follows.

Fix (a, b) with $f(a) = g(b) = p$. The tangent space of the diagonal is $T_{(p,p)}\Delta_M = \{(u, u) \mid u \in T_pM\}$, and $\operatorname{im}D_{(a,b)}(f \times g) = \{(D_af\,v,\ D_bg\,w) \mid v \in T_aK,\ w \in T_bL\}$. Given an arbitrary $(A, B) \in T_pM \times T_pM$, transversality of f, g provides v, w with $D_af\,v - D_bg\,w = A - B$. Set $u = A - D_af\,v = B - D_bg\,w$. Then

$$(A, B) = (D_af\,v,\ D_bg\,w) + (u, u) \in \operatorname{im}D_{(a,b)}(f \times g) + T_{(p,p)}\Delta_M,$$

so $f \times g \pitchfork \Delta_M$. Hence $K \times_M L$ is a regular submanifold of codimension $\dim M$. The ambient projections $K \times L \to K$ and $K \times L \to L$ restrict to the smooth maps π_K, π_L, and $f \circ \pi_K = g \circ \pi_L$ by the defining equation, completing the proof.

Example 7.4.7: Fibered Products

1. The name is explained by the trivial case. For $\pi_1\colon M \times Z_1 \to M$ and $\pi_2\colon M \times Z_2 \to M$ the projections (submersions, hence transverse), the fibered product is

$$(M \times Z_1) \times_M (M \times Z_2) \cong M \times Z_1 \times Z_2,$$

   the family of product fibers $Z_1 \times Z_2$ over M.

2. Let $p\colon S^3 \to S^2$ be the Hopf map. It is a submersion, so transverse to itself, and the fibered product $S^3 \times_{S^2} S^3$ is a smooth 4-manifold whose map to S^2 has fiber $S^1 \times S^1$ over each point, a torus bundle pulled back from the Hopf fibration $S^1 \to S^3 \to S^2$, the archetypal nontrivial *fiber bundle* of chapter 11.

3. Let S^1 lie in the xy-plane and $(S^1)'$ in the yz-plane of $\mathbb{R}^3$. Wherever they meet, $\dim(TS^1 + T(S^1)') \le 2 < 3$, so they can *never* meet transversally in $\mathbb{R}^3$, even when $S^1 \cap (S^1)'$ happens to be a manifold. Transversality is a property of the pair *together with the ambient space*, not of the intersection alone.

Transversality to the fibers of a product recovers, and sharpens, the description of a graph.

Theorem 7.4.8: Global Characterization of Graphs

Let $S \subseteq M \times N$ be an embedded submanifold, with π_M, π_N the projections. The following are equivalent:

1. S is the graph of a smooth map $f\colon M \to N$;
2. $\pi_M|_S\colon S \to M$ is a diffeomorphism;
3. for every $p \in M$, S meets $\{p\} \times N$ transversally in exactly one point.

When they hold, $f = \pi_N \circ (\pi_M|_S)^{-1}$.

***Proof*:**
(1) $\Rightarrow$ (2). If $S = \{(x, f(x)) \mid x \in M\}$, then $\pi_M|_S(x, f(x)) = x$ is a bijection with smooth inverse $x \mapsto (x, f(x))$, hence a diffeomorphism.

(2) $\Rightarrow$ (3). Since $\pi_M|_S$ is a bijection, exactly one point (p,q) of S lies over each p, so $S \cap (\{p\} \times N) = \{(p,q)\}$. At (p,q), the isomorphism $D(\pi_M|_S)\colon T_{(p,q)}S \to T_pM$ shows $D\pi_M$ is injective on $T_{(p,q)}S$, so $T_{(p,q)}S \cap (\{0\} \times T_qN) = T_{(p,q)}S \cap \ker D\pi_M = 0$. As $\dim S = \dim M$, dimensions give $T_{(p,q)}S \oplus (\{0\} \times T_qN) = T_pM \times T_qN = T_{(p,q)}(M \times N)$, which is transversality to $\{p\} \times N$.

(3) $\Rightarrow$ (1). Exactly one point over each p makes $\pi_M|_S$ a bijection, defining a set map $f\colon M \to N$ by $S = \{(p, f(p)) \mid p \in M\}$. At the point (p,q) over p, the transverse intersection $S \cap (\{p\} \times N)$ is a submanifold of dimension $\dim S - \dim M$; being the single point (p,q), it is 0-dimensional, forcing $\dim S = \dim M$ and $T_{(p,q)}S \cap (\{0\} \times T_qN) = 0$. Hence $D(\pi_M|_S)$ is an isomorphism at every point, and $\pi_M|_S$ is a smooth bijective local diffeomorphism, so a diffeomorphism. Then $f = \pi_N \circ (\pi_M|_S)^{-1}$ is smooth and S is its graph, completing the proof.

Dropping to one point gives the local statement, the transversality form of the implicit function theorem.

Corollary 7.4.9: Local Characterization of Graphs

Let $S \subseteq M \times N$ be an immersed submanifold with $\dim S = \dim M$, and $(p,q) \in S$. If S meets $\{p\} \times N$ transversally at (p,q), then a neighbourhood of (p,q) in S is the graph of a smooth map $f\colon U \to N$ defined on a neighbourhood U of p.

***Proof*:**
Since $\dim S = \dim M$, transversality $T_{(p,q)}S + (\{0\} \times T_qN) = T_pM \times T_qN$ forces $\dim \big(T_{(p,q)}S \cap (\{0\} \times T_qN)\big) = \dim S + \dim N - (\dim M + \dim N) = 0$, so $D(\pi_M|_S)\colon T_{(p,q)}S \to T_pM$ is an isomorphism of equidimensional spaces. By the inverse function theorem $\pi_M|_S$ restricts to a diffeomorphism from a neighbourhood V of (p,q) in S onto a neighbourhood U of p; then $f = \pi_N \circ (\pi_M|_S)^{-1}\colon U \to N$ is smooth and V is its graph, completing the proof.

7.5 Stability and Genericity

Two features single transversality out as the right notion of clean intersection. It is *stable*, unchanged by small perturbations of the maps, and it is *generic*, achievable by an arbitrarily small perturbation of any map. Stability follows from compactness and genericity from Sard's theorem; together they say that among all smooth maps the transverse ones form a large, robust class. The perturbations are packaged as homotopies.

Definition 7.5.1: Smooth Homotopy

A *smooth homotopy* between smooth maps $f_0, f_1 \colon M \to N$ is a smooth map $F \colon M \times [0,1] \to N$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$; write $f_t = F(\cdot, t)$.

Definition 7.5.2: Stability

A property of a smooth map $f \colon M \to N$ is *stable* if for every smooth homotopy f_t with $f_0 = f$ there is an $\varepsilon > 0$ such that f_t has the property for all $t < \varepsilon$.

On a compact source, being an immersion is stable, because injectivity of the derivative is the non-vanishing of a minor.

Proposition 7.5.3: Stability of Immersions and Submersions

On a compact manifold M, the properties of $f \colon M \to N$ being an immersion, and of being a submersion, are each stable.

***Proof*:**
Take the immersion case; the submersion case is identical with rows and columns exchanged. Let f_t be a homotopy with $f_0 = f$ an immersion. At each $p \in M$, $D_p f$ is injective, so in local coordinates its Jacobian has a nonvanishing $m \times m$ minor ($m = \dim M$). That minor is a continuous function of (p, t) and is nonzero at $(p, 0)$, so it is nonzero on a neighbourhood $U_p \times [0, \varepsilon_p)$, where $D_q f_t$ is injective. The sets $U_p \times \{0\}$ cover the compact $M \times \{0\}$; a finite subcover has a smallest ε, and f_t is an immersion for all $t < \varepsilon$, completing the proof.

Theorem 7.5.4: Stability of Transversality on Compact Sources

Let K be compact and $L \subseteq M$ a closed embedded submanifold. If $f \colon K \to M$ is transverse to L, then transversality to L is stable under homotopies of f.

***Proof*:**
Let f_t be a homotopy with $f_0 = f$, and $F \colon K \times [0,1] \to M$ the homotopy map. Cover K by neighbourhoods on which f_t stays transverse for a short time, then use compactness.

Where f misses L: for $p \notin f^{-1}(L)$, $f(p)$ lies in the open set $M \setminus L$ (here L closed is used), so $F^{-1}(M \setminus L)$ is open and contains $(p, 0)$; choose $U_p \times [0, \varepsilon_p)$ inside it, on which f_t avoids L and is vacuously transverse.

Where f meets L: for $p \in f^{-1}(L)$, pick near $f(p)$ a submersion φ with $L = \varphi^{-1}(0)$. By the transversality criterion (box 7.4.2), transversality of f_t at a point of $(\varphi \circ f_t)^{-1}(0)$ is exactly $\varphi \circ f_t$ being a submersion there, i.e. a nonvanishing maximal minor of $D(\varphi \circ f_t)$. That minor is continuous in (p, t) and nonzero at $(p, 0)$, so nonzero on some $U_p \times [0, \varepsilon_p)$. The U_p cover the compact K; a finite subcover with smallest ε makes f_t transverse to L for all $t < \varepsilon$, completing the proof.

Genericity is the deeper half, and it is where Sard's theorem enters. The tool is a whole family of perturbations at once.

Definition 7.5.5: Smooth Family of Maps

A *smooth family* of maps $N \to M$ parametrized by a manifold S is a smooth map $F\colon N \times S \to M$; write $F_s = F(\cdot, s)$.

The next theorem is the engine of genericity: if the *total* family is transverse to X, then almost every member is. It is often called the parametric transversality, or Thom transversality, theorem.

Theorem 7.5.6: Parametric Transversality

Let $X \subseteq M$ be an embedded submanifold and $F\colon N \times S \to M$ a smooth family transverse to X. Then F_s is transverse to X for almost every $s \in S$.

***Proof*:**
Since $F \pitchfork X$, the preimage $W = F^{-1}(X)$ is an embedded submanifold of $N \times S$ (proposition 7.4.3). Let $\pi\colon N \times S \to S$ be the projection and restrict it to $\pi|_W\colon W \to S$. By Sard's theorem (theorem 7.2.1) almost every $s \in S$ is a regular value of $\pi|_W$; it suffices to show that each such s makes F_s transverse to X.

Fix a regular value s, a point $p \in F_s^{-1}(X)$, and $q = F_s(p) \in X$. The goal is

$$T_q M = T_q X + \operatorname{im} D_p F_s.$$

Three facts are in hand. First, $F \pitchfork X$ gives $T_q M = T_q X + \operatorname{im} D_{(p,s)} F$. Second, s a regular value of $\pi|_W$ gives $D_{(p,s)}(\pi|_W)(T_{(p,s)} W) = T_s S$. Third, $T_{(p,s)} W = (D_{(p,s)} F)^{-1}(T_q X)$ (corollary 7.4.5). Now take any $w \in T_q M$. By the first fact,

$$w = a + D_{(p,s)} F(v_1, e_1), \qquad a \in T_q X,\ (v_1, e_1) \in T_p N \times T_s S.$$

By the second fact there is $(v_2, e_1) \in T_{(p,s)} W$ with the same S-component e_1 (the projection π reads off that component). By the third fact $D_{(p,s)} F(v_2, e_1) \in T_q X$. Splitting off the S-direction, and writing $\iota_s\colon N \to N \times S$, $\iota_s(x) = (x, s)$, so that $F \circ \iota_s = F_s$,

$$D_{(p,s)} F(v_1, e_1) = \underbrace{D_{(p,s)} F(v_2, e_1)}_{\in T_q X} + D_{(p,s)} F(v_1 - v_2, 0), \qquad D_{(p,s)} F(v_1 - v_2, 0) = D_p F_s(v_1 - v_2).$$

Hence $w = a + D_{(p,s)} F(v_2, e_1) + D_p F_s(v_1 - v_2) \in T_q X + \operatorname{im} D_p F_s$, which is the required transversality, completing the proof.

To perturb a *given* map one needs a family through it that is itself transverse to X. When the target is $\mathbb{R}^n$ this is immediate: $F(x,s) = f(x) + s$ on $N \times \mathbb{R}^n$ is a submersion, hence transverse to everything, and $F_0 = f$. For a general target the family is built inside a tubular neighbourhood, which is where the embedding theory of section 7.6 is borrowed.

Theorem 7.5.7: Transversality Homotopy Theorem

Let $X \subseteq M$ be an embedded submanifold. Every smooth map $f\colon N \to M$ is homotopic to a smooth map transverse to X.

***Proof*:**
Embed M in some $\mathbb{R}^K$ (theorem 7.6.1) and let $\pi\colon T \to M$ be the retraction of a tubular neighbourhood $T \subseteq \mathbb{R}^K$ of M (theorem 7.8.1). For each x, $f(x) + b \in T$ once $b \in \mathbb{R}^K$ is small, so on the open unit ball $B \subseteq \mathbb{R}^K$ set

$$F\colon N \times B \to M, \qquad F(x,b) = \pi\big(f(x) + \varepsilon\, b\big),$$

with $\varepsilon > 0$ chosen (using compact exhaustion, or a smooth $\varepsilon(x)$) so that $f(x) + \varepsilon b \in T$ throughout. For fixed x the map $b \mapsto f(x) + \varepsilon b$ is a submersion onto a neighbourhood of $f(x)$ in $\mathbb{R}^K$, and π is a submersion on T, so F is a submersion; in particular $F \pitchfork X$. By parametric transversality (theorem 7.5.6) there is b, arbitrarily close to 0, with F_b transverse to X. Finally $F_0(x) = \pi(f(x)) = f(x)$ since $f(x) \in M$ and π fixes M, so $t \mapsto F_{tb}$ is a homotopy from f to the transverse map F_b, completing the proof.

Transversality of two maps $f\colon N \to M$ and $g\colon N' \to M$ is transversality of $f \times g$ to the diagonal $\Delta_M \subseteq M \times M$, so the compact-source stability and the genericity theorems apply to pairs of maps as well, perturbing either factor.

Exercise 7.5.1

1. Let $C \subseteq \mathbb{R}^n$ be closed and $f(x) = d(x, C)$ the distance to C, a continuous function with $f^{-1}(0) = C$. Let $M = \Gamma(f) \subseteq \mathbb{R}^{n+1}$ be its graph and $N = \mathbb{R}^n \times \{0\}$. Show M and N are topological manifolds with $M \cap N = C \times \{0\} \cong C$, so that the transverse-intersection theorem fails outright for merely topological manifolds: C can be any closed set.

7.6 Whitney's Immersion and Embedding Theorems

The definition of a manifold names no ambient space. Whitney's theorem shows none is lost by demanding one: every n-manifold is a submanifold of a Euclidean space whose dimension depends only on n. The proof is the first real dividend of Sard's theorem. Build some embedding into a large $\mathbb{R}^N$ by hand, then project it to lower and lower dimensions, discarding at each step only a measure-zero set of bad directions.

Theorem 7.6.1: Whitney Immersion and Embedding

Every smooth n-manifold M admits a smooth immersion into $\mathbb{R}^{2n}$ and a proper smooth embedding into $\mathbb{R}^{2n+1}$. In particular M is diffeomorphic to a closed submanifold of $\mathbb{R}^{2n+1}$.

***Proof*:**
Take M compact first.

Step 1: an embedding into some $\mathbb{R}^N$. Choose a finite atlas (U_i, φ_i), $i = 1, \ldots, k$, and a partition of unity $\{\rho_i\}$ with $\operatorname{supp}\rho_i \subseteq U_i$ (theorem 1.5.5). Each $\rho_i\varphi_i$, extended by 0 off U_i, is a smooth map $M \to \mathbb{R}^n$, and

$$F = (\rho_1\varphi_1, \ldots, \rho_k\varphi_k, \rho_1, \ldots, \rho_k)\colon M \to \mathbb{R}^{k(n+1)}$$

is an embedding. It is injective: if $F(p) = F(q)$, pick i with $\rho_i(p) > 0$; then $\rho_i(q) = \rho_i(p) > 0$, so $p, q \in U_i$, and $\rho_i(p)\varphi_i(p) = \rho_i(q)\varphi_i(q)$ forces $\varphi_i(p) = \varphi_i(q)$, hence $p = q$. It is an immersion: where $\rho_i > 0$ the coordinates recover $\varphi_i = (\rho_i\varphi_i)/\rho_i$, so dF_p is injective. An injective immersion of a compact manifold is an embedding (proposition 3.1.17).

Step 2: lowering the dimension. Let $M \subseteq \mathbb{R}^N$ with $N > 2n+1$, and for a unit vector v let π_v be orthogonal projection onto $v^\perp \cong \mathbb{R}^{N-1}$. As $\ker \pi_v = \mathbb{R}v$, the restriction $\pi_v|_M$ is an immersion unless $v \in T_pM$ for some p, and is injective unless $v = \pm(p-q)/|p-q|$ for distinct $p, q \in M$. The tangent directions are the image of the unit tangent bundle $UM = \{(p, w) \mid w \in T_pM,\ |w| = 1\}$, a compact $(2n-1)$-manifold, under $(p, w) \mapsto w$; the secant directions are the image of the $2n$-manifold $(M \times M) \setminus \Delta$ under $(p, q) \mapsto (p-q)/|p-q|$. Both are smooth images of manifolds of dimension $< N-1 = \dim S^{N-1}$, hence have measure zero in S^{N-1} (corollary 7.2.3). A unit vector outside their union makes $\pi_v|_M$ an injective immersion, so an embedding, into $\mathbb{R}^{N-1}$; iterating drops the target to $\mathbb{R}^{2n+1}$. Ignoring injectivity, a v avoiding only the $(2n-1)$-dimensional tangent directions exists whenever $2n-1 < N-1$, so the projection continues to $\mathbb{R}^{2n}$ and gives an immersion there. On a compact manifold every embedding is proper.

The noncompact case. For general second-countable M a smooth proper function exists: for a partition of unity $\{\rho_i\}$ subordinate to a locally finite cover by precompact charts, $f = \sum_i i\,\rho_i$ satisfies $f(x) > m$ whenever $x \notin \bigcup_{i \le m} U_i$, so each sublevel set $\{f \le m\}$ is compact. Because an n-manifold has covering dimension n, its open covers refine into $n+1$ locally finite families with pairwise-disjoint members ([5]); running Step 1 on each family yields a smooth injective immersion $G\colon M \to \mathbb{R}^L$, so (G, f) is a proper injective immersion, hence a closed embedding $M \hookrightarrow \mathbb{R}^{L+1}$. Replacing f by a generic distance-squared function $x \mapsto |x-a|^2$ on the image (proper, with compact sublevel sets, and Morse) keeps (G, f) a proper closed embedding, so we may take the last coordinate f to be Morse, with $\operatorname{Crit} f$ discrete. Projecting $\mathbb{R}^{L+1}$ along directions orthogonal to the f-axis retains f, hence properness, and can lose injectivity only through a secant with $f(p) = f(q)$. Away from the discrete set $\operatorname{Crit} f \times \operatorname{Crit} f$ the locus $\{(p, q) \mid f(p) = f(q),\ p \ne q\}$ is a $(2n-1)$-manifold, since $(p, q) \mapsto f(p) - f(q)$ is a submersion there; its secant image, the countably many critical-to-critical secants, and the $(2n-1)$-dimensional tangent directions together have measure zero in each sphere. The projection therefore reduces to a proper embedding in $\mathbb{R}^{2n+1}$, and, in one further unrestricted projection, to an immersion in $\mathbb{R}^{2n}$, completing the proof.

Corollary 7.6.2: Manifolds Are Euclidean Submanifolds

Every smooth n-manifold is diffeomorphic to a closed embedded submanifold of $\mathbb{R}^{2n+1}$.

***Proof*:**
The embedding of theorem 7.6.1 is proper, so its image is closed (a proper map into a locally compact Hausdorff space is closed) and it is a diffeomorphism onto that image.

Whitney later sharpened the bounds by one, to an immersion in $\mathbb{R}^{2n-1}$ and an embedding in $\mathbb{R}^{2n}$, through the far subtler *Whitney trick* for cancelling double points; the elementary projection above gives the $2n$ and $2n+1$ bounds, which suffice throughout this book.

7.7 The Normal Bundle

To record how a submanifold sits inside the ambient manifold, one keeps at each point the directions transverse to it. Assembled over the submanifold these form the normal bundle, the object the tubular neighbourhood theorem identifies with an actual neighbourhood.

Definition 7.7.1: Normal Bundle

Let $S \subseteq M$ be an embedded submanifold. The *normal space* to S at p is the quotient $N_pS = T_pM/T_pS$, and the *normal bundle* is $NS = \bigsqcup_{p\in S} N_pS$ with the projection $\nu\colon NS \to S$ sending N_pS to p.

The quotient needs no metric; a Riemannian metric, when present, identifies N_pS with the orthogonal complement $(T_pS)^\perp \subseteq T_pM$.

Proposition 7.7.2: The Normal Bundle Is a Vector Bundle

NS is a smooth manifold of dimension $\dim M$, and $\nu\colon NS \to S$ is a smooth vector bundle of rank $\mathrm{codim} S$: every $p \in S$ has a neighbourhood V in S with $\nu^{-1}(V) \cong V \times \mathbb{R}^{\mathrm{codim} S}$ linearly on fibers. The zero section $p \mapsto 0 \in N_pS$ is an embedding of S.

***Proof*:**
Around p take a slice chart (W, ψ), $\psi(W \cap S) = \psi(W) \cap (\mathbb{R}^s \times 0)$ with $s = \dim S$. For each $q \in W \cap S$ the coordinate vectors $\partial_{s+1}, \dots, \partial_n$ at q project to a basis of $N_qS = T_qM/T_qS$, giving a bijection

$$\Phi\colon \nu^{-1}(W \cap S) \to (W \cap S) \times \mathbb{R}^{\mathrm{codim} S}, \qquad (q, [w]) \mapsto (q, w^{s+1}, \dots, w^n),$$

linear on each fiber. On the overlap of two slice charts the transition is smooth, and the change in the last-coordinate identification is the induced isomorphism of quotients T_qM/T_qS, whose matrix is a smooth invertible function of q; so the Φ form a smooth vector-bundle atlas. The total dimension is $s + \mathrm{codim} S = \dim M$, and the zero section is S. This is the first instance of the general vector bundles of chapter 11.

7.8 Tubular Neighbourhoods

An embedded submanifold sits inside the ambient manifold no more tightly than its normal directions allow: a neighbourhood of it looks like a neighbourhood of the zero section of its normal bundle, and collapses smoothly back onto it. This retraction is the tool the transversality homotopy theorem (theorem 7.5.7) uses to perturb a map into general position, and it underlies the Thom class and Poincaré duality later.

Theorem 7.8.1: Tubular Neighbourhood Theorem

Let $S \subseteq M$ be an embedded submanifold. Then S admits a *tubular neighbourhood*: an open set $U \subseteq M$ containing S, diffeomorphic to the total space of the normal bundle of S by a diffeomorphism fixing S pointwise, together with a smooth retraction $\pi\colon U \to S$ (a submersion, with $\pi|_S = \mathrm{id}_S$; under the diffeomorphism it is the normal-bundle projection). In particular an embedded submanifold $M \subseteq \mathbb{R}^K$ has an open neighbourhood $T \subseteq \mathbb{R}^K$ carrying a smooth submersive retraction $\pi\colon T \to M$.

***Proof*:**

The Euclidean case $S \subseteq \mathbb{R}^K$. Identify N_pS with the orthogonal complement $(T_pS)^\perp \subseteq \mathbb{R}^K$ and form

$$E\colon NS \to \mathbb{R}^K, \qquad E(p,v) = p + v.$$

At a zero-section point $(p,0)$ the derivative is the identity on $T_pS \oplus N_pS = \mathbb{R}^K$ (tangent directions include, normal directions map by $v \mapsto v$), hence an isomorphism; by the inverse function theorem E is a local diffeomorphism near the zero section. For S compact it is injective on $\{(p,v) \mid |v| < \varepsilon\}$ for small ε: otherwise there would be $(p_i, v_i) \neq (p_i', v_i')$ with $E(p_i, v_i) = E(p_i', v_i')$ and $|v_i|, |v_i'| \to 0$, and a convergent subsequence would produce a point of the zero section at which E is not locally injective, against the previous line. Then $T = E(\{(p,v) \mid |v| < \varepsilon\})$ is open, E maps the ε-disc bundle diffeomorphically onto T, and $\pi = \nu \circ E^{-1}\colon T \to S$ is the nearest-point retraction, a submersion because ν is. For noncompact S a positive smooth function $\varepsilon(p)$ replaces the constant, obtained from a partition of unity.

The general case $S \subseteq M$. Embed M in some $\mathbb{R}^K$ (theorem 7.6.1) and let $r_M\colon T_M \to M$ be the Euclidean tubular retraction of M from the case just proved. Realise the normal bundle of S in M inside $\mathbb{R}^K$ as $N_pS = (T_pS)^\perp \cap T_pM$, and set

$$\Psi\colon NS \to M, \qquad \Psi(p,v) = r_M(p+v),$$

defined for $|v|$ small enough that $p + v \in T_M$. Then $\Psi(p,0) = r_M(p) = p$, and for $w \in T_pS$ and $u \in N_pS$,

$$D\Psi_{(p,0)}(w,u) = D(r_M)_p(w+u) = w + u,$$

since $w + u \in T_pM$ and $D(r_M)_p$ is the orthogonal projection onto T_pM; thus $D\Psi_{(p,0)} = \mathrm{id}_{T_pM}$. As in the Euclidean case the inverse function theorem and a compactness (or smooth $\varepsilon(p)$) argument make Ψ a diffeomorphism from an ε-disc bundle in NS onto an open $U \subseteq M$ containing S, fixing S; the retraction is $\pi = \nu \circ \Psi^{-1}$, a submersion, completing the proof.

7.9 Whitney's Approximation Theorem

Sard and the tubular neighbourhood together smooth continuous maps. The mechanism is first visible for maps into $\mathbb{R}^k$, where a partition of unity averages local constant approximations; a general target is reduced to this by embedding it and retracting.

Theorem 7.9.1: Whitney Approximation

Let M, N be smooth manifolds and $f\colon M \to N$ continuous. Then f is homotopic to a smooth map, and the homotopy may be taken relative to any closed set $A \subseteq M$ on a neighbourhood of which f is already smooth. When $N = \mathbb{R}^k$, the smooth map can be taken within any prescribed positive continuous function of f in norm.

***Proof*:**

Target $\mathbb{R}^k$. Let $\varepsilon\colon M \to (0,\infty)$ be continuous. Each p has a neighbourhood U_p with $|f(x)-f(p)| < \varepsilon(x)$ for $x \in U_p$ (continuity of f and ε). Take a partition of unity $\{\rho_i\}$ subordinate to a locally finite refinement $\{U_{p_i}\}$ (theorem 1.5.5) and set $g = \sum_i \rho_i\, f(p_i)$, a smooth map. Then

$$|g(x) - f(x)| = \Big|\sum_i \rho_i(x)\big(f(p_i) - f(x)\big)\Big| \le \sum_i \rho_i(x)\,|f(p_i) - f(x)| < \varepsilon(x),$$

the last step because $\rho_i(x) \neq 0$ only for $x \in U_{p_i}$. The straight-line homotopy $(1-t)f + tg$ stays within ε of f, so $f \simeq g$. If f is smooth on a neighbourhood W of the closed set A, choose a smooth χ that is 1 near A and supported in W, and replace g by $\chi f + (1-\chi)g$: still smooth and ε-close, and equal to f near A, so the homotopy is relative to A.

General target. Embed N in some $\mathbb{R}^K$ (theorem 7.6.1) with tubular neighbourhood T and retraction $r\colon T \to N$ (theorem 7.8.1). Pick a continuous $\delta\colon N \to (0,\infty)$ so that the δ-ball around each $y \in N$ lies in T. Viewing f as a map into $\mathbb{R}^K$, the first part gives a smooth $g_0\colon M \to \mathbb{R}^K$ with $|g_0(x) - f(x)| < \delta(f(x))$, so $g_0(x) \in T$; set $g = r \circ g_0$, a smooth map into N. The segment from $f(x)$ to $g_0(x)$ lies in T, so $t \mapsto r\big((1-t)f(x) + tg_0(x)\big)$ is a homotopy in N from $r(f(x)) = f(x)$ to $g(x)$. Taking $g_0 = f$ near A as above makes it relative to A, completing the proof.

Exercise 7.9.1

1. No compact n-manifold embeds in $\mathbb{R}^n$.
2. Compute the normal bundle of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$, and show it is trivial.
3. Deduce from theorem 7.9.1 that two smooth maps $f_0, f_1\colon M \to N$ that are continuously homotopic are *smoothly* homotopic.

8

Intersection and Degree Theory

Transversality (section 7.4) turns intersection into a numerical invariant. When a map $f\colon M \to N$ is transverse to a closed submanifold $Z \subseteq N$ of complementary dimension, $f^{-1}(Z)$ is a finite set of points; counting them, with signs once everything is oriented, gives an integer unchanged under homotopy. This *intersection number* is the common source of the *degree* of a map, the Euler characteristic, and the Lefschetz fixed-point count. Its invariance is the parity of section 5.11: the parameter space of a homotopy is one dimension up, and its boundary carries the two ends of the homotopy, where the count must balance. From the signed count the chapter builds the degree, then the Lefschetz number and the index of a vector field, closing with the Poincaré-Hopf theorem, which equates the total index of a vector field with the Euler characteristic $\chi(M) = I(\Delta, \Delta)$.

8.1 Mod-2 Intersection Theory

The transversality theory of chapter 7 makes the preimage $f^{-1}(Z)$ of a submanifold a manifold of predictable dimension whenever $f \pitchfork Z$. This section extracts the first topological invariant from that fact. When M is compact and Z is closed of complementary dimension, $f^{-1}(Z)$ is a finite set, and the parity of its size is unchanged as f varies through a homotopy. The mechanism is the Boundary Theorem: a homotopy fills in a compact one-dimensional cobordism between the two preimages, and a compact 1-manifold has an even number of boundary points (corollary 5.11.2). Counting preimages modulo 2 therefore descends to homotopy classes, and the resulting invariants are the mod-2 intersection number and, when Z is a single point, the mod-2 degree.

Throughout, manifolds are boundaryless unless a boundary is named, and a map $f\colon M \to N$ and a closed submanifold $Z \subseteq N$ are *complementary* when $\dim M + \dim Z = \dim N$; the source M is compact. For such data a transverse f has $f^{-1}(Z)$ a compact 0-manifold, a finite set.

The Boundary Theorem rests on one geometric fact: the preimage of a transverse submanifold under a map on a manifold with boundary is again a manifold with boundary, and its boundary is exactly the part lying over the boundary of the source. The transverse-preimage theorem of chapter 7 handles interior points; boundary points need one added transversality hypothesis, after which they contribute a genuine boundary.

Lemma 8.1.1: Preimage Under a Map on a Manifold with Boundary

Let W be a manifold with boundary, N a boundaryless manifold, and $Z \subseteq N$ a closed embedded submanifold. Suppose $F\colon W \to N$ is transverse to Z and its restriction $\partial F = F|_{\partial W}\colon \partial W \to N$ is also transverse to Z. Then $F^{-1}(Z)$ is an embedded submanifold with boundary of W, with

$$\partial\big(F^{-1}(Z)\big) = F^{-1}(Z) \cap \partial W, \qquad \mathrm{codim} F^{-1}(Z) = \mathrm{codim} Z$$

the codimensions being taken in W and N respectively.

***Proof*:**
Write $k = \mathrm{codim} Z$ and $d = \dim W$. The statement is local, so fix $a \in F^{-1}(Z)$ and, using that Z is embedded, a neighbourhood U of $F(a)$ together with a submersion $\varphi\colon U \to \mathbb{R}^k$ with $Z \cap U = \varphi^{-1}(0)$. On $F^{-1}(U)$ the preimage is the zero set of $g = \varphi \circ F$, and transversality of F to Z at a point p is exactly surjectivity of $D_p g$, that is 0 being a regular value of g (the computation is that of proposition 7.4.3).

Step 1: Interior points. If $a \in \mathrm{int}\, W$, then near a the map F is defined on a boundaryless manifold and $F \pitchfork Z$, so proposition 7.4.3 makes $F^{-1}(Z)$ an embedded submanifold of codimension k near a, with no boundary contributed.

Step 2: Boundary points. Let $a \in F^{-1}(Z) \cap \partial W$. Choose a boundary chart carrying a neighbourhood of a to the half-space $H^d = \big\{x \in \mathbb{R}^d \;\big|\; x_d \geq 0\big\}$ with $a \mapsto 0$ and $\partial W \mapsto \{x \mid x_d = 0\}$. In this chart F is, by the definition of smoothness on a manifold with boundary, the restriction of a smooth map defined on an open neighbourhood of 0 in $\mathbb{R}^d$; hence $g = \varphi \circ F$ extends to a smooth $\widetilde{g}\colon V \to \mathbb{R}^k$ on an open $V \ni 0$ in $\mathbb{R}^d$. Transversality of F to Z at a makes $D_0\widetilde{g}$ surjective, so 0 is a regular value of $\widetilde{g}$ and $C = \widetilde{g}^{-1}(0)$ is, near 0, an embedded submanifold of dimension $d - k$ (proposition 7.4.3).

Transversality of ∂F to Z at a says that the restriction of $D_0\widetilde{g}$ to the hyperplane $H_0 = \{x \mid x_d = 0\} = T_0\partial W$ is surjective onto $\mathbb{R}^k$. Since $T_0 C = \ker D_0\widetilde{g}$, surjectivity of $D_0\widetilde{g}|_{H_0}$ forces $T_0 C \cap H_0$ to have dimension $\dim T_0 C - 1$, that is $T_0 C \not\subseteq H_0$; concretely the coordinate x_d has nonzero differential along C at 0. Thus $x_d|_C$ is a submersion near 0, and

$$F^{-1}(Z) \cap H^d = C \cap \{x \mid x_d \geq 0\} = (x_d|_C)^{-1}\big([0,\infty)\big)$$

near a is the preimage of a half-line under a submersion from a $(d-k)$-manifold, hence an embedded $(d-k)$-manifold with boundary whose boundary is $(x_d|_C)^{-1}(0) = C \cap \partial W = F^{-1}(Z) \cap \partial W$.

Steps 1 and 2 exhibit $F^{-1}(Z)$ locally as a submanifold with boundary of the stated codimension, with boundary points precisely those of $F^{-1}(Z) \cap \partial W$, completing the proof.

The codimension-complementary case specializes this to what the next result needs: when Z is complementary to ∂W, the preimage is one dimensional, and a compact 1-manifold has a controlled boundary.

Theorem 8.1.2: The Boundary Theorem

Let W be a compact manifold with boundary and N a boundaryless manifold, and let $Z \subseteq N$ be a closed submanifold with $\dim \partial W + \dim Z = \dim N$. If $F\colon W \to N$ is transverse to Z and its boundary restriction ∂F is transverse to Z, then $F^{-1}(Z) \cap \partial W$ is a finite set of even cardinality.

***Proof*:**
By lemma 8.1.1, $F^{-1}(Z)$ is a submanifold with boundary of W of dimension

$$\dim F^{-1}(Z) = \dim W - \operatorname{codim} Z = \dim W - (\dim N - \dim Z) = 1$$

where the last equality uses $\dim \partial W = \dim W - 1$ and $\dim \partial W + \dim Z = \dim N$. Its boundary is $F^{-1}(Z) \cap \partial W$. As Z is closed and F continuous, $F^{-1}(Z)$ is closed in the compact W, hence compact. A compact 1-manifold with boundary has an even number of boundary points (corollary 5.11.2), and these are exactly the points of $F^{-1}(Z) \cap \partial W$, so that set is finite of even cardinality, as we sought to show.

The Boundary Theorem is the source of the homotopy invariance below. Read it as a conservation law: the points of $F^{-1}(Z)$ lying over the boundary of W are the endpoints of a compact system of arcs and circles inside W, and endpoints of arcs come in pairs. Whatever a homotopy does in the interior, the count it leaves on the two ends must balance modulo 2. Both transversality hypotheses are used: $F \pitchfork Z$ makes the interior a 1-manifold, and $\partial F \pitchfork Z$ makes the ends genuine boundary points rather than tangencies.

Fix now the setting of a compact source. For $f \pitchfork Z$ with complementary dimensions the preimage $f^{-1}(Z)$ is a compact 0-manifold (proposition 7.4.3), a finite set, and its parity is the invariant to isolate.

Definition 8.1.3: Mod-2 Intersection Number of a Transverse Map

Let M be compact, N boundaryless, and $Z \subseteq N$ a closed submanifold with $\dim M + \dim Z = \dim N$. For a smooth map $f\colon M \to N$ transverse to Z, the *mod-2 intersection number* is

$$I_2(f, Z) = \#\, f^{-1}(Z) \bmod 2 \;\in\; \mathbb{Z}/2$$

the number of preimage points counted modulo 2.

The definition asks only for a count, so it is completely explicit once a transverse map is in hand. The content, deferred to the next theorem, is that the count does not depend on which transverse map in a homotopy class is used.

Example 8.1.4: A Curve Meeting a Curve on the Torus

Let $N = T^2 = S^1 \times S^1$, let $M = S^1$, and let $f\colon S^1 \to T^2$, $f(z) = (z, 1)$, parametrize the horizontal circle $S^1 \times \{1\}$. Take $Z = \{1\} \times S^1$, a closed 1-submanifold, so $\dim M + \dim Z = 1 + 1 = 2 = \dim N$. A point $f(z) = (z, 1)$ lies in Z exactly when $z = 1$, so $f^{-1}(Z) = \{1\}$ is a single point.

At $z = 1$ the image $Df(T_1 S^1)$ is the tangent line to $S^1 \times \{1\}$, while $T_{(1,1)}Z$ is the tangent line to $\{1\} \times S^1$; these are the two coordinate directions of $T_{(1,1)}T^2$ and together span it, so $f \pitchfork Z$. Hence $I_2(f, Z) = 1$. The two circles cross once, and modulo 2 they cannot be separated: no matter how f is deformed, an odd number of crossings with Z persists.

Homotopy invariance is where the count becomes a topological invariant. A homotopy of maps $M \to N$ is a map on the cylinder $W = M \times [0, 1]$, whose boundary is the two ends; if the homotopy is transverse to Z, the Boundary Theorem forces the two end-counts to agree modulo 2. The only technical point is that a homotopy may be perturbed to be transverse to Z without disturbing its ends, which are already transverse.

Lemma 8.1.5: Boundary-Fixing Transversality

Let W be a compact manifold with boundary, N boundaryless, and $Z \subseteq N$ a closed submanifold, and let $F\colon W \to N$ be smooth with $\partial F = F|_{\partial W}$ transverse to Z. Then F is homotopic, through maps agreeing with F on ∂W, to a smooth map F' transverse to Z; the boundary restriction $\partial F' = \partial F$ is then still transverse to Z.

***Proof*:**

Embed N in some $\mathbb{R}^K$ (theorem 7.6.1) and let $\pi\colon T \to N$ be the retraction of a tubular neighbourhood $T \subseteq \mathbb{R}^K$ (theorem 7.8.1). Choose a smooth function $\varepsilon\colon W \to [0, \infty)$ vanishing exactly on ∂W and positive on the interior: patch the boundary-chart functions $x_d \geq 0$ (the last half-space coordinate, which vanishes on ∂W) and the constant 1 on interior charts with a subordinate partition of unity (chapter 1); the result is nonnegative, vanishes on ∂W, and is positive at every interior point, where some chart contributes a positive value. Shrinking ε by a small constant (using compactness of W), arrange $F(x) + \varepsilon(x)\, b \in T$ for all $x \in W$ and all b in the open unit ball $B \subseteq \mathbb{R}^K$, and set

$$F_b(x) = \pi\big(F(x) + \varepsilon(x)\, b\big)$$

a smooth family with $F_0 = F$.

Step 1: The family is a submersion over the interior. For $x \in \operatorname{int} W$ one has $\varepsilon(x) > 0$, so $b \mapsto F(x) + \varepsilon(x)b$ is a submersion onto a neighbourhood of $F(x)$, and π is a submersion; hence the map $(x, b) \mapsto F_b(x)$ on $\operatorname{int} W \times B$ is a submersion, in particular transverse to Z.

Step 2: A generic parameter works. By parametric transversality (theorem 7.5.6) there is b, arbitrarily close to 0, with $F_b|_{\operatorname{int} W}$ transverse to Z. On ∂W, where $\varepsilon = 0$, one has $F_b(x) = \pi(F(x)) = F(x)$ since $F(x) \in N$ and π fixes N; thus F_b agrees with F on ∂W. At a boundary point a, transversality of F_b to Z is automatic once $\partial F_b = \partial F$ is transverse there, because $\operatorname{im} D_a(\partial F_b) \subseteq \operatorname{im} D_a F_b$ enlarges the image only further.

Set $F' = F_b$. Then $F' \pitchfork Z$ on all of W: on the interior by Step 2, and at each boundary point because $\partial F' = \partial F$ is transverse to Z by hypothesis, which as noted forces transversality of F' there. The homotopy $s \mapsto F_{sb}$ is stationary on ∂W, so F' is homotopic to F rel ∂W with $\partial F' = \partial F$, completing the proof.

With a homotopy available transverse to Z and fixed on its already-transverse ends, the invariance of the count is now a direct reading of the Boundary Theorem.

Theorem 8.1.6: Homotopy Invariance of the Mod-2 Intersection Number

Let M be compact, N boundaryless, and $Z \subseteq N$ a closed submanifold with $\dim M + \dim Z = \dim N$. If $f_0, f_1 \colon M \to N$ are homotopic and each transverse to Z, then $I_2(f_0, Z) = I_2(f_1, Z)$.

***Proof*:**
Choose a homotopy $F \colon W \to N$ on the cylinder $W = M \times [0,1]$, with $F(x,0) = f_0(x)$ and $F(x,1) = f_1(x)$. Then W is compact with $\partial W = M \times \{0\} \sqcup M \times \{1\}$, and its boundary restriction is $\partial F = f_0 \sqcup f_1$, transverse to Z by hypothesis. Also $\dim \partial W + \dim Z = \dim M + \dim Z = \dim N$, so Z is complementary to ∂W.

By lemma 8.1.5 the homotopy may be taken transverse to Z without altering its already-transverse boundary values: replace F by a map, homotopic rel ∂W, with $F \pitchfork Z$ and $\partial F = f_0 \sqcup f_1$ still transverse to Z. The Boundary Theorem (theorem 8.1.2) applies, so $F^{-1}(Z) \cap \partial W$ has even cardinality. This set is

$$\left(f_0^{-1}(Z) \times \{0\}\right) \sqcup \left(f_1^{-1}(Z) \times \{1\}\right)$$

of size $\# f_0^{-1}(Z) + \# f_1^{-1}(Z)$. Evenness of the sum forces $\# f_0^{-1}(Z) \equiv \# f_1^{-1}(Z) \pmod 2$, that is $I_2(f_0, Z) = I_2(f_1, Z)$, as we sought to show.

The count is a homotopy invariant, so it measures the map's homotopy class, not the map. This is the promised conservation law made precise: crossings with Z can be created or destroyed only in pairs, because each pair is born or dies at the two ends of an arc in the cylinder. The invariant is coarse, valued in $\mathbb{Z}/2$, and the price of that coarseness is that no orientation is needed anywhere; the finer $\mathbb{Z}$-valued theory, which does need orientations, is taken up in section 8.3.

Homotopy invariance also removes the transversality restriction from the definition. An arbitrary continuous map is not transverse to anything in particular, but it is homotopic to one that is, and the invariant of the transverse representative can be transferred back.

Every continuous map $f \colon M \to N$ is homotopic to a smooth map (smooth approximation, chapter 7), and every smooth map is homotopic to one transverse to Z (theorem 7.5.7). So f is homotopic to a smooth map transverse to Z, and the following definition and proposition make the transfer legitimate.

Definition 8.1.7: Mod-2 Intersection Number of a Continuous Map

Let M be compact, N boundaryless, and $Z \subseteq N$ closed with $\dim M + \dim Z = \dim N$. For any continuous $f \colon M \to N$, choose a smooth map g homotopic to f and transverse to Z, and set

$$I_2(f, Z) = I_2(g, Z) \ \in \ \mathbb{Z}/2$$

Proposition 8.1.8: Well-Definedness of the Extended Intersection Number

The value $I_2(f, Z)$ of definition 8.1.7 is independent of the chosen transverse representative g.

***Proof*:**
Let g and g' both be smooth, homotopic to f, and transverse to Z. Then g is homotopic to g' (through f), and both are transverse to Z, so theorem 8.1.6 gives $I_2(g, Z) = I_2(g', Z)$. The common value is thus independent of the representative, completing the proof.

By construction $I_2(f, Z)$ depends only on the homotopy class of f, and for a smooth transverse f it recovers $\#f^{-1}(Z) \bmod 2$; homotopy invariance now holds for all continuous maps at once. A continuous but nowhere-differentiable loop $f\colon S^1 \to T^2$, for instance, still has a well-defined $I_2(f, Z)$ with the meridian Z of example 8.1.4, computed from any smooth approximation.

When both objects are submanifolds of N, the intersection number becomes a symmetric pairing. For a compact submanifold $X \subseteq N$ the inclusion $\iota_X\colon X \hookrightarrow N$ is a map from a compact manifold, and applying the intersection number to it counts how X meets Z.

Definition 8.1.9: Mod-2 Intersection Number of Submanifolds

Let $X, Z \subseteq N$ be submanifolds of a boundaryless manifold N with X compact, Z closed, and $\dim X + \dim Z = \dim N$. The *mod-2 intersection number* of X and Z is

$$I_2(X, Z) = I_2(\iota_X, Z) \ \in \ \mathbb{Z}/2$$

where $\iota_X\colon X \hookrightarrow N$ is the inclusion. When $X \pitchfork Z$ (transverse as submanifolds), $I_2(X, Z) = \#(X \cap Z) \bmod 2$.

Proposition 8.1.10: Symmetry of the Mod-2 Intersection Number

Let $X, Z \subseteq N$ be compact submanifolds of complementary dimension. Then $I_2(X, Z) = I_2(Z, X)$.

***Proof*:**
Two submanifolds are transverse as submanifolds precisely when the inclusion of either is transverse to the other, and this condition is symmetric in X and Z. When $X \pitchfork Z$ the intersection $X \cap Z$ is a finite set, and

$$I_2(X, Z) = \#(X \cap Z) \bmod 2 = \#(Z \cap X) \bmod 2 = I_2(Z, X)$$

the count being literally the same set of points read from either side.

In general neither inclusion need be transverse. The mod-2 intersection number is unchanged under an isotopy of *either* factor: for the map being intersected this is theorem 8.1.6, and for the target submanifold the same Boundary-Theorem argument applies to the trace of the isotopy in $N \times [0, 1]$. Now perturb ι_X to an embedding onto a copy X' of X transverse to Z (theorem 7.5.7); moving X to X' in whichever slot it occupies,

$$I_2(X, Z) = I_2(X', Z) = \#(X' \cap Z) \bmod 2 = \#(Z \cap X') \bmod 2 = I_2(Z, X') = I_2(Z, X),$$

the inner equalities because $X' \pitchfork Z$ is a transverse pair counted the same from either side, completing the proof.

Symmetry says the pairing does not remember which factor was perturbed. It is the mod-2 shadow of the fact that an intersection point belongs to both submanifolds at once, and it has immediate geometric bite.

Example 8.1.11: Homology of Curves on the Torus

On $T^2 = S^1 \times S^1$ take the two coordinate circles $A = S^1 \times \{1\}$ and $B = \{1\} \times S^1$, each a compact 1-submanifold complementary to the other.

1. A and B meet transversally in the single point $(1, 1)$, exactly as in example 8.1.4, so $I_2(A, B) = 1$. Symmetry gives $I_2(B, A) = 1$ as well.
2. Consequently neither circle bounds. If $A = \partial W$ for a compact surface $W \subseteq T^2$ with boundary, the inclusion of A would extend over W, forcing $I_2(A, B) = 0$ (the extension argument is theorem 8.1.2); but it is 1. For the same reason A and B cannot be pulled apart: any homotopy leaves an odd number of crossings.
3. The self-pairing $I_2(A, A)$ is 0. The circle $A = S^1 \times \{1\}$ is disjoint from its pushoff $A' = S^1 \times \{-1\}$, and ι_A is homotopic to the inclusion of A', which is transverse to A with empty intersection; thus $I_2(A, A) = I_2(A', A) = 0$.

The two circles are the generators of the first homology of the torus, and the pairing I_2 reproduces the mod-2 intersection form: A and B pair to 1, each pairs with itself to 0.

The last invariant of the section is the mod-2 degree, the intersection number against a single point. Its definition presupposes that the count $I_2(f, \{y\})$ does not depend on the point y. When N is connected this is true, and the reason is that any point of N can be slid to any other by a diffeomorphism isotopic to the identity, along which the count cannot change.

Lemma 8.1.12: Homogeneity of a Connected Manifold

Let N be a connected boundaryless manifold. For any two points $y_0, y_1 \in N$ there is a diffeomorphism $h \colon N \to N$, smoothly isotopic to the identity, with $h(y_0) = y_1$.

***Proof*:**

Step 1: A local push in Euclidean space. Let $a, c \in \mathbb{R}^n$ with $|c| < 1$, and let $b = a + c$. Choose a bump function $\rho \colon \mathbb{R}^n \to [0, 1]$ (chapter 1) equal to 1 on the closed ball $\overline{B}(a, |c|)$ and supported in $B(a, 1)$, and set $X = \rho\, c$, a vector field with the constant direction c. Since X vanishes outside the compact ball $\overline{B}(a, 1)$ it is complete (chapter 5), so its flow $\{\varphi_t\}_{t \in \mathbb{R}}$ consists of diffeomorphisms of $\mathbb{R}^n$, each equal to the identity outside $B(a, 1)$, with $\varphi_0 = \mathrm{id}$. For $t \in [0, 1]$ the point $a + tc$ lies in $\overline{B}(a, |c|)$, where $\rho \equiv 1$, so the trajectory through a solves $\dot{x} = c$ there: $\varphi_t(a) = a + tc$, and $\varphi_1(a) = a + c = b$. Thus $\{\varphi_t\}_{t \in [0,1]}$ is an isotopy from id to a diffeomorphism carrying a to b.

Step 2: An equivalence relation. Declare $y \sim y'$ when some diffeomorphism of N isotopic to the identity carries y to y'. This is reflexive (the identity), symmetric (an inverse isotopy), and transitive (concatenate isotopies), so it is an equivalence relation on N.

Step 3: The classes are open. Fix $y \in N$ and a chart carrying a neighbourhood of y diffeomorphically onto $\mathbb{R}^n$. Given y' near y, the chart sends y, y' to points a, b with $|b - a| < 1$, and Step 1 produces an isotopy of $\mathbb{R}^n$ supported in a ball inside the chart image, carrying a to b. Because it is the identity outside a compact subset of the chart image, it extends by the identity to an isotopy of N carrying y to y'. Hence every point near y is equivalent to y, and each class is open.

Step 4: Connectedness. The classes partition N into open sets, and N is connected, so there is a single class. In particular $y_0 \sim y_1$, which is the assertion, completing the proof.

Proposition 8.1.13: The Point-Count Is Independent of the Point

Let M be compact, N connected, both boundaryless, with $\dim M = \dim N$, and let $f\colon M \to N$ be continuous. Then $I_2(f, \{y\})$ is independent of $y \in N$. For a smooth f and a regular value y, it equals $\#f^{-1}(y) \bmod 2$.

***Proof*:**
A point $\{y\} \subseteq N$ is a closed submanifold of dimension 0, complementary to M since $\dim M + 0 = \dim N$, so $I_2(f, \{y\})$ is defined. It suffices to treat smooth f, replacing a continuous map by a smooth homotopic one without changing any value (proposition 8.1.8). For smooth f a point y is a regular value exactly when $f \pitchfork \{y\}$, and then $I_2(f, \{y\}) = \#f^{-1}(y) \bmod 2$.

Fix a regular value y_0 of f, which exists by Sard's theorem (theorem 7.2.1), and let $y_1 \in N$ be arbitrary. By lemma 8.4.3 there is a diffeomorphism $h\colon N \to N$, isotopic to the identity, with $h(y_0) = y_1$; write h_t for the isotopy, $h_0 = \mathrm{id}$, $h_1 = h$. Then $h_t \circ f$ is a homotopy from f to $h \circ f$, so homotopy invariance gives

$$I_2(f, \{y_1\}) = I_2(h \circ f, \{y_1\})$$

Since h is a diffeomorphism with $h(y_0) = y_1$, the map $h \circ f$ is transverse to $\{y_1\}$ exactly where f is transverse to $\{y_0\}$, and $(h \circ f)^{-1}(y_1) = f^{-1}(y_0)$. As y_0 is a regular value of f, both sides are transverse counts and

$$I_2(h \circ f, \{y_1\}) = \#(h \circ f)^{-1}(y_1) \bmod 2 = \#f^{-1}(y_0) \bmod 2$$

Combining, $I_2(f, \{y_1\}) = \#f^{-1}(y_0) \bmod 2$ for every $y_1 \in N$. The right side does not involve y_1, so $I_2(f, \{y\})$ is the same for all y, and it equals $\#f^{-1}(y) \bmod 2$ at any regular value, completing the proof.

Definition 8.1.14: Mod-2 Degree

Let M be compact, N connected, both boundaryless, with $\dim M = \dim N$. The *mod-2 degree* of a continuous map $f\colon M \to N$ is

$$\deg_2 f = I_2(f, \{y\}) \ \in \ \mathbb{Z}/2$$

for any $y \in N$, well defined by proposition 8.1.13. For smooth f and a regular value y, $\deg_2 f = \#f^{-1}(y) \bmod 2$.

The mod-2 degree records a single bit: whether f hits a generic point an even or odd number of times. It is a homotopy invariant, since $I_2(f, \{y\})$ is, so homotopic maps have the same degree, and a map of odd degree is never homotopic to a constant, whose degree is 0.

Example 8.1.15: The Degree of $z \mapsto z^k$ on the Circle

View $S^1 \subseteq \mathbb{C}$ and let $f\colon S^1 \to S^1$, $f(z) = z^k$ for an integer k. For $k \neq 0$ the derivative in the angular coordinate $z = e^{i\theta}$, $f(z) = e^{ik\theta}$, multiplies velocities by $k \neq 0$, so every point of S^1 is a regular value. The fibre over $w = e^{i\alpha}$ is the set of k-th roots $\left\{e^{i(\alpha+2\pi j)/k} \;\middle|\; 0 \leq j < |k|\right\}$, of size $|k|$. Hence

$$\deg_2(z \mapsto z^k) = |k| \bmod 2 = k \bmod 2$$

For $k = 0$ the map is constant; a point y other than its value is a regular value with empty fibre, so $\deg_2 = 0$, again $k \bmod 2$. Thus $z \mapsto z^k$ has mod-2 degree 1 for k odd and 0 for k even. In particular the identity ($k = 1$) has degree 1 and the squaring map ($k = 2$) has degree 0, so squaring is homotopic to a constant through maps $S^1 \to S^1$ only if its degree vanishes, which it does; the identity, of degree 1, is not.

Exercise 8.1.1

1. Compute the mod-2 degree of the antipodal map $a\colon S^n \to S^n$, $a(x) = -x$, and of a constant map $S^n \to S^n$ (for $n \geq 1$). Which pairs of these maps must fail to be homotopic?

2. Let M be compact and N connected, both boundaryless with $\dim M = \dim N$, and let $f\colon M \to N$ be continuous. Show that if f is not surjective then $\deg_2 f = 0$. Deduce that any map homotopic to a constant has mod-2 degree 0.

3. Let W be a compact manifold with boundary $\partial W = X$, let N be boundaryless, and let $Z \subseteq N$ be closed with $\dim X + \dim Z = \dim N$. Suppose $G\colon W \to N$ is smooth and its restriction $g = G|_X$ is transverse to Z. Show that $I_2(g, Z) = 0$.

4. On $T^2 = S^1 \times S^1$ let $A = S^1 \times \{1\}$ and $B = \{1\} \times S^1$. Using $I_2(A, B) = 1$, show that neither A nor B is the boundary of a compact surface-with-boundary in T^2, and that A and B cannot be made disjoint by an isotopy. Contrast the situation of two disjoint embedded circles on S^2.

5. Let $f\colon M \to N$ and $g\colon N \to P$ be continuous maps between compact connected boundaryless manifolds of one common dimension. Prove the multiplicativity $\deg_2(g \circ f) = \deg_2(g) \cdot \deg_2(f)$ in $\mathbb{Z}/2$.

8.2 Winding Numbers and the Jordan-Brouwer Theorem

Mod-2 intersection theory (section 8.1) turns a geometric count into a homotopy invariant. Its first payoff is a number older than topology itself: whether a closed hypersurface encloses a point off it. The mod-2 winding number decides this by counting, modulo two, how a direction map wraps around the point, and the answer proves the Jordan-Brouwer separation theorem in every dimension. The same parity bookkeeping, applied to maps that commute with the antipodal involution, yields the Borsuk-Ulam theorem and, as a corollary, the Ham Sandwich theorem. Every result in this section is the mod-2 degree of a single sphere-valued map, read in a different geometric disguise.

The one tool imported from section 8.1 is the mod-2 degree $\deg_2 g \in \mathbb{Z}/2$ of a map $g\colon M \to S^k$ from a compact k-manifold: it is the number of preimages of a regular value counted modulo two, is independent of that value, and is a homotopy invariant. The one tool imported from earlier chapters is the parity of the boundary of a compact 1-manifold, $\#\partial N \equiv 0 \pmod 2$ (corollary 5.11.2).

Definition 8.2.1: Mod-2 Winding Number

Let M be a compact $(n-1)$-manifold with $n \geq 2$, and let $f\colon M \to \mathbb{R}^n$ be a smooth map whose image avoids a point $z \in \mathbb{R}^n$. The *direction map* of f about z is

$$u_{f,z}\colon M \to S^{n-1}, \qquad u_{f,z}(x) = \frac{f(x) - z}{|f(x) - z|}$$

The *mod-2 winding number* of f about z is the mod-2 degree of the direction map,

$$W_2(f, z) := \deg_2\left(u_{f,z}\right) \in \mathbb{Z}/2$$

When $X \subseteq \mathbb{R}^n$ is a compact hypersurface and $z \notin X$, its winding number about z is $W_2(X, z) := W_2(\iota_X, z)$, the winding number of the inclusion $\iota_X\colon X \hookrightarrow \mathbb{R}^n$; the direction map is written $u_z(x) = (x - z)/|x - z|$.

The direction map records where each point of M sits as seen from z, and its mod-2 degree records how many times, modulo two, this view sweeps around the whole sphere of directions. For M a hypersurface enclosing z, the view covers every direction an odd number of times; for z far outside, the surface subtends only part of the sky and no direction is covered an odd number of times. The circle makes both cases explicit.

Example 8.2.2: Winding of the Circle

Take $n = 2$, $M = S^1$, and $X = S^1 \subseteq \mathbb{R}^2$ the unit circle, so $u_z\colon S^1 \to S^1$.

For $z = 0$ the direction map is $u_0(x) = x$, the identity of S^1, whose only regular value v has the single preimage v; hence $\deg_2 u_0 = 1$ and $W_2(X, 0) = 1$. The origin is enclosed.

For z with $|z| > 1$, the circle lies to one side of z: every point of X is within angular distance $\arcsin(1/|z|) < \pi/2$ of the direction $-z/|z|$ pointing from z toward the origin, so u_z misses the open half of S^1 centred on $z/|z|$. A missed value is a regular value with empty preimage, so $\deg_2 u_z = 0$ and $W_2(X, z) = 0$. Points outside the circle are not enclosed, and the winding number sees the difference.

The dividing line between the two computations is that the outside point is joined to infinity by a path missing X, while the inside point is not. The next lemma turns this into the mechanism behind every winding-number vanishing: a map that extends across a filling of its domain winds trivially.

Lemma 8.2.3: Winding Numbers and Bounding

Let W be a compact manifold with boundary $\partial W = M$, and let $F\colon W \to \mathbb{R}^n$ be smooth with $z \notin F(W)$. Then $W_2(F|_M, z) = 0$.

***Proof*:**
Because $z \notin F(W)$, the direction map extends over W: the smooth map

$$U\colon W \to S^{n-1}, \qquad U(y) = \frac{F(y) - z}{|F(y) - z|}$$

restricts on $\partial W = M$ to $u := u_{F|_M, z}$. By theorem 7.2.1 choose a point $p \in S^{n-1}$ that is a regular value of both U and $u = U|_{\partial W}$. Then $U^{-1}(p)$ is a compact 1-manifold with boundary

$$\partial\big(U^{-1}(p)\big) = U^{-1}(p) \cap \partial W = u^{-1}(p)$$

so $\#u^{-1}(p)$ is even (corollary 5.11.2). Hence $\deg_2 u = 0$, that is $W_2(F|_M, z) = 0$, as we sought to show.

This is the winding-number face of the Boundary Theorem of section 8.1: a count taken on the boundary of a compact region vanishes because it balances across the region. The circle example is the case $W = \overline{B^2}$, F the inclusion, and z outside the disc. Read the other way, the lemma is a criterion: if the winding number is nonzero, then f does *not* extend to a z-avoiding map on any filling, and in particular z cannot be pushed to infinity without crossing the image. This is what makes the winding number decide the inside-outside question, once its dependence on z is understood. The dependence is the mildest possible.

Lemma 8.2.4: Winding by Ray Counting

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface, $z \notin X$, and $v \in S^{n-1}$. If the open ray $R_{z,v} = \{z + tv \mid t > 0\}$ is transverse to X, then v is a regular value of u_z and

$$W_2(X, z) \equiv \#\big(X \cap R_{z,v}\big) \pmod 2$$

***Proof*:**
A point $x \in X$ satisfies $u_z(x) = v$ if and only if $x - z = |x - z|\, v$, that is $x = z + tv$ with $t = |x - z| > 0$, so $u_z^{-1}(v) = X \cap R_{z,v}$. The map u_z is the radial projection $y \mapsto (y - z)/|y - z|$ restricted to X; its differential at such an x has kernel the radial line $\mathbb{R}\,(x - z)$ and is an isomorphism on any complement onto $T_v S^{n-1}$. Thus $d(u_z)_x$ is injective on $T_x X$ precisely when $T_x X$ does not contain the radial direction v, that is when $T_x X + \mathbb{R}v = \mathbb{R}^n$, which is transversality of the ray to X at x. So v is a regular value, and $W_2(X, z) = \deg_2 u_z = \#u_z^{-1}(v) \bmod 2 = \#(X \cap R_{z,v}) \bmod 2$ (section 8.1), completing the proof.

The ray count makes the winding number computable and exposes both how it stays constant and how it jumps. Sliding z without crossing X deforms the direction map continuously and cannot change its degree; sliding z across X adds or removes exactly one intersection on a suitable ray.

Proposition 8.2.5: The Winding Number Is Locally Constant

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface. The function $z \mapsto W_2(X, z)$ is locally constant on $\mathbb{R}^n \setminus X$, hence constant on each connected component. It equals 0 for every z outside a ball containing X.

***Proof*:**
If z_0 and z_1 lie in a ball disjoint from X, the segment $z_s = (1-s)z_0 + sz_1$ stays in $\mathbb{R}^n \setminus X$, and $(x, s) \mapsto (x - z_s)/|x - z_s|$ is a homotopy from u_{z_0} to u_{z_1}. Homotopic maps have equal mod-2 degree (section 8.1), so $W_2(X, z_0) = W_2(X, z_1)$; thus $W_2(X, -)$ is locally constant, and constant on components. If $X \subseteq B_r(0)$ and $|z| > r$, choose $v = z/|z|$: the ray $R_{z,v}$ points away from the ball and misses X, so $\#(X \cap R_{z,v}) = 0$ and $W_2(X, z) = 0$ by lemma 8.2.4, completing the proof.

Lemma 8.2.6: Crossing a Hypersurface Flips the Winding Number

Let $X \subseteq \mathbb{R}^n$ be a compact hypersurface and ℓ a line transverse to X. If $z_0, z_1 \in \ell \setminus X$ are chosen so that the segment between them meets X in exactly one point, then

$$W_2(X, z_0) + W_2(X, z_1) = 1 \quad \text{in } \mathbb{Z}/2$$

***Proof*:**
Let v be the unit direction of ℓ pointing from z_1 toward z_0, and set $t_0 = |z_0 - z_1| > 0$, so $z_0 = z_1 + t_0 v$. Because ℓ is transverse to X, so is each ray along it, and v is a regular value of both u_{z_0} and u_{z_1} (lemma 8.2.4). The two rays are collinear:

$$R_{z_1,v} = \{z_1 + tv \mid t > 0\} = \big(z_1, z_0\big] \sqcup R_{z_0,v}, \qquad R_{z_0,v} = \{z_1 + tv \mid t > t_0\}$$

so $X \cap R_{z_1,v} = \big(X \cap (z_1, z_0]\big) \sqcup \big(X \cap R_{z_0,v}\big)$. The half-open segment $(z_1, z_0]$ is the part of $[z_0, z_1]$ not equal to z_1, and since $z_0 \notin X$ while the segment meets X in its single interior crossing, $X \cap (z_1, z_0]$ is that one point. Hence $\#(X \cap R_{z_1,v}) = \#(X \cap R_{z_0,v}) + 1$, and reducing modulo two, $W_2(X, z_1) = W_2(X, z_0) + 1$ by lemma 8.2.4, completing the proof.

Local constancy and the crossing rule are the two halves of a separation theorem. The winding number is constant on each region cut out by X and changes by one on stepping across X, so it can take at most two values, and it takes both. That X cuts $\mathbb{R}^n$ into exactly two regions, and not more, is the content of Jordan-Brouwer; the crossing rule supplies the lower bound of two, and a tubular neighbourhood supplies the upper bound.

Theorem 8.2.7: Jordan-Brouwer Separation

Let $X \subseteq \mathbb{R}^n$ ($n \geq 2$) be a compact connected smooth hypersurface. Then $\mathbb{R}^n \setminus X$ has exactly two connected components,

$$D_0 = \{z \mid W_2(X,z) = 0\}, \qquad D_1 = \{z \mid W_2(X,z) = 1\}$$

of which D_0 is unbounded and D_1 is bounded. Each D_i is open, and X is the topological boundary of each: $\partial D_0 = \partial D_1 = X$.[a]

[a] Camille Jordan stated the case $n = 2$ (the closed curve) in 1887, and L. E. J. Brouwer proved the general-dimensional separation around 1911; the winding-number proof given here is the smooth-category streamlining of Brouwer's.

***Proof*:**
By proposition 8.2.5 the function $W_2(X,-)$ is constant on each component of $\mathbb{R}^n \setminus X$, so D_0 and D_1 are unions of components. Both are nonempty: the region outside a ball containing X has $W_2 = 0$, and applying lemma 8.2.6 to a line transverse to X (a generic line, by theorem 7.2.1) produces a point with $W_2 = 1$. Thus $\mathbb{R}^n \setminus X$ has at least two components.

Step 1: X is two-sided. By the tubular neighbourhood theorem (theorem 7.8.1) there is an open neighbourhood N of X and a diffeomorphism of N with the total space of the normal bundle $\nu \to X$, carrying X to the zero section; ν is a real line bundle, since $\dim \mathbb{R}^n - \dim X = 1$. Then $N \setminus X$ is diffeomorphic to ν minus its zero section, which fibers over X with fiber the two rays $\mathbb{R} \setminus \{0\}$. If $N \setminus X$ were disconnected, fiberwise negation, a homeomorphism exchanging the two rays in each fiber, would exchange its components, so each component would meet every fiber in exactly one ray; choosing that ray would be a nowhere-vanishing section of ν, forcing ν to be trivial. Suppose then, for contradiction, that ν is nontrivial; the above shows $N \setminus X$ is connected, so $W_2(X,-)$ is constant on it. But for any $x \in X$ a short segment through x transverse to X has its two endpoints in $N \setminus X$ on opposite sides, and by lemma 8.2.6 they carry different winding numbers, a contradiction. Hence ν is trivial: $N \setminus X \cong X \times (\mathbb{R} \setminus \{0\})$ splits into two connected collars N_+ and N_-, each diffeomorphic to $X \times \mathbb{R}$.

Step 2: at most two components. Let C be any component of $\mathbb{R}^n \setminus X$; it is open. Its closure meets X: otherwise C would be open and closed in the connected space $\mathbb{R}^n$, forcing $C = \mathbb{R}^n$, impossible as $X \neq \emptyset$. Pick $x \in \overline{C} \cap X$. Since N is a neighbourhood of x and C is disjoint from X, the set C meets $N \setminus X = N_+ \sqcup N_-$; as $N_\pm$ are connected and disjoint from X, whichever one C meets is contained in C. So every component contains N_+ or N_-, and with only two collars there are at most two components. Combined with the lower bound, $\mathbb{R}^n \setminus X$ has exactly two components, distinguished by $W_2 \in \{0,1\}$, so they are D_0 and D_1.

Step 3: bounded, unbounded, and boundary. Choose r with $X \subseteq B_r(0)$. The set $\mathbb{R}^n \setminus \overline{B_r(0)}$ is connected (as $n \geq 2$) and has $W_2 = 0$, so it lies in D_0; hence D_0 is unbounded, and $D_1 \subseteq B_r(0)$ is bounded. Finally, the two collars N_+, N_- carry different winding numbers (their points near a common $x \in X$ are on opposite sides of X, lemma 8.2.6), so one lies in D_0 and the other in D_1. Thus every $x \in X$ is a limit of points of D_0 and of points of D_1, giving $X \subseteq \partial D_0 \cap \partial D_1$. Conversely $\partial D_i \subseteq X$, because D_i is open and its complement $X \sqcup D_{1-i}$ has the open set D_{1-i}

contributing no boundary points. Hence $\partial D_0 = \partial D_1 = X$, completing the proof.

A compact connected smooth hypersurface therefore behaves exactly like the sphere: it has a well-defined inside and outside, and it is the full common boundary of both. The proof extracts more than separation. Step 1 shows such a hypersurface is *two-sided*, its normal bundle trivial, and hence (with the standard orientation of $\mathbb{R}^n$) orientable, a fact that fails for hypersurfaces in other ambient manifolds, where the normal bundle may be nontrivial, as for the one-sided $\mathbb{RP}^{n-1} \subseteq \mathbb{RP}^n$. The smoothness is doing real work: the topological Jordan-Brouwer theorem for continuous embeddings holds as well, but its proof runs through the homology of the complement rather than a tubular neighbourhood, and belongs to algebraic topology ([7]). The winding number gives the separation and, at the same time, names the two pieces.

Definition 8.2.8: Inside and Outside

For a compact connected smooth hypersurface $X \subseteq \mathbb{R}^n$, the bounded component $D_1 = \{z \mid W_2(X, z) = 1\}$ of $\mathbb{R}^n \setminus X$ is the *inside* of X, and the unbounded component $D_0 = \{z \mid W_2(X, z) = 0\}$ is the *outside*. A point $z \notin X$ is inside X if and only if $W_2(X, z) = 1$.

This inside-outside criterion is a decision procedure: to test whether z lies inside X, shoot a generic ray from z and count its transverse crossings with X; the point is inside precisely when the count is odd (lemma 8.2.4). It is the higher-dimensional even-odd rule familiar for polygons in the plane, and it is independent of the ray because the winding number is.

The same parity idea, transferred from a point in $\mathbb{R}^n$ to the antipodal involution of a sphere, proves the Borsuk-Ulam theorem. The bridge is a single fact about maps that anticommute with the antipode.

Definition 8.2.9: Antipodal Map

A map $g \colon S^n \to S^k$ is *antipodal* (or *odd*) if $g(-x) = -g(x)$ for all x.

The mod-2 degree of an antipodal self-map is forced to be odd.

Lemma 8.2.10: Antipodal Self-Maps Have Odd Degree

Every smooth antipodal map $A \colon S^m \to S^m$ has $\deg_2 A = 1$, for all $m \geq 0$.

***Proof*:**
Induction on m.

Base $m = 0$. An antipodal map $A \colon S^0 \to S^0$ sends the two points $\{1, -1\}$ to $A(1) = a$ and $A(-1) = -a$, a bijection; each point of S^0 is a regular value with a single preimage, so $\deg_2 A = 1$.

Reduction to the equator ($m \geq 1$). Let $\Sigma = S^{m-1} = S^m \cap \{x_{m+1} = 0\}$ be the equator, with closed hemispheres $D_\pm = S^m \cap \{\pm x_{m+1} \geq 0\}$, so $\partial D_+ = \Sigma$ and $D_- = -D_+$. The restriction $A|_\Sigma \colon \Sigma \to S^m$ has image of measure zero (its domain has dimension $m - 1 < m$, theorem 7.2.1), so it misses a point q, and by antipodal symmetry of its image it misses $-q$ too. The complement $S^m \setminus \{q, -q\}$ deformation-retracts, antipodally and equivariantly, onto the great sphere $\{x \perp q\}$ along meridians. Composing $A|_\Sigma$ with this retraction and then with an antipodal-preserving rotation carrying $\{x \perp q\}$ to Σ, then extending the resulting homotopy of $A|_\Sigma$ over a symmetric collar of Σ (theorem 7.8.1), produces a smooth antipodal map homotopic to A that carries Σ into Σ. Since homotopic maps have equal mod-2 degree, assume from now on $A(\Sigma) \subseteq \Sigma$, and write $\widehat{A} := A|_\Sigma : \Sigma \to \Sigma$, itself antipodal, with $\deg_2 \widehat{A} = 1$ by the inductive hypothesis.

A parity count. Set $B := A|_{D_+} : D_+ \to S^m$. Fix a point y in the open upper hemisphere and an arc $\gamma \subseteq S^m$ from y to $-y$ meeting the equator Σ in exactly one point w, chosen generically so that B is transverse to γ, that $y, -y$ are regular values of B, and that w is a regular value of $\widehat{A}$ (theorem 7.2.1). Then $N := B^{-1}(\gamma)$ is a compact 1-manifold with boundary

$$\partial N = B^{-1}(y) \ \sqcup \ B^{-1}(-y) \ \sqcup \ \big(B^{-1}(\gamma) \cap \Sigma\big)$$

and $B^{-1}(\gamma) \cap \Sigma = \widehat{A}^{-1}(\gamma \cap \Sigma) = \widehat{A}^{-1}(w)$, since $\widehat{A}$ maps Σ into Σ and $\gamma \cap \Sigma = \{w\}$. By corollary 5.11.2 the total count $\#\partial N$ is even, so

$$\#B^{-1}(y) + \#B^{-1}(-y) \equiv \#\widehat{A}^{-1}(w) \pmod 2$$

Conclusion. Because $A(\Sigma) \subseteq \Sigma$ avoids y and $-y$, the preimages $A^{-1}(y)$ lie in the open hemispheres. Those in D_+ are exactly $B^{-1}(y)$; each x in the open lower hemisphere with $A(x) = y$ corresponds under $x \mapsto -x$ to a point $-x \in D_+$ with $A(-x) = -y$, that is to a point of $B^{-1}(-y)$. Hence, at the regular value y,

$$\deg_2 A = \#A^{-1}(y) \equiv \#B^{-1}(y) + \#B^{-1}(-y) \equiv \#\widehat{A}^{-1}(w) = \deg_2 \widehat{A} = 1 \pmod 2$$

completing the induction.

The lemma says an odd symmetry cannot be unwound: no antipodal self-map of a sphere is mod-2 nullhomotopic. Its immediate corollary is that the antipodal symmetry cannot be pushed down a dimension.

Corollary 8.2.11: No Antipodal Map to a Lower Sphere

For $n \geq 1$ there is no continuous antipodal map $g : S^n \to S^{n-1}$.

***Proof*:**
Suppose $g : S^n \to S^{n-1}$ is continuous and antipodal. Smooth approximation followed by symmetrization replaces it by a smooth antipodal map: if g' is a smooth approximation with $|g' - g|$ small, then $g'(x) - g'(-x)$ is close to $2g(x) \neq 0$, so $x \mapsto (g'(x) - g'(-x))/|g'(x) - g'(-x)|$ is smooth, antipodal, and defined everywhere. Assume then g smooth. Its restriction to the equator $\Sigma = S^{n-1} \subseteq S^n$ is an antipodal self-map $h = g|_\Sigma : S^{n-1} \to S^{n-1}$, so $\deg_2 h = 1$ by lemma 8.2.10. But h extends over the upper hemisphere disc D^n (via $g|_{D^n} : D^n \to S^{n-1}$, with $\partial D^n = \Sigma$): for a common regular value p of $g|_{D^n}$ and h, the preimage $(g|_{D^n})^{-1}(p)$ is a compact 1-manifold with boundary $h^{-1}(p)$, so $\#h^{-1}(p)$ is even (corollary 5.11.2) and $\deg_2 h = 0$. This contradicts $\deg_2 h = 1$, so no such g exists, completing the proof.

Borsuk-Ulam is the geometric restatement of this impossibility. If a map to $\mathbb{R}^n$ separated every antipodal pair, the normalized difference would be exactly the forbidden antipodal map to S^{n-1}.

Theorem 8.2.12: Borsuk-Ulam

Every continuous map $f\colon S^n \to \mathbb{R}^n$ ($n \geq 1$) identifies some antipodal pair: there exists $x \in S^n$ with $f(x) = f(-x)$.

***Proof*:**
Suppose $f(x) \neq f(-x)$ for every $x \in S^n$. Then

$$u\colon S^n \to S^{n-1}, \qquad u(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

is a well-defined continuous map, and it is antipodal, since $u(-x) = (f(-x) - f(x))/|f(-x) - f(x)| = -u(x)$. This contradicts corollary 8.2.11. Hence some x has $f(x) = f(-x)$, as we sought to show.

Borsuk-Ulam says a continuous field of n real quantities on the n-sphere cannot separate every antipodal pair: somewhere the two ends of a diameter read identically. In the proof, the failure of the conclusion manufactures exactly the antipodal map to a lower sphere that odd degree forbids. Two consequences follow: an existence theorem it hands to measure theory, and a concrete reading on the 2-sphere.

The Ham Sandwich theorem is the statement that a single cut bisects several masses at once. It is Borsuk-Ulam with the sphere of directions reinterpreted as the sphere of affine hyperplanes.

Corollary 8.2.13: Ham Sandwich Theorem

Let $A_1, \ldots, A_n$ be Lebesgue-measurable subsets of $\mathbb{R}^n$, each of finite measure. Then there is an affine hyperplane $H \subseteq \mathbb{R}^n$ that simultaneously bisects every A_i: each of the two closed half-spaces bounded by H contains exactly half the measure of each A_i.[a]

[a] Named for the problem of halving a sandwich of bread, ham, and bread with one straight cut; the general statement and this Borsuk-Ulam proof are due to Stone and Tukey (1942).

***Proof*:**
Parametrize oriented affine hyperplanes by the sphere $S^n \subseteq \mathbb{R}^{n+1} = \mathbb{R}^n \times \mathbb{R}$. To a unit vector (a, b) with $a \in \mathbb{R}^n$, $b \in \mathbb{R}$, associate the closed half-space $H^+_{(a,b)} = \{x \in \mathbb{R}^n \mid a \cdot x \geq b\}$, and define

$$f\colon S^n \to \mathbb{R}^n, \qquad f(a,b) = \big(\mu_1(a,b), \ldots, \mu_n(a,b)\big), \quad \mu_i(a,b) = \operatorname{vol}\big(A_i \cap H^+_{(a,b)}\big)$$

Each μ_i is continuous: as (a, b) varies, the half-space varies, and dominated convergence applies because a hyperplane has Lebesgue measure zero, so no mass sits on the boundary in the limit. By theorem 8.2.12 there is (a, b) with $f(a, b) = f(-(a, b))$. Negating (a, b) replaces the half-space by its closure-complementary one (the other side of the same hyperplane), so $\mu_i(-(a, b)) = |A_i| - \mu_i(a, b)$, again because the bounding hyperplane carries no measure. The equality $\mu_i(a, b) = \mu_i(-(a, b))$ therefore reads $\mu_i(a, b) = |A_i|/2$ for every i. Finally $a \neq 0$: if $a = 0$ then $(a, b) = (0, \pm 1)$ and each half-space is all of $\mathbb{R}^n$ or empty, forcing $\mu_i \in \{0, |A_i|\}$, incompatible with $\mu_i = |A_i|/2$ unless every A_i is null (in which case any hyperplane bisects). Thus $H = \{x \mid a \cdot x = b\}$ is a genuine hyperplane bisecting all the A_i, completing the proof.

The measurable sets need not be connected or convex; the one hyperplane splits each in half regardless. The proof is a template: any n continuous mass distributions on $\mathbb{R}^n$ are simultaneously bisected, because the bisection condition is exactly the antipodal coincidence that Borsuk-Ulam guarantees. The planar case is the sandwich itself, two regions and one line.

Example 8.2.14: Equal Temperature and Pressure at Antipodes
Model the surface of the Earth as S^2 and suppose temperature $T\colon S^2 \to \mathbb{R}$ and barometric pressure $P\colon S^2 \to \mathbb{R}$ vary continuously. The pair $f = (T, P)\colon S^2 \to \mathbb{R}^2$ satisfies the hypotheses of theorem 8.2.12 with $n = 2$, so there is a point $x \in S^2$ with $f(x) = f(-x)$: at x and its antipode $-x$ the temperature and the pressure agree simultaneously. No smoothness or genericity of T and P is needed, only continuity; the conclusion is forced by the topology of the sphere, not by any feature of the weather. The same statement with a single scalar (temperature alone) is the $n = 1$ case applied to a great circle: some pair of antipodal points on any circle has equal temperature.

Exercise 8.2.1

1. Let $S^{n-1} \subseteq \mathbb{R}^n$ be the unit sphere. Show directly that $W_2(S^{n-1}, z) = 1$ when $|z| < 1$ and $W_2(S^{n-1}, z) = 0$ when $|z| > 1$, using the bounding lemma for the outside and the ray count for the inside.
2. Let M be a compact $(n-1)$-manifold and $f\colon M \to \mathbb{R}^n$ smooth. Show that if z_0 and z_1 lie in the same connected component of $\mathbb{R}^n \setminus f(M)$, then $W_2(f, z_0) = W_2(f, z_1)$. (This is why the winding number depends on z only through its component.)
3. Deduce from the proof of theorem 8.2.7 that a compact connected smooth hypersurface $X \subseteq \mathbb{R}^n$ is orientable, by exhibiting a consistent orientation of T_xX from a global unit normal.
4. Use the Borsuk-Ulam theorem to prove that no continuous map $g\colon S^n \to \mathbb{R}^n$ is injective, and conclude that S^n does not embed in $\mathbb{R}^n$.
5. State and prove the planar Ham Sandwich theorem: any two bounded measurable regions $A_1, A_2 \subseteq \mathbb{R}^2$ are simultaneously bisected by a line. Then show separately that a single bounded measurable region can be bisected by a line in every prescribed direction.

8.3 Oriented Intersection Theory

The mod-2 theory of section 8.1 counts transverse preimages, but only their parity survives. Orienting M, N, and Z promotes that count to a signed integer, the oriented intersection number $I(f, Z) \in \mathbb{Z}$, which refines $I_2(f, Z)$ and carries strictly more information: from it come a degree, a Lefschetz number, and an Euler characteristic. This section fixes the orientation conventions that attach a sign to each transverse intersection, defines $I(f, Z)$, and proves it a homotopy invariant by the same cobordism argument as before, now tracking signs through the oriented Boundary Theorem. The number then transfers to submanifolds, where it acquires a symmetry $I(X, Z) = \pm I(Z, X)$ and, on the diagonal, computes the self-intersection that section 8.6 will name the Euler characteristic.

Everything below rests on a single bookkeeping device: an orientation of a vector space is a choice of positive ordered basis up to positive determinant, and for a direct sum $U = V \oplus W$ the notation $[V] \oplus [W] = [U]$ means that a positive basis of V followed by a positive basis of W is a positive basis of U. Reordering the summands, or reversing one of them, changes $[U]$ by the sign of the corresponding permutation or by -1. The three conventions the theory needs are all instances of the same rule, and they are chosen once so that they never fight one another.

8.3.1 Convention: Orientation Sign Rules Fix oriented manifolds throughout, with orientations recorded fibrewise as above.

1. *Product orientation.* $M \times N$ is oriented so that a positive frame of T_pM followed by a positive frame of T_qN is a positive frame of $T_{(p,q)}(M \times N)$; that is, $[T_pM] \oplus [T_qN] = [T_{(p,q)}(M \times N)]$.
2. *Boundary orientation.* A frame of $T_x(\partial M)$ is positive exactly when prefixing it with an outward-pointing vector gives a positive frame of T_xM (outward normal first).
3. *Normal orientation.* For an oriented submanifold $Z \subseteq N$ of the oriented N, the normal space $\nu_y(Z) = T_yN/T_yZ$ is oriented so that $[\nu_y(Z)] \oplus [T_yZ] = [T_yN]$ (normal direction first).

The normal orientation is the pattern to remember: the complementary directions come first, the submanifold second. The boundary orientation is the case $Z = \partial M$ with the outward normal as the one complementary direction, and the sign rule below is the same choice applied to $df_x(T_xM)$ as the complementary piece. With the conventions fixed, a transverse map orients its preimage.

Definition 8.3.2: Orientation of a Preimage

Let M and N be oriented, let $Z \subseteq N$ be a closed oriented submanifold without boundary, and let $f : M \to N$ be transverse to Z. By proposition 7.4.3, $S = f^{-1}(Z)$ is a submanifold of M with $\operatorname{codim}_M S = \operatorname{codim}_N Z$, and $T_xS = (df_x)^{-1}(T_{f(x)}Z)$ at each $x \in S$ (corollary 7.4.5). Transversality makes the induced map

$$df_x : \nu_x(S) = T_xM/T_xS \longrightarrow \nu_{f(x)}(Z) = T_{f(x)}N/T_{f(x)}Z$$

an isomorphism. The *preimage orientation* of S is defined by giving $\nu_x(S)$ the orientation pulled back from the normal orientation of $\nu_{f(x)}(Z)$ through this isomorphism, and then orienting T_xS by the direct-sum rule $[\nu_x(S)] \oplus [T_xS] = [T_xM]$.

The construction reads off in one line for two oriented curves crossing in an oriented surface. If $\dim M = \dim Z = 1$ and $\dim N = 2$, then at a transverse crossing x the space T_xS is zero, $\nu_x(S) = T_xM$, and the orientation of the point is a sign: it records whether the isomorphism $df_x : T_xM \to \nu_{f(x)}(Z)$ matches the two chosen normal orientations. That sign is the object the whole theory counts, so it is worth a name of its own.

Definition 8.3.3: The Sign of a Transverse Intersection

Let M, N, Z be oriented with $Z \subseteq N$ closed and without boundary, let $f\colon M \to N$ be transverse to Z, and suppose the dimensions are *complementary*, $\dim M + \dim Z = \dim N$. For $x \in f^{-1}(Z)$ transversality gives the direct sum

$$df_x(T_xM) \oplus T_{f(x)}Z = T_{f(x)}N$$

and df_x is injective on T_xM. The *intersection sign* of f at x is

$$\operatorname{sign}(x) = \begin{cases} +1 & \text{if } [df_x(T_xM)] \oplus [T_{f(x)}Z] = [T_{f(x)}N] \\ -1 & \text{otherwise} \end{cases}$$

where $df_x(T_xM)$ carries the orientation pushed forward from T_xM. This is the preimage orientation of definition 8.3.2 in the zero-dimensional case, read as an element of $\{\pm 1\}$.

The sign compares two ways of building the orientation of $T_{f(x)}N$: the one carried into N along f and the one already on Z. Where the map's image reinforces the ambient orientation the crossing counts positively, where it opposes it the crossing counts negatively. Summing these signs over a compact source gives a finite integer.

Definition 8.3.4: Oriented Intersection Number

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary, with $\dim M + \dim Z = \dim N$. For $f\colon M \to N$ transverse to Z, the set $f^{-1}(Z)$ is a compact 0-manifold, hence finite, and the *oriented intersection number* is

$$I(f, Z) = \sum_{x \in f^{-1}(Z)} \operatorname{sign}(x) \in \mathbb{Z}$$

Reducing each sign modulo 2 collapses ± 1 to 1 and recovers the mod-2 count of section 8.1, so $I(f, Z) \equiv I_2(f, Z) \pmod 2$. The integer sees more: it distinguishes a map that crosses Z twice with cancelling signs from one that misses Z altogether, whereas I_2 cannot. The next example makes both the computation and the refinement concrete.

Example 8.3.5: Winding on the Torus

Write the torus as $T^2 = S^1 \times S^1$ with $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ and coordinates (x, y), oriented by $[\partial_x] \oplus [\partial_y]$. Fix the meridian $Z = \{0\} \times S^1$, a closed oriented 1-submanifold with positive tangent ∂_y, complementary to any map from a 1-manifold. For an integer $k \neq 0$ let

$$f_k\colon S^1 \to T^2, \qquad f_k(\theta) = (k\theta, 0)$$

with S^1 oriented by ∂_θ. Then $f_k(\theta) \in Z$ iff $k\theta \equiv 0 \pmod{2\pi}$, giving the $|k|$ points $\theta_j = 2\pi j/k$, $j = 0, \ldots, |k| - 1$. At each, $df_k(\partial_\theta) = k\,\partial_x$, transverse to $T_{f_k(\theta_j)}Z = \mathbb{R}\,\partial_y$ because $k \neq 0$. The intersection sign compares $[k\,\partial_x] \oplus [\partial_y]$ with the torus orientation $[\partial_x] \oplus [\partial_y]$: for $k > 0$ this is orientation-preserving and the sign is $+1$, for $k < 0$ it is -1. All $|k|$ points share the sign of k, so

$$I(f_k, Z) = k$$

The map wraps the first circle k times, and the oriented count records the winding with its sense.

Modulo 2 this is k mod 2, exactly $I_2(f_k, Z)$: the mod-2 number cannot tell f_2 (two cancelling-in-parity crossings) from a constant map, but $I(f_2, Z) = 2 \neq 0$ does. This integer is the degree of the circle map $\theta \mapsto k\theta$, which section 8.4 recovers as intersection with a point.

Homotopy invariance is what makes $I(f, Z)$ a topological rather than an incidental quantity, and its proof is the oriented refinement of the cobordism argument behind section 8.1. The engine is the signed count on a compact 1-manifold, where the classification of section 5.11 forces the signs to cancel. For a compact oriented 1-manifold C and a boundary point $x \in \partial C$, the boundary orientation of box 8.3.1 assigns x the sign $\sigma_\partial(x) = +1$ if the positive unit tangent of C points out of C at x, and $\sigma_\partial(x) = -1$ if it points inward.

Theorem 8.3.6: Oriented Boundary Theorem

Let C be a compact oriented 1-manifold with boundary. Then the boundary signs sum to zero,

$$\sum_{x \in \partial C} \sigma_\partial(x) = 0$$

***Proof*:**
By the classification of compact 1-manifolds (section 5.11), C is a finite disjoint union of components, each diffeomorphic to the circle S^1 or to the closed interval $[0,1]$. A circle component has empty boundary and contributes nothing. On an interval component, the orientation is a single coherent choice of positive tangent direction along the arc; that direction points outward at one endpoint and inward at the other, since the two outward directions at the ends of an arc are opposite. So each interval contributes $(+1) + (-1) = 0$. Summing over components gives 0, as we sought to show.

This refines the parity statement of corollary 5.11.2, which records only that $\#\partial C$ is even: here the signed sum vanishes on the nose. To run the homotopy argument the boundary signs of a preimage must be matched to intersection signs, and the two differ by a single universal sign, computed once in the following lemma.

Lemma 8.3.7: Boundary of an Oriented Preimage

Let W be an oriented manifold with boundary, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary. Let $F\colon W \to N$ be transverse to Z with $\partial F = F|_{\partial W}$ also transverse to Z, and suppose $\dim F^{-1}(Z) = 1$. Then $S = F^{-1}(Z)$ is a 1-manifold with $\partial S = (\partial F)^{-1}(Z)$, and at each $x \in \partial S$ the boundary sign of S (with its preimage orientation) and the intersection sign of ∂F are related by

$$\sigma_\partial(x) = (-1)^{\operatorname{codim} Z} \operatorname{sign}_{\partial F}(x)$$

***Proof*:**
Write $c = \operatorname{codim} Z$, so $\dim W = c + 1$ and $\dim \partial W = c$. By the preimage theorem for manifolds with boundary (lemma 8.1.1), the two transversality hypotheses make S a 1-manifold meeting ∂W transversally in $\partial S = (\partial F)^{-1}(Z)$. Fix $x \in \partial S$, set $y = F(x)$, and let $q = (dF_x \bmod T_y Z)\colon T_x W \to \nu_y(Z)$; transversality makes q surjective with kernel $T_x S$.

Step 1: Adapted bases. Choose a positive basis $u_1, \dots, u_c$ of $\nu_y(Z)$ and lift it to $b_1, \dots, b_c \in T_xW$ with $q(b_i) = u_i$; let t be the positive unit tangent of S at x, so $q(t) = 0$. The preimage orientation is defined by $[\nu_x(S)] \oplus [T_xS] = [T_xW]$, which says exactly that $(b_1, \dots, b_c, t)$ is a positive basis of T_xW. Let $n \in T_xW$ be outward at x and $(w_1, \dots, w_c)$ a positive basis of $T_x(\partial W)$; the boundary convention makes $(n, w_1, \dots, w_c)$ positive as well, so the change of basis P from $(b_1, \dots, b_c, t)$ to $(n, w_1, \dots, w_c)$ has $\det P > 0$.
Step 2: The two signs. Writing $t = t_0 + \alpha n$ with $t_0 \in T_x(\partial W)$, the positive tangent points outward exactly when $\alpha > 0$, so $\sigma_\partial(x) = \operatorname{sign}(\alpha)$. The map ∂F is transverse to Z with complementary dimensions, and its intersection sign is $+1$ iff $q|_{T_x(\partial W)}$ carries the boundary orientation of ∂W to the normal orientation of $\nu_y(Z)$; hence $\operatorname{sign}_{\partial F}(x) = \operatorname{sign}(\det A)$, where A is the matrix of $q|_{T_x(\partial W)}$ from (w_j) to (u_i). Because $\partial F \pitchfork Z$, the map $q|_{T_x(\partial W)}$ is onto, so A is invertible.
Step 3: One determinant. Expand the new basis in the old one: $n = \sum_i B_i b_i + Dt$ and $w_j = \sum_i A_{ij} b_i + c_j t$ (the coefficient of b_i in w_j is A_{ij} because $q(w_j) = \sum_i A_{ij} u_i$). Reading these as the columns of P and moving the first column past the remaining c,

$$\det P = (-1)^c \det \begin{pmatrix} A & B \\ c^\mathsf{T} & D \end{pmatrix} = (-1)^c \det(A)\,(D - c^\mathsf{T} A^{-1} B)$$

by the Schur complement. Solving $t = \sum_j \beta_j w_j + \alpha n$ for its b_i- and t-coefficients gives $A\beta + \alpha B = 0$ and $c^\mathsf{T}\beta + \alpha D = 1$, whence $\alpha^{-1} = D - c^\mathsf{T} A^{-1} B$. Therefore $\det P = (-1)^c \det(A)/\alpha$, and $\det P > 0$ forces $\operatorname{sign}(\alpha)\operatorname{sign}(\det A) = (-1)^c$, that is $\sigma_\partial(x) = (-1)^{\operatorname{codim} Z} \operatorname{sign}_{\partial F}(x)$, completing the proof.

The universal sign $(-1)^{\operatorname{codim} Z}$ is the same at every boundary point, so it factors out of any sum and cannot obstruct a cancellation. With it in hand the invariance is immediate.

Theorem 8.3.8: Homotopy Invariance of the Oriented Intersection Number

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary, with $\dim M + \dim Z = \dim N$. If $f_0, f_1 \colon M \to N$ are homotopic and each transverse to Z, then $I(f_0, Z) = I(f_1, Z)$.

***Proof*:**
Let $m = \dim M$.
Step 1: A transverse cobordism. Extend a homotopy from f_0 to f_1 to a smooth map $F \colon M \times [0,1] \to N$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$. The boundary values are already transverse to Z, so by boundary-fixing transversality (lemma 8.1.5) we may homotope F rel $M \times \{0,1\}$ to make $F \pitchfork Z$ while fixing f_0 and f_1. Then $S = F^{-1}(Z)$ is compact ($M \times [0,1]$ is compact and Z closed) of dimension $(m+1) - \operatorname{codim} Z = 1$, a compact oriented 1-manifold with

$$\partial S = f_1^{-1}(Z) \times \{1\} \;\sqcup\; f_0^{-1}(Z) \times \{0\}$$

Step 2: Summing the boundary. Give $M \times [0,1]$ the product orientation and S its preimage orientation. The oriented Boundary Theorem (theorem 8.3.6) gives $\sum_{x \in \partial S} \sigma_\partial(x) = 0$, and lemma 8.3.7 with $\operatorname{codim} Z = m$ turns this into

$$\sum_{x \in \partial S} \operatorname{sign}_{\partial F}(x) = 0$$

Step 3: The two ends. The outward normals of $M \times \{1\}$ and $M \times \{0\}$ are $+\partial_t$ and $-\partial_t$, so with the product orientation the boundary orientation of $M \times \{1\}$ is $(-1)^m$ times that of M and the boundary orientation of $M \times \{0\}$ is $-(-1)^m$ times it. Reversing the orientation of the source negates every intersection sign, so $\operatorname{sign}_{\partial F} = (-1)^m \operatorname{sign}_{f_1}$ on the top face and $\operatorname{sign}_{\partial F} = (-1)^{m+1} \operatorname{sign}_{f_0}$ on the bottom. Substituting,

$$0 = (-1)^m \sum_{f_1^{-1}(Z)} \operatorname{sign}_{f_1} + (-1)^{m+1} \sum_{f_0^{-1}(Z)} \operatorname{sign}_{f_0} = (-1)^m \big(I(f_1, Z) - I(f_0, Z)\big)$$

Since $(-1)^m \neq 0$, $I(f_0, Z) = I(f_1, Z)$, as we sought to show.

Invariance does two things at once. It says $I(f, Z)$ depends only on the homotopy class of f, and it lets the number be defined for maps that are not transverse to Z at all: perturb them until they are. The same argument, run with Z moved instead of f, shows $I(f, Z)$ is unchanged when Z is carried across an ambient isotopy, since sliding Z by an orientation-preserving diffeomorphism h_1 isotopic to the identity is the same as replacing f by $h_1^{-1} \circ f \simeq f$. Both freedoms are used below.

Definition 8.3.9: Oriented Intersection Number of an Arbitrary Map

Let M be compact and oriented, N oriented, and $Z \subseteq N$ a closed oriented submanifold without boundary of complementary dimension. For *any* smooth $f: M \to N$, choose $g \simeq f$ with $g \pitchfork Z$ (possible by theorem 7.5.7) and set

$$I(f, Z) = I(g, Z)$$

The choice does not matter: two transverse maps homotopic to f are homotopic to each other, so theorem 8.3.8 makes their intersection numbers equal. A merely continuous f is first replaced by a smooth homotopic map, exactly as in the mod-2 theory of section 8.1, so $I(f, Z)$ extends to continuous maps and remains a homotopy invariant. In particular the number attaches to submanifolds through their inclusions.

Definition 8.3.10: Oriented Intersection Number of Submanifolds

Let X be a compact oriented submanifold and Z a closed oriented submanifold without boundary of the oriented manifold N, with $\dim X + \dim Z = \dim N$. The *oriented intersection number* of X and Z is

$$I(X, Z) = I(i_X, Z), \qquad i_X : X \hookrightarrow N$$

When $X \pitchfork Z$ the inclusion is transverse and

$$I(X, Z) = \sum_{x \in X \cap Z} \operatorname{sign}(x), \qquad \operatorname{sign}(x) = +1 \iff [T_x X] \oplus [T_x Z] = [T_x N]$$

When X and Z are both compact, each may be counted against the other, and the two counts differ only by the price of reordering a frame of X past a frame of Z.

Theorem 8.3.11: Symmetry of the Intersection Number

Let X and Z be compact oriented submanifolds of the oriented manifold N with $\dim X + \dim Z = \dim N$. Then

$$I(X,Z) = (-1)^{(\dim X)(\dim Z)} I(Z,X)$$

***Proof*:**
Transversality is symmetric ($X \pitchfork Z$ iff $Z \pitchfork X$) and generic, so after a small ambient isotopy of X we may assume $X \pitchfork Z$; by the isotopy invariance noted after theorem 8.3.8 this changes neither $I(X,Z)$ nor $I(Z,X)$. Now X and Z meet in the same finite set $X \cap Z$ from both points of view. At a point x of it, transversality gives $T_xX \oplus T_xZ = T_xN$, and the two signs are

$$\operatorname{sign}_{I(X,Z)}(x) = +1 \iff [T_xX] \oplus [T_xZ] = [T_xN], \qquad \operatorname{sign}_{I(Z,X)}(x) = +1 \iff [T_xZ] \oplus [T_xX] = [T_xN]$$

Passing a positive frame of T_xX (of length $\dim X$) across a positive frame of T_xZ (of length $\dim Z$) is a permutation of sign $(-1)^{(\dim X)(\dim Z)}$, so $\operatorname{sign}_{I(X,Z)}(x) = (-1)^{(\dim X)(\dim Z)} \operatorname{sign}_{I(Z,X)}(x)$ at every x. Summing over $X \cap Z$ gives $I(X,Z) = (-1)^{(\dim X)(\dim Z)} I(Z,X)$, completing the proof.

The symmetry has teeth even when $X = Z$. Taking $Z = X$ forces $I(X,X) = (-1)^{(\dim X)^2} I(X,X)$, so a compact oriented submanifold of odd dimension whose double fills the ambient dimension has intersection number zero with itself. To make sense of $I(X,X)$ at all, note that X is never transverse to itself (it meets itself in all of X); the number is defined only through the perturbation of definition 8.3.9, and computing it is the first place the tubular neighbourhood theorem does real work.

Definition 8.3.12: Self-Intersection Number

Let X be a compact oriented submanifold of the oriented manifold N with $2\dim X = \dim N$. The *self-intersection number* of X is $I(X,X) = I(i_X, X)$, the intersection number of X with itself computed by perturbing the inclusion to a transverse map.

The perturbation has a canonical shape. A tubular neighbourhood identifies a neighbourhood of X in N with the normal bundle $\nu(X)$, and a section of $\nu(X)$ pushes X off itself; the pushed-off copy meets X exactly at the zeros of the section, and the signs agree. So the self-intersection number is a count of zeros, the invariant a bundle carries in its own right.

Proposition 8.3.13: Self-Intersection via the Normal Bundle

Let X be a compact oriented submanifold of the oriented N with $2\dim X = \dim N$, and let $\nu(X)$ be its normal bundle, an oriented bundle of rank $\dim X$ over X. Then $I(X,X)$ equals the number of zeros, counted with signs, of any section $s\colon X \to \nu(X)$ transverse to the zero section. In particular $I(X,X)$ depends only on $\nu(X)$; it is the *Euler number* of $\nu(X)$.

***Proof*:**
Step 1: Push X off itself. By the tubular neighbourhood theorem (theorem 7.8.1) there is an open set $U \supseteq X$ in N and a diffeomorphism $\Phi\colon \nu(X) \to U$ carrying the zero section to X by the identity. Let s be a section transverse to the zero section; rescaling by a small constant

leaves its zeros and their transversality untouched and places its image in the domain of Φ, so $g = \Phi \circ s \colon X \to U \subseteq N$ is defined. Sliding s to the zero section, $\Phi \circ (\tau s)$ for $\tau \in [0,1]$, is a homotopy from g to i_X, so $I(X,X) = I(g,X)$ by definition 8.3.9.

Step 2: The count. Since Φ is a diffeomorphism carrying the zero section to X, $g(x) \in X$ iff $s(x) = 0$, and $g \pitchfork X$ there iff s is transverse to the zero section. Hence $g^{-1}(X)$ is the zero set of s, and $I(g,X) = \sum_{s(x)=0} \operatorname{sign}_g(x)$.

Step 3: Signs match. At a zero x of s the vertical derivative $ds_x \colon T_xX \to \nu_x(X)$ is a well-defined isomorphism (transversality to the zero section), and dg_x sends T_xX to the graph of ds_x inside $T_xN = T_xX \oplus \nu_x(X)$. Deforming that graph to its vertical part through subspaces still complementary to T_xX leaves the intersection sign unchanged, so $\operatorname{sign}_g(x)$ equals the sign of $[\nu_x(X)] \oplus [T_xX] = [T_xN]$ read through ds_x, which is $+1$ iff ds_x preserves orientation. This is exactly the sign of the zero x. Therefore $I(X,X) = \sum_{s(x)=0} \operatorname{sign}(\det ds_x)$. Any two transverse sections are homotopic through sections, and small rescalings keep them inside U, so theorem 8.3.8 makes the signed count independent of s: it is an invariant of $\nu(X)$, the Euler number, completing the proof.

The zero section is the model case, and the diagonal is the case that matters for the rest of the chapter.

Example 8.3.14: The Zero Section and the Diagonal

1. *Zero section of a bundle.* For an oriented rank-k bundle $E \to X$ over a compact oriented X of dimension k, the zero section $X_0 \subseteq E$ is a compact oriented submanifold with $2\dim X_0 = \dim E$ and normal bundle $\nu(X_0) \cong E$. So $I(X_0, X_0)$ is the Euler number of E: the signed count of zeros of a generic section of E. Taking $E = TS^2$ and $s = \operatorname{grad}(\text{height})$, the only zeros are the north and south poles, each nondegenerate with two eigenvalues of one sign (a source and a sink), so each contributes $\operatorname{sign}(\det ds) = +1$ and the Euler number is 2.

2. *The diagonal.* For a compact oriented M of dimension m, the diagonal $\Delta = \{(p,p) \mid p \in M\} \subseteq M \times M$ is a compact oriented submanifold with $\dim \Delta = m$ and $2\dim\Delta = \dim(M \times M)$. Its tangent space at (p,p) is $\{(v,v) \mid v \in T_pM\}$, and the map $(v,w) \mapsto w - v$ identifies $\nu_{(p,p)}(\Delta) = T_{(p,p)}(M \times M)/T_{(p,p)}\Delta$ with T_pM; hence $\nu(\Delta) \cong TM$. So $I(\Delta,\Delta)$ is the Euler number of TM, the signed count of zeros of a vector field on M. For $M = S^2$ this is 2 by part (1), so $I(\Delta,\Delta) = 2$ in $S^2 \times S^2$. This integer, the self-intersection of the diagonal, is the quantity section 8.6 will name the Euler characteristic $\chi(M)$ and equate with the total index of any vector field.

The two computations are the same computation: the tubular neighbourhood turns the diagonal into the zero section of TM, and the Euler number of TM is what a vector field measures. Poincaré-Hopf is then the statement that this self-intersection equals the sum of indices, which section 8.6 proves. One sign convention deserves a closing remark, because it explains why so many geometric intersection numbers turn out to be positive.

8.3.15 Note: Complex Intersections Are Positive A complex vector space has a preferred orientation: any complex basis $e_1, \dots, e_n$, read as the real basis $e_1, ie_1, \dots, e_n, ie_n$, is positive, and any two complex bases give the same real orientation. For complex subspaces $V \oplus W = U$ this preferred orientation satisfies $[V] \oplus [W] = [U]$. Consequently, if X and Z are complex submanifolds of a complex manifold, of complementary real dimension and meeting transversally, then every intersection sign is $+1$ and $I(X, Z) = \#(X \cap Z) \geq 0$. Two distinct lines in the complex projective plane, for instance, meet transversally in a single point, so their intersection number is $+1$ rather than -1.

Exercise 8.3.1

1. On the torus $T^2 = S^1 \times S^1$, oriented by $[\partial_x] \oplus [\partial_y]$, let $a = S^1 \times \{q\}$ (oriented by ∂_x) and $b = \{p\} \times S^1$ (oriented by ∂_y). Show that a and b meet transversally in one point, compute $I(a, b)$ and $I(b, a)$, and verify theorem 8.3.11 directly.
2. Show that for $f\colon M \to N$ with M compact oriented and Z closed oriented of complementary dimension, the oriented intersection number reduces to the mod-2 number, $I(f, Z) \equiv I_2(f, Z) \pmod 2$. Deduce that a nonzero $I_2(f, Z)$ forces $I(f, Z) \neq 0$, but not conversely.
3. Let X be a compact oriented submanifold of an oriented N with $2 \dim X = \dim N$. Using theorem 8.3.11, show that $I(X, X) = 0$ whenever $\dim X$ is odd. Illustrate with a circle embedded in a surface.
4. Determine how $I(f, Z)$ changes when the orientation of M is reversed, when that of Z is reversed, and when that of N is reversed. (Hold the other two fixed each time and track the intersection sign of a single transverse point.)
5. Compute the self-intersection of the diagonal $\Delta \subseteq T^2 \times T^2$. Identify $\nu(\Delta) \cong T(T^2)$, exhibit a nowhere-zero section, and conclude $I(\Delta, \Delta) = 0$. Reconcile this with exercise 8.3.1 (3) (does it apply?) and with the value $\chi(T^2) = 0$.

8.4 The Degree of a Map

The oriented intersection number of section 8.3 measures a map $f\colon M \to N$ against a fixed target Z. Taking Z to be a single point strips the target of all geometry: what survives is an integer attached to f alone, recording how many times f covers a typical value of N, counted with sign. That integer is the *degree*. This section extracts it from the intersection theory already in hand, records its formal properties, and spends them: the fundamental theorem of algebra and the Brouwer fixed-point theorem fall out immediately, the Hopf theorem shows the degree is the *only* homotopy invariant of a self-map of a sphere, and the same construction applied to a pair of submanifolds produces their linking number.

Fix compact oriented n-manifolds M and N with N connected, and a smooth map $f\colon M \to N$. A point $y \in N$ is a regular value exactly when f is transverse to the 0-dimensional submanifold $\{y\}$, so the machinery of section 8.3 applies verbatim with $Z = \{y\}$. The point y carries its positive orientation as a 0-manifold, and $f^{-1}(y)$ is a compact 0-manifold, hence a finite set of points, each of which receives a sign.

Definition 8.4.1: Brouwer Degree

Let M and N be compact oriented n-manifolds with N connected, and let $f\colon M \to N$ be smooth. For a regular value $y \in N$ the preimage $f^{-1}(y)$ is finite (by corollary 3.2.10 together with compactness of M), and the *degree of f at y* is

$$\deg(f, y) \;=\; \sum_{x \in f^{-1}(y)} \operatorname{sign}\det df_x$$

where $\operatorname{sign}\det df_x = +1$ if $df_x\colon T_xM \to T_yN$ is orientation-preserving and -1 if it is orientation-reversing. This is the oriented intersection number $I(f, \{y\})$ of section 8.3. Once theorem 8.4.4 shows the value is independent of y, it is written $\deg f$ and called the *degree* of f.

The sign of a point $x \in f^{-1}(y)$ is exactly the intersection sign of the standing orientation convention (definition 8.3.3): the isomorphism df_x is $+1$ when it carries the orientation of T_xM to that of T_yN. When M and N are connected of the same dimension and f is a diffeomorphism, every value is regular and the single preimage of each point contributes ± 1 uniformly, so the degree is $+1$ for an orientation-preserving diffeomorphism and -1 for an orientation-reversing one. The first concrete computation is this diffeomorphism case.

Example 8.4.2: Degree of the Identity and a Reflection

The identity $\mathrm{id}_N\colon N \to N$ has every point as a regular value, with a single preimage on which $d(\mathrm{id}_N)_y$ is the identity of T_yN; hence $\deg(\mathrm{id}_N, y) = +1$ for every y, so $\deg \mathrm{id}_N = 1$.

Let $r\colon S^n \to S^n$ be the reflection $r(x_0, x_1, \ldots, x_n) = (-x_0, x_1, \ldots, x_n)$, the restriction of the linear reflection R of $\mathbb{R}^{n+1}$ in the hyperplane $x_0 = 0$. It is a diffeomorphism, so every $y \in S^n$ is regular with one preimage. Orient S^n as the boundary of the unit disc with the outward-normal-first convention: a basis $(v_1, \ldots, v_n)$ of $T_xS^n = x^\perp$ is positive precisely when $(x, v_1, \ldots, v_n)$ is positive in $\mathbb{R}^{n+1}$. Since $\det R = -1$, applying R to the positive frame $(x, v_1, \ldots, v_n)$ yields a *negative* frame $(r(x), Rv_1, \ldots, Rv_n)$ of $\mathbb{R}^{n+1}$, so $(dr_x v_1, \ldots, dr_x v_n)$ is a negative basis of $T_{r(x)}S^n$. Thus dr_x reverses orientation, every point contributes -1, and $\deg r = -1$.

That $\deg(f, y)$ does not depend on the regular value y is the first substantive claim, and it is what promotes $\deg(f, y)$ to an invariant $\deg f$ of the map. The argument moves a chosen regular value to any other by a diffeomorphism of N that can be deformed to the identity, so the following lemma is the engine.

Lemma 8.4.3: Homogeneity Lemma

Let N be a connected smooth manifold (without boundary) and let $y, y' \in N$. Then there is a diffeomorphism $h\colon N \to N$, smoothly isotopic to id_N through diffeomorphisms equal to the identity outside a compact set, with $h(y) = y'$. In particular h is orientation-preserving.

***Proof*:**
Step 1: A compactly supported push in $\mathbb{R}^n$. Fix $a \in \mathbb{R}^n$ with $|a| < 1$ and a bump function $\varphi\colon \mathbb{R}^n \to [0,1]$ (chapter 1) with $\varphi \equiv 1$ on the closed unit ball $\overline{B_1}$ and $\varphi \equiv 0$ outside B_2. The vector field $V(x) = \varphi(x)\, a$ has compact support, hence is complete, and generates a flow ψ_t of diffeomorphisms of $\mathbb{R}^n$ (chapter 5) with $\psi_0 = \mathrm{id}$ and each ψ_t equal to the identity outside B_2. On $\overline{B_1}$, where $\varphi \equiv 1$, the integral curve from the origin solves $\gamma'(t) = a$, $\gamma(0) = 0$, so $\gamma(t) = ta$; since $|ta| \le |a| < 1$ for $t \in [0,1]$, this curve stays in B_1 and $\psi_1(0) = a$. The path $t \mapsto \psi_t$ is an isotopy from id to ψ_1 through diffeomorphisms fixing everything outside B_2.

Step 2: Globalize by connectedness. Declare $y \sim y'$ if some diffeomorphism of N, isotopic to id_N through compactly supported diffeomorphisms, carries y to y'. This is an equivalence relation: reflexivity uses id_N, symmetry the inverse isotopy, and transitivity the composition of two such isotopies. Given $y \in N$, choose a chart $\varphi\colon U \to \mathbb{R}^n$ with $\varphi(y) = 0$, and shrink so that some ball B_ρ lies in $\varphi(U)$. Rescaling the construction of Step 1 to be supported in B_ρ, every point y' with $|\varphi(y')|$ small is $\sim y$: transport ψ_1 through φ and extend by the identity outside U, which is smooth because ψ_1 is the identity near ∂B_ρ. Hence each equivalence class is open, and since N is connected there is one class, so $y \sim y'$.

Orientation. For any h so produced, $\det dh_x$ is continuous in the isotopy parameter, never zero, and equal to $+1$ at id_N; it therefore stays positive, so h preserves orientation, completing the proof.

Theorem 8.4.4: The Degree Is Well Defined and Homotopy Invariant

Let M, N be compact oriented n-manifolds with N connected, and $f\colon M \to N$ smooth.

1. The value $\deg(f, y)$ is the same for every regular value $y \in N$; call it $\deg f$.
2. If $f \simeq g$ are smoothly homotopic maps $M \to N$, then $\deg f = \deg g$.

***Proof*:**
Part (2) is the homotopy invariance of the oriented intersection number $I(f, \{y\})$ established in section 8.3, applied to the fixed target $Z = \{y\}$; it rests, as there, on the signed boundary count of a compact oriented 1-manifold vanishing (theorem 8.3.6), the signed refinement of the mod-2 parity used for $\deg_2$.

For part (1), let y and y' be two regular values. By the Homogeneity Lemma (lemma 8.4.3) there is an orientation-preserving diffeomorphism $h\colon N \to N$, isotopic to id_N, with $h(y') = y$. Two computations combine.

Naturality. The point x lies in $f^{-1}(y) = f^{-1}(h(y'))$ if and only if $h^{-1}(f(x)) = y'$, i.e. $x \in (h^{-1} \circ f)^{-1}(y')$, and since h^{-1} is a diffeomorphism, y' is a regular value of $h^{-1} \circ f$ with the same preimage set. At each such x,

$$\operatorname{sign}\det d(h^{-1}\circ f)_x = \operatorname{sign}\big(\det dh^{-1}_{f(x)}\cdot\det df_x\big) = \operatorname{sign}\det df_x$$

because h^{-1} preserves orientation. Summing gives $\deg(h^{-1}\circ f,\, y') = \deg(f, y)$.

Homotopy. Because h is isotopic to id_N, the maps $h^{-1}\circ f$ and f are homotopic (compose f with the isotopy), so part (2) gives $\deg(h^{-1}\circ f, y') = \deg(f, y')$.

Combining, $\deg(f, y) = \deg(f, y')$, as we sought to show.

The degree turns a homotopy problem into arithmetic. Two maps with different degrees cannot be homotopic, and a single integer, computed at one convenient regular value, decides the question for all of them. The Homogeneity Lemma also isolates the one geometric fact that makes this possible: a connected manifold looks the same near any point, so no regular value is distinguished, and the count cannot depend on which one is used.

The algebraic behaviour of the degree under composition and under reversal of orientation is recorded next; each part is a direct reading of the signed count.

Proposition 8.4.5: Basic Properties of the Degree

Let M, N, P be compact oriented n-manifolds with N, P connected, and let $f\colon M \to N$, $g\colon N \to P$ be smooth.

1. $\deg(g\circ f) = \deg(g)\,\deg(f)$.
2. Reversing the orientation of M (or of N) negates the degree: with $-M$ denoting M reoriented, $\deg(f\colon -M \to N) = -\deg f$.
3. If f is not surjective, then $\deg f = 0$.
4. $\deg \mathrm{id}_N = 1$.

***Proof*:**
(1). By theorem 7.2.1 choose $z \in P$ that is a regular value of both g and $g\circ f$. Write $g^{-1}(z) = \{y_1, \dots, y_m\}$. If $x \in (g\circ f)^{-1}(z)$ then $d(g\circ f)_x = dg_{f(x)}\circ df_x$ is invertible, forcing df_x invertible; thus every y_j is a regular value of f, and $(g\circ f)^{-1}(z) = \bigsqcup_j f^{-1}(y_j)$. Now

$$\deg(g\circ f) = \sum_j \sum_{x\in f^{-1}(y_j)} \operatorname{sign}\big(\det dg_{y_j}\cdot\det df_x\big) = \sum_j \operatorname{sign}(\det dg_{y_j}) \sum_{x\in f^{-1}(y_j)} \operatorname{sign}\det df_x$$

The inner sum is $\deg(f, y_j) = \deg f$ by theorem 8.4.4, so this equals $\big(\sum_j \operatorname{sign}\det dg_{y_j}\big)\deg f = \deg(g)\,\deg(f)$.

(2). Reversing the orientation of M reverses the ambient frame at every x, hence flips each $\operatorname{sign}\det df_x$; reversing that of N has the same effect. Either negates the whole sum.

(3). If $y \notin \operatorname{im} f$, then y is vacuously a regular value with $f^{-1}(y) = \emptyset$, so $\deg(f, y) = 0$; by theorem 8.4.4, $\deg f = 0$.

(4). Immediate from example 8.4.2, completing the proof.

Property (3) is the one exploited most often: a map of nonzero degree is forced to hit every value, so degree computations are existence proofs in disguise. Property (1) makes the degree a homomorphism from composition to multiplication, and it computes the antipodal map at a stroke.

Example 8.4.6: Degree of the Antipodal Map

The antipodal map $a\colon S^n \to S^n$, $a(x) = -x$, is the restriction of $-I$ on $\mathbb{R}^{n+1}$, which factors as the product of the $n+1$ coordinate reflections $r_0, r_1, \ldots, r_n$, where r_i negates the i-th coordinate. Each r_i has degree -1 by the computation in example 8.4.2, so by proposition 8.4.5(1),

$$\deg a = (\deg r_0)(\deg r_1)\cdots(\deg r_n) = (-1)^{n+1}$$

Thus the antipodal map is orientation-preserving on odd-dimensional spheres and orientation-reversing on even-dimensional ones. This parity is the seed of the hairy ball phenomenon: on S^{2n} the antipodal map is not homotopic to the identity, since their degrees -1 and $+1$ differ.

The first payoff is algebraic. A complex polynomial extends to a self-map of the sphere, and its degree there is visibly its algebraic degree, so a map of positive degree cannot avoid the value 0.

Theorem 8.4.7: Fundamental Theorem of Algebra

Every complex polynomial $p(z) = a_k z^k + a_{k-1} z^{k-1} + \cdots + a_0$ with $k \geq 1$ and $a_k \neq 0$ has a root in $\mathbb{C}$.

***Proof*:**

Identify S^2 with the one-point compactification $\mathbb{C} \cup \{\infty\} = \mathcal{P}^1$, using the two stereographic charts z and $w = 1/z$. The polynomial p defines a map $P\colon S^2 \to S^2$ with $P(\infty) = \infty$: in the chart w near ∞, writing the target coordinate $\zeta = 1/P$,

$$\zeta = \frac{w^k}{a_k + a_{k-1}w + \cdots + a_0 w^k}$$

which is smooth at $w = 0$ because the denominator equals $a_k \neq 0$ there. So P is smooth, and its degree is defined.

Step 1: Homotope to the leading term. For $t \in [0,1]$ set $p_t(z) = a_k z^k + t\,(a_{k-1}z^{k-1} + \cdots + a_0)$. Each p_t is a polynomial of degree exactly k with leading coefficient a_k, so it extends as above to a smooth map $P_t\colon S^2 \to S^2$ fixing ∞, and the same denominator computation shows $(z,t) \mapsto P_t(z)$ is a smooth homotopy. Since $P_1 = P$ and $P_0(z) = a_k z^k$, homotopy invariance (theorem 8.4.4) gives $\deg P = \deg P_0$.

Step 2: Compute the degree of $z \mapsto a_k z^k$. Take the regular value $w_0 = a_k$. The equation $a_k z^k = a_k$ reads $z^k = 1$, whose solutions are the k distinct k-th roots of unity (elementary in polar form, no appeal to the theorem being proved); the point ∞ is not a preimage since $P_0(\infty) = \infty$. At each root z, the map P_0 is holomorphic with derivative $k a_k z^{k-1} \neq 0$, and a nonzero complex-linear map has real determinant $|k a_k z^{k-1}|^2 > 0$, so dP_0 preserves orientation and each of the k preimages contributes $+1$. Hence $\deg P_0 = k$.

Therefore $\deg P = k \geq 1$, so P is surjective by proposition 8.4.5(3). In particular $0 \in \mathrm{im} P$, and since $P(\infty) = \infty \neq 0$ the point mapping to 0 lies in $\mathbb{C}$, giving $z_0 \in \mathbb{C}$ with $p(z_0) = 0$, completing the proof.

The proof says something sharper than existence: counted with multiplicity, p takes each generic value exactly k times, because the degree is k and every preimage sign is $+1$. Holomorphicity is doing the orientation bookkeeping for free, which is why the complex degree never cancels. The same surjectivity mechanism, run on a manifold rather than a polynomial, yields the Brouwer fixed-point theorem.

Lemma 8.4.8: No Smooth Retraction of the Disc onto its Boundary

There is no smooth map $\rho\colon D^n \to S^{n-1}$ with $\rho(x) = x$ for all $x \in S^{n-1} = \partial D^n$.

***Proof*:**
Suppose such a retraction ρ exists. Define $H\colon S^{n-1} \times [0,1] \to S^{n-1}$ by $H(x,t) = \rho(tx)$. This is smooth, and $H(x,1) = \rho(x) = x$ while $H(x,0) = \rho(0)$ is a constant. Thus $\mathrm{id}_{S^{n-1}}$ is homotopic to a constant map. But $\deg \mathrm{id}_{S^{n-1}} = 1$ (example 8.4.2) while a constant map is not surjective and so has degree 0 (proposition 8.4.5(3)); homotopy invariance (theorem 8.4.4) forces $1 = 0$, a contradiction. For $n = 1$ the target S^0 is disconnected: a retraction $\rho\colon D^1 \to S^0$ would fix the two endpoints ± 1, so its image contains both, yet $\rho(D^1)$ is connected as the continuous image of the interval and $\{\pm 1\}$ is not connected, again a contradiction. Hence no retraction exists, completing the proof.

Theorem 8.4.9: Brouwer Fixed-Point Theorem

Every smooth map $g\colon D^n \to D^n$ has a fixed point.

***Proof*:**
Suppose $g(x) \neq x$ for every $x \in D^n$. For each x, the ray starting at $g(x)$ and passing through x meets S^{n-1} in a unique point beyond x; let $\rho(x)$ be that point,

$$\rho(x) = x + s(x)\,\frac{x - g(x)}{|x - g(x)|}$$

where $s(x) \geq 0$ is the nonnegative root of $|\rho(x)|^2 = 1$. The discriminant of this quadratic in s is strictly positive (since $|x| \leq 1$ and the direction is a unit vector), so s is a smooth function of x, and ρ is smooth. If $x \in S^{n-1}$ then $|x| = 1$ gives $s(x) = 0$, so $\rho(x) = x$. Thus ρ is a smooth retraction $D^n \to S^{n-1}$, contradicting lemma 8.4.8. Therefore g has a fixed point, as we sought to show.

The disc has no interesting self-maps to speak of, yet it cannot be continuously retracted onto its edge, and that single obstruction pins every self-map to a fixed point. The smoothness hypothesis is removable: a continuous self-map is uniformly approximated by a smooth one (theorem 7.5.7 and the approximation theorems of chapter 7), and a limit of near-fixed-points is a fixed point. The degree has now proved two classical theorems by the same move, non-vanishing forces surjectivity. The Hopf theorem asserts the converse organizing principle for spheres: the degree is a complete invariant.

Theorem 8.4.10: Hopf Degree Theorem

Two smooth maps $f, g\colon S^n \to S^n$ are smoothly homotopic if and only if $\deg f = \deg g$. Consequently the degree is a bijection from the set of homotopy classes of self-maps of S^n to $\mathbb{Z}$.

***Proof*:**
The forward implication is homotopy invariance (theorem 8.4.4). For the converse, the case $n = 1$ admits a self-contained argument, and the general case is deferred to the framed cobordism construction of chapter 10, whose machinery is not yet available.

The case $n = 1$. Write $S^1 = \mathbb{R}/\mathbb{Z}$ with covering map $\pi\colon \mathbb{R} \to S^1$, $\pi(s) = e^{2\pi i s}$. For a smooth $f\colon S^1 \to S^1$, the composite $f \circ \pi\colon \mathbb{R} \to S^1$ has simply connected domain, so it lifts to a smooth $F\colon \mathbb{R} \to \mathbb{R}$ with $\pi \circ F = f \circ \pi$. Since $\pi(F(s + 1)) = f(\pi(s)) = \pi(F(s))$, the difference $F(s + 1) - F(s)$ is a continuous integer, hence a constant $d \in \mathbb{Z}$. Counting the signed solutions of $F(s) \equiv a \pmod 1$ for a regular value $\pi(a)$ shows $\deg f = F(1) - F(0) = d$: as s runs over $[0, 1]$, F climbs from $F(0)$ to $F(0) + d$, crossing each level $a + \mathbb{Z}$ a net d times with sign $\operatorname{sign} F'$.

Now suppose $\deg f = \deg g = d$, with lifts F, G satisfying $F(s + 1) = F(s) + d$ and $G(s + 1) = G(s) + d$. The straight-line homotopy $F_u = (1 - u)F + uG$ satisfies $F_u(s + 1) = F_u(s) + d$, so each F_u descends through π to a smooth map $f_u\colon S^1 \to S^1$; this is a homotopy from f to g.

The general case. The complete argument is the Pontryagin framed cobordism correspondence developed in chapter 10: a regular value of f has a finite preimage which, framed by the derivative of f, represents a framed cobordism class, and $\deg f$ is exactly that class in the framed cobordism group of a point, $\mathbb{Z}$. Two maps of equal degree have framed-cobordant preimages, and a framed cobordism between them is precisely a homotopy. The verification of this dictionary requires the tubular-neighbourhood surgery of that construction, so it is deferred there rather than sketched here.

For S^1 the argument is concrete: the degree is the number of times the circle wraps around itself, read off from the lift, and two maps that wrap the same number of times are slid into one another by interpolating their lifts. The general statement identifies $\pi_n(S^n)$ with $\mathbb{Z}$ by the degree, the differential-topological counterpart of the Hurewicz isomorphism $\pi_n(S^n) \cong \mathbb{Z}$ of algebraic topology ([7]); the two agree because both count preimages of a point. That the degree classifies self-maps of spheres is what makes it the natural target for obstruction and index arguments, the vector-field index of section 8.6 among them.

The final construction of the section is a degree attached not to a single map but to a pair of disjoint submanifolds. When their dimensions are one short of complementary in the ambient space, the unit vector pointing from one to the other sweeps out a self-map of a sphere, and its degree measures how the two are intertwined.

Definition 8.4.11: Linking Number

Let X and Y be disjoint compact oriented boundaryless submanifolds of $\mathbb{R}^n$ with $\dim X + \dim Y = n-1$. The *Gauss map* (or direction map)

$$u\colon X \times Y \to S^{n-1}, \qquad u(x,y) = \frac{y-x}{|y-x|}$$

is well defined because $X \cap Y = \emptyset$ forces $y \neq x$. Its domain $X \times Y$ is a compact oriented manifold (product orientation) of dimension $\dim X + \dim Y = n-1 = \dim S^{n-1}$, so the *linking number* is

$$\operatorname{lk}(X,Y) = \deg u \ \in \ \mathbb{Z}$$

Because the degree is a homotopy invariant, $\operatorname{lk}(X,Y)$ is unchanged when X and Y are moved by any isotopy of $\mathbb{R}^n$ that keeps them disjoint: deforming X or Y deforms u through Gauss maps, and the degree cannot jump. The linking number is therefore an invariant of the isotopy class of the disjoint pair, the simplest numerical measure of entanglement. The Hopf link and its unlinked counterpart show it is nonzero.

Example 8.4.12: The Hopf Link

Work in $\mathbb{R}^3$, so $n=3$ and $\dim X = \dim Y = 1$. Take the unit circle in the xy-plane and a unit circle in the xz-plane centred at $(1,0,0)$,

$$X = \{(\cos s, \sin s, 0) \mid s \in \mathbb{R}/2\pi\mathbb{Z}\}, \qquad Y = \{(1+\cos t, 0, \sin t) \mid t \in \mathbb{R}/2\pi\mathbb{Z}\}$$

which are disjoint. To find the preimage under u of the direction $v = (0,0,1)$, solve $Y(t) - X(s) = \lambda v$ with $\lambda > 0$. The y-coordinate gives $\sin s = 0$, so $s \in \{0, \pi\}$; the x-coordinate gives $1 + \cos t = \cos s$. For $s = \pi$ this forces $\cos t = -2$, impossible; for $s = 0$ it forces $\cos t = 0$, and then $\lambda = \sin t > 0$ selects $t = \pi/2$. So $u^{-1}(v) = \{(0, \pi/2)\}$ is a single point, at which a direct check shows du is an isomorphism, so v is a regular value and $|\operatorname{lk}(X,Y)| = 1$.

By contrast, place two small circles in disjoint balls about $(-2,0,0)$ and $(2,0,0)$. Every difference $y - x$ then has positive first coordinate, so u misses the open hemisphere $\{v_0 < 0\}$ of S^2; being non-surjective, u has degree 0 (proposition 8.4.5(3)), so this unlinked pair has $\operatorname{lk} = 0$. Since the linking number is an isotopy invariant, the two configurations cannot be deformed one into the other through disjoint pairs: the Hopf link does not come apart.

The direction u hides a more geometric description: $\operatorname{lk}(X,Y)$ equals the signed number of times Y crosses any compact oriented surface bounded by X, as the Hopf example illustrates with Y piercing the disc spanning X exactly once. The two viewpoints are reconciled by the extension principle behind homotopy invariance, that a Gauss map extending over a bounding manifold has degree zero. The linking number is also symmetric up to a sign fixed by the ambient dimension.

Proposition 8.4.13: Symmetry of the Linking Number

For disjoint compact oriented submanifolds $X, Y \subseteq \mathbb{R}^n$ with $\dim X + \dim Y = n - 1$,

$$\operatorname{lk}(Y, X) = (-1)^{n+(\dim X)(\dim Y)} \operatorname{lk}(X, Y)$$

In particular, for two disjoint circles in $\mathbb{R}^3$ the linking number is symmetric, $\operatorname{lk}(Y, X) = \operatorname{lk}(X, Y)$.

***Proof*:**
The Gauss map for (Y, X) is $u_{Y,X}(y, x) = (x-y)/|x-y| = -\,u_{X,Y}(x, y)$. Thus $u_{Y,X} = \alpha \circ u_{X,Y} \circ \sigma$, where $\sigma\colon Y \times X \to X \times Y$ is the swap and $\alpha\colon S^{n-1} \to S^{n-1}$ is the antipodal map. The swap has degree $(-1)^{(\dim X)(\dim Y)}$ (reordering the two blocks of an oriented product basis), and the antipodal map of S^{n-1} has degree $(-1)^n$ by example 8.4.6. By proposition 8.4.5(1), $\operatorname{lk}(Y, X) = \deg \alpha \cdot \deg u_{X,Y} \cdot \deg \sigma = (-1)^{n+(\dim X)(\dim Y)} \operatorname{lk}(X, Y)$. For $n = 3$ and $\dim X = \dim Y = 1$ the exponent is $3 + 1 = 4$, even, so the linking number is symmetric, completing the proof.

The linking number closes the circle of this section: the degree, born as an intersection count against a point, reappears as an invariant of two submanifolds against each other, computed by the direction map into a sphere. The same integer will return in section 8.6 as the index of a vector field, where the sphere is swept out not by the direction between two manifolds but by the direction of the field near one of its zeros.

Exercise 8.4.1

1. Compute the degree of the power map $f\colon S^1 \to S^1$, $f(z) = z^k$ (for $k \in \mathbb{Z}$), directly from the definition by choosing a regular value and summing the signs. Confirm the answer against the lift $F(s) = ks$ under $\pi(s) = e^{2\pi i s}$.
2. Let $f\colon S^n \to S^n$ be smooth with no fixed point. Show that f is homotopic to the antipodal map, and deduce that any smooth self-map of S^n whose degree differs from $(-1)^{n+1}$ has a fixed point.
3. Show that a smooth map $f\colon S^n \to S^n$ that factors through the equator, i.e. $f = \iota \circ g$ for some $g\colon S^n \to S^{n-1}$ and the inclusion $\iota\colon S^{n-1} \hookrightarrow S^n$, has degree 0.
4. Prove that if $\deg f \neq 0$ then f is surjective, and use this to give a one-line proof that a polynomial map $p\colon \mathcal{P}^1 \to \mathcal{P}^1$ of positive degree is onto.
5. Using the Hopf degree theorem for $n = 1$, show that there are infinitely many homotopy classes of smooth maps $S^1 \to S^1$, and that a map of odd degree is not homotopic to any map of even degree.
6. Let $X, Y \subseteq \mathbb{R}^n$ be disjoint compact oriented submanifolds with $\dim X + \dim Y = n - 1$. Show that if $X = \partial W$ for a compact oriented submanifold $W \subseteq \mathbb{R}^n$ (with boundary) disjoint from Y, then $\operatorname{lk}(X, Y) = 0$.

8.5 The Lefschetz Fixed-Point Theorem

The oriented intersection number of the previous sections counts how two submanifolds, or a map and a submanifold, meet. Applied to a single self-map $f: M \to M$ it answers a question with no intersections in sight: does f return some point to itself? The device is to replace f by a pair of submanifolds of $M \times M$ whose meetings are exactly the fixed points of f, the diagonal and the graph, and to read the fixed-point question off their oriented intersection number. That number is a homotopy invariant, it obstructs fixed-point-free maps, and for a map whose fixed points are nondegenerate it is a signed count computed one point at a time.

Throughout, M is a compact oriented n-manifold without boundary and $f: M \to M$ is smooth. The product $M \times M$ is a $2n$-manifold, oriented by the product orientation. Two compact n-dimensional submanifolds sit inside it: the *diagonal* $\Delta = \{(x, x) \mid x \in M\}$ and the *graph* $\Gamma_f = \{(x, f(x)) \mid x \in M\}$. Each is the image of an embedding of M, namely $x \mapsto (x, x)$ and $x \mapsto (x, f(x))$, and each is oriented by transporting the orientation of M along that embedding. Their dimensions are complementary, $n + n = 2n$, so the oriented intersection number $I(\Delta, \Gamma_f)$ of section 8.3 is defined.

Definition 8.5.1: Lefschetz Number

The *Lefschetz number* of a smooth map $f: M \to M$ on a compact oriented manifold is

$$L(f) = I(\Delta, \Gamma_f)$$

the oriented intersection number in $M \times M$ of the diagonal Δ with the graph Γ_f.

Because oriented intersection numbers are homotopy invariant (section 8.3) and a homotopy f_t drags the graph Γ_{f_t} along with it, $L(f)$ depends only on the homotopy class of f. It is an integer attached to f that cannot change as f is deformed.

The whole construction is engineered around one incidence.

Proposition 8.5.2: Fixed Points as an Intersection

The assignment $x \mapsto (x, x)$ is a bijection from the fixed-point set of f onto $\Delta \cap \Gamma_f$. In particular Δ and Γ_f meet if and only if f has a fixed point.

***Proof*:**
A point of Δ has the form (x, x) and a point of Γ_f has the form $(x', f(x'))$. They coincide precisely when $x = x'$ and $x = f(x') = f(x)$, that is, when $f(x) = x$. The assignment $x \mapsto (x, x)$ therefore sends fixed points bijectively to $\Delta \cap \Gamma_f$, completing the proof.

The oriented intersection number vanishes when the two submanifolds are disjoint, so a nonzero Lefschetz number forces Δ and Γ_f to meet.

Theorem 8.5.3: Lefschetz Fixed-Point Theorem

Let M be compact and oriented and $f: M \to M$ smooth. If $L(f) \neq 0$, then f has a fixed point.

***Proof*:**
Suppose f has no fixed point. By proposition 8.5.2 the diagonal and the graph are then disjoint, $\Delta \cap \Gamma_f = \emptyset$. Disjoint submanifolds are vacuously transverse, and the oriented intersection number is the signed count of intersection points (section 8.3), an empty sum. Hence $L(f) = I(\Delta, \Gamma_f) = 0$. The contrapositive is the assertion, as we sought to show.

The theorem converts a fixed-point problem into the computation of an integer insensitive to deformation. Where the Brouwer fixed-point theorem (section 8.4) forced a fixed point on the disc from the impossibility of retracting it to its boundary, the Lefschetz number performs the same service for a closed manifold: a single nonzero invariant rules out every fixed-point-free representative of the homotopy class. The converse fails, a map with a fixed point can still have $L(f) = 0$, but the obstruction is often computable when the map itself is not. The antipodal map $a\colon S^2 \to S^2$, $a(x) = -x$, has no fixed point, since $-x = x$ has no solution on the sphere; proposition 8.5.2 makes $\Delta \cap \Gamma_a$ empty, so $L(a) = 0$.

The map that meets itself everywhere is the identity, and its Lefschetz number is forced to be an intersection number of the diagonal with itself.

Corollary 8.5.4: The Identity Map

For the identity map, $\Gamma_{\mathrm{id}_M} = \Delta$, so

$$L(\mathrm{id}_M) = I(\Delta, \Delta)$$

the self-intersection number of the diagonal.

***Proof*:**
The graph of the identity is $\{(x, x) \mid x \in M\} = \Delta$, and the orientation transported along $x \mapsto (x, \mathrm{id}_M(x)) = (x, x)$ is the orientation of Δ. Thus $L(\mathrm{id}_M) = I(\Delta, \Gamma_{\mathrm{id}_M}) = I(\Delta, \Delta)$, as we sought to show.

The self-intersection number $I(\Delta, \Delta)$ is the integer that section 8.6 adopts as the definition of the *Euler characteristic* $\chi(M)$; the identity map is the bridge, $L(\mathrm{id}_M) = \chi(M)$. Homotopy invariance then reads: every map homotopic to the identity shares the Lefschetz number $\chi(M)$, so if $\chi(M) \neq 0$, every such map has a fixed point. The computation in example 8.5.10 below gives $\chi(S^2) = 2$, and the antipodal map, with $L(a) = 0 \neq 2$, cannot be homotopic to the identity on S^2.

To turn $L(f)$ from an obstruction into a number one can compute, the fixed points must be countable and each must carry a well-defined sign. Both hold when the graph meets the diagonal transversally, and transversality has a linear-algebra criterion at a fixed point.

Definition 8.5.5: Lefschetz Fixed Point

A fixed point p of $f\colon M \to M$ is a *Lefschetz fixed point* (or *nondegenerate*) if the differential $df_p\colon T_pM \to T_pM$ has no eigenvalue equal to 1, equivalently if $df_p - \mathrm{id}$ is invertible. The map f is a *Lefschetz map* if all its fixed points are Lefschetz.

At a fixed point $f(p) = p$, so df_p is an endomorphism of T_pM and the eigenvalue condition makes sense. It is exactly transversality in $M \times M$.

Proposition 8.5.6: Transversality of the Graph and the Diagonal

Let p be a fixed point of f. Under the canonical identification $T_{(p,p)}(M \times M) = T_pM \oplus T_pM$,

$$T_{(p,p)}\Delta = \{(v, v) \mid v \in T_pM\}, \qquad T_{(p,p)}\Gamma_f = \{(v, df_pv) \mid v \in T_pM\}$$

and Γ_f is transverse to Δ at (p, p) if and only if p is a Lefschetz fixed point.

***Proof*:**
The diagonal is the image of $x \mapsto (x, x)$, whose differential at p is $v \mapsto (v, v)$; the graph is the image of $x \mapsto (x, f(x))$, whose differential at p is $v \mapsto (v, df_pv)$. This gives the two tangent spaces displayed, each of dimension n.
Transversality at (p, p) means $T_{(p,p)}\Delta + T_{(p,p)}\Gamma_f = T_pM \oplus T_pM$. The two summands have dimensions adding to $n + n = 2n = \dim(T_pM \oplus T_pM)$, so the sum is everything if and only if it is direct, that is, if and only if $T_{(p,p)}\Delta \cap T_{(p,p)}\Gamma_f = 0$. A nonzero vector in the intersection is a pair $(v, v) = (v, df_pv)$ with $v \neq 0$, which forces $df_pv = v$: a nonzero 1-eigenvector of df_p. Thus the intersection is nonzero exactly when 1 is an eigenvalue of df_p, and transversality holds exactly when it is not, which is the definition of a Lefschetz fixed point (definition 8.5.5), completing the proof.

Transversality of complementary-dimensional submanifolds makes their intersection a 0-manifold, so the fixed points of a Lefschetz map are isolated; compactness of M, hence of Δ, then makes them finite in number. Each contributes to $I(\Delta, \Gamma_f)$ its local intersection sign, and that sign is computed by the same matrix that detected transversality.

Definition 8.5.7: Local Lefschetz Number

The *local Lefschetz number* of f at a Lefschetz fixed point p is

$$L_p(f) = \operatorname{sign} \det(df_p - \mathrm{id}) \in \{+1, -1\}$$

The determinant is that of an endomorphism of T_pM, so $L_p(f)$ needs no choice of basis or orientation of the tangent space; it is intrinsic to f at p. The next lemma identifies it with the oriented intersection sign that $I(\Delta, \Gamma_f)$ assigns to (p, p), which is where the orientation of M enters.

Lemma 8.5.8: The Local Intersection Sign

Let p be a Lefschetz fixed point of f. The oriented intersection sign of Δ and Γ_f at (p, p), taken in the order (Δ, Γ_f), equals $L_p(f) = \operatorname{sign} \det(df_p - \mathrm{id})$.

***Proof*:**
Fix a positively oriented basis $e_1, \ldots, e_n$ of T_pM, and let A be the matrix of df_p in this basis, with I the $n \times n$ identity matrix. The product orientation of $T_pM \oplus T_pM$ has the positively oriented basis $(e_1, 0), \ldots, (e_n, 0), (0, e_1), \ldots, (0, e_n)$. By proposition 8.5.6, a positively oriented basis of $T_{(p,p)}\Delta$ is $(e_1, e_1), \ldots, (e_n, e_n)$ and a positively oriented basis of $T_{(p,p)}\Gamma_f$ is $(e_1, df_pe_1), \ldots, (e_n, df_pe_n)$, these being the images of $e_1, \ldots, e_n$ under the orienting embeddings. The intersection sign in the order (Δ, Γ_f) is the sign of the determinant of the frame obtained by

concatenating these two bases, expressed in the product-orientation basis (section 8.3). Written in $n \times n$ blocks, that frame is

$$\begin{pmatrix} I & I \\ I & A \end{pmatrix}$$

whose first block column lists the Δ vectors (e_i, e_i) and whose second lists the Γ_f vectors (e_i, Ae_i). Subtracting the first block row from the second leaves $\begin{pmatrix} I & I \\ 0 & A - I \end{pmatrix}$, whose determinant is $\det(A - I) = \det(df_p - \mathrm{id})$. Its sign is the intersection sign, as we sought to show.

Summing the local signs over the finitely many fixed points recovers the global invariant.

Theorem 8.5.9: The Lefschetz-Hopf Theorem

Let M be compact and oriented and let $f\colon M \to M$ be a Lefschetz map. Then f has finitely many fixed points, and

$$L(f) = \sum_{p\,:\,f(p)=p} \operatorname{sign}\det(df_p - \mathrm{id}) = \sum_{p\,:\,f(p)=p} L_p(f)$$

***Proof*:**
By proposition 8.5.6, f being a Lefschetz map means $\Gamma_f \pitchfork \Delta$ at every point of $\Delta \cap \Gamma_f$. Two complementary-dimensional submanifolds meeting transversally do so in a 0-manifold; as a closed subset of the compact manifold Δ, the intersection $\Delta \cap \Gamma_f$ is compact, hence finite. Proposition 8.5.2 identifies it with the fixed-point set of f, which is therefore finite.
For a transverse intersection of complementary dimension, the oriented intersection number is the sum of the local intersection signs over the intersection points (section 8.3):

$$L(f) = I(\Delta, \Gamma_f) = \sum_{(p,p) \in \Delta \cap \Gamma_f} (\text{sign at } (p,p))$$

By lemma 8.5.8 the sign at (p,p) is $\operatorname{sign}\det(df_p - \mathrm{id}) = L_p(f)$. Summing over the fixed points gives the stated formula, completing the proof.

The theorem is the fixed-point analogue of counting the preimages of a regular value: a global homotopy invariant is pinned down by purely local data, the linearization of f at each fixed point. The identity map is excluded, since every point is fixed and none is nondegenerate, which is exactly why computing $\chi(M) = L(\mathrm{id}_M)$ requires perturbing the identity to a Lefschetz map first, the route through vector fields that section 8.6 takes.

Example 8.5.10: Rotations of the Sphere
Let $R_\theta\colon S^2 \to S^2$ be the rotation by a fixed angle $\theta \notin 2\pi\mathbb{Z}$ about the vertical axis. Its fixed points are the two poles N and S, where the axis meets the sphere. At each pole the differential is the rotation by θ of the tangent plane, so in an orthonormal basis

$$df_N = df_S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Hence
$$\det(df_p - \mathrm{id}) = (\cos\theta - 1)^2 + \sin^2\theta = 2(1-\cos\theta) > 0$$
for $\theta \notin 2\pi\mathbb{Z}$, so both poles are Lefschetz fixed points with local number $L_N(R_\theta) = L_S(R_\theta) = +1$. By theorem 8.5.9,
$$L(R_\theta) = 1 + 1 = 2$$
Each R_θ is homotopic to id_{S^2} through the rotations $R_{s\theta}$, $s \in [0,1]$, so homotopy invariance gives $L(\mathrm{id}_{S^2}) = 2$. By corollary 8.5.4 this is the self-intersection number of the diagonal, so $\chi(S^2) = I(\Delta, \Delta) = 2$.

The computation recovers a fact about the sphere from data at two points, and it exposes the role of degeneracy: as $\theta \to 0$ the two fixed points remain the poles, but $\det(df_p - \mathrm{id}) = 2(1-\cos\theta) \to 0$, so nondegeneracy is lost in the limit, where the identity is degenerate at every point. The invariant $L = 2$ survives the degeneration.

The identity is not the only degenerate map, but degeneracy is never forced: an arbitrarily small perturbation makes every fixed point nondegenerate.

Proposition 8.5.11: Genericity of Lefschetz Maps

Every smooth map $f\colon M \to M$ on a compact manifold is homotopic to a Lefschetz map. Consequently $L(f)$ is always the finite signed count of theorem 8.5.9 for a suitable representative of its homotopy class.

***Proof*:**
Embed M as a closed submanifold of some $\mathbb{R}^N$ (Whitney, chapter 7), and let $\pi\colon U \to M$ be the retraction of a tubular neighbourhood $U \subseteq \mathbb{R}^N$ of M (theorem 7.8.1); it is a submersion with $\pi|_M = \mathrm{id}_M$. Since M is compact, there is an open ball $B \subseteq \mathbb{R}^N$ about 0 with $f(x) + a \in U$ for all $x \in M$ and $a \in B$. Define
$$F\colon M \times B \to M \times M, \qquad F(x,a) = \big(x,\ \pi(f(x)+a)\big)$$
and write $f_a = \pi(f(\cdot)+a)\colon M \to M$, so that $F(\cdot, a) = \gamma_{f_a}$ is the graph map of f_a and $f_0 = f$.
Step 1: F is transverse to Δ. Fix (x,a) with $F(x,a) \in \Delta$; then $f_a(x) = x$. Holding x fixed and varying a, the differential of $a \mapsto \pi(f(x)+a)$ is $d\pi_{f(x)+a}$ restricted to the $\mathbb{R}^N$ directions, which is surjective onto T_xM because π is a submersion; these directions contribute all of $\{0\} \oplus T_xM$ to the image of $dF_{(x,a)}$. Holding a fixed and varying x contributes vectors of the form $(w, *)$ with w ranging over T_xM. Subtracting a vertical vector from such a $(w,*)$ yields $(w,0)$, so the image of $dF_{(x,a)}$ contains both $\{0\} \oplus T_xM$ and $T_xM \oplus \{0\}$, hence is all of $T_xM \oplus T_xM$. Thus F is a submersion at (x,a), and in particular $F \pitchfork \Delta$.
Step 2: perturb. By the parametric transversality theorem (theorem 7.5.6), for almost every $a \in B$ the map $F(\cdot, a) = \gamma_{f_a}$ is transverse to Δ. Choose such an a; then $\Gamma_{f_a} \pitchfork \Delta$, so by proposition 8.5.6 every fixed point of f_a is Lefschetz, that is, f_a is a Lefschetz map. The straight-line homotopy $t \mapsto \pi(f(\cdot)+ta)$, $t \in [0,1]$, connects $f_0 = f$ to f_a within the smooth maps $M \to M$, so $f \simeq f_a$. Homotopy invariance of L together with theorem 8.5.9 gives the final clause, completing the proof.

Every homotopy class of self-map thus has a representative whose Lefschetz number is a genuine count, and the count is the same across the class. The invariant carries no more information than the homotopy class, but it extracts from that class a computable integer.

8.5.12 Remark: The Homological Reading A second Lefschetz number is built from the action of f on homology rather than from intersection theory, as the alternating sum $\sum_i (-1)^i \operatorname{tr}\big(f_*\colon H_i(M) \to H_i(M)\big)$ of the traces of the induced maps. The homological Lefschetz fixed-point theorem of [7] evaluates it, for a Lefschetz map, as a signed count over the fixed points of f, so it vanishes exactly when $I(\Delta, \Gamma_f)$ does and carries the same fixed-point obstruction. The two are not identical, however. The homological count sums the fixed-point index sign $\det(\mathrm{id} - df_p)$, whereas lemma 8.5.8 computes the intersection sign in the order (Δ, Γ_f) as $\operatorname{sign}\det(df_p - \mathrm{id})$; since $\det(\mathrm{id} - df_p) = (-1)^n \det(df_p - \mathrm{id})$, these differ by a global sign set by the dimension:

$$I(\Delta, \Gamma_f) = (-1)^n \sum_i (-1)^i \operatorname{tr}(f_*)$$

The factor $(-1)^n$ is the price of ordering the diagonal before the graph and is invisible for even n. For $f = \mathrm{id}_M$ the maps f_* are the identity, their traces are the Betti numbers b_i, and the alternating sum is $\sum_i (-1)^i b_i$; a closed odd-dimensional manifold has vanishing Euler characteristic, so the sign drops out and this is the homological form of the identity $\chi(M) = I(\Delta, \Delta)$ recorded in section 8.6.

Exercise 8.5.1

1. Let M be compact and oriented and suppose $f\colon M \to M$ is homotopic to a map with no fixed points. Show that $L(f) = 0$. Deduce that the antipodal map $a\colon S^2 \to S^2$, $a(x) = -x$, is not homotopic to the identity.
2. Let M be compact and oriented of dimension n, and let $f \equiv q$ be a constant map. Show that q is the only fixed point, that it is a Lefschetz fixed point, and that $L(f) = (-1)^n$. In particular $L(f) = 1$ when M is even-dimensional.
3. A matrix $A \in M_2(\mathbb{Z})$ with $\det(A - I) \neq 0$ induces a smooth map $f_A\colon T^2 \to T^2$ on the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, $f_A(x) = Ax \bmod \mathbb{Z}^2$. Show that every fixed point of f_A is Lefschetz, that f_A has exactly $|\det(A - I)|$ fixed points, and that $L(f_A) = \det(A - I)$. Evaluate this for the "cat map" $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.
4. Let p be a Lefschetz fixed point of f, and let k be the number of real eigenvalues of df_p (counted with algebraic multiplicity) that are less than 1. Show that $L_p(f) = (-1)^k$.
5. Let M be compact and oriented and $f \simeq \mathrm{id}_M$. Show that $L(f) = I(\Delta, \Delta) = \chi(M)$. Exhibit a fixed-point-free self-map of the torus T^2 that is homotopic to the identity, and explain why this is consistent with the Lefschetz fixed-point theorem.

8.6 Vector Fields and the Poincaré-Hopf Theorem

Oriented intersection theory attached a signed integer to a transverse pair of complementary submanifolds, and the sections before this one turned that integer into the degree of a map and the Lefschetz number of a self-map. One transverse pair is left to count. A vector field is a section of the tangent bundle, and it meets the zero section exactly at its zeros; counting those zeros with signs produces a local invariant, the *index*, while summing the indices over the manifold produces a single global integer. The Poincaré-Hopf theorem identifies that sum with the self-intersection number of the diagonal, the Euler characteristic, and so ties the local behaviour of a field near its equilibria to the global topology of the space it lives on. The hairy ball theorem and a topological Gauss-Bonnet formula fall out at once.

Throughout, M is a compact boundaryless manifold and, where signs are counted, oriented (per the standing conventions of this chapter).

A vector field v vanishes at its *equilibria*. Away from a zero the normalized field $v/|v|$ is a well-defined unit vector, and as one circles a small sphere around an isolated zero this unit vector sweeps out a map of spheres. The number of times it wraps around is the local invariant this section is built on.

Definition 8.6.1: Index of an Isolated Zero

Let v be a smooth vector field on an n-manifold M with an *isolated zero* at p: that is, $v(p) = 0$ and v has no other zero on some neighbourhood of p. Choose a chart (U, φ) with $\varphi(p) = 0$; the coordinate frame $\partial/\partial x^1, \dots, \partial/\partial x^n$ trivializes TM over U and expresses v as a map $\tilde{v} \colon \varphi(U) \to \mathbb{R}^n$ with $\tilde{v}(0) = 0$ and no other zero on a closed ball $\overline{B}_\varepsilon$. The *local Gauss map*

$$g_\varepsilon \colon S^{n-1}_\varepsilon \to S^{n-1}, \qquad g_\varepsilon(x) = \frac{\tilde{v}(x)}{|\tilde{v}(x)|}$$

is a smooth map between compact oriented manifolds of equal dimension with connected target, so it has a degree (section 8.4). The *index* of v at p is

$$\operatorname{ind}_p(v) := \deg(g_\varepsilon) \in \mathbb{Z}$$

The index measures the winding of the direction field around the zero, and the sign records whether the field turns the same way as, or opposite to, the standard sphere. The definition names two choices, the chart and the radius; the next proposition shows the index depends on neither, so it is an invariant of the field near the point.

Before proving that, here are the indices that occur for planar fields, the pictures worth carrying.

Example 8.6.2: Indices of Planar Vector Fields

Identify $\mathbb{R}^2$ with $\mathbb{C}$, so a planar field is a map $\mathbb{C} \to \mathbb{C}$ with an isolated zero at the origin, and its index is the degree of the induced self-map of S^1.

1. *Source* $v(x, y) = (x, y)$, that is $v(z) = z$. The Gauss map is the identity, of degree $+1$, so $\operatorname{ind}_0(v) = +1$.
2. *Sink* $v(x, y) = (-x, -y)$, that is $v(z) = -z$. The Gauss map is the antipodal map of S^1, of degree $(-1)^2 = +1$, so $\operatorname{ind}_0(v) = +1$.
3. *Saddle* $v(x, y) = (x, -y)$. The linear part has determinant -1; the Gauss map has degree -1, so $\operatorname{ind}_0(v) = -1$.
4. *Rotation* $v(x, y) = (-y, x)$, that is $v(z) = iz$. The determinant is $+1$ and $\operatorname{ind}_0(v) = +1$.
5. *Higher order* $v(z) = z^k$ for $k \geq 1$. On S^1 this is $\theta \mapsto k\theta$, of degree k, so $\operatorname{ind}_0(v) = k$; the conjugate field $v(z) = \bar{z}^{\,k}$ has index $-k$.

The first four have determinant ± 1 and their index is the sign of that determinant; the last shows that a degenerate zero can carry any integer index.

The pattern in the first four items is general and is the computational heart of the whole theory. Say the zero p is *nondegenerate* if the *linearization* $Dv_p \colon T_pM \to T_pM$ is invertible. This linearization is well-defined at a zero: since $v(p) = 0$, a change of coordinates multiplies the first-order term of v by conjugation, leaving $\det(Dv_p)$ and its sign unambiguous.

Proposition 8.6.3: The Index Is Well-Defined

The index $\mathrm{ind}_p(v)$ of an isolated zero does not depend on the radius ε or on the chart used to compute it. If the zero is nondegenerate, then

$$\mathrm{ind}_p(v) = \operatorname{sign}\det(Dv_p)$$

***Proof*:**
Work in the chart, so $\tilde{v} \colon \overline{B}_\varepsilon \to \mathbb{R}^n$ has 0 as its only zero.

Step 1: Independence of the radius. For $0 < r \le \varepsilon$ the field $\tilde{v}$ is nonzero on the closed annulus $r \le |x| \le \varepsilon$. Pulling each sphere back to the unit sphere by $x \mapsto \rho x$, the maps $x \mapsto \tilde{v}(\rho x)/|\tilde{v}(\rho x)|$ for $\rho \in [r, \varepsilon]$ give a homotopy $S^{n-1} \to S^{n-1}$ between the normalized fields on the radius-r and radius-ε spheres. Since $x \mapsto \rho x$ preserves orientation, these represent g_r and g_ε; homotopic maps have equal degree, so $\deg(g_r) = \deg(g_\varepsilon)$.

Step 2: Independence of the chart. A change of chart carries $\tilde{v}$ to its pushforward $w = Dg \circ \tilde{v} \circ g^{-1}$ under a diffeomorphism g of neighbourhoods of 0 in $\mathbb{R}^n$ fixing 0, and it suffices to show $\mathrm{ind}_0(w) = \mathrm{ind}_0(\tilde{v})$. Set $A = Dg_0$ and $g_s(x) = g(sx)/s$ for $s \in (0, 1]$, with $g_0 := A$; this is a smooth family of diffeomorphisms fixing 0, interpolating from $g_1 = g$ to the linear map A. The pushforwards $w_s = (g_s)_*\tilde{v}$ form a continuous family, each with 0 as its only zero on a small ball; by compactness they are bounded away from 0 on a small enough sphere, so their Gauss maps are all homotopic and $\mathrm{ind}_0(w) = \mathrm{ind}_0(A_*\tilde{v})$. Finally A and, when g preserves orientation, the identity lie in the same path component of $\mathrm{GL}(n, \mathbb{R})$; a path through $\mathrm{GL}(n, \mathbb{R})$ from A to id homotopes $A_*\tilde{v}$ to $\tilde{v}$ without introducing zeros on a small sphere, giving $\mathrm{ind}_0(A_*\tilde{v}) = \mathrm{ind}_0(\tilde{v})$. If instead g reverses orientation, it reverses the orientations of both the domain sphere and the target sphere of the Gauss map at once, and the degree is unchanged; the index is therefore independent of the chart with no orientation hypothesis at all.

Step 3: The nondegenerate formula. If $A = Dv_p$ is invertible, then $\tilde{v}(x) = Ax + o(|x|)$, so on a small sphere g_ε is homotopic to $x \mapsto Ax/|Ax|$. The linear map $x \mapsto Ax$ carries S^{n-1} to an ellipsoid, and radial projection back to S^{n-1} has degree $+1$ if $\det A > 0$ and -1 if $\det A < 0$, since $\mathrm{GL}^+(n, \mathbb{R})$ is path-connected and every element of $\mathrm{GL}^-(n, \mathbb{R})$ is a reflection composed with an element of $\mathrm{GL}^+(n, \mathbb{R})$. Hence $\mathrm{ind}_p(v) = \operatorname{sign}\det(Dv_p)$, completing the proof.

Step 2 does more than tidy the definition: it shows the index makes sense on any manifold, orientable or not, because an orientation-reversing coordinate change flips the two spheres together. The index is thus a genuine local invariant of the field, and the nondegenerate formula reduces its computation to a determinant. The general index of a possibly degenerate zero is recovered from nondegenerate ones by perturbation, which is the mechanism the Poincaré-Hopf proof will exploit.

To sum indices into a statement, the target of that sum must be named. The right target is a single integer attached to M itself, and this chapter has already built the object that carries it: the self-intersection number of the diagonal.

Definition 8.6.4: Euler Characteristic

Let M be a compact oriented manifold and $\Delta \subseteq M \times M$ its diagonal. The *Euler characteristic* of M is the self-intersection number of the diagonal,

$$\chi(M) := I(\Delta, \Delta)$$

Equivalently, since the graph of the identity is the diagonal, $\Gamma_{\mathrm{id}_M} = \Delta$, it is the Lefschetz number of the identity map, $\chi(M) = L(\mathrm{id}_M)$ (section 8.3, section 8.5).

Defining $\chi(M)$ this way makes it visibly a homotopy-invariant integer, and one intrinsic to M: it needs no triangulation, no cell structure, and no metric, only the diagonal and the intersection pairing. The Lefschetz formula $\chi(M) = L(\mathrm{id}_M)$ is the special case of the fixed-point theory of the previous section where every point is a fixed point, so the local fixed-point indices of the identity, once Poincaré-Hopf is proved, will turn out to be exactly the indices of a vector field.

8.6.5 Remark: The Homological Euler Characteristic The reader who knows singular homology will recognize $\chi(M)$ as the alternating sum of Betti numbers,

$$\chi(M) = \sum_{i=0}^{\dim M} (-1)^i b_i, \qquad b_i = \operatorname{rank} H_i(M)$$

The identification of $I(\Delta, \Delta)$ with this sum runs through the Künneth theorem and Poincaré duality and is carried out in [7]; it is not needed here, where the intersection-theoretic definition is self-contained. Because the index sum below is defined from local charts alone, both the definition of χ and the Poincaré-Hopf theorem extend verbatim to compact manifolds that are not orientable, where the homological sum (with field coefficients) is the invariant they compute.

The circle is the first check, and it can be run directly from the definition.

Example 8.6.6: The Euler Characteristic of the Circle

For $M = S^1$ the diagonal $\Delta \subseteq S^1 \times S^1 = T^2$ is the $(1,1)$-curve $\{(\theta, \theta)\}$. Push it off itself to $\Delta' = \{(\theta, \theta + \pi)\}$; the pushed-off curve is homotopic to Δ and disjoint from it, since $\theta = \theta + \pi$ has no solution on S^1. Hence $\chi(S^1) = I(\Delta, \Delta) = I(\Delta', \Delta) = 0$. The circle has a nowhere-vanishing vector field, the unit tangent $\partial/\partial\theta$, and its vanishing Euler characteristic is exactly what will let that field exist.

Everything is now in place. The Poincaré-Hopf theorem says the local index sum of any vector field with isolated zeros equals this global invariant, so in particular the sum does not depend on the field.

Theorem 8.6.7: Poincaré-Hopf

Let M be a compact oriented manifold and v a smooth vector field on M with only isolated zeros. Then

$$\sum_{p\colon v(p)=0} \operatorname{ind}_p(v) = \chi(M)$$

In particular the sum is independent of the field v.

***Proof*:**
The compactness of M makes the zero set of v finite, so the sum is finite. The argument computes $\chi(M) = I(\Delta, \Delta)$ by pushing the diagonal off itself along v.

Step 1: The normal bundle of the diagonal is the tangent bundle. At a point $(p,p) \in \Delta$,

$$T_{(p,p)}(M \times M) = T_pM \oplus T_pM, \qquad T_{(p,p)}\Delta = \{(u,u) \mid u \in T_pM\}$$

so the normal space $N_{(p,p)}\Delta = T_{(p,p)}(M \times M)/T_{(p,p)}\Delta$ is carried isomorphically to T_pM by $[(u,w)] \mapsto w - u$. These isomorphisms assemble to a vector bundle isomorphism $N\Delta \cong TM$ over the identification $\Delta \cong M$, $(p,p) \mapsto p$. By the tubular neighbourhood theorem (theorem 7.8.1) there is a diffeomorphism Φ from a neighbourhood of the zero section in TM onto a neighbourhood of Δ in $M \times M$, restricting to $\Delta \cong M$ on the zero section. Replacing v by εv for small $\varepsilon > 0$ (which changes neither the zeros nor the indices, as $v/|v|$ is unchanged) puts the image of $v\colon M \to TM$ inside this neighbourhood, so

$$\Delta_v := \Phi\big(\{(p, v(p)) : p \in M\}\big) \subseteq M \times M$$

is a submanifold, a pushoff of Δ, with $\Delta_v \cap \Delta = \Phi(\{v = 0\})$: the pushed-off diagonal meets the diagonal exactly over the zeros of v. As v is homotopic to the zero field through tv, the submanifold Δ_v is homotopic to Δ, and homotopy invariance of the oriented intersection number (section 8.3) gives $I(\Delta, \Delta) = I(\Delta, \Delta_v)$.

Step 2: A nondegenerate zero contributes its index. Suppose first that every zero of v is nondegenerate. Then v, as a section of TM, is transverse to the zero section, so Δ is transverse to Δ_v and $I(\Delta, \Delta_v)$ is the signed count of the finitely many intersection points. Fix a zero p and an oriented chart, trivializing TM near p as $B \times \mathbb{R}^n$ with the zero section $B \times \{0\}$; to first order Δ_v is the graph $\{(x, Dv_p\, x) \mid x\}$. The tangent spaces $\mathbb{R}^n \times \{0\}$ and $\{(u, Dv_p u) \mid u\}$ span the ambient $\mathbb{R}^n \times \mathbb{R}^n$ precisely because Dv_p is invertible, and with the orientation conventions fixed in section 8.3 the sign of the intersection at p is $\operatorname{sign}\det(Dv_p)$. By proposition 8.6.3 this sign is $\operatorname{ind}_p(v)$. Summing,

$$\chi(M) = I(\Delta, \Delta_v) = \sum_{p\colon v(p)=0} \operatorname{sign}\det(Dv_p) = \sum_{p\colon v(p)=0} \operatorname{ind}_p(v)$$

Step 3: The general case by perturbation. Let v have isolated but possibly degenerate zeros $p_1, \dots, p_k$, and choose disjoint closed balls B_j around them, in coordinates, on which v has no other zero. Inside a ball, v reads as $\tilde{v}\colon B_j \to \mathbb{R}^n$; take a bump function β equal to 1 near the centre and 0 near ∂B_j, and (by Sard's theorem, theorem 7.2.1) a regular value ξ of $\tilde{v}$ with

$0 < |\xi| < \min\{|\tilde{v}(x)| \mid \beta(x) < 1\}$, the minimum positive because $\tilde{v}$ vanishes only at the centre, where $\beta = 1$. The field v' agreeing with v outside the balls and equal to $\tilde{v} - \beta\xi$ inside is smooth, equals v on each ∂B_j, is nonzero wherever $\beta < 1$ (there $|\tilde{v}| > |\xi| \geq |\beta\xi|$), and has only nondegenerate zeros in the cores (there $\tilde{v} = \xi$ is a regular value, so $D\tilde{v}$ is invertible). The degree of $v/|v| = v'/|v'|$ on ∂B_j equals $\mathrm{ind}_{p_j}(v)$ and, by additivity of the degree over the interior zeros, also equals $\sum_{q \in B_j : v'(q)=0} \mathrm{ind}_q(v')$. Summing over j,

$$\sum_{p:\, v(p)=0} \mathrm{ind}_p(v) = \sum_{q:\, v'(q)=0} \mathrm{ind}_q(v') = \chi(M)$$

by Step 2 applied to the nondegenerate field v', as we sought to show.

The theorem is a bridge between two registers. On the left is differential data, the linearizations of a field at its equilibria, computed determinant by determinant; on the right is a topological invariant of M that does not mention any field. That the left side is forced to equal the right regardless of the field is the content, and it is what makes the index sum useful: to compute $\chi(M)$ one may exhibit any convenient field, and to constrain the fields on M one may compute $\chi(M)$ by any means. The running example of the sphere shows both directions at work.

Example 8.6.8: The Euler Characteristic of Spheres

On $S^n \subseteq \mathbb{R}^{n+1}$ take the field $v(p) = e_{n+1} - \langle e_{n+1}, p\rangle\, p$, the tangential part of the constant vector e_{n+1} (the gradient of the height function). Its zeros are the points where e_{n+1} is normal to the sphere, namely the poles $N = e_{n+1}$ and $S = -e_{n+1}$. Near S the field points radially outward, a source, so $\mathrm{ind}_S(v) = +1$; near N it points radially inward, a sink, whose Gauss map is the antipodal map of S^{n-1}, so $\mathrm{ind}_N(v) = (-1)^n$. By theorem 8.6.7,

$$\chi(S^n) = \mathrm{ind}_S(v) + \mathrm{ind}_N(v) = 1 + (-1)^n = \begin{cases} 2, & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$$

So even spheres have Euler characteristic 2 and odd spheres have 0, matching $\chi(S^1) = 0$ from example 8.6.6.

Example 8.6.9: The Torus and the Genus-g Surface

The n-torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ carries the nowhere-vanishing field $\partial/\partial\theta^1$, which has no zeros, so theorem 8.6.7 gives $\chi(T^n) = 0$ for every $n \geq 1$.

For a closed orientable surface Σ_g of genus g, stand it upright in $\mathbb{R}^3$ and take the downhill field of the height function. On a surface a source (a minimum) and a sink (a maximum) each have index $+1$, and a saddle has index -1, following example 8.6.2. The height function on Σ_g has one minimum, one maximum, and $2g$ saddles (one at the top and one at the bottom of each of the g holes), so

$$\chi(\Sigma_g) = 1 + 1 - 2g = 2 - 2g$$

The case $g = 1$ recovers $\chi(T^2) = 0$, and $g = 0$ recovers $\chi(S^2) = 2$.

The even-sphere computation has an immediate consequence that predates the general theorem and gave the subject one of its names. If $\chi(M) \neq 0$, then no vector field on M can avoid vanishing, for a nowhere-vanishing field has no zeros and its (empty) index sum is 0.

Theorem 8.6.10: Hairy Ball Theorem

The even-dimensional sphere S^{2n} admits no nowhere-vanishing smooth vector field. Equivalently, every smooth vector field tangent to S^{2n} vanishes somewhere.

***Proof*:**
A nowhere-vanishing field has isolated zeros vacuously, so theorem 8.6.7 applies and gives $0 = \sum \mathrm{ind}_p(v) = \chi(S^{2n})$. But $\chi(S^{2n}) = 2$ by example 8.6.8, a contradiction. Hence no such field exists, as we sought to show.

The name is the two-dimensional picture: one cannot comb the hair flat on a sphere without leaving a cowlick. The obstruction is entirely the nonvanishing of χ, and the odd spheres show it is sharp. On $S^{2n-1} \subseteq \mathbb{R}^{2n} = \mathbb{C}^n$ the field $v(z) = iz$ is tangent, since $\langle iz, z\rangle = 0$, and nowhere zero, since $|iz| = |z| = 1$; in real coordinates it is $(x_1, \dots, x_{2n}) \mapsto (-x_2, x_1, -x_4, x_3, \dots)$. Thus odd spheres are combable, in agreement with $\chi(S^{2n-1}) = 0$. The general principle behind both cases is the converse direction, due to Hopf: a compact manifold admits a nowhere-vanishing field if and only if $\chi(M) = 0$. Poincaré-Hopf is the forward half, and the converse is developed in [7].

For a hypersurface in Euclidean space the index sum can be read off a single map, the field of unit normals, and this yields the topological core of the Gauss-Bonnet theorem. Let $M^n \subseteq \mathbb{R}^{n+1}$ be a compact boundaryless hypersurface, oriented by a choice of unit normal.

Definition 8.6.11: Gauss Map

The *Gauss map* of an oriented hypersurface $M^n \subseteq \mathbb{R}^{n+1}$ is

$$G\colon M \to S^n, \qquad G(p) = \nu(p)$$

sending each point to its outward unit normal.

Since M and S^n are compact oriented n-manifolds and S^n is connected, G has a degree (section 8.4). For the round sphere $S^n \subseteq \mathbb{R}^{n+1}$ itself the outward normal at p is p, so G is the identity and $\deg G = 1$. The theorem below reads this against the Euler characteristic.

Theorem 8.6.12: Topological Gauss-Bonnet

Let $M^n \subseteq \mathbb{R}^{n+1}$ be a compact boundaryless hypersurface with Gauss map G. Then

$$\chi(M) = \big(1 + (-1)^n\big) \deg G$$

In particular, for n even, $\deg G = \frac{1}{2}\,\chi(M)$; for n odd both sides vanish.

***Proof*:**
Choose a unit vector $a \in S^n$ that is a regular value of G and for which $-a$ is also a regular value; almost every a qualifies by Sard's theorem (theorem 7.2.1). On M consider the height function

$h_a(x) = \langle a, x \rangle$ and its gradient field

$$w(p) = a - \langle a, \nu(p) \rangle \, \nu(p)$$

the tangential projection of a. Then $w(p) = 0$ exactly when a is normal to M at p, that is when $\nu(p) = \pm a$, so the zeros of w are the finite set $G^{-1}(a) \cup G^{-1}(-a)$.

Differentiating and projecting to T_pM at a zero p (the normal component does not affect the index), the linearization is

$$Dw_p = -\langle a, \nu(p) \rangle \, dG_p \colon T_pM \to T_pM$$

where $dG_p \colon T_pM \to T_{G(p)}S^n$ is identified with an endomorphism of T_pM through $T_{G(p)}S^n = \nu(p)^\perp = T_pM$. Because a is a regular value of G, dG_p is invertible at each such p, so every zero of w is nondegenerate and, by proposition 8.6.3,

$$\operatorname{ind}_p(w) = \operatorname{sign}\det(Dw_p) = \big(-\langle a, \nu(p) \rangle\big)^n \operatorname{sign}\det(dG_p)$$

At a zero with $\nu(p) = -a$ the scalar $-\langle a, \nu(p) \rangle$ is $+1$, so $\operatorname{ind}_p(w) = \operatorname{sign}\det(dG_p)$; at a zero with $\nu(p) = a$ it is -1, so $\operatorname{ind}_p(w) = (-1)^n \operatorname{sign}\det(dG_p)$. The degree of G, computed at the regular value a and again at the regular value $-a$, is

$$\deg G = \sum_{p \in G^{-1}(a)} \operatorname{sign}\det(dG_p) = \sum_{p \in G^{-1}(-a)} \operatorname{sign}\det(dG_p)$$

since the degree is independent of the regular value (section 8.4). Summing the indices and applying theorem 8.6.7,

$$\chi(M) = \sum_{p \in G^{-1}(a)} (-1)^n \operatorname{sign}\det(dG_p) + \sum_{p \in G^{-1}(-a)} \operatorname{sign}\det(dG_p) = (-1)^n \deg G + \deg G$$

which is $(1 + (-1)^n) \deg G$, completing the proof.

For n even the formula reads $\chi(M) = 2 \deg G$; the sphere checks it, with $\chi(S^{2m}) = 2$ and $\deg G = 1$. For n odd it says only $0 = 0$, consistent with the vanishing of the Euler characteristic of an odd-dimensional closed manifold. The content is that the degree of the normal map, an integer computed by counting how the tangent planes turn, is half a homotopy invariant of M.

8.6.13 Foreshadowing: The Curvature Form of Gauss-Bonnet The degree of the Gauss map is also an integral: for a surface, $\deg G = \frac{1}{4\pi} \int_M K \, dA$ with K the Gaussian curvature, so $\int_M K \, dA = 2\pi \, \chi(M)$, the classical Gauss-Bonnet theorem. The passage from the topological count above to the curvature integral requires a metric and its second fundamental form, which this book does not develop; the metric form is proved in [4]. What is established here is the topological skeleton on which that theorem hangs: the total curvature is a homotopy invariant because it computes the degree of a map of spheres.

Exercise 8.6.1

1. Compute the index at the origin of the planar vector fields $v(z) = z^k$ and $v(z) = \bar{z}^k$ for $k \geq 1$, and of $v(x, y) = (x^2 - y^2,\, 2xy)$. Deduce that a single isolated zero can carry any integer index.

2. Let M and N be compact oriented manifolds. Show that if v on M and w on N have isolated zeros, then the field (v, w) on $M \times N$ has isolated zeros with $\mathrm{ind}_{(p,q)}(v, w) = \mathrm{ind}_p(v) \cdot \mathrm{ind}_q(w)$, and conclude the product formula $\chi(M \times N) = \chi(M)\,\chi(N)$.

3. Using that $\mathrm{ind}_p(-v) = (-1)^n\,\mathrm{ind}_p(v)$ for a zero of a field on an n-manifold, prove that every compact odd-dimensional manifold has $\chi(M) = 0$.

4. Let f be a smooth function on M with a nondegenerate critical point at p, and let λ be its Morse index (the number of negative eigenvalues of the Hessian). Show that the gradient field ∇f (for any Riemannian metric) has a nondegenerate zero at p with $\mathrm{ind}_p(\nabla f) = (-1)^\lambda$.

5. Use theorem 8.6.7 and the product formula of exercise 8.6.1 (2) to decide, for each pair, whether $S^{2n} \times S^{2m}$ and $S^{2n} \times S^{2m+1}$ admit a nowhere-vanishing vector field, exhibiting one where it exists.

6. The torus T^2 embeds in $\mathbb{R}^3$ as the surface of revolution of a circle. Compute $\deg G$ for its Gauss map two ways: directly from theorem 8.6.12 using $\chi(T^2)$, and by identifying the points where the outward normal equals a fixed generic direction.

9

Morse Theory

10

Cobordism and the Pontryagin-Thom Construction

10.1 Cobordism

This section is a prelude to future studies. Compact $(n+1)$ manifolds with boundary provide us with a natural equivalence relation between compact n-manifolds known as *cobordism*

Definition 10.1.1: Cobordism

Two compact n-manifolds M_1, M_2 are called *cobordant* when there exists N, a compact $n+1$ compact manifold with boundary, such that ∂N is isomorphic to $M_1 \sqcup M_2$. All manifolds cobordant to M form the *cobordism class of* M. We say that M is *null-cobordant* if $M = \partial N$ for N a compact $n+1$ manifold with boundary

It is important to assume compactness, or else all manifolds are null-cobordant by taking the Cartesian product with the noncompact manifold with boundary $[0, 1)$. All n-manifolds that can be embedded into $\mathbb{R}^{n+1}$ are null-cobordant. AN example of a non nullcobordant manifold is $\mathbb{R}\mathrm{P}^2$. Let Ω^n be the set of cobordism classes of compact n-manifolds, including $\emptyset$. Then notice that:

$$[M_1] + [M_2] = [M_1 \sqcup M_2]$$

defines a binary operation on Ω^n that makes it an abelian group (with identity $[\emptyset]$). In fact, the additive inverse of any $[M]$ is $[M]$, making it 2-torsion:

Proposition 10.1.2: Additive Torsion

for any $x \in \Omega^n$, $x + x = 0$

***Proof*:**
For any manifold $M<$ the manifold with boundary $M \times [0,1]$ has boundary $M \sqcup M$. Hence $[M] + [M] = [\emptyset] = 0$, as required.

If we take the direct sum $\Omega^\bullet = \bigoplus_{n\in\mathbb{N}} \Omega^n$, then we can endow this structure with multiplication:

$$[M_1] \cdot [M_2] = [M_1 \times M_2]$$

this makes $\Omega^\bullet$ a commutative ring known as the *cobordism ring* with multiplicative unit $[*]$, the class of 0-manifold consisting of a single point. Naturally, this ring is graded by dimension.

Example 10.1.3: Cobordism

1. The n-sphere S^n is null-cobordant (i.e. cobordant to $\emptyset$) since $\partial B_{n+1}(0,1) \cong S^n$.
2. Any oriented compact 2-manifold is null cobordant: we may embed it in $\mathbb{R}^3$ and the "inside" is a 3-manifold with boundary

We may wonder what is the ring structure of the cobordism ring. To get some regularity, we will require our manifolds to be *smooth*. For now, we shall state the result by Thom:

Theorem 10.1.4: Cobordism Ring

The cobordism ring is a (countably generated) polynomial ring over $\mathbb{F}_2$, with generators in every dimensin $n \neq 2^k - 1$, that is

$$\Omega^\bullet = \mathbb{F}_2[x_2, x_4, x_5, x_6, x_8, ...]$$

***Proof*:**
see (this) paper

Part III

Bundles, Forms, and Integration

In this part, we will start exploring another that can be put on manifolds that will give information at each point which shall be glued together in a local manner similarly to how we consructed the tangnet bundle. In fact, it is a generalization of the tangent bundle, in . So far, we have structure that are locally $\mathbb{R}^n$ with regularity C^k for $k \in \{0, 1, ..., n, ..., \infty\}$ that are also Hausdorff and paracompact. Equivalently, we studied structures that can be constructed by taking open subsets, and gluing parts of them via C^k maps ($k \in \{0, 1, ..., n, ..., \infty\}$) in such a way that the spaces remain Hausdorff. In this section, we will now take a look at spaces that are locally the Cartesian product of an open subset of our manifold M and a *fiber*. We will get to a formal definition in the next chapter, we will for now simply give the high-level intuition on this new type of structure. Take for example $E = S^1 \times (-1, 1)$. This space comes equipped with a natural projection $\pi : E \to S^1$. Then for any open $U \subseteq S^1$ and consider $\pi^{-1}(U)$, we want it to be isomorphic to $U \times (-1, 1)$ in such a way that fibers of U remain above U, that is we want this diagram to commute:

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr] & & U \times (-1,1) \arrow[dl] \\
& U \arrow[ul] &
\end{tikzcd}$$

The isomorphism in question will be part of the regularity information. The space E is called the *fiber bundle*. The type of fiber will determine the type of fiber bundles we are working with: for example they may be vector-spaces (giving us vector-bundles), or groups (giving us group bundles), or circle (giving us Hopf bundles), and so on. Fiber bundles will turn out give us a lot of information about manifolds and allow us to not only define many new manifolds, but define many concepts on manifolds like length and distance, measure, curvature, angle, and so forth.

11

Vector and Cotangent Bundles

A vector bundle is a fiber bundle whose fibers are vector spaces glued by linear transition maps. This chapter builds the general notion first, a space that is locally a product $U \times F$ but globally twisted, and then specializes to the vector bundles that carry the rest of the book: their sections, the maps between them, and their orientations.

11.1 Fiber Bundles

Definition 11.1.1: Fiber Bundle

A *fiber bundle* is a 4-tulpe (B, F, E, π) consisting of

1. a Manifold B (the base space)
2. a fibre type F (some topological space or more generally an object in a category)
3. E, the *total space* of the fibre bundle (a topological space)
4. $\pi : E \to \pi$ the *bundle projection*

where the bundle projection satisfies the following commutative diagram: for each $p \in B$, there exists a $U_p \subseteq B$ containing B such that

$$\begin{tikzcd} \pi^{-1}(U_p) \arrow[rr, "\Psi_p"] \arrow[dr, "\pi"'] & & U_p \times F \arrow[dl, "\pi_1"] \\ & U_p & \end{tikzcd}$$

with the added condition that if $U, V \subseteq B$ such that $U \cap V \neq \emptyset$, then there exists a transition map such that

$$\begin{tikzcd} (U \cap V) \times F \arrow[rr, "\varphi_{UV}"] \arrow[dr, "\pi_1"'] & & (U \cap V) \times F \arrow[dl, "\pi_1"] \\ & U \cap V & \end{tikzcd}$$

commutes and $\varphi_{UV}(p \in U \cap V) : F \to F$ is an automomorphism (i.e. there is a continuous map $\varphi_{UV} : U \cap V \to \mathrm{Aut}(F)$)
the set $\pi^{-1}(p) = E_p$ are called the *fiber over* p. A collection (U_p, Ψ_p) that covers E is called a *bundle Atlas*, and the maps Ψ_p are called *local trivializations*.

Just like manifold, we can construct fibre bundles:

Proposition 11.1.2: Constructing Fiber Bundles

here, in Marco Notes

***Proof*:**

11.2 Application: Jet Bundles and Intersection Theory

As mentioned on page **??**, the tangent space can be thought of as a first order linear approximation of a map $f : M \to N$, i.e. a first order Taylor series approximation in a coordinate free way. In this section, we generalize this notion, leading to a coordinate-free version of the taylor approximation of a function known as a *Jet* and *Jet bundle*.

11.3 Vector Bundles

We saw that when we have two manifolds M_1 and M_2, we can define the product manifold $M_1 \times M_2$ for which the universal property of products hold. These shapes are particularly easy to study, as their tangent bundle can be done component-wise, and so we reduce the study of manifolds to the study of it's components. However, there are manifolds that are not products of manifolds, however they *locally* look like product of manifolds. Perhaps the most famous example is the *mobius band*, which is $S^1 \times (0, 1)$, but with a twist. The fact that it's a twist means we can simply identify the mobius band with $S^1 \times (0, 1)$, however we can define a structure on the mobius band such that the mobius band is covered in a special atlas that shows that the mobius band is always *locally isomorphic* to an open subset of $S^1 \times (0, 1)$! Thus, the mobius band is locally a product manifold (up to diffeomorphism).

In this chapter, explore this idea in a more general context, and show how this concept of being "locally a product" differs from that of a manifold.

Definition 11.3.1: Vector Bundles

Let M be a topological space. Then a (real) *vector bundle of rank k over M* is a topological space E together with a surjective homomorphism $\pi : E \to M$ satisfying

1. For each $p \in M$, $\pi^{-1}(p) = E_p$ is a k-dimensional real vector space
2. For each $p \in M$, there exists a neighborhood U of p in M and a homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ satisfying:
   (a) $\pi_U \circ \Phi = \pi$, where $\pi_U : U \times \mathbb{R}^k \to \mathbb{R}^k$ is the projection map
   (b) For each $q \in U$, the restriction of Φ to E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k$

   The map Φ is called the *local trivialization*

Usually, E is called the *total space*, M the *base space*, and π the *projection*

As usual, if we make the vector spaces $\mathbb{C}^k$, thne we will call such a bundle a *complex vector bundle*. If π is smooth, we call the resulting bundle a *smooth vector-bundle*. We will sometimes say "E is a vector-bundle over M", or "$E \to M$ is a vector bundle", or "$\pi : E \to M$ is a vector bundle". Soon, we will define what it means to have morphisms between local trivializations, which will allow us to *bundle atlases*.

The condition on the bundle map can be written as the diagram:

$$\begin{tikzcd}
\pi^{-1}(U_p) \arrow[r, "\Psi_p"] \arrow[d, "\pi"] & U_p \times F \arrow[dl, "\pi_I"] \\
U_p &
\end{tikzcd}$$

The diagram can be thought of as representing the fact that Ψ_p is a *fiber-preserving map.* in particular, if we just had the diagram

$$\begin{tikzcd} \pi^{-1}(U_p) \arrow[r, "\Psi_p"] \arrow[d, "\pi"] & U_p \times F \\ U_p & \end{tikzcd}$$

Then we can imagine that there could be homeomorphisms that do not send fibers to fibers and yet is a homeomorphism. That extra map closing the diagram forces this type of mapping. (Another cool think Marco said: You can think of this map sending verticals to verticals in the cartesian product, but on the gluing region it does not sent horizontals to horizontals, hence the non-cartesian nature of the entire fibre bundle)

To make condition (a) and the diffeomorphism condition more intutiive, I like to think about (a) as making sure that the tangent bundle patch $\pi^{-1}(U)$ "stays at the same place". For exmaple, you can imagine take $\pi^{-1}(S^1)$, and making all the tangent spaces "orthogonal" so that you get a big cylander. In the process of doing so, we must make sure not to "rotate" the circle, since if we do then $\pi_U \circ \Phi \neq \pi$. The diffeomorphism condition insures that when we "straighten it out", we do so smoothly (i.e. can't rip or bend or do so in an non locally linear way).

If there exists a local trivialization for all of M (called the *global trivialization*), then E we give E a special name:

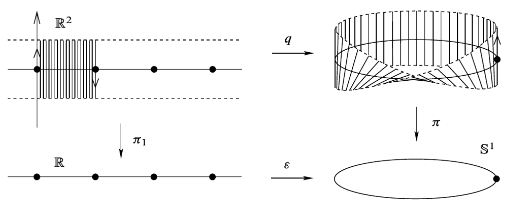
Definition 11.3.2: Trivial Bundle

Let E be a vector bundle over M. Then E is called a *trivial vector bundle* there exists a local trivialization over all of M. Such a trivialization is called the *global trivialization.* We say that E is *smoothly traivial*

The image of E that is a trivial bundle I keep in mind is a circle with $\mathbb{R}$ attached at every point that is wiggled a bit. The tiny wiggling doesn't change that E is just the infinite cylander $S^1 \times \mathbb{R}$ up to diffeomorphism.

Example 11.3.3: Vector Bundles

1. For any topological space M, $E = M \times \mathbb{R}^k$ with the usual projection map is a vector-bundle. This is called the *product bundle*, and is always a trivial bundle (the identity map being the global trivialization). If M is a smooth manifold (with or without boundary) then $M \times \mathbb{R}^k$ is smooth trivial
2. We will define the mobius strip and show it is the total space for a vector bundle over S^1.



3. $\mathbb{R}^{n+1}$ is a vector bundle over $\mathbb{RP}^n$, mapping a line in $\mathbb{R}^{n+1}$ to the appropriate equivalence

class in $\mathbb{R}\mathrm{P}^n$. The map $\pi : \mathbb{R}^{n+1} \to \mathbb{R}\mathrm{P}^n$ sending $x \mapsto [x]$ is the desired projection. If we

Proposition 11.3.4: Tangent Bundles Are Vector Bundles

Let M be a smooth n-manifold (with or without boundary), and let TM be its tangent bundle. Given the standard projection map $\pi : TM \to M$, the natural vector space structure of each T_pM, and the manifold sructure on TM, TM is a smooth vector bundles of rank n over M

***Proof*:**
Let $(U, \mathbf{x})$ be an admissible chart for M. Using this, define the map $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ by:

$$\Phi\left(v^i \left.\frac{\partial}{\partial x^i}\right|_p\right) = (p, (v^1, v^2, ..., v^n))$$

This map is naturally linear on fibers (being the identity map with respect to the appropriate basis), and $\pi_1 \circ \Phi = \pi$. It remains to show that this map is a diffeomorphism. Since $\mathbf{x}$ is a diffeomorphism, the map:

$$\pi^{-1}(U) \xrightarrow{\Phi} U \times \mathbb{R}^n \xrightarrow{\mathbf{x}\times\mathrm{id}_{\mathbb{R}^n}} \mathbf{x}(U) \times \mathbb{R}^n$$

is equal to $\tilde{\mathbf{x}}$ that we have constructed in proposition 2.4.5. Since $\tilde{\mathbf{x}}$ and $\mathbf{x} \times \mathrm{id}_{\mathbb{R}^n}$ are diffeomorphisms, so is Φ by proposition 1.3.16. Thus, Φ is a smooth local trivialization, as we sought to show.

Naturally, most interesting vector bundles are non-trivial, and so will require more than one local trivialization. The following lemma tells how the composition of two local trivialization behave. Essentially, we will show that there is a *transition function*[1] that is satisfied by the local trivializations. This will motivate the definition of a *bundle atlas* momentarily:

Lemma 11.3.5: transition maps for bundle atlas

Let $\pi : E \to M$ be a smooth vector bundle of rank k over M. Suppose $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ and $\Psi : \pi^{-1}(V) \to V \times \mathbb{R}^k$ are two smooth localizations with E with $U \cap V = \emptyset$. Then there exists a smooth map $\tau : (U \cap V) \to \mathbb{R}^k$ such that the composition $\Phi \circ \Psi^{-1} : (U \cap V) \to \mathbb{R}^k \to (U \cap V) \to \mathbb{R}^k$ has the form:

$$\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p)v)$$

where $\tau(p)v$ denotes the usual action of the $k \times k$ matrix $\tau(p)$ on the vector $v \in \mathbb{R}^k$

***Proof*:**
Essentially, notice that the following diagram commutes:

$$\begin{tikzcd}
U \cap V \times \mathbb{R}^k \arrow[dr, "\pi_1"'] & \pi^{-1}(U \cap V) \arrow[l, "\Psi"'] \arrow[r, "\Phi"] \arrow[d, "\pi"] & U \cap V \times \mathbb{R}^k \arrow[dl, "\pi_1"] \\
& U \cap V &
\end{tikzcd}$$

[1] The use of the word "function" here instead of map is due to historical legacy

with the appropriate restriction's of the maps. It follows that $\pi_1 \circ (\Phi \circ \Psi^{-1}) = \pi_1$, i.e.

$$\Phi \circ \Psi^{-1}(p, v) = (p, \sigma(p, v))$$

where $\sigma : (U \cap V) \times \mathbb{R}^k \to \mathbb{R}^k$ is some appropriate smooth map.

Now, for each fixed $p \in U \cap V$, the map $v \mapsto \sigma(p, v)$ is bijective linear map $\mathbb{R}^k$ to itself. Thus, there is a nonsingular matrix $\tau(p)$ such that $\sigma(p, v) = \tau(p)v$. It remains to show that $\tau : U \cap V \to \mathrm{GL}_k(\mathbb{R})$ is smooth. Treating $\mathrm{GL}_k(\mathbb{R})$ as an open subset of $\mathbb{R}^{n^2}$, it suffices to assume that $U \cap V$ is the domain of the chart (φ, U) when the chart is restricted. With this, we want to show that

$$\tau \circ \varphi^{-1} : A \subseteq \mathbb{R}^k \to B \subseteq \mathbb{R}^{n^2}$$

is smooth, which is equivalent to showing every component function is smooth, which comes down to showing that every component function of τ is smooh. However, this is evident: Since the image of any $\tau(p)$ is a matrix, we simply combine τ with the appropriate inclusion and projection maps (which are both smooth maps) to get the appropriate component maps τ_i^j to be smooth. But then $\tau_i^j \circ \varphi^{-1}$ would be smooth for each i, j, and so τ is smooth, completing the proof.

Example 11.3.6: Bundle Transition Map

The easiest example of τ would be in the case of the tangent bundle TM over M. Then if Φ and Ψ are two local trivializations of TM, each associated assocaited to two different charts $(\pi^{-1}(U), \tilde{\mathbf{x}})$ and $(\pi^{-1}(V), \tilde{\mathbf{y}})$ of TM, the transition function between them is the jacobian matrix of the coordinate transition map, $\tau = D(\varphi \circ \psi^{-1})$:

$$\Psi \circ \Phi^{-1} : U \cap V \times \mathbb{R}^n \to U \cap V \times \mathbb{R}^n$$

$$\begin{aligned} &\Psi \circ \Phi^{-1}(p_1, p_2, ..., p_n, v^1, ..., v^n) \\ &\quad = \left(p_1, p_2, ..., p_n, \sum_j \frac{\partial \tilde{x}^1}{\partial x^j}(x(p))v^j, ..., \sum_j \frac{\partial \tilde{x}^n}{\partial x^j}(x(p))v^j \right) \end{aligned}$$

which is clearly smooth and has the form given in the previous lemma with $\tau(p)$ being the Jacobian matrix at p

If two local trivializations respect above property, we will call them compatible. Using this, we have the following definnition:

Definition 11.3.7: Bundle Atlas

Let $\pi : E \to M$ be a vector bundle. Then the collection of compatible local trizializations for E is called the *bundle atlas* for E.

We have shown that tangent bundle's are manifolds, but what about general vector bundles? If we want a vector bundle E to be a smooth vector bundle, we would require defining a smooth structure on E that induces the manifold topology (or give a manifold topology and then show we can create a smooth structure on it), and then find local trivializations for every point (i.e. some collection of local trivializations). The next lemma tells us that we just need to construct the local trivializations as long as they overlap with smooth transition functions

(can also call lemma Vector Bundle Chart Lemma)

Lemma 11.3.8: Bundle Atlas inducing Manifold

Let M be a smooth manifold (with or without boundary), and for each $p \in M$, let E_p be real vector space of some fixed dimension k. Let $E = \coprod_{p \in M} E_p$, and $\pi : E \to M$ the map mapping any E_p to p. Then if we have the following "data":

1. There exists an open cover $\{(U_\alpha, \Phi_\alpha)\}_{\alpha \in A}$ for M
2. For each $\alpha \in A$, there exists a bijective map $\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{R}^k$ whose restriction to each E_p is a vector space isomorphism from E_p to $(\{p\} \times \mathbb{R}^k) \cong \mathbb{R}^k$
3. For each $\alpha, \beta \in A$ where $U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $\tau_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_k(\mathbb{R})$ such that $\Phi_\alpha \circ \Phi_\beta^{-1}$ from $(U_\alpha \cap U_\beta) \times \mathbb{R}^k$ to itself is of the form:
$$\Phi_\alpha \circ \Phi_\beta^{-1}(p, v) = (p, \tau_{\alpha\beta} v)$$

Then E has a unique topology and smooth structure making it into a smooth manifold (with or without boundary) and a smooth rank k vector bundler over M, with π as projection and $\{(U_\alpha, \Phi_\alpha)\}_{\alpha \in A}$ as bundle atlas

***Proof*:**
p. 253 John Lee

Using this lemma, we can constrct some interesting examples

Example 11.3.9: Further Example Of Vector Bundles
p. 254

1. (Whitney Sums)
2. (restriction of a vector bundle)

11.4 Local and Global sections of Vector Bundles

As we've seen in section 5.2, we can define a *frame* on a tangent bundle. This concept readily generlizes to vector bundles:

Definition 11.4.1: Local And Global Frame

Let $\pi : E \to M$ be a vector bundle and $U \subseteq M$ and open subset. Then an ordered set of local sections $(\sigma_1, \sigma_2, ..., \sigma_n)$ is said to be *linearly inpendent* on U if every $(\sigma_1|_p, ..., \sigma_n|_p)$ is linearly independent on E_p for each $p \in U$, and is said to *span the vector bundle* U if $(\sigma_1|_p, ..., \sigma_n|_p)$ spans E_p for each $p \in U$.

If an ordered set of vector-fields has both of these properties for some $U \subseteq M$, then it is called a *local frame* on M. If $U = M$, then it is called a *global frame*. If each σ_i is smooth, then it is called a *smooth frame*.

Notice a terminological difference between frames defined using vector-fields and on vector-bundles: A frame on a tangent bundles with the old definition would be "a frame on M", while in our new definition it would be "a frame on TM". We will use both interchangibally with the undersatnding that these to definition match $E = TM$.

Just like for local frames on manifolds, local farmes on vector bundles are relatively easy to find:

Proposition 11.4.2: Completion Of Local Frames On Vector Bundles

p. 258 John Lee

***Proof*:**
Given as on the spot exercise in John Lee

The easiet example of a vector bundle admitting a global frame is the prouct bundle $E = M \times \mathbb{R}^n \to \mathbb{R}^n$, with standard basis $(e_1, e_2, ..., e_n)$ for $\mathbb{R}^n$ an a frame $(\tilde{e}_i)$ for E defined by:

$$\tilde{e}_i(p) = (p, e_i)$$

If M is a smooth manifold, this frame is smooth.

Example 11.4.3: Local Frames and Local Trivialization

Let $\pi : E \to M$ be a vector bundle and $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ a local trivialization. Then we can use this local trivialization to construct a local frame on U. Inspired by the case for the product bundle, let

$$\sigma_i : M \to E \qquad \sigma_i(p) = \Phi^{-1} \circ \tilde{e}_i(p)$$

If M is smooth, then σ_i is the composition of smooth maps, and hence smooth. Furthermore:

$$\pi \circ \sigma_i(p) = \pi(\pi^{-1}(p)) = p$$

and hence each σ_i is indeed a section. They are naturally a frame, being a basis at every p, and so (σ_i) is a local frame.

Thus, given a local trivialization on U, we can construct a local frame on U. This frame is called the *local frame associated with* Φ, since $\tilde{e}_i$ is the cannonical basis for $\mathbb{R}^k$

In the above example, we took a local trivialization and found a smooth local frame. The converse is alos possible:

Proposition 11.4.4: Local Frame Gives Local Trivialization

Let $\pi : E \to M$ be a [smooth] vector bundle and (σ_i) a [smooth] local frame on U. Then (σ_i) is associated with a [smooth] local trivialization Φ.

***Proof*:**

($\rightarrow$) Let E be trivial, so that $\Phi : \pi^{-1}(M) \to M \times \mathbb{R}^k$ is a global trivialization. Define the section maps:

$$s_i : M \to E \qquad s_i(p) = \Phi^{-1} \circ \overline{e}_i(p)$$

where $\overline{e}_i : M \to M \times \mathbb{R}^k$ maps $p \mapsto (p, e_i)$. Since Φ^{-1} is smoooth and each $\overline{e}_i$ is smooth (being in inclusion map), each s_i is smooth. Furthermore, Since Φ^{-1} is a diffeomorphism, the elements $(s_1(p), s_2(p), ..., s_k(p))$ form a linearly independent set for E_p for all $p \in M$.

($\Leftarrow$) Let the smooth section maps $s_1, s_2, ..., s_k$ be everywhere linearly independent. Since E is of rank k, $(s_1(p), ..., s_k(p))$ forms a basis for each E_p. Define the map:

$$\Phi : \pi^{-1}(M) \to M \times \mathbb{R}^k \qquad (v_i s_p(p)) \mapsto (p, v_1, ..., v_k)$$

This map is bijective since each $\Phi|_p$ is bijective, where we see that each $\Phi|_p$ is bijective since $(s_1(p), ..., s_k(p))$ form a basis for $\pi^{-1}(p)$. Clearly, $\Phi|_p$ is a linear isomorphism, so it remains to show that Φ is a diffeomorphism, or in particular since Φ is bijective, by the constant rank theorem, it suffices to show it's a local diffeomorphism.

The key is to take advantage of the fact that E is a smooth vector-bundle, since this implies E is covered in smooth local trivializations: there exists a subset $U \subseteq M$ containing p such that $\Psi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ is a diffeomorphism. Thus, given that Ψ is a diffeomrophism, if we show that $\Psi \circ \Phi^{-1}|_{U \times \mathbb{R}^k}$ is a diffeomorphism (from $U \times \mathbb{R}^k$ to itself), Then $\Phi^{-1}|_{U \times \mathbb{R}^k}$ would be a diffeomrophism (by homework 1), and so Φ on the appropariate domain would be a diffeomorphism. From now on, we will assume the domain is appropriately restricted.

We show now that $\Psi \circ \Phi^{-1}$ is smooth: since Ψ is smooth, so is $\Psi \circ \sigma_i$ for each $1 \leq i \leq k$. If we let

$$\sigma_i^j = \pi^j \circ \Psi \circ \sigma_i|_U : U \to \pi^{-1}(U) \to U \times \mathbb{R}^k \to \mathbb{R}$$

represent the components of σ_i with respect to to the coordinates given by Ψ, we get that:

$$\Psi \circ \Phi^{-1}(p, x_1, ..., x_k) = \left(p, \sum_i x_i \sigma_i^1(p), ..., \sum_i x_i \sigma_i^k(p) \right)$$

which is certainly smooth. Next, to show that $(\Psi \circ \Phi^{-1})^{-1}$ is smooth, note that we could have represented the above map as:

$$\Psi \circ \Phi^{-1}(p, x) = (p, S(p)x)$$

where $x = (x_1, x_2, ..., x_k)$ and

$$S(p) = \begin{pmatrix} \sigma_1^1(p) & \cdots & \sigma_k^1(p) \\ \vdots & \ddots & \vdots \\ \sigma_1^k(p) & \cdots & \sigma_k^k(p) \end{pmatrix}$$

Notice that $S(p)$ is invertible at each p since $\sigma_i(p)$ is linearly independent at each p (even better, they form a basis at each E_p since E_p is k-dimensional). Hence, $S(p) \in \mathrm{GL}_k(\mathbb{R})$. By the previous homework, the function ι^{-1} mapping $A \mapsto A^{-1}$ is smooth. Then the form of $(\Psi \circ \Phi^{-1})^{-1}$ is given by replacing S with S^{-1}, and so is certainly smooth.

Thus, we showed that Φ is a local diffeomorphsim, and so as a consequence of the constant rank theorem a diffeomorphism, and so combining with the fact that the image of Φ is $M \times \mathbb{R}^k$ and Φ_p is a linear isomorphism, Φ is a global trivalization, completing the proof.

Corollary 11.4.5: Global Frame Then Global Trivialization

Let $\pi : E \to M$ be a [smooth] vector bunle and (σ_i) a [smooth] global frame. Then (σ_i) is associated with a [smooth] global trivialization Φ.

***Proof*:**
This is a direct result of the last proposition.

11.5 Bundle Homomorphisms

It is perhaps clear that every vector bundle E over S^1 of rank 1 that is a trivial bundle simple wiggles the lines a bit – these are all equivalent bundles, we would really want to treat these as the same. This motivates the following definition

Definition 11.5.1: Bundle Homomorhpism

Let $\pi : E \to M$ and $\pi' : E' \to M'$ be two vector bundles. Let $F : E \to E'$ be a continuous map. Then F is called a *bundle homomorphism* if there exists a map $f : M \to M'$ such that $\pi' \circ F = f \circ \pi$, i.e. the following diagram commutes

$$\begin{tikzcd} E \arrow[r, "F"] \arrow[d, "\pi"] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "f"] & M' \end{tikzcd}$$

with the property that for each $p \in M$, the restriction $F|_{E_p} : E_p \to E'_{f(p)}$ is a linear map. If F is a bundle homomorhpism and f is the induced map between M and M' satisfying the aforementioned condition, we say that F *covers* f.

Naturally, id is a bundle homomorphism, and if there exists a bundle homomorphism G such that $F \circ G = G \circ F = \mathrm{id}$, then F is said to be a *bundle isomorphism* and $G = F^{-1}$.

Example 11.5.2: Bundle Homomorphisms

1. Let M be a smooth manifold and $M_{p\times n}(\mathbb{R})$ be the set of $p\times n$ matrices. Let $T : M \to M_{p\times n}(\mathbb{R})$

be a smooth map between these spaces. Then

$$\begin{array}{ccc} M \times \mathbb{R}^n & \longrightarrow & M \times \mathbb{R}^p \\ \downarrow & & \downarrow \\ M & \longrightarrow & M \end{array}$$

where $(m, v) \mapsto (m, T(m)(v))$ is a bundle homomorphism. It is a bundle isomorphism if $p = n$.

2. TS^1 is a trivial bundle. To see this, notice this diagram commutes:

$$\begin{tikzcd} TS^1 \arrow[rr, "F"] \arrow[dr, "\pi"'] & & S^1 \times \mathbb{R} \arrow[dl, "\pi"] \\ & S^1 & \end{tikzcd}$$

where $(a, t_{v_a}) \mapsto (a, t)$ (t_{v_a} being the element t with respect to the basis element v_a). As an interesting remark, TS^2 is not a trivial bundle. In fact, the only other two trivial sphere bundles are TS^3 and TS^7 (which we don't prove here)

3. If G is a Lie group, then TG is always a trivial bundle (it's in your homework, but as a hint remember that the binary operation is invertible...)

11.6 Oriented Bundle

We first recall what it means for a vector-space to be oriented. The idea of orientation start when it was realised that in $\mathbb{R}^3$, there is a notion of "left-handedness" and "right-handedness", where a something like your hand or a molecule cannot simply be rotated to create the other object. In chemistry, this is called *chirality*, and can have major effects on many chemical processes (ex. one chirality can be harmless to the neuro-chemsitry of the brain, while the other is cocaine). We can with not too much difficulty imagine the analogous scenario happening in $\mathbb{R}^2$ or $\mathbb{R}$, where in $\mathbb{R}^2$ we can turn anti-clockwise or clockwise (with many mathematicians have the ingrained preference of going anti-clockwise when choosing a basis), while in $\mathbb{R}$ there is a preference for going positive, which is usually going on the right of a real number line.

In the case of $\mathbb{R}$, $\mathbb{R}^2$, and $\mathbb{R}^3$, there is a natural way of choosing a basis that satisfies these conditions: for $\mathbb{R}$, it is (1), for $\mathbb{R}^2$, it is usually $((1,0),(0,1))$, while for $\mathbb{R}^3$ it is $((1,0,0),(0,1,0),(0,0,1))$. In terms of orientation, the first one is going to the positive, the second one going anti-clockwise on the plane to get to the second one, and for he first one, if we have a right-hand that curls it's fingers from the first basis to the second, the thumb points towards the third basis vector. We will call all of these the *positive orientation*. To generlize this pattern into $\mathbb{R}^n$, we will define a basis to be *positivly oriented* if the determinant of the basis is positive (remember that a basis is ordered, and that the sign of the determinant is sensitive to changes in order).

This same principle can be generlized for any vector space V. If $\dim(V) \geq 1$, given an (ordered) basis for V, the basis can have a positive or negative orientation depending on it's determinant. In fact, we can define an equivalence class of basis where two basis are in the same equivalence class if and only if the transition matrix between them has positive determinant. If that is the case, we will say that the two bases are *consistenly oriented*. Since the determinant of a transition matrix can only either be positive or negative, there are only 2 possible equivalence classe. These two equivalence classes on the bases of V are called the *orienations of* V. Because of this, if $[E_1, E_2, ..., E_n]$ is an orientation of V, then we can unambiguously denote the opposite orientation of V by $-[E_1, E_2, ..., E_n]$. A vector-sapce V along with an orientation is called a *oriented vector-space*, and we say that V is *oriented*. With this terminology, if V is oriented and $(E_1, ..., E_n)$ is a basis for V, then if $(E_1, E_2, ..., E_n)$ is in the orienation of V, we would say that $(E_1, E_2, ..., E_n)$ is *positively oriented*, and *negatively oriented* otherwise. If $\dim(V) = 0$, then we will assign either ± 1 to be it's orientation. With this new way of phrasing orientation, we will say that the orienation of $[e_1, e_2, ..., e_n]$ for $\mathbb{R}^n$ is the *standard orientation*.

When we had the change of basis transformation $B : V \to V$, we were taking the determinant of of the change of basis matrix B. We can do a similar process for arbitrary maps: a linear map between oriented vector spaces $T : V \to W$ is called *orientation preserving* if for some choice of basis in the orietnation of V and W, $\det(T) > 0$, and *orientation-reversing* if $\det(T) < 0$.

Given oriented vector spaces (V, μ), (W, ν), we call a linear transformation $T : V \to W$ *orientation preserving* if $[T(v_1), ..., T(v_n)]\nu$ where $\mu[v_1, v_2, ..., v_n]$.

We now consider vector bundles. We can have each vector-space in a vector bundle have an orientation. This is not too intereting, since these orientation can vary wildely. We will want a restriction on a vector-bundle so that the orienation's varry consistenly:

Definition 11.6.1: Oriented Vector Bundle

Let E be a vector bundle over M. Then E is called an *oriented vector bundle*, if there exists a collection of orientations $[\mu_x]$ for each $x \in M$ (collectively labeled $[\mu]$) such that for a local trivialization:

$$\Phi : E|_U \to U \times \mathbb{R}^k$$

If $\mathbb{R}^k$ is given the standard orientation of $[e_1, e_2, ..., e_k]$, each Φ_x is orientation preserving on each $x \in U$

Note that thse if we consider a trivial vector bundle $E = M \times \mathbb{R}^n$ and a bundle isomorphism from E to itself, then E is an oriented vector bundle if and only if it is *connected* (use Intermediate Value Theorem?). Note that if E has an orientation μ, it also has an inverse orientation by considering $-\mu = \{-\mu_x\}$; however, if it has no orientation, then E also cannot have an inverse orientation. A bundle is called *orientable* if it has an orientation.

Definition 11.6.2: Oriented Manifold

A manifold is called an *oriented manifold* if TM is orientable as a vector bundle. If μ is an orientation of TM, then μ is also called the orientation of M.

This condition can be simplified: If TM is orientable, then for any two local trivializations Φ, Ψ induced by charts $(U, \mathbf{x})$ and $(\mathbf{y}, V)$, we have:

$$\begin{tikzcd}
\mathbf{x}(U\cap V) & U\cap V\times\mathbb{R}^k \arrow[l, "\mathbf{x}\circ\pi_1"'] \arrow[dr, "\pi_1"'] & \pi^{-1}(U\cap V) \arrow[l, "\Phi"'] \arrow[r, "\Psi"] \arrow[d, "\pi"] & U\cap V\times\mathbb{R}^k \arrow[r, "\mathbf{y}\circ\pi_1"] \arrow[dl, "\pi_1"] & \mathbf{y}(U\cap V) \\
& & U\cap V & &
\end{tikzcd}$$

where every map on top of the diagram is a diffeomrophism. Thus, we have the diffeomophism from $\mathbf{x}(U \cap V)$ to $\mathbf{y}(U \cap V)$, which by definition of Φ and Ψ is the transition map. Let's say the maps from $\mathbf{x}(U \cap V)$ to $\pi^{-1}(U \cap V)$ is labeled as φ^{-1} and the maps from $\pi^{-1}(U \cap V)$ to $\mathbf{y}(U \cap V)$ is labeled as ψ. These two maps are Then since TM is orientable:

$$\det D(\psi \circ \varphi^{-1})(x) > 0$$

Example 11.6.3: Oriented Manifold

1. S^n is orientable.
2. The mobius band is not orientable.

11.7 Cotangent Bundle

Recall that the derivative takes smooth functions (more generaly n-differentiable functions), and gives back a function that is a linear approximation at every point

$$D(-) : C^\infty(\mathbb{R}^n, \mathbb{R}^m) \to C^\infty(R^n, \mathcal{L}(\mathbb{R}^n \to \mathbb{R}^m))$$

In the case where $m = 1$, we get

$$D(-) : C^\infty(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n, (\mathbb{R}^n)^*)$$

This required that we attach a copy of $\mathbb{R}^n$ at each tangent space. As we saw with tangent bundles, we require a more sophisticated structure. In this chapter we work on defining the appropriate structure.

What is the point of defining the cotangent bundle when we already have the tangent bundle? This is because integration ought to be coordinate independent. The point of using functional's is that they allow for integration to be done in a coordinate-independent way.

More generally, some of the most important constructs in mathematics are maps on dual spaces, the integral being perhaps the most famous such example. In particular, if a function $f : \mathbb{R} \to \mathbb{R}$, then we can take $\int_a^b \_\_\_ : \hom_{\mathbf{Meas}}(\mathbb{R}, \mathbb{R}) \to \mathbb{R}$, where $f \mapsto \int_a^b f(t)dt$.

When integrating in higher dimension, we need to keep track of two pieces of information: the dimension of the ambient space we are in (which in elementary calculus is always $\mathbb{R}^n$) and the dimension of the object we are integrating (ex. a curve, a plane, etc.). As an example of integrating over 1-dimensional objects in an arbitrary dimensional $\mathbb{R}^n$, we can define some curve $\gamma : [a, b] \to \mathbb{R}^n$, and then have $f : \mathbb{R}^n \to \mathbb{R}$. We might wish to define $\int_\gamma f(t)dt$ which would give the "length" of the curve. it if it is a straight line, it should be the same as just measuring it with a ruler, and if it is curved, it should be the same as taking a piece of string and measuring out the length, and then pulling it straight and then measuring it with a ruler. You may have seen before that:

$$\int_\gamma f(s)ds = \int_a^b f(\gamma(t))t\sqrt{1 + \gamma'(t)^2}dt$$

The intuitive formula was given by the fact that the squareroot information inside the integral was a mini-pythagoras theorem. But how would we come to this conclusion more readily?

More generally, what if we had a general n-dimensional M instead of $\mathbb{R}^n$, how would we find the area along a curve? what about if we are trying to integrate something more complicated than a curve? The more general discussion for objects that are not curves will be done in chatper 13, which will be a generalization of our discussion of the case for curves.

Definition 11.7.1: Cotangent Bundle

Let M be a smooth manifold and TM its tangent bundle. Then define:

$$T_p^*(M) = (T_pM)^* \qquad T^*M = \coprod_{p\in M} T_p^*M$$

The space T_p^*M is called the *cotangent space* of M at p, and T^*M is called the *cotangent bundle* of M. An element $v \in T_p^*M$ is called a *tangent covector at* p or just *covector at* p.

Given a chart $(U, \mathbf{x})$, if $\left(\frac{\partial}{\partial x^i}\Big|_p\right)$ are coordinates for T_pM, there exists a dual basis:

$$(\lambda^i|_p)$$

and as we've just seen, if $\omega \in T_p^*M$, then $\omega = \omega_i\lambda^i|_p$, where

$$\omega_i = \omega\left(\frac{\partial}{\partial x^i}\bigg|_p\right) \in \mathbb{R}$$

We now have to deal with what happens if we have another chart $(V, \mathbf{y})$ containing p. We would thus get another basis $(\mu^i|_p)$ for T_p^*M. Then we know from our computation in equation (1.2) on page 35 that:

$$\frac{\partial}{\partial x^i}\bigg|_p = \sum_j \frac{\partial y^j}{\partial x^i}(p)\,\frac{\partial}{\partial y^j}\bigg|_p$$

Thus, writing ω in both coordinates: $\omega = \omega_i\lambda^i|_p = \tilde{\omega}_i\tilde{\lambda}^i|_p$, we get:

$$\omega_i = \omega\left(\frac{\partial}{\partial x^i}\bigg|_p\right) = \omega\left(\sum_j \frac{\partial y^j}{\partial x^i}(p)\,\frac{\partial}{\partial y^i}\bigg|_p\right) = \sum_j \frac{\partial y^j}{\partial x^i}(p)\tilde{\omega}_j$$

Hence, we have the following two equations: one is the components of a vectors after change of variables on a charts of a tangent bundle, another is a change of variables on the charts of a cotangent bundle:

$$\tilde{v}^i = \sum_j \frac{\partial y^i}{\partial x^j}(p)v^j \qquad\qquad w_i = \sum_j \frac{\partial y^j}{\partial x^i}(p)\tilde{\omega}_j$$

Notice how the indices are a bit different, that is, the change of variables for cotangent bundles has the "dual" effect as the change of variables for the tangent bundle.

(got to build-up these computations earlier to motivate the terms *covariant vector* and *contravariant vector*)

We now turn out attention to T^*M and covector-fields. Given any chart $(U, \mathbf{x})$ where $\mathbf{x} = (x^1, x^2, ..., x^n)$, we would like as with vector fields to make a "covector field" using $x^1, x^2, ..., x^n$:

$U \to T^*M$ as *coordinate covector fields.* To do so, we must make sure that T^*M has a vector-bundle structure. The proof of this is essentially the same for all dual of vector-bundles, and so we prove the slightly more general result and show as as corrollary that cotangent bundle's are vector-vundles

Lemma 11.7.2: Dual Bundles Are Vector Bundles

Let $\pi : E \to M$ be any smooth vector bundle. Then the dual E^* is a well-defined vector bundle over M i.e., there exits bundle charts and transition mappings corresponding to those for E.

***Proof*:**
First, since all our vector bundles are over the reals, they are inner products so each E_p is naturally isomorphic to E_p^*. Using this, define $\pi^* : E^* \to M$ to be the map sending

$$\sum_i c_i e_{i,p}^* \mapsto p$$

where $e_{i,p}^*$ is the dual basis of E_p. Since each E_p is naturally isomorphic to E_p^* (given the inner product), then bundle atlas for E can be made to a bundle atlas for E^*, as we'll now demonstrate. It is clear that for any Φ in the bundle atlas of E, then Φ^* sending:

$$\Phi^* : \pi^{-1}(U) \to U \times \mathbb{R}^n \qquad \Phi^*\left(c_i e_{i,p}^*\right) = (p, c_1, ..., c_n)$$

is still smooth and induces a linear isomorphism for each E_p^*. It remains to show that the transition functions are well-defined and smooth. For E, we have that for any patches U, V of the vector-bundle, there exists a $\tau : U \cap V \to \mathrm{GL}(n)$ such that

$$\Psi^{-1} \circ \Phi : U \cap V \times \mathbb{R}^n \to U \cap V \times \mathbb{R}^n \qquad (p, v) \mapsto (p, \tau(p)v)$$

where $\tau(p)$ is the change of basis matrix and τ varries smoothly. Thus, we just need to find how a change of basis works for dual spaces. If $v_p \in E_p$, then represented in two different bases, we get:

$$v_p = \sum_i c_i e_i^* = \sum_i d_i f_i^*$$

Computing the ceofficients, we get:

$$\begin{aligned} c_i &= v(e_i) \\ &= v\left(\tau(p)^{-1} f_i\right) \\ &= v\left(\sum_j \tau(p)_{ji}^{-1} f_j\right) \\ &= \sum_j \tau(p)_{ji}^{-1} v(f_j) \\ &= \sum_j \tau(p)_{ji}^{-1} d_i \end{aligned}$$

Thus, we get that

$$(\Psi^*)^{-1} \circ \Phi^* : U \cap V \times \mathbb{R}^n \to U \cap V \times \mathbb{R}^n \qquad (p, v*) \mapsto (p, (\tau^{-1})^t(p)v*)$$

since τ^{-1} is smooth with smooth inverse (as we've shown the homework on Lie Groups) it is a diffeomorphism, and since the transpose is a linear transormation, $(\tau^{-1})^t$ is also a diffeomorphism. Hence:

$$\tau^* = (\tau^{-1})^t$$

is the desired transition function, showing that $\pi^* : E^* \to M$ is still a vector-bundle, as we sought to show.

Corollary 11.7.3: Cotangent Bundle Is Vector-Bundle

Let M be a n-manifold and T^*M be the associated cotangent bundle. Then there is a unique topological and smooth structure on T^*M making it a smooth vector bundle of rank n for which all coordinate vectors are smooth local sections.

***Proof*:**
This comes from the fact that T^*M is the dual of TM, and so we apply the previous lemma.

If $(U, \mathbf{x})$ is a chart for M, then we can define a chart from $\pi^{-1}(U)$ to $\mathbb{R}^{2n}$ by:

$$(\omega_i \lambda^i|_p) \mapsto (x^1(p), ..., x^n(p), \omega_1, ..., \omega_n)$$

This chart is called the *natural coordinate chart* for T^*M with respect to $\mathbf{x}$, just as it was for TM.

Definition 11.7.4: Covector-Fields: Differential 1-Forms

Let M be a manifold and T^*M be its cotangent bundle. Then a section map $\sigma_i : M \to T^*M$ is called a *covector field* or a *differential* 1*-form* (or just *1-form* for short).

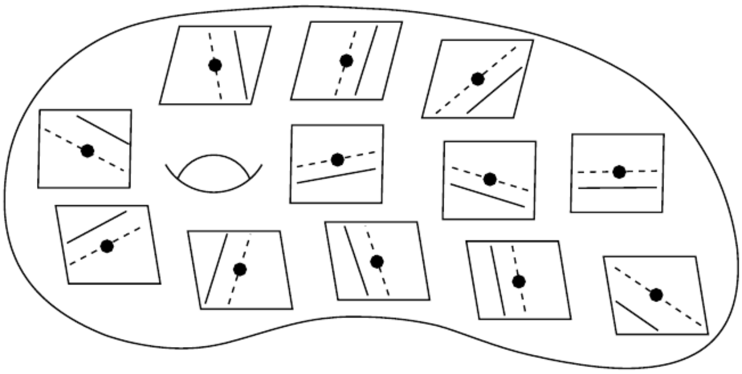
Proposition 11.7.5: Smoothness Criterion For Covector Fields

Let M be a smooth manifold and ω a rough covectorfield. then the following are equivalent:

1. ω is smooth
2. In every coordinate chart, each component function is smooth
3. each point of M is contained in some coordinate chart in which ω has smooth coordinate functions
4. For every smooth vector field $X \in \mathfrak{X}(M)$, the function $\omega(X)$ is smooth on M
5. For every open subset $U \subseteq M$ and every smooth vector-field X on U, the function $\omega(U) : U \to \mathbb{R}$ is smooth on U.

***Proof*:**
was left as exercise

Vector-fields are relatively easy to picture, since one of their equivalent constructions was as tangent vectors to curves. Covector fields are not so easy to visualize since they are functions, however since they are linear functions we may geometrically represent them via the location of the fibers of 0 and 1. Consider the image:



where for for any $\omega_p \in T_p^*M$, imagine taking it's kernel (which will be have co-dimension 1, dimension 1 in the above image), and the fiber above $\omega_p^{-1}(1)$, which is parallel to the kernel since ω_p is linear. The point of the kernel is to see the "direction", so to speak, that we care about, while the fiber above 1 tell's give us an idea of the "size" of ω_p: when the covector-field is "small" (which I'll take as the values of $\omega_p(e_i)$ are small), then lines of points that map to $\omega_p(v) = 1$ becomes far from the kernel (since if $\omega_p(e_i) \ll 1$, by linearity there would have to be a $r \in \mathbb{R}$, $1 \ll r$ such that $\epsilon_p(r \cdot e_i) \gg 1$). If we have a covector-field then, these hyperlines vary continuously, or smoothly if the covector-field is smooth.

(coframes)

Definition 11.7.6: Collection Of Covector Fields

Let M be a manifold. Then $\mathfrak{X}^*(M)$ is the collection of all smooth covectorfields on M Then $\mathfrak{X}^*(M)$ is the collection of all smooth covectorfields on M

As with $\mathfrak{X}$, $\mathfrak{X}^*$ is naturally a $C^\infty(M)$-module with the action of $f \cdot \omega$ being defined pointwise as:

$$(f\omega)_p = f(p)\omega_p$$

11.8 Differential of Functions

Recall in elementary calculus that we have defined the gradient of a function. In the current notation, we would write it as:

$$\nabla f = \sum_i^n \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}$$

In other words, we have a map $\nabla : C^\infty(\mathbb{R}^n) \to TM$, or $\nabla f : \mathbb{R}^n \to TM$ (i.e. a vector-field!). This definition worked well when we were not concerned about being coordinate independent. However, this will now pose a problem for us:

Example 11.8.1: Gradient In Different Coordinates
Let $f(x, y) = x^2$ on $\mathbb{R}^2$ and let X be the vector space defined by:

$$X = \nabla f = 2x \frac{\partial}{\partial x}$$

Computing X in polar coordinates (using the change of variables formula), we see that it will *not* be equal to:

$$\frac{\partial f}{\partial r} \frac{\partial}{\partial r} + \frac{\partial f}{\partial \theta} \frac{\partial}{\partial \theta}$$

Although partial derivative of a smooth function cannot be intepreted in a coodinate-independent way as the components of a vector field, it turns out they *can* be interpreted in a coordinate-*independent* as the components of a *covector field*. The following takes the example of ∇f and gives us how we can generlize it to give us a covector field:

Definition 11.8.2: Differential Of Smooth Function

Let $f : M \to \mathbb{R}$ be a smooth function on a manifold M (with or without boundary). Then we define the covector field $df : M \to T^*M$ where $df(p) \in T_p^*M$ as:

$$df(p) : T_pM \to \mathbb{R} \qquad df(p)(v) = v(f)$$

The covector field df is called the *differential* of f.

Recall that $T_pM = \mathrm{der}(C_p^\infty(M)) \subseteq (C_p^\infty(M))^\vee$, and so $v : C_p^\infty(M) \to \mathbb{R}$, and hence $df_p(v) = v(f)$ is well-defined. We will often write df_p and $df(p)$ interchangeably. We claimed that df is a smooth covector field, we should actually prove it:

Proposition 11.8.3: Differential Is Smooth Covector Field

Let $f : M \to \mathbb{R}$ be a smooth function and df the differential of f. Then df is a smooth covectorfield.

***Proof*:**
It is straight-forward to see that for any $p \in M$ and $v_p \in T_pM$ that $df_p(v_p)$ is indeed a linear functional, and so $df_p(v) \in T_p^*M$. For smoothness of df, by proposition 11.7.5, we have that for any smooth vector-field X on M, $df(X)$ is smooth, since Xf is smooth, and so df is smooth, as

we sought to show.

To see that df is the proper generlization of the gradient, consider $(U, \mathbf{x})$ to be a chart on M, so $x = (x^1, x^2, ..., x^n)$. Let (λ^i) be the corresponding coframe on U. Then we can write

$$df_p = \sum_i A_i(p)\lambda^i|_p$$

for appropriate functions functional $A_i : U \to \mathbb{R}$ representing the coeficients as p varries, where

$$A_i(p) = df_p\left(\left.\frac{\partial}{\partial x^i}\right|_p\right) = \left.\frac{\partial}{\partial x^i}\right|_p f = \frac{\partial f}{\partial x^i}(p)$$

Note that if the covector field is smooth, then each $A_i \in C^\infty(U)$. We get that df in coordinates is represented as:

$$df_p = \sum_i \frac{\partial f}{\partial x^i}(p)\lambda_i|_p \tag{11.1}$$

Hence, the component functions of df in any smooth coordinate chart are the partial derivaties of f with respect to those coordinates. Notice how this is exactly the formula for the gradient, but translted into a co-vectorfield. Hence df is indeed the appropraite generlization of the gradient.

Applying equation (11.1) to the special case where $f = x^i$ is a component function of some chart $\mathbf{x}$, we get

$$dx^j|_p = \sum_i \frac{\partial x^j}{\partial x^i}\lambda^i|_p = \sum \delta^j_i \lambda^i|_p = \lambda^j|_p$$

hence, $dx^i|_p = \lambda^i|_p$! We can thus re-write equation (11.1) as:

$$df_p = \sum_i \frac{\partial f}{\partial x^i}(p)dx_i|_p$$

so, as an equation between covector *fields*, we get:

$$df = \sum_i \frac{\partial f}{\partial x^i}(p)dx_i$$

and in the 1-dimensional case, we get:

$$df = \frac{\partial f}{\partial x}(p)dx$$

Which is now *exactly* the definition of the gardient with $\frac{\partial}{\partial x^i}$ replaced with dx^i. From now on, we replace λ^i with dx^i.

Example 11.8.4: Differential Of Function

Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x, y) = x^2 \cos(x)$. Then Given any chart on $\mathbb{R}^2$ (ex. the identity chart), we get:

$$\begin{aligned} df &= \frac{\partial x^2 y\cos(x)}{\partial x}dx + \frac{\partial x^2 y \cos(x)}{\partial y}dy \\ &= (2xy\cos(x) - x^2 y\sin(x))dx + x^2\cos(x)dy \end{aligned}$$

Proposition 11.8.5: Properties Of Differential

Let M be a smooth manifold and $f, g \in C^\infty(M)$. Then:

1. for any constants $a, b \in \mathbb{R}$, $d(af + bg) = ad(f) + bd(g)$
2. $d(fg) = fd(g) + d(f)g$
3. $d(f/g) = (fd(g) + d(f)g)/g^2$ on the set where $g \neq 0$
4. If $J \subseteq \mathbb{R}$ is an interval containing the image of f, and $h : J \to \mathbb{R}$ is a smooth function, then $d(h \circ f) = (h' \circ f)df$
5. If f is constant, then $df = 0$

***Proof*:**
John left as exercise. For (4), use coordinate representation.

Since df generlizes ∇f, we would hope that the information that ∇f gave us is still given by df, in paricular the information given when $\nabla f = 0$. The following proposition shows us that it does:

Proposition 11.8.6: Functions With Vanishing Differential

Let $f : M \to \mathbb{R}$ be a smooth real valued function, then $df = 0$ if and only if f is constant on each component of M

***Proof*:**
It suffices to check in the case where M is connected. If f is constant, then by proposition 11.8.5 $df = 0$

Conversely, suppose $df = 0$. Let $p \in M$ be some pont, and consider $\mathcal{C} = \{q \in M \mid f(q) = f(p)\}$. Choose some $q \in M$, an let U be some smooth coordinate ball centered at q with coordinate chart $\mathbf{x} = (x^1, x^2, ..., x^n)$. Since

$$0 = df = \frac{\partial f}{\partial x^i} dx^i$$

we get that $\frac{\partial f}{\partial x^i} \equiv 0$ on U, then it is not difficult using elementary caluculs that f is constant on U. This shows that $\mathcal{C}$ is open. Furthermore, $\mathcal{C} = f^{-1}(p)$, and hence is clsoed by continuity, making $\mathcal{C}$ clopen. Hence, $\mathcal{C} = M$, since M is connected, as we sought to show.

Just like how ∇f represent the small change with respect to the "independent" variables x^i, we get a similar meaning for df with the right interprtation. (df_p best linaer approximation of ∇f at p)

You may have noticed that we have already encounted something like df_p before: If $f : M \to \mathbb{R}$, then $T_p f : T_p M \to T_p \mathbb{R}$, and since $T_p \mathbb{R} \cong \mathbb{R}$ we have $T_p f : T_p M \to \mathbb{R}$, so $T_p f \in T_p^* M$! In fact, thse are indeed identical: notice how they are exactly the same in coordinate represetnations! The usefulness of distinguishing $T_p f$ and df_p with two different symols will come from the role df_p will play in integrating functions over manifolds!

(motivation for the following)

Proposition 11.8.7: Derivative Of A Function Along A Curve

Let M be a smooth manifold, $\gamma : [a,b] \to M$ a smooth curve, and $f : M \to \mathbb{R}$ a smooth function. Then the derivative of $f \circ \gamma : [a,b] \to \mathbb{R}$ is given by:

$$(f \circ \gamma)'(t) = df_{\gamma(t)}(\gamma'(t))$$

***Proof*:**
Visually, what we are working towards is:

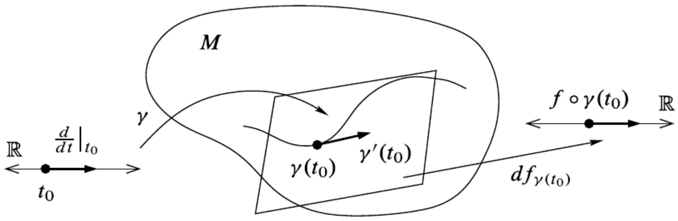


Fig. 11.3 Derivative of a function along a curve

This is definition unwinding, for any $t_0 \in [a,b]$:

$$\begin{aligned}
df_{\gamma(t_0)}(\gamma'(t_0)) &= \gamma'(t_0)f && \text{definition of } df \\
&= d\gamma_{t_0}\left(\left.\frac{d}{dt}\right|_{t_0}\right) f && \text{definition of } \gamma'(t) \\
&= \left.\frac{d}{dt}\right|_{t_0} (f \circ \gamma) && \text{definition of } d\gamma \\
&= (f \circ \gamma)'(t_0) && \text{definition of } d/dt|_{t_0}
\end{aligned}$$

11.9 Pullback Of Covectors

Given $f : M \to N$, we get $T_p f : T_p M \to T_{f(p)} N$, the dual of which is $T_p f^* : T_{f(p)} N^* \to T_p M^*$, the collection of which is $T f^* : TN^* \to TM^*$. We give a name to this function:

Definition 11.9.1: Pullback of tangent Space

Let $f : M \to N$ be a smooth map. Then the map Tf^* is called the *pullback* of f, or the *cotangent map* of f. Unravelling the definition, we get $T_p f^*(\omega) \in T_p^* M$:

$$T_p f^*(\omega)(v) = \omega(T_p f(v))$$

Naturally, T_p^* is a contravariant functor from the category of pointed smooth manifolds to the category of real vector spaces, while T^* is a contravariant functrom from the cateogry of smooth manifolds to itself. It is thus pretty inconvinient to call the elemnts of T_p^*M "covariant vectors", but it is now a convention entrenched in differential geometry.

(a word on covector field pullback. Won't be able to do yet because haven't discussed vector field pushforward because that requries lie group homomorhpisms to do successfully)

Definition 11.9.2: Pullback Of Covector Field

Let $f : M \to N$ be a smooth map, $Tf^* : TN^* \to TM^*$ be the pullback of f, and $\omega : N \to T^*N$ a covector-field on N (i.e. a 1-form). Then the *pullback* of ω by f is a rough covector field $(f^*\omega) : M \to T^*M$ given by:

$$(f^*\omega)_p(v) = \omega_{f(p)}(T_pf(v))$$

which we can represent as:

$$(f^*\omega)_p = T_pf^*(\omega_{f(p)})$$

We do not yet know that $(f^*\omega)$ is continuous or that it is smooth if ω is smooth. The following proposition gives us that:

Proposition 11.9.3: Properties of Pullback of Covector Fields

Let $f : M \to N$ be a smooth map, $u : N \to \mathbb{R}$ a continuous real-valued function on N, $\omega : N \to T^*N$ a covector field on N. Then:

$$f^*(u \cdot \omega) = (u \circ f) \cdot f^*\omega$$

where $(u \cdot \omega)_p = u(p)\omega$ is the action defined on $\mathfrak{X}^*$. If u is smooth, then:

$$f^*du = d(u \circ f)$$

***Proof*:**
To prove the first equation, we see that:

$$\begin{aligned}
(f^*(u \cdot w))_p &= T_pf^*((u \cdot \omega)_{f(p)}) && \text{pullback definition} \\
&= T_pf^*(u(f(p))\omega_{f(p)}) && \text{module action} \\
&= u(f(p))T_pf^*(\omega_{f(p)}) && \text{Linearity of } T_pf^* \\
&= u(f(p))(f^*\omega)_p && \text{pullback definition} \\
&= (u \circ f) \cdot (f^*\omega)_p && \text{module action}
\end{aligned}$$

Givign equality. If u is furthermore smooth, then du is a covector field on N. For any $v \in T_pM$,

$$\begin{aligned}
(f^*du)_p(v) &= T_pf^*(du_{f(p)})(v) && \text{pullback definition} \\
&= du_{f(p)}(T_pf(v)) && \text{evaluate } T_pf^* \\
&= T_pf(v)(u) && \text{evaluate } du \\
&= v(u \circ f) && \text{evaluate } T_pf(v) \\
&= d(u \circ f)_p(v) && \text{pullback definition}
\end{aligned}$$

Proposition 11.9.4: Continuous And Smooth 1-Forms

Let $f : M \to N$ be a smooth map and $\omega : N \to T^*N$ a 1-form on N. Then $f^*\omega$ is a continuous 1-form n M. If ω is smooth, so is $f^*\omega$.

***Proof*:**
Let $f : M \to N$ be smooth, $p \in M$ be arbitrary, and choose smooth coordinate charts (y^j) for N in a neighborhood V of $f(p)$. Let $U = f^{-1}(V)$ which is a neighborhood of p. Writting ω as $\omega = \omega_j dx^j$ for continuous functions ω_j on V, and using the above proposition twice, (on $f|_U$), we get:

$$f^*\omega = f^*(\omega_j dx^j) = (\omega_j \circ f)f^*dy^j = (\omega_j \circ f)d(y^j \circ f)$$

Often, the above formula is written as:

$$f^*\omega = (\omega_i \circ f)d(f^i) \tag{11.2}$$

where f^i is the component function of f.

Example 11.9.5: Pullback
Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be $(u, v) = f(x, y, z) = (x^2y, y\sin z)$. Let $\omega \in \mathfrak{X}^*(\mathbb{R}^2)$ be:

$$\omega = udv + vdu$$

Then $f^*\omega$ would be:

$$\begin{aligned}
f^*\omega &= (u \circ f)d(v \circ f) + (v \circ f)d(u \circ f) \\
&= (x^2y)d(y\sin z) + y\sin(z)d(x^2y) \\
&= 2xy^2 \sin z dx + 2x^2y \sin z dy + x^2y^2 \cos z dz
\end{aligned}$$

(1-form restriciton to submanifold)

11.10 Line Integrals

As mentioned, covector-fields (which are 1-forms) allow us to define integration on manifolds, in particular line integrals (hence the name 1-form).

In the simplest case, we can take $[a, b] \subseteq \mathbb{R}$ and consider a smooth covector field $\omega : [a, b] \to T^*[a, b]$ on $[a, b]$ (smooth here means that it extens to a smooth covector field to some open neighborhood of $[a,b]$). If we let t denote the standard coordinate on $\mathbb{R}$ (i.e. the coordinate chart $(t, \mathbb{R})$), Then we can represent ω in terms of coordinates as: $\omega_t = f(t)dt$ for some smooth function $f : [a,b] \to \mathbb{R}$. Using this, we can finally define the *integral of ω over* $[a,b]$ to

$$\int_{[a,b]} \omega = \int_a^b f(t)dt$$

This seems to be dependent on the particular choice of basis to represent ω and hence not well-defined; the following proposition shows that is not the case *given* that we preserve orientation!

Proposition 11.10.1: Diffeomrophsim Invariance Of The Integral

Let ω be a smooth covector-field on a compact interval $[a,b] \subseteq \mathbb{R}$. If $\varphi : [c,d] \to [a,b]$ is an increasing diffeomorphism, then

$$\int_{[c,d]} \varphi^* \omega = \int_{[a,b]} \omega$$

***Proof*:**
Lettting s denote the standard coordinate on $[c,d]$ and t on $[a,b]$, then by proposition 11.9.3, we see that the pull-back $\varphi^*\omega$ has the coordinate expression $(\varphi^*\omega)_s = f(\varphi(s))\varphi'(s)ds$. Putting this into the definition of the ordinary integarl and using change of variables, we get:

$$\int_{[c,d]} \varphi^* \omega = \int_c^d f(\varphi(s))\varphi'(s)ds = \int_a^b f(t)dt = \int_{[a,b]} \omega$$

Note (or prove) that if φ was a decresing diffeomorphism, we would get:

$$\int_{[c,d]} \varphi^* \omega = -\int_{[a,b]} \omega$$

Thus, we see that it is essentially well-defined up to a minus sign. Thus, we will keep track of this information in order to be able to define integration that change the curve even if it introduces a negative:

Definition 11.10.2: Reparamaterization

Let $\gamma : [a,b] \to M$, $\tilde{\gamma} : [c,d] \to M$ be curves into a smooth manifold. If $\tilde{\gamma} = \gamma \circ \varphi$ for some diffeomorphism $\varphi : [c,d] \to [a,b]$, then if φ is an *increasing* function, it is called a *forward re-paramaterization*, and if φ is a decreasing function, it is called a *bacwkards reparamaterization.*

Proposition 11.10.3: Parameter Independent Line Integral

p. 290 John lee

***Proof*:**
here

(now gies a thing about piecwise line integral results, skipped for now, p. 288, I believe it is to show linaerity of integral. Make sure to amend the definition of reparamaterization to be aware of piecewise segemnts)

Definition 11.10.4: Line Integral On Manifold

Let $\gamma : [a, b] \to M$ be a smooth curve segment and ω a smooth covector field on M. Then we define the *line integral* of ω over γ to be the real number:

$$\int_\gamma \omega = \int_{[a,b]} \gamma^* \omega$$

Proposition 11.10.5: Properties of Line Integral

Let M be a smooth manifold (with or without boundary). Suppose $\gamma : [a, b] \to M$ is a piecewise smooth curve-segment and $\omega_1, \omega_2 \in \mathfrak{X}^*(M)$. Then:

1. For any $c_1, c_2 \in \mathbb{R}$:
$$\int_\gamma (c_1\omega_1 + c_2\omega_2) = c_1 \int_\gamma \omega_1 + c_2 \int_\gamma \omega_2$$
2. If γ is a constant map, then:
$$\int_\gamma \omega = 0$$
3. If $\gamma_1 = \gamma|_{[a,b]}$ and $\gamma_2 = \gamma|_{[c,d]}$ with $a < c < b$, then:
$$\int_\gamma \omega = \int_{\gamma_1} \omega + \int_{\gamma_2} \omega$$
4. If $f : M \to N$ is any smooth map and $\eta \in \mathfrak{X}^*(M)$, then:
$$\int_\gamma f^* \eta = \int_{f \circ \gamma} \eta$$

***Proof*:**
exercise

Example 11.10.6: Computing Line Integral

Let $M = \mathbb{R}^2 \setminus \{0\}$ given by

$$\omega = \frac{xdy - ydx}{x^2 + y^2}$$

with $\gamma : [0, 2\pi] \to M$ with $\gamma(t) = (\cos(t), \sin(t))$. The answer should be 2π. Check John Lee p. 290 if you're stuck.

This next proposition gives another way we can see the line integral

Proposition 11.10.7: Concrete Line Integral

Let $\gamma : [a, b] \to M$ be a piecewise smooth curve segment. Then the line integral of ω over γ can be expressed as:

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\gamma'(t))dt$$

***Proof*:**
unwinding definition, John Lee p.291

Theorem 11.10.8: Ftc For 1-Forms

Let M be a smooth manifold (with or without boundary). Suppose $f : M \to \mathbb{R}$ is smoot hand $\gamma : [a, b] \to M$ is a piecewise smooth curve segment in M. Then:

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$$

***Proof*:**
By proposition 11.9.3 and proposition 11.10.7, we get:

$$\int_\gamma df = \int_a^b df_{\gamma(t)}(\gamma'(t))dt = \int_a^b (f \circ \gamma)'dt$$

which by the normal fundamental theorem of calculus is equal to:

$$f(\gamma(b)) - f(\gamma(a))$$

If we have a curve, the mid-points cancel.

11.11 Conservative Covector Field

The FTC shows that integarls of forms of the form df are particularly easy once f is known.

Definition 11.11.1: Exact Differnetial

Let ω be a 1-form. Then ω is called *exact* if there exists a f such that

$$\omega = df$$

In this case, f is called the *potential* of ω.

Thus, it is meaningful to wonder which forms are exact, if not all. The FTC provides a hint: the integral of a form should be $f(\gamma(b)) - f(\gamma(a))$. Thus, any *closed curve* (so $\gamma(a) = \gamma(b)$) should integrate to zero. We give a name to form with this proeprty:

Definition 11.11.2: Conservative Form

Let ω be a 1-form. Then ω is called *conservative* if every line integral over a piecewise smooth closed curve is 0.

Proposition 11.11.3: Line Integral Independent On Coservative Form

A smooth 1-from ω is conservative if and only if its line integrals are path independent in the senes that if $\int_\gamma \omega = \int_{\tilde{\gamma}} \omega$ whenever γ and $\tilde{\gamma}$ are piecewise smooth curve-segments with the same starting and ending point

***Proof*:**
Remember that we defined integration on *piece-wise* curve segments, making this proposition relatively easy!

12

Tensor Bundles and Differential Forms

Knowing how to integrate one-dimensional manifolds, it is natural to upgrade this to higher dimensional manifolds. Furthermore, in higher dimensions, the notion of an inner product (and indeed other forms such as quadratic forms) become much more interesting. In this chapter, we define the right object to use to keep track of this information locally so that it can be glued together to give more global information. Importantly, a lot of the interesting maps we are working with are multi-linear (symmetric maps, determinant, non-degenerate forms, and so forth), and hence the object we construct shall be able to be the right fiber bundle to hold all this information.

12.1 Tensor Bundle

Definition 12.1.1: Tensor Bundle

Let M be a smooth manifold, TM it's tangent bundle, and $\mathcal{T}^k T_pM$ the k-tensors on T_pM. Then:

$$\mathcal{T}^k(T^*M) = \coprod \mathcal{T}^k(T_p^*M)$$

is called the *[covariant] tensor bundle* of M. Analogously, we define:

$$\mathcal{T}^k(TM) = \coprod \mathcal{T}^k(T_pM)$$

to be the *[contravariant] tensor bundle* of M, and

$$\mathcal{T}^{(k,l)}(TM) = \coprod \mathcal{T}^{k,l}(T_pM)$$

to be the *mixed tensor bundle* of M of type (k,l)

Notice that the tangent and cotangent bundles are both tensor bundles, and hence they are all special cases.

Definition 12.1.2: Tensor Field

Let M be a smooth manifold. Then ω is a (covariant, contravariant, mixed) *tensor field* if it is a section map from M to the (covariant, contravariant, mixed) tensor bundle. A tensor field is *smooth* in the same way a section map for a vector-bundle is smooth

Thus, contravariant 1-tensor fields are vector-fields, covariant 1-tensor field are covector-fields. Since 0-tensors are scalars (i.e. $\mathbb{R}$), 0-tensors fields are continuous real valued functions. The space of smooth sections on these tensor bundles, which we'll denote $\Gamma(\mathcal{T}^k(T^*M)), \Gamma\left(\mathcal{T}^k(TM), \Gamma(\mathcal{T}^{(k,l)}(TM)\right)$ is an infinite dimensional vector-space over $\mathbb{R}$, and a module over $C^\infty(M)$. For any smooth coordinate chart $(U, \mathbf{x})$, these bundles can be written as:

$$A = \begin{cases} A_{i_1\ldots i_k}\, dx^{i_1} \otimes \cdots \otimes dx^{i_k}, & A \in \Gamma\big(T^k T^* M\big); \\ A^{i_1\ldots i_k} \dfrac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \dfrac{\partial}{\partial x^{i_k}}, & A \in \Gamma\big(T^k TM\big); \\ A^{i_1\ldots i_k}_{j_1\ldots j_l} \dfrac{\partial}{\partial x^{i_1}} \otimes \cdots \otimes \dfrac{\partial}{\partial x^{i_k}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_l}, & A \in \Gamma\big(T^{(k,l)} TM\big). \end{cases}$$

The most important tensor fields for us are *covariant k-tensor* fields, and so we single-out a notation for them:

$$\mathfrak{T}^k(M) = \Gamma(\mathcal{T}^k(T^*M))$$

(some useful lemmas and propositions)

(define symmetric and alternating tensor fields)

12.2 Pullback Of Tensor Fields

Just like covector fields can be pulled back along smooth maps, so can covariant tensor fields (which should be excitting, since we were able to define line integrals on manifolds due to this). The fact that we can pull back like this is probably the main reason covarian tensor fields are prioritized in study

Definition 12.2.1: Tensor-Field Pullback

Let $f : M \to N$ be a smooth map. Then for any $p \in M$ and any k-tensor $\alpha \in \mathcal{T}^k(T^*_{f(p)}N)$, define $T_pf^*(\alpha) \in \mathcal{T}^k(T^*_pM)$ to be:

$$(T_pf^*(\alpha))(v_1, ..., v_k) = \alpha(T_pf(v_1), ..., T_pf(v_k))$$

for any $v_1, v_2, ..., v_k \in T_pM$. This is called the *pointwise pullback* of α by f at p. If $A : N \to \mathcal{T}^k(T^*_{f(p)}N)$ is a covariant k-tensor field on N, define the rough k-tensor field f^*A on M, called the *pullback* of A by f by:

$$(f^*A)_p = T_pf^*(A_{f(p)})$$

where this tensor fields acts on vectors $v_1, v_2, ..., v_k \in T_pM$ by:

$$T^*_p(f^*A)(v_1, ..., v_k) = \alpha(T_pf(v_1), ..., T_pf(v_k))$$

Proposition 12.2.2: Properties Of Pullbacks

Let $f : M \to N$, $g : N \to P$ be smoth maps, A, B covariant tensor fields on N, and $\tau : N \to \mathbb{R}$ a real-valued function on N. Then:

1. $f^*(\tau B) = (\tau \circ f)f^*B$
2. $f^*(A \otimes B) = f^*A \otimes f^*B$
3. $f^*(A + B) = f^*A + f^*B$
4. f^*B is a *continuous* tensor field, and is smooth if B is smooth
5. $(g \circ f)^*B = f^*(g^*B)$
6. $(\mathrm{id}_N)^*B = B$

***Proof*:**
John left as exercise, since it repeats many previous things we've seen in covector fields

Note that if τ is a 0-tensor field an B a k-tensor field, if we interpret $\tau \otimes B$ as fB $f^*\tau$ as $\tau \circ f$, then (1) in the previous propostion is just a special case of (b).

Corollary 12.2.3: Computing Pullbacks

Let $f : M \to N$ be a smooth map an B a covariant k-tensor field on N. Picking $p \in M$, if $(V, \mathbf{y})$ is a coordinate chart containing $f(p)$ wit $\mathbf{y} = (y^1, y^2, ..., y^n)$, then we can write f^*B as:

$$f^* \left(B_{i_1 \dots i_n}) dy^{i_1} \otimes \cdots \otimes dy^{i_k}\right)$$
$$= (B_{i_1 \dots i_n} \circ f) d(y^{i_1} \circ f) \otimes \cdots \otimes d(y^{i_k} \circ f))$$

***Proof*:**
exercsie

Example 12.2.4: Pullback Of Tensorfield
p. 321 John Lee

12.3 Differential Forms

(motivation: alternating tensors are universal with respect to determinants, so they naturally capture volume informatin. An alternating tensor field thus captures volume information locally. The goal of this section is to find a way to use that information to integrate on manifolds)

The first thing to realize is that there is no way of integrating functions without being coordinate depenent[1]. For example, Let $C \subseteq \mathbb{R}^n$ be a closed ball, $f : C \to \mathbb{R}$ be $f(x) \equiv 1$. The intuitively:

$$\int_C f dV = \text{Vol}(C)$$

However, this is clearly dependent on some choice of coordinates, which we know for arbitrary manifolds there is no natural choice. On the other hand, we showed that a covector field could be integrated over curves in a natural way. Let's look at the geometry of covector fields a little more closely. Given a smooth manifold M, a covector field $\omega : M \to TM^*$ assigns a number to *each* tangent vector in TM (contrast this to how a vector-field simply chooses *a* tangent vector at every point). Since $\omega_p : TM \to \mathbb{R}$ is a linaer function, scaling the tangent vectors has the effect of scaling the output by the same scalar. Due to this, we only need to think of the magnitude and direction of $\omega_p(e_i)$. Thus, ω assigned a sort of "signed length meter" to each 1-dimensional subspace of each tangent space T_pM. When computing the line integral of a co-vector field, we are in effect assigning a "length" to a curve by using this varying measure of scale along points in a curve.

We know see a generlization of this, a sort of "field" that can be integrated on a k-submanifold of an n-manifold. The value at each point of this field whould give a sort of "signed volume meter" on k-dimensional subspaces of the tangent space. John Lee phrases it this way:

> It is a machine ω that accept any k tangent vectors $(v_1, v_2, ..., v_k)$ and returns a number $\omega(v_1, v_2, ..., v_k)$ that we can think of as a "signed volume" of the parrallelepiped spanned by those vectors, measured according to the scale determined by ω

[1]At least without adding any extra structure like a Riemann Metric TBD

The most famous such tool we've leanrt in linear algebra to be det on $\mathbb{R}^n$. For example, if $n = 2$ and $v_1, v_2 \in \mathbb{R}^2$, then $\det(v_1, v_2)$ gives the area of the square these two vectors give, along with an orientation determined by whether the determinant is positive or negative. The function det in higher dimensions give signed parrallelepipeds.

Let us now consider what properties such a "signed k-dimensional volume meter" ω should have. First, we know if we multiply any edge of an n-dimensional volume, we should scale the volume of the entire area by that scalar, and if we add two parrallelepipeds where all the edges are of the same length, except two, then we should add the two parrallelepipeds together i.e. ω should be multi-linear given the edges as input,

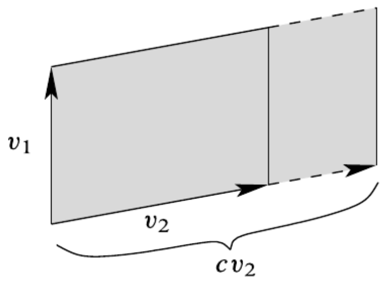


Fig. 16.2 Scaling by a constant

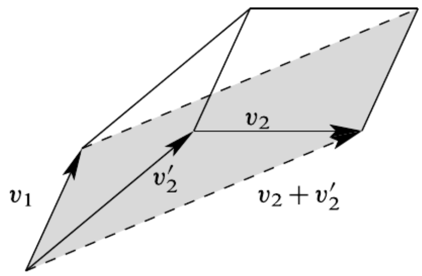


Fig. 16.3 Sum of two vectors

Finally, if two of the vectors of a parrallelepipeds are the same, then in the given dimension we would expect the resulting volume to be zero. These tools in this chapter

12.4 Algebra of Alternating Tensor

Lemma 12.4.1: Alternating Tensor Equivalence

Let α be a covariant k-tensor on a finite dimensional vector space V. Then the following are equivalent:

1. α is alternating
2. $\alpha(v_1, v_2, ..., v_k) = 0$ whenever the k-tuple $(v_1, v_2, ..., v_k)$ is linearly depenent
3. α gives a zero whenver 2 argumetns are equal:

$$\alpha(v_1, ..., w, ..., w, ..., v_k) = 0$$

***Proof*:**
p.350 John

We can defin the *alternization* to be the map $\text{Alt} : T^k(V^*) \to \Lambda^k(V^*)$,

$$\alpha \mapsto \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn}\sigma)({}^{\sigma}\alpha)$$

Example 12.4.2: Alternating Tensor

A 1-tensor is always alternating. For a general alternating 2-tensor or 3-tensor from a general tensor (do it as an exercsie)

If α is already alrernating, then $\text{Alt}(\alpha) = \alpha$.

To keep track of computations of alternating tensors the following convention is useful: If $I = (i_1, i_2, ..., i_k)$ is a collection of positive integers, it is called a *multi-index* of length k. If I is such a multi-index, and $\sigma \in S_k$, then:

$$I_\sigma = (i_{\sigma(1)}, ..., i_{\sigma(k)})$$

Note that with this definiton that $(I_\sigma)_\tau = I_{\sigma\tau}$.

Definition 12.4.3: Elementary Alternating Tensor

Let V be a finite dimensional vector space, and $(\epsilon^1, \epsilon^2, ..., \epsilon^n)$ is any basis for V^*. For each multi-index set $I = (i_1, i_2, ..., i_k)$ where $1 \leq i_1 \leq \cdots \leq i_k \leq n$, define the *alternating covariant k-tensor* $\epsilon^I = \epsilon^{i_1 \cdots i_k}$ by:

$$\epsilon^I(v_1, v_2, ..., v_k) = \det \begin{pmatrix} \epsilon^{i_1}(v_1) & \cdots & \epsilon^{i_1}(v_k) \\ \vdots & \ddots & \vdots \\ \epsilon^{i_k}(v_1) & \cdots & \epsilon^{i_k}(v_k) \end{pmatrix} = \det \begin{pmatrix} (v_1)^{i_1} & \cdots & (v_k)^{i_1} \\ \vdots & \ddots & \vdots \\ (v_1)^{i_k} & \cdots & (v_k)^{i_k} \end{pmatrix}$$

To streamline computations, it is nice to generlize the Kronecker delta to:

$$\delta_I^J = \det \begin{pmatrix} \delta_{j_1}^{i_1} & \cdots & \delta_{j_k}^{i_1} \\ \vdots & \ddots & \vdots \\ \delta_{j_1}^{i_k} & \cdots & \delta_{j_k}^{i_k} \end{pmatrix}$$

In particular, we have that:

$$\delta_I^J = \begin{cases} \text{sgn}\sigma & \text{neither } I \text{ nor } J \text{ has repeated index and } J = I_\sigma \text{ for some } \sigma \in S_k \\ 0 & \text{either } I \text{ or } J \text{ has repeated index or } J \text{ is not a permutation of } I \end{cases}$$

Proposition 12.4.4: Properties Of Elementary k-Tensors

Let (E_i) be a basis for V and (ϵ^i) a basis for V^*. Let ϵ^I be an elementary k-tensor. Then:

1. If I has a repeated index , then $\epsilon^I = 0$
2. If $J = I_\sigma$, then $\epsilon^I = (\text{sgn}\sigma)\epsilon^J$
3. The result of evaluating ϵ^I on a sequence of basis vectors is:

$$\epsilon^I(E_{j_1}, ..., E_{j_k}) = \delta^J_I$$

***Proof*:**
exercsise

This proposition suggests that to form a basis for alternating k-tensors, we should be looking at *increasing* multi-indexes. We will start using the following notation to say we are summing over all *increasing* multi-inexes:

$$\sum_I{}' \alpha_I \epsilon^I = \sum_{I:i_1<\cdots<i_k} \alpha_I \epsilon^I$$

Proposition 12.4.5: Basis For $\Lambda^k(V^*)$

Let V be an n dimensional vector space, V^* the corresponding dual with a basis (ϵ^i). Then for each positive integer $k \leq n$, the collection of k-covectors:

$$\mathcal{E} = \left\{\epsilon^I \mid I \text{ is an increasing multi-index of length } k\right\}$$

forms a basis for $\Lambda^k(V^*)$. Thus:

$$\dim(\Lambda^k(V^*)) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

***Proof*:**
p. 353-354 Jonn Lee

Next, we define tensor product that keeps alternating tensors alteranting, that is $v \otimes w \in \Lambda^k(V^*)$

Definition 12.4.6: Wedge Product

Let V be a finite dimensional vector space, $\omega \in \Lambda^K(V^*)$ and $\eta \in \Lambda^l(V^*)$. Then the *wedge product* or *exterior product*, as deifne as:

$$\omega \wedge \eta = \frac{(k+l)!}{k!l!} \text{Alt}(\omega \otimes \eta)$$

The complicated coefficient is for the following purpose:

Proposition 12.4.7: Wedging Elementary Alternating Tensors

Let V be an n-dimensional vector space, (ϵ^i) a basis for V^*. Then for any multi-indexes I, J:

$$\epsilon^I \wedge \epsilon^J = \epsilon^{IJ}$$

***Proof*:**
p.355 John Lee

Proposition 12.4.8: Properties Of Wedge Product

Let $\omega, \omega', \eta, \eta'$, an ζ be multicovectors on a finite dimensional vector space V. Then: Go over bilinearity, anti-commutativity, associativity, and

1. (Bilinearity) for all $a, a' \in \mathbb{R}$:

$$(a\omega + a'\omega') \wedge \eta = a(\omega \wedge \eta) + a'(\omega' \wedge \eta)$$
$$\eta \wedge \eta(a\omega + a'\omega') = a(\eta \wedge \omega) + a'(\eta \wedge \omega)$$

2. (Associativity)

$$\omega \wedge (\eta \wedge \zeta) = (\omega \wedge \eta) \wedge \zeta$$

3. If (ϵ^i) is any basis for V^*, and $I = (i_1, i_2, ..., i_k)$, then:

$$\epsilon^{i_1} \wedge \cdots \wedge \epsilon^{i_k} = \epsilon^I$$

4. For any covectors $\omega^1, \omega^2, ..., \omega^k$ and vectors $v_1, v_2, ..., v_k$,

$$\omega^1 \wedge \cdots \wedge \omega^k(v_1, ..., v_k) = \det(\omega^j(v_i))$$

***Proof*:**
p. 356

Note that not all k-covectors η can is the wedge product of covectors:

$$\eta = \omega^1 \wedge \cdots \wedge \omega^k$$

If it is, then η is called *decomposable*. Any k-covector is the lienar combination of decoposable k-covectors.

Corollary 12.4.9: Uniqueness Of Wedge Product

Show that if $f : \Lambda^k(V^*) \times \Lambda^\ell(V^*) \to \Lambda^{k+\ell}(V^*)$ is associative, bilinear, and anti-commutative, then $f \equiv \wedge$.

***Proof*:**
exercise

Definition 12.4.10: Exterior Algebra

Define:

$$\Lambda(V^*) = \oplus_{k=0}^{n} \Lambda^k(V^*)$$

Then this, along with the wedge product, is an anti-commutative, graded, associative algebra, called the *extior algebra* or *Grassmanian Algebara*. Naturally,

$$\dim \Lambda(V^*) = 2^n$$

12.5 Interior Multiplication

I ommited for now John Lee p. 358

12.6 Differential Forms on Manifolds

Now that we have built-up the properties we desire for our "volume metric", we next show how to attach this to a manifold:

Definition 12.6.1: Alternating Tensor Bundle

Let M be a manifold and $T^k(TM)$ be the k-tensor bundle on M. Then the subbundle consisting of all alternating k-tensors is denoted:

$$\Lambda^k(T^*M) = \coprod_{p \in M} \Lambda^k(T_p^*M)$$

Definition 12.6.2: Differential Form

Let M be a manifold and $\Lambda^k T^*M$ the alternating tensor bundle on M. Then any section $\omega : M \to \Lambda^k T^*M$ is called a *differential k-form*.

Definition 12.6.3: Collection of k-Forms

Let M be a manifold. Then the collection of all k-forms on M is denoted:

$$\Omega^k(M)$$

which is a vector-space over $\mathbb{R}$. We will consider $\Omega^0(M) = C^\infty(M)$.

We can define the wedge product between a k-form and l-form pointwise:

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p$$

which produces a $(k+l)$-form. If f is a zero form, then we interpred $f \wedge \omega$ as $f\omega$. We define:

$$\Omega^*(M) = \bigoplus_{k \in \mathbb{N}} \Omega^k(M)$$

Then the wedge product on $\Omega^*(M)$ turns it into an associative, anticommutative graded algebra. Given a smooth chart, we can write ω as:

$$\omega = \sum_I{}' \omega_I dx^{i_1} \wedge \cdots \wedge dx^{i_k} = \sum_I{}' \omega_I dx^I$$

In terms of (this lemma), we get:

$$dx^{i_1} \wedge \cdots \wedge dx^{i_n} \left(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n} \right) = \delta^I_J$$

Example 12.6.4: k-form

1. A 0-form is just a continuous real valued function $f : M \to \mathbb{R}$.
2. A 1-form is a co-vector field as we've seen before; for example if $M = \mathbb{R}$ with coordinate dt then $dt : M \to T^*M$ is the 1-form where $dt(p) = dt_p \in T_p^*(M)$. As $T_p(M)$ is spanned by $\frac{\partial}{\partial t}$, have

   $$dt_p : T_p(M) \to \mathbb{R} dt_p(v) = dt_p \left(\frac{\partial}{\partial t} \right) = 1$$

   Furthermore, there is a $C^\infty(\mathbb{R})$-action on $\Lambda^1 T^*(\mathbb{R})$ where $f \cdot \omega$ is defined to return a form ω' which is point-wise to evaluate to

   $$\omega'_p(a) = f(p)\omega_p(a)$$

   And as $\Lambda^1 T^*(\mathbb{R})$ is a 1-dimensional bundle, all 1-forms are of the form $f\, dt$.

   To define more complicated differential forms (i.e. on manifolds that require more than one chart), we shall need to show how to insure forms defined on the open patches are compatible, see example 12.6.10.

   These functionals on the tangent space give you a notion of "size" at any point. Thus, given any vector in T_pM, a linear functional will tell you the "volume metric" at the moment for that vector.
3. If $M = \mathbb{R}^3$, here are 2 examples of smooth 2-forms:

   $$\begin{aligned} \omega &= (\cos xy) dy \wedge dz \\ \eta &= dx \wedge dy + dx \wedge dz + dy \wedge dz \end{aligned}$$

   These two functions both give different "2d" volume metrics on a 3 dimensional subspace. For any point, we have $\omega, \eta \in \Lambda^2 T_p^* M$ are bilinear alternative maps; in particular given a choice of coordinates (say x, y, z) we have a basis:

   $$\left.\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}\right|_p \qquad \left.\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial z}\right|_p \qquad \left.\frac{\partial}{\partial y} \wedge \frac{\partial}{\partial z}\right|_p$$

For a concrete example, take $p = (\pi, 1, 1)$ and ω as above so that we have a map:

$$\omega_p : T_pM \times T_pM \to \mathbb{R}$$

Say we plug in $\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$, then:

$$\begin{aligned}
\omega_{(\pi,1,1)}\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) &= (\cos(xy)dy \wedge dz)_{(\pi,1,1)}\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \\
&= (\cos(1 \cdot \pi)dy \wedge dz)_{(\pi,1,1)}\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \\
&= (-1\, dy \wedge dz)_{(\pi,1,1)}\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \\
&= -1\left(dy\left(\frac{\partial}{\partial y}\right) \cdot dz\left(\frac{\partial}{\partial z}\right) - dy\left(\frac{\partial}{\partial z}\right) \cdot dz\left(\frac{\partial}{\partial y}\right)\right) \\
&= -1(1 \cdot 1 - 0 \cdot 0) \\
&= -1
\end{aligned}$$

4. Every 3-form on $\mathbb{R}^3$ is a real-valued continuous functions times $dx \wedge dy \wedge dz$. This is the most "elementary" example. Notice how for any point $p \in \mathbb{R}^3$ the vector space is spanned by $dx \wedge dy \wedge dz|_p$; more generally $\Lambda^3 T^*M$ is a $C^\infty(\mathbb{R}^3)$-module where the action of multiplying by a smooth function f is defined point-wise as $f \cdot dx \wedge dy \wedge dz|_p = f(p)dx \wedge dy \wedge dz|_p$.

   Given a point $p \in \mathbb{R}^3$, we get all functionals are of the form $(f(p)dx \wedge dy \wedge dz)_p : (T_pM)^3 \to \mathbb{R}$ for $f \in C^\infty(\mathbb{R}^3)$. Notice that if we plug in the basis, the determinant is simply the diagonal. By letting $f = 1$ be the constant one function, we get:

$$(dx \wedge dy \wedge dz)_p\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) = dx\left(\frac{\partial}{\partial x}\right) dy\left(\frac{\partial}{\partial y}\right) dz\left(\frac{\partial}{\partial z}\right) = 1 \cdot 1 \cdot 1 = 1$$

   telling us that a 1 by 1 by 1 cube should have volume 1 in this metric.

Like with co-vector fields, the pull-back will play a central role in being able to integrate on manifolds independent of coordinate chart:

Definition 12.6.5: Differential Form Pullback

Let $f : M \to N$ be a smooth map and ω a k-form on N. Then $F^*\omega$ is a differential form on M defined by:

$$(f^*\omega)_p(v_1, ..., v_k) = \omega_{F(p)}(T_pf(v_1), ..., T_pf(v_k))$$

Proposition 12.6.6: Properties Of k-Form Pullback

Let $f : M \to N$ be a smooth map. Then:

1. $f^* : \Omega^k(N) \to \Omega^k(M)$ is a linear map
2. $f^*(\omega \wedge \eta) = (f^*\omega) \wedge (f^*\eta)$
3. In any smooth chart:

$$f^*\left(\sum_I{}' \omega_I dy^{i_1} \wedge \cdots \wedge dy^{i_k}\right) = \sum_I{}' (\omega_I \circ f)\left(d(y^{i_1} \circ f) \wedge \cdots \wedge d(y^{i_k} \circ f)\right)$$

***Proof*:**
left as exercise

Example 12.6.7: Pullback

1. Let $f : \mathbb{R}^2 \to \mathbb{R}^3$, $f(u,v) = (u, v, u^2 - v^2)$ and let ω be the 2-form, $\omega = ydx \wedge dz + xdy \wedge dz$ on $\mathbb{R}^3$. Then the pullback $f^*\omega$ is:

$$\begin{aligned} f^*(ydx \wedge dz + xdy \wedge dz) &= vdu \wedge d(u^2 - v^2) = udv \wedge d(u^2 - v^2) \\ &= vu \wedge (2udu - 2vdv) = udv \wedge (2udu - 2vdv) \\ &= -2v^2 du \wedge dv + 2u^2 dv \wedge du \end{aligned}$$

   which simplies to:

$$f^*\omega = -2(u^2 + v^2) du \wedge dv$$

2. Let $\omega = dx \wedge dy$ on $\mathbb{R}^2$. We will show how change of coordinates is naturally captured in pullbacks: thinking of $x = r\cos(\theta)$ and $y = r\sin(\theta)$ as the change of coordinates from cartiain to polar coordinates, we get:

$$\begin{aligned} dx \wedge dy &= d(r\cos\theta) \wedge (r\sin\theta) \\ &= (\cos(\theta)dr - r\sin\theta(d\theta) \wedge (\sin(\theta)dr + r\cos(\theta)d\theta) \\ &= rdr \wedge d\theta \end{aligned}$$

   Notice that this is exactly what we would put in the change of variable substituion!

The following result generlizes the 2nd example in the above, in particular the change of variables requires that the dimensions of the domain and codomain match:

Proposition 12.6.8: Pullback Formula for Top-degree form

Let $f : M \to N$ be a smooth map between smooth manifolds that are each n-dimensional. If (x^i) and (y^i) are smooth coordinates on open subsets $U \subseteq M$ and $V \subseteq N$ respectively, and $u : V \to \mathbb{R}$ is a real continuous valued function on V, then the following holds on $U \cap f^{-1}(V)$:

$$f^*(u\,dy^1 \wedge \cdots \wedge dy^n) = (u \circ f)(\det Df)(dx^1 \wedge \cdots \wedge dx^n)$$

***Proof*:**
Since the fiber of each M in $\Lambda^n T^*M$ is spanned by $dx^1 \wedge \cdots \wedge dx^n$, it suffices to show that both sides of the equatoin in the proposition give the same result when evaluated on $(\partial/\partial x^1, ..., \partial/\partial x^n)$. By proposition 12.6.6:

$$f^*(u\,dy^1 \wedge \cdots \wedge dy^n) = (u \circ f)(df^1 \wedge \cdots \wedge df^n)$$

and using properties of the wedge product, we have that:

$$df^1 \wedge \cdots \wedge df^n \left(\frac{\partial}{\partial x^1}, ..., \frac{\partial}{\partial x^n}\right) = \det\left(df^j\left(\frac{\partial}{\partial x^i}\right)\right) = \det\left(\frac{\partial f^j}{\partial x^i}\right)$$

Thus, the left hand side gives $(u \circ f)\det(Df)$ when applied to $(\partial/\partial x^i, ..., \partial/\partial x^n)$. On the right hand side, we get the same thing since:

$$dx^1 \wedge \cdots \wedge dx^n \left(\frac{\partial}{\partial x^1}, ..., \frac{\partial}{\partial x^n}\right) = 1$$

completing the proof

Corollary 12.6.9: Change Of Variables

Let M be a smooth manifold and $(U, \mathbf{x})$, $(V, \mathbf{y})$ be overlapping patches. Then the following identity holds on $U \cap V$):

$$dy^1 \wedge \cdots \wedge dy^n = \det\left(\frac{\partial y^j}{\partial x^i}\right) dx^1 \wedge \cdots \wedge dx^n$$

***Proof*:**
Let $f = \text{id}$ in proposition 12.6.8 on $U \cap V$

(a word about interior multiplication)

Example 12.6.10: Differential Form on Circle
Let $M = S^1$ equipped with the stereographic projection:

1. (U_N, φ_N), where $U_N = S^1 \setminus \{N\}$ and $\varphi_N : U_N \to \mathbb{R}$ is the stereographic projection from the north pole
$$\varphi_N(x, t) = \frac{x}{1 - t}$$

2. (U_S, φ_S), where $U_S = S^1 \setminus \{S\}$ and $\varphi_S : U_S \to \mathbb{R}$ is the stereographic projection from the south pole
$$\varphi_S(x,t) = \frac{x}{1+t}$$

The overlap between these charts $U_N \cap U_S = S^1 \setminus \{N, S\}$ consists of the circle minus the north and south poles, and the transition map $\varphi_S \circ \varphi_N^{-1} : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ is given by:

$$\varphi_S \circ \varphi_N^{-1}(u) = \frac{1}{u}$$

Hence, if u, v are the coordinates given by there charts, we have $u = \frac{1}{v}$.

A differential 1-form ω on S^1 is a smooth assignment of a linear functional $\omega_p : T_pS^1 \to \mathbb{R}$ to each point $p \in S^1$. In local coordinates, a 1-form can be expressed as:

$$\omega|_{U_N} = f_N(u)du$$
$$\omega|_{U_S} = f_S(v)dv$$

where $u = \varphi_N(p)$, $v = \varphi_S(p)$, and f_N, f_S are smooth functions on the respective domains. Importantly, for these local expressions to define a global 1-form, they must be compatible on the overlap. Given that $v = \frac{1}{u}$, we have by corollary 12.6.9 that $dv = -\frac{1}{u^2}du$. Thus, for compatibility we must have

$$f_S(v)dv = f_S\left(\frac{1}{u}\right) \cdot \left(-\frac{1}{u^2}du\right) := f_N(u)du$$

In particular, we must have
$$f_S\left(\frac{1}{u}\right) \cdot \left(-\frac{1}{u^2}\right) = f_N(u)$$

or equivalently:

$$f_S(v) = -v^2 \cdot f_N\left(\frac{1}{v}\right)$$

Let's construct some specific 1-forms:

1. The most basic 1-form is defined by $f_N(u) = 1$ and $f_S(v) = -v^2$:

$$\omega|_{U_N} = du$$
$$\omega|_{U_S} = -v^2 dv$$

   To verify compatibility: $-v^2dv = -\left(\frac{1}{u}\right)^2 \cdot \left(-\frac{1}{u^2}du\right) = du$

2. Another 1-form can be defined using $f_N(u) = \frac{1}{1+u^2}$ and $f_S(v) = \frac{v^2}{1+v^2}$:

$$\omega|_{U_N} = \frac{1}{1+u^2}du$$
$$\omega|_{U_S} = \frac{v^2}{1+v^2}dv$$

This 1-form has a special property: it's precisely $d\theta$, where θ is the angular coordinate on S^1. We can see this by expressing $u = \tan(\theta/2)$ for $\theta \in (-\pi, \pi)$ in the U_N chart, which gives $du = \frac{1}{2}\sec^2(\theta/2)d\theta = \frac{1+u^2}{2}d\theta$, so $\frac{1}{1+u^2}du = \frac{1}{2}d\theta$.

This 1-form $d\theta$ is closed but not exact. It's closed because $d(d\theta) = 0$. To see it's not exact, we can integrate it around the circle:

$$\oint_{S^1} d\theta = 2\pi \neq 0$$

If $d\theta$ were exact, i.e., $d\theta = df$ for some global function f, then this integral would equal zero by Stokes' theorem. Since it doesn't, $d\theta$ cannot be exact.

3. We can construct exact 1-forms as differentials of globally defined functions. For example, let f be defined by:

$$f|_{U_N}(p) = \frac{u}{1+u^2}$$
$$f|_{U_S}(p) = \frac{v}{1+v^2}$$

This is a well-defined global function because in the overlap:

$$\frac{v}{1+v^2} = \frac{\frac{1}{u}}{1+\frac{1}{u^2}} = \frac{1}{u+\frac{1}{u}} = \frac{u}{u^2+1}$$

The 1-form df is:

$$df|_{U_N} = \frac{1-u^2}{(1+u^2)^2}du$$
$$df|_{U_S} = \frac{1-v^2}{(1+v^2)^2}dv$$

The non-exactness of $d\theta$ reflects the fact that the first de Rham cohomology group of S^1 is non-trivial: $H^1_{dR}(S^1) \cong \mathbb{R}$, with $[d\theta]$ as its generator. This corresponds to the topological fact that S^1 has a non-trivial fundamental group $\pi_1(S^1) \cong \mathbb{Z}$. This shall be explored in depth in 14.

12.7 Exterior Derivative

(John Lee p.362)

We've seen in section ref:HERE not all 1-forms arise as the differential of a function, $df = \omega$, and so we surmised properties necessary for this to happen, in particular we found that for any coordinate chart:

$$\frac{\partial w_j}{\partial x^i} - \frac{\partial w_i}{\partial x^j} = 0$$

and called a form that satisfies this property a *closed form* (which we showed was coordinate independent in proposition ref:HERE). How would we then ask of such a property for k-forms? The key to the generization comes down to the antisymmetric properties of indices i and j in a k-form, so "localizing" at any two indices, we may imagine them as ij-components of 2-form. Defining a 2-form $d\omega$ "locally" in each smooth chart, we would get:

$$d\omega = \sum_{i<j} \left(\frac{\partial w_j}{\partial x^i} - \frac{\partial \omega_i}{\partial x^j} \right) dx^i \wedge dx^j$$

and with this, we would say that ω is closed if $d\omega = 0$. TBD

Definition 12.7.1: Exterior Derivative on $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$ be an open subset. Then if $\omega \in \Omega^k(U)$ is represented as:

$$\omega = \sum_J{}' \omega_J dx^J$$

Then define the *exterior derivative* $d : \Omega^k(U) \to \Omega^{k+1}(U)$ to be:

$$d\omega = d\left(\sum_J{}' \omega_J dx^J \right) = \sum_J{}' d\omega_J \wedge dx^J$$

Essentially, we take the differential of the function in front, and then wedge the result. Breaking down the definition, we get:

$$d\left(\sum_J{}' \omega_J dx^{j_1} \wedge \cdots \wedge dx^{j_k} \right) = \sum_J{}' \sum_i \frac{\partial \omega_J}{\partial x^i} dx^1 \wedge dx^{j_1} \cdots dx^{j_k}$$

If ω is a 1-form, then:

$$\begin{aligned} d(\omega_j dx^j) &= \sum_{i,j} \frac{\partial \omega^j}{\partial x^i} dx^i \wedge dx^j \\ &= \sum_{i<j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j + \sum_{i>j} \frac{\partial \omega_j}{\partial x^i} dx^i \wedge dx^j \\ &= \sum_{i<j} \left(\frac{\partial w_j}{\partial x^i} - \frac{\partial w_i}{\partial x^j} \right) dx^i \wedge dx^j \end{aligned}$$

and if ω is a 0-form, so that $\omega = f$ for some smooth real valued function, then:

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i$$

showing that the exterior derivative is related to the derivative. However, a key property that is critically different from the derivative is that $d(df) = 0$! We thus really shouldn't think of d as generalizing taking the derivative; the fact that it involves the wedge product mean it qualitatively should be interpreted differently. First, some key property of the exterior derivative ought to be proven:

Proposition 12.7.2: Properties Of Exterior Derivative On $\mathbb{R}^n$

Let $M = \mathbb{R}^n$, $\Omega^k(\mathbb{R})$ be the differential k-forms over $\mathbb{R}^n$, and d the exterior derivative. Then:

1. d is linear over $\mathbb{R}$
2. If on $U \subseteq \mathbb{R}^n$, ω is a smooth k-form and η is a smooth ℓ-form, then:
$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$
3. $d \circ d \equiv 0$
4. d commutes with pulbacks: if $U \subseteq R^n$, $V \subseteq \mathbb{R}^m$, $f : U \to V$ a smooth map, and $\omega \in \Omega^k(V)$, then:
$$f^*(d\omega) = d(f^*\omega)$$

***Proof*:**

1. by definition of the wedge product, d is $\mathbb{R}$-linear
2. It suffices to show this on "simple tensors", or "simple k-forms", so let $\omega = udx^I \in \Omega^k(M)$ and $\eta = vdx^J \in \Omega^l(M)$. The first thing we'll show is that $d(udx^I) = du \wedge dx^I$ even for an *arbitrary* multi-index set I. If I has repeated index, then it is clear that $d(udx^I) = 0 = du \wedge dx^I$. If not, let σ be a permutation sending I to an increasing multi-index set J. Then:
$$d(udx^I) = (\operatorname{sgn}\sigma) d(udx^J) = (\operatorname{sgn}\sigma) du \wedge dx^J = du \wedge dx^I$$
Using this, we may compute:
$$\begin{aligned} d(\omega \wedge \eta) &= d((udx^I) \wedge (vdx^J)) \\ &= d(uvdx^I \wedge dx^J) \\ &= (vdu \wedge udv) \wedge dx^I \wedge dx^J \\ &= (du \wedge dx^I) \wedge (vx^J) + (-1)^k (udx^I) \wedge (dv \wedge dx^J) \\ &= d\omega \wedge \eta + (-1)^k \omega \wedge d\eta \end{aligned}$$
3. Do it for the 0-form case first, then go through the computation's, it will work out
4. It is sufficeint to show it for simple k-forms, $\omega = udx^{i_1} \wedge \cdots \wedge dx^{i_k}$. Computing the left hand side:
$$\begin{aligned} f^*(d(udx^{i_1} \wedge \cdots \wedge dx^{i_k})) &= f^*(du \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \\ &= d(u \circ f) \wedge d(x^1 \circ f) \wedge \cdots \wedge d(x^{i_i} \wedge f) \end{aligned}$$
and right hand side:
$$\begin{aligned} d(f^*(udx^{i_1} \wedge \cdots \wedge dx^{i_k})) &= d((u \circ f) d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f)) \\ &= d(u \circ f) \wedge d(x^1 \circ f) \wedge \cdots \wedge d(x^{i_i} \wedge f) \end{aligned}$$
completing the proof

From this, let us interpret $d \circ d = 0$. The key idea is that taking the d of a form should tell you the oriented variation of the form, and (key to distinguish it from the derivative operator) whether this variation is "consistent". For example, we know that any closed loop on the graph of f must give return to the same spot, so

$$\int_\gamma f' = \int_\gamma f'(x)dx = \int_\gamma df = 0$$

If we can define a different form ω that *doesn't* have this property, it means that all the little variations are not "consistent". We shall see that differential k-forms, like 1-forms, can be integrated to quantify volumes of manifolds. If the variation is "consistent" like a function, Then we would expect that the integral of a closed k-dimensional manifold is zero if it is the boundary of a $k+1$-dimensional manifold, which is equivalent to $d\omega = 0$. If $d\omega \neq 0$ (or it is not zero on some open subset), that means that $\omega \neq df$ (i.e. the form is not itself representing some changing. Now, we think of $d(df) = 0$ as meaning that df is "consistent" with its change, that integrating along any closed path of df should give 0, or alternatively that it should be path-independent.

We will now show that we can show there exists a unique map $d : \Omega^k(M) \to \Omega^{k+1}(M)$ where we now replaced $\mathbb{R}^n$ with M. The part (4) of this question can be in the special case of f being a diffeomorphism: Then this says that the exterior derivative is "invariant" under pullback!

Proposition 12.7.3: Exterior Derivative On Manifold

Let M be a smooth manifold (with or without boundary). Then there exists a unique operator $d : \Omega^k(M) \to \Omega^{k+1}(M)$ for all k such that

1. d is linear over $\mathbb{R}$
2. If $\omega \in \Omega^k(M)$ an $\eta \in \Omega^l(M)$, then:
$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$$
3. $d \circ d \equiv 0$
4. For $f \in \Omega^0(M) = C^\infty(M)$, df is the differential of f, given by $df(X) = Xf$.

such a map for any k is called the *exterior derivative.*

***Proof*:**
p.365

(word on anti-derivation)

Proposition 12.7.4: Commuting With Pullback

Let $f : M \to N$ be a smooth map. Then for each k, the pullback map $f^* : \Omega^k(N) \to \Omega^k(M)$ commutes with d, that is for all $\omega \in \Omega^k(N)$

$$f^*(d\omega) = df^*(\omega)$$

***Proof*:**
p.366

12.7.1 Exterior Derivative and Vector Calculus in $\mathbb{R}^3$

12.8 Invariant formula for the External Derivative

(Requires Lie Brakets)

12.9 Orienting Manifolds Using Forms

chap 15 John Lee

13

Integration on Manifolds

We have now constructed our "volume metric" that will give information at each point on a manifold. We can now show how this leads to integrating manifolds!

13.1 Integration of Differential Forms

Just like for line integarls, we will first treat the case of $\mathbb{R}^n$. We will say that $D \subseteq \mathbb{R}^n$ is a *domain of integration* if it's a bounded set with a measure zero boundary.

Definition 13.1.1: Integration Of Differential Form

Let $D \subseteq \mathbb{R}^n$ be a domain of integration an ω a (continuous) n-form on $\overline{D}$, which in terms of coordinates can be written as $\omega = f dx^1 \wedge \cdots \wedge dx^n$ for some (continuous) function $f : \overline{D} \to \mathbb{R}$. Then we define the *integral of* ω *over* D to be

$$\int_D \omega = \int_D f dV$$

which in terms of coordinates can be written as:

$$\int_D f dx^1 \wedge \cdots \wedge dx^n = \int_D f dx^1 \cdots dx^n$$

Essentially, we can think of integrating an n-form in $\mathbb{R}^n$ by erasing the wedges! A slight generalization of the above is if $U \subseteq \mathbb{R}^n$ (or $\mathbb{H}^n$) is an open subset and ω is compactly supported n-form on U. Then we define:

$$\int_U \omega = \int_D \omega$$

where $D \subseteq U$ is any domain of integration containing $\operatorname{supp}\omega$ (ex a rectangle), and ω is extended to 0 on the compliment of it's support.

Naturally, we must check that this definition is invariant of the domain of integration:

Proposition 13.1.2: Domain Independent

Let D and E be open domains of integration in $\mathbb{R}^n$ (or $\mathbb{H}^n$). Let $G : \overline{D} \to \overline{E}$ be a smooth map that restricts to an orientation-preserving or orientation-reversing diffeomorphism. Then if ω is an n-form on $\overline{E}$, then:

$$\int_D G^*\omega = \begin{cases} \int_E \omega & G \text{ is orientation preserving} \\ -\int_E \omega & G \text{ is orientation reversing} \end{cases}$$

***Proof*:**
Let $(y^1, y^2, ..., y^n)$ be the standard coordinates on E and $(x^1, x^2, ..., x^n)$ the standard coordinates on D. Let's first let G be orientation-preserving. Putting ω into coordinate form $\omega f dx^1 \wedge \cdots \wedge dx^n$, we can use the change of variables formula along with ref:HERE for pullback of n-forms to get:

$$\begin{aligned} \int_E \omega &= \int_E f dV \\ &= \int_D (f \circ G)|\det DG| dV \end{aligned}$$

$$\int_D (f \circ G)(\det DG) dV$$

$$\int_D (f \circ G)(\det DG) dx^1 \wedge \cdots \wedge dx^n$$

$$\int_D G^*\omega$$

If G is orientation-reversing, we do the exact same computation, except a negative sign will be introduced.

We next want to extend this theorem to compactly supported n-forms defiend on open sets. Since arbitrary open subsets or arbitrary compact subsets are not guaranteed to be domains of integration, we need the following lemma:

Lemma 13.1.3: Finding Domain Of Integration

Let $U \subseteq \mathbb{R}^n$ (or $\mathbb{H}^n$) be an open subset and let $K \subseteq U$ be a compact subset. Then there exists an open domain of integration D such that

$$K \subseteq D \subseteq U \qquad \overline{D} \subseteq U$$

***Proof*:**
p. 403

Proposition 13.1.4: Domain Independence Extended

Let $U, V \subseteq \mathbb{R}^n$ (or $\mathbb{H}^n$) be open subsets, $G : U \to V$ an orientation-preserving or orientation-reversing diffeomorphism. If ω is a compactly supported n-form on V, then:

$$\int_V \omega = \pm \int_U G^* \omega$$

where it is positive if G is orientation-perserving, and negative otherwise.

***Proof*:**
p. 404

13.1.1 Integration on Manifolds

Now that we know how to integarte a form on $\mathbb{R}^n$, we look at integrating a form in one chart:

Definition 13.1.5: Integration On Manifolds on One Chart

Let M be an oriented smooth n-manifold, ω a smooth n-form on M. Suppose that ω is compactly supported in the domain of one smooth chart (U, φ) that is either positively or negatively oriented. Then the *integral of* ω *over* M is:

$$\int_M \omega = \pm \int_{\varphi(U)} (\varphi^{-1})^* \omega$$

where the sign depends on the orietnation of φ

Proposition 13.1.6: Chart Independent

Let ω be a n-form that is defined as in definition 13.1.5. Then $\int_M \omega$ does not depend on the choice of smooth chart whose domain contains $\operatorname{supp} \omega$

***Proof*:**
p. 405

Using this, we can now extend the definition of integration using partitions of unity:

Definition 13.1.7: Integartion On Manifolds

Let M be an oriented smooth n-manifold, an ω a compactly suppored smooth n-form on M. Let $\{U_i\}$ be a finite open cover of $\operatorname{supp} \omega$ by domains of positively or negatively oriented smooth charts. Let $\{\psi_i\}$ be a subordinate smooth partition of unity. Then the *integral of* ω *over* M is:

$$\int_M \omega = \sum_i \int_M \psi_i(\omega)$$

The reason we allow for negatively oriented charts is that it may not be possible to find positively oriented boundary charts on a 1-manifold with boundary, as noted in the proof of proposition ref:HERE. Since for each i, the n-form $\psi_i(\omega)$ is compactly supported in U_i, each of these terms well-defined by our definition for one chart. The final key step to do is show the independence of choice of open cover and parition of unity:

Proposition 13.1.8: Integration On Manifolds Well-Defined

The definition $\int_M \omega$ given does not depend on the choice of open cover or partition of unity

***Proof*:**
p. 405

Thus, we now have a way to (at least theoretically) have a notion of integration on a manifold! For the special case of a of a compactly supported 0-form (i.e, a real-valued function) f over an oriented 0-manifold M, we would define:

$$\int_M f := \sum_{p \in M} \pm f(p)$$

Since f is compactly supported, there are only finitely many nonzero terms, and hence the sum is well-defined without any requirement for convergene conditions.

If $S \subseteq M$ is an oriented immerse k-submanifold (with or without boundary), and ω is a k-form on M, whose restriction to S is compactly supported, then we can interpret $\int_S \omega$ as $\int_S \iota_S^* \omega$ where $\iota_S^* : S \hookrightarrow M$ is the natural inclusion map. If M is also compact n manifold with boundary and ω is an $(n-1)$-form on M, then we can interpret $\int_{\partial M} \omega$ unambiguously as $\int_{\partial M} \iota_{\partial M}^* \omega$, with ∂M having the induced (Stokes) orientation.

It should be noted that we can extend this definition to non-compactly suppored forms, and this can be useful in many cases, hower we would need to pay attention to convergence issues (i.e. define an analogous improper integral), and so such a notion will be left for later

Proposition 13.1.9: Properties Of Integrating Forms

Suppose that M and N are non-empty oriented smooth n-manifolds (with or wihtout boundary), and ω, η are compactly supported n-forms on M.

1. If $a, b \in \mathbb{R}$, then:
$$\int_M a\omega = b\eta = a\int_M \omega + b\int_M \eta$$
2. If $-M$ denotes the opposite-oriented manifold, then:
$$\int_{-M} \omega = -\int_M \omega$$
3. If ω is a positively-oriented form, then
$$\int_M \omega > 0$$
4. (Diffeomorphism Invariance) If $f : M \to N$ is an orietnation-preserving or orietnation-reversing diffeomorphism, then:
$$\int_M \omega = \begin{cases} \int_N f^*\omega & f \text{ is orientation-preserving} \\ -\int_N f^*\omega & f \text{ is orientation-reversing} \end{cases}$$

***Proof*:**
p. 408 John Lee

Though this is nice theoretically, this would be a huge hassle to actually integrate (imagine trying to find a partition of unity which can be nicely integarble). It is much better to chop-op the manifold and integarte in each peace. The following gives a way of doing this through parameterization using a pullback:

Proposition 13.1.10: Integration Over Parameterization

Let M be an oriented smooth n-manifold (with or without boudary), and let ω be a compactly supported n-form on M. Suppose that $D_1, D_2, ..., D_k$ are open domains of integration in $\mathbb{R}^n$, and for $i = 1, ..., k$, we are given smooth maps $f_i : \overline{D_i} \to M$ satisfying:

1. f_i restricts to an orientation-preserving diffeomrophism from D_i onto an open subsets $W_i \subseteq M$
2. $W_i \cap W_j = \emptyset$ when $i \neq j$
3. $\operatorname{supp} \omega \subseteq \overline{W_1} \cup \cdots \cup \overline{W_k}$

Then:

$$\int_M \omega = \sum_{i=1}^{k} \int_{D_i} f_i^* \omega$$

***Proof*:**
As before, it is enough to assume that ω is compactly supported in the domain of one oriented smooth chart (U, φ). Restricting to sufficeintly nice maps, we may assume that U is pre-compact (closure is compact), $Y = \varphi(U)$ is a domain of integration in $\mathbb{R}^n$ (or $\mathbb{H}^n$), and φ extends to a diffeomorphism from $\overline{U}$ to $\overline{Y}$

(the rest is on p.409)

Example 13.1.11: Integrating On Manifold

We will use this result to compute the integral of a 2-form over S^2, oriented as the boundary of $\overline{D}^2$. Let ω be the two form:

$$\omega = x dy \wedge dz + y dz \wedge dx + z dx \wedge dy$$

Let $D = (0, \pi) \times (0, 2\pi)$ be an open rectangle, and let $f : \overline{D} \to S^2$ be the spherical coordinate parameterization $f(\psi, \theta) = (\sin\psi \cos\theta, \sin\psi \sin\theta, \cos\psi)$. In example ref:HERE, we showed that f is orientation preserving, and thus it satisfies the hypotheses of proposition 13.1.10. Next, note that:

$$\begin{aligned} f^* dx &= \cos\psi \cos\theta d\psi - \sin\psi \sin\theta d\theta \\ f^* dy &= \cos\psi \sin\theta d\psi - \sin\psi \cos\theta d\theta \\ f^* dz &= -\sin\psi d\psi \end{aligned}$$

Thus:

$$\begin{aligned}
\int_{S^2} \omega &= \int_D (-\sin^3\varphi\cos^2\theta d\theta\wedge d\varphi + \sin^3\varphi\sin^2\theta d\varphi\wedge d\theta) \\
&\quad + \cos^2\varphi\sin\varphi\cos^2\theta d\varphi\wedge d\theta - \cos^2\varphi\sin\varphi\sin^2\theta d\theta\wedge d\varphi) \\
&\overset{!}{=} \int_D \sin\varphi d\varphi\wedge d\theta \\
&= \int_0^{2\pi}\int_0^{\pi} \sin\varphi d\varphi d\theta \\
&= 4\pi
\end{aligned}$$

where the $\overset{!}{=}$ comes from repeated use of the $\sin^2+\cos^2=1$ identity.

(comment on how the hypothesis can be slightly relaxed and still work well, p. 410)

13.2 Stoke's Theorem

One of the big results we learn in calculus is that that:

$$\int_a^b f'(x)dx = f(b) - f(a)$$

considering f as a form, we can convert this into:

$$\int_{[a,b]} df = \int_{\partial[a,b]} f$$

This is one of the key powerful results of calculus: since df holds all the variation information within a space, then f can be re-interpreted as holding this "cumulative" information! Perhaps the most important first result in differential geometry is that this same idea can be massively generalized for arbitrary n-froms!

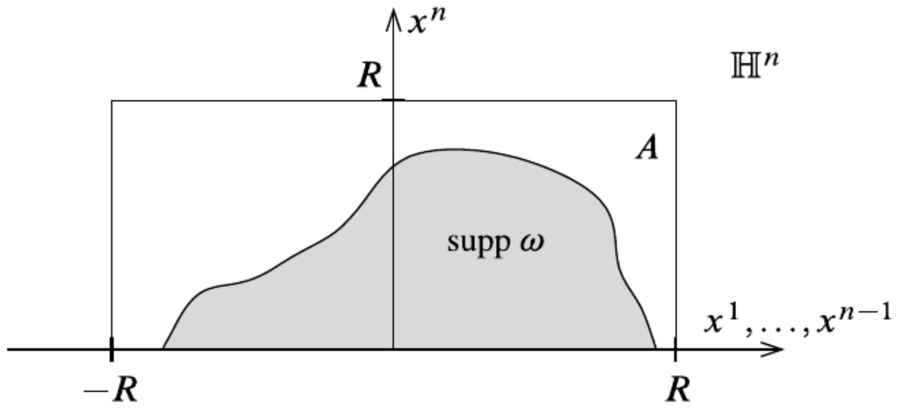
Theorem 13.2.1: Stokes Theorem

Let M be an oriented smooth n-manifold with boundary, and let ω be a compactly supported smooth $(n-1)$-form on M. Then:

$$\int_M d\omega = \int_{\partial M}\omega$$

where ∂M has the induced Stokes orientation

***Proof*:**
We start with a special case: suppose that $M = \mathbb{H}^n$. Since ω has compact support, there exists an $R > 0$ st $\operatorname{supp}\omega$ is contained in $A = [-R,R]^{n-1}\times[0,R]$:



Since $M = \mathbb{H}^n$, we only need one chart. Hence, write ω in standard coordinates:

$$\omega = \sum_{i=1}^{n} \omega_i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$$

where $\widehat{dx^i}$ is the ommited component. Thus, computing $d\omega$:

$$\begin{aligned} d\omega &= \sum_{i=1}^{n} d\omega_i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n \\ &= \sum_{i,j=1}^{n} \frac{\partial \omega_i}{\partial x^j} \wedge dx^j \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n \\ &= \sum_{i=1}^{n} (-1)^{i-1} \frac{\partial \omega_i}{\partial x^i} \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n \end{aligned}$$

Then, computing the integral:

$$\int_{\mathbb{H}^n} d\omega = \sum_{i=1}^{n} (-1)^{i-1} \int_A \frac{\partial w_i}{\partial x^i} dx^1 \wedge \cdots \wedge \quad = \sum_{i=1}^{n} (-1)^{i-1} \int_0^R \int_{-R}^R \int \cdots \int_{-R}^R \frac{\partial w_i}{\partial x^i}(x) dx^1 \cdots dx^n$$

We can now by Fubini-Tonelli's theorem, we can switch the order of the integrals, and so we will move the dx^i to the front

$$\begin{aligned} &= \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \int \cdots \int_{-R}^R \frac{\partial w_i}{\partial x^i}(x) dx^1 \cdots dx^n \\ &= \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \int \cdots \int_{-R}^R \frac{\partial w_i}{\partial x^i}(x) dx^i \cdots \widehat{dx^i} \cdots dx^n \\ &= \sum_{i=1}^{n-1} (-1)^{i-1} \int_0^R \int_{-R}^R \int \cdots \int_{-R}^R [w_i(x)]_{x^i=-R}^{x^i=R} dx^1 \cdots \widehat{dx^i} \cdots dx^n \\ &= 0 \end{aligned}$$

where the last line comes from the fact that we have chosen R so that $\omega = 0$ when $x^i = \pm R$. The only term which might not be zero is the one for which $i = n$. For that term, we have:

$$\begin{aligned}
&= (-1)^{i-1} \int_{-R}^{R} \int \cdots \int_{-R}^{R} \int_{0}^{R} \frac{\partial w_i}{\partial x^i}(x) dx^n \cdots \cdots dx^{n-1} \\
&= (-1)^{i-1} \int_{-R}^{R} \int \cdots \int_{-R}^{R} [w_i(x)]_{x^n=0}^{x^i=R} \, dx^1 \cdots \cdots dx^{n-1} \\
&= (-1)^{i-1} \int_{-R}^{R} \int \cdots \int_{-R}^{R} [w_i(x)]_{x^n=0}^{x^i=R} \, dx^1 \cdots \cdots dx^{n-1} \\
&= (-1)^{i-1} \int_{-R}^{R} \int \cdots \int_{-R}^{R} \omega_n(x^1, ..., x^{n-1}, 0) dx^1 \cdots \cdots dx^{n-1}
\end{aligned}$$

since $\omega_n = 0$ when $x^n = R$.

Let's now compare this to the other side:

$$\int_{\partial \mathbb{H}^n} \omega = \sum_i \int_{A \cap \partial \mathbb{H}^n} \omega_i(x^1, ..., x^{n-1}, 0) dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$$

since x^n vanishes on $\partial \mathbb{H}^n$, the pullback of dx^n to the boundary is identically 0 (John Lee give sthis as an exercsie 11.30). Thus, the only term that is non-zero is the one for which $i = n$, which becomes:

$$\int_{\partial \mathbb{H}^n} \omega = \int_{A \cap \partial \mathbb{H}^n} \omega_n(x^1, ..., x^{n-1}, 0) dx^1 \wedge \cdots \wedge dx^{n-1}$$

Since the coordinates $(x^1, x^2, ..., x^{n-1})$ are positively oriented for $\partial \mathbb{H}^n$ when n is even and negatively when n is odd, see that the left hand side is equal to the right hand side.

The next special case is for $M = \mathbb{R}^n$. In this case, we have that $\operatorname{supp} \omega \subseteq [-R, R]^n$. Doing the same computation, we see now that that the terms of $i = n$ both vanish, and so the left hand side is zero. Since there is no boundary, the right hand side is zero as well.

Now, let M be an arbitrary smooth manifold with boundary, but ω is an $(n-1)$-form that is compactly supported in the domain of a single positively or negatively oriented smooth chart (U, φ). Assuming that φ has a positively oriented boundary chart, the definition gives us:

$$\int_M d\omega = \int_{(\mathbb{H}^n)} (\varphi^{-1})^* d\omega = \int_{\mathbb{H}^n} d\left((\varphi^{-1})^* \omega\right)$$

But then, by the computation we've just done, this gives:

$$\int_{\partial \mathbb{H}^n} (\varphi^{-1})^* \omega$$

with $\partial \mathbb{H}^n$ having the induced orientation. Since $d\varphi$ take outwards pointing vectors on ∂M to outwards pointing vectors on $\mathbb{H}^n$ (by proposition 5.41 on p. 118 of John Lee), it follows that $\varphi|_{U \cap \partial M}$ is an orientation-preserving diffeomorphism onto $\varphi(U) \cap \partial \mathbb{H}^n$, and thus the above equation is equal to:

$$\int_{\partial M} \omega$$

For negatively oriented smooth boundary charts, the same argument applies where we would have negative signs show up on both sides. For interior charts, we will replace $\mathbb{H}^n$ with $\mathbb{R}^n$ and get the same result as before.

Finally, let ω be an arbitrary compactly supported smooth $(n-1)$-form. Choose some cover of $\operatorname{supp}\omega$ by finitely many domains of positively or negative oriented smooth charts $\{U_i\}_{i=1}^n$, and choose a subordinate smooth partition of unity $\{\psi_i\}$. Then we can apply the preceeding argument to $\psi_i\omega$ for each i to obtain:

$$\begin{aligned}\int_{\partial M}\omega &= \sum_i \int_{\partial M}\psi_i\omega \\ &= \sum_i \int_M d(\psi_i\omega) \\ &= \sum_i \int_M d\psi\wedge\omega + \psi_i d\omega \\ &= \int_M d\left(\sum_i \psi\right)\wedge\omega + (\sum_i \psi_i)d\omega \\ &= 0 + \int_M d\omega \\ &= \int_M d\omega\end{aligned}$$

since $\sum_i \psi_i \equiv 1$

Example 13.2.2: Special Case: Line Integrals

Let M be a smooth manifold and let $\gamma : [a,b] \to M$ be a smooth embedding so that $S = \gamma([a,b])$ is an embedding 1-submanifold with boundary in M. If we give S the orientation such that γ is orientation-preserving, then for any smooth function $f \in C^\infty(M)$, Stoke's Theorem tells us that:

$$\int_\gamma df = \int_{[a,b]} \gamma^* df = \int_S df = \int_{\partial S} f = f(\gamma(b)) - f(\gamma(a))$$

Thus, we get back the Fundamental theorem for line integarls! When we take $\gamma : [a,b] \to \mathbb{R}$, then Stok'es theorem is just our ordinary fundamental theorem of calculus.

Corollary 13.2.3: Integrals Of Exact Forms

Let M be a compact oriented smooth manifold without boundary. Then the integral of every exact form over M is zero:

$$\int_M d\omega = 0 \quad \text{if } \partial M = 0$$

***Proof*:**
Let M be a compact manifold where $\partial M = \emptyset$. Since η be an exact form so that $\eta = d\omega$ for some ω. Then:

$$\int_M \eta = \int_M d\omega = \int_{\partial M} \omega = \int_{\emptyset} \omega = 0$$

Corollary 13.2.4: Integarls Over Closed Form With Boundary

Let M be a smooth compact oriented manifold with boundary. If ω is a closed form on M, then the integral of ω over ∂M is zero:

$$\int_{\partial M} \omega = 0 \qquad \text{if } d\omega = 0 \text{ on } M$$

***Proof*:**
Let ω be a closed form so that $d\omega = 0$. Then:

$$\int_{\partial M} \omega = \int_M d\omega = \int_M 0 = 0$$

Notice that if ω is a closed form that is also exact (so $\omega = d\eta$), we would get:

$$\int_{\partial M} \omega = \int_{\partial M} d\eta = \int_M d(d\eta) = \int_M 0 = 0$$

Thus, if $\int_S \omega \neq 0$ for some manifold S, we can conclude these two powerful results:

Corollary 13.2.5: Characterizting Exact Forms

Let M be a smooth manifold (with or without boundary), $S \subseteq M$ is a compact smooth k-submanfiold *without boundary*, and ω any closed k-form on M. If $\int_S \omega \neq 0$, then both of the following are true:

1. ω is not exact on M
2. S is not the boundary of an oriented compact smoth submanifold with boundary in M

***Proof*:**
If ω was exact, then $\int_S \omega = 0$. Furthermore, if $S = \partial M$ for some manifold, then:

$$\int_S \omega = \int_{\partial M} \omega = \int_M d\omega = \int_M 0 = 0$$

Example 13.2.6: Non Exact Form
p. 415 with $\mathbb{R}^2 \setminus \{(0,0)\}$

We naturaly get back classical results from vector-calculus:

Theorem 13.2.7: Green's Theorem

Suppose that D is a compact regular domain on $\mathbb{R}^2$, and P, Q are smooth real valued functions on D. Then:

$$\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy = \int_{\partial D} Pdx + Qdy$$

***Proof*:**
Apply Stokes theorem to any 1-form $Pdx + Qdy$.

13.3 Divergence Theorem

The divergence theorem applies to an n-dimensional manifold with boundary in $\mathbb{R}^n$. You can imagine that on the boundary, there is an outward facing normal unit vector. Recall that if F is a vector-field in $\mathbb{R}^n$, then:

$$\text{div}(F) = \frac{\partial f}{\partial x^1} + \cdots + \frac{\partial f}{\partial x^n}$$

Theorem 13.3.1: Divergence Theorem

Let F be a smooth (or C^1) vector-field and $M \subseteq \mathbb{R}^n$. Then:

$$\int_M \text{div} F dV_{\mathbb{R}^n} = \int_{\partial M} \langle F, n \rangle dV_{\partial M}$$

where $dV_{\mathbb{R}^n}$ is a volume element of $\mathbb{R}^n$, i.e. $dx_1 \wedge \cdots \wedge dx_n$.

***Proof*:**
not covered??

The other volume form (i.e. $n-1$-form on ∂M) presented is a bit ambiguous, i.e. $dV_{\partial M}$. We define it so that:

$$dV(x)(v_1, ..., v_{n-1}) = 1$$

if $(v_1, v_2, ..., v_{n-1})$ is in ∂M and $[v_1, v_2, ..., v_{n-1}] = \partial \mu_x$. Thus, this differential form is always just the volume of an $n-1$ parrallelepiped spanned by $v_1, v_2, ..., v_{n-1}$. Then, since we have coordinates, we can compute it directly:

$$dV(x)(v_1, v_2, ..., v_{n-1}) = \det \begin{pmatrix} n(x) \\ v_1 \\ \vdots \\ v_{n-1} \end{pmatrix} = (-1)^{n-1} \det \begin{pmatrix} v_1 \\ \cdots \\ v_{n-1} \\ n(x) \end{pmatrix} = (-1)^n \langle (v_1, ..., v_{n-1}), n(x) \langle$$

(Now I'm not sure what's going on..) By Riesz's Representation Theorem, if $z = (v_1, ..., v_{n-1})$, then:

$$\det \begin{pmatrix} v_1 \\ \cdots \\ v_{n-1} \\ n(x) \end{pmatrix} = \langle w, z \rangle$$

actually, he just goes on the prove the theorem, which atm I have no interest in doing so.

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De Rham Cohomology

One of the most impactful discoveries of modern differential geometry is that holes (of various dimensions) on a smooth manifold can be detected differential forms, namely by computing the failure for closed forms to be exact. On a very abstract level, this mirrors a prevailing theme in mathematics where a geometric concept (ex. a whole, a line, curvature, etc.) is captures by a collection of some functions in one way or another. As shall be proven in theorem ref:HERE, differential forms through the use of De Rham Cohomology shall be able to detect whole.

More concretely, we can think of the failure of closed forms as being exact as the failure of the local information

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What Next?

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